

AI Pricing & Revenue Optimization Engine

Full Technical Report

*Dominick's Finer Foods Cereals scanner panel
Kilts Center for Marketing, University of Chicago Booth*

Report version	1.0
Report generated	2026-08-27
Dataset	Dominick's Cereals — 4,707,776 UPC × store × week rows
Selected model	M1 ridge log-log — test WAPE 0.4565
Pricing elasticity	pooled -2.029, two-way clustered 95% CI [-2.24, -1.82]
Headline audit finding	2.9% of recommendations come from an interior model optimum
Quality gates	538 tests passed · ruff clean · 0 unsupported claims
Companion files	...REPORT_SOURCES.md (evidence trail) · REPORT_QA.md (QA record)

Nothing in this report is a causal claim. Every counterfactual figure is a
MODEL-INTERNAL estimate: the same fitted price-response model both proposes
and scores the candidate prices.

Contents

Parts and numbered sections are listed below; the 690 third-level headings are omitted to keep this to a few pages. Entries are clickable, and the contents can be rebuilt at any time by right-clicking it and choosing Update Field.

How to read this report.....	7
PART I — EXECUTIVE AND BUSINESS CONTEXT	8
1. Executive summary	8
2. Business problem	13
3. Why pricing is a difficult data-science problem	15
PART II — DATA	18
4. Dataset selection.....	18
5. Raw data acquisition	19
6. Dominick's data model.....	20
7. Price and quantity semantics	21
8. Gross margin and AAC.....	22
9. Data quality audit	23
10. Zero-price investigation	26
11. Promotion coding.....	28
12. Canonical processed dataset.....	30
PART III — EXPLORATORY PRICING ANALYSIS.....	31
13. Overall retail performance	31
14. Product concentration	31
15. Historical price variation	32
16. Cross-store pricing variation	33
17. Cost / AAC stability	34
18. Demand, time and promotions	35
PART IV — FEATURE ENGINEERING AND LEAKAGE.....	38
19. Prediction target.....	38
20. Feature availability framework	38
21. Temporal features	40
22. Lag and rolling features	40
23. Leakage tests	43
PART V — PRICE ELASTICITY	45
24. Economic definition.....	45
25. Arc elasticity	45
26. Log-log regression	46
27. Naive elasticity result	47
28. Fixed-effects specifications	48

29. Robust and clustered inference	50
30. Product-level elasticities	52
31. Empirical-Bayes shrinkage.....	54
32. Elasticity stability over time	58
33. Causal limitations	59
PART VI — DEMAND FORECASTING	62
34. Modelling objective.....	62
35. Temporal train / validation / test split	62
36. Baselines.....	63
37. Ridge log-log model (M1 — selected)	64
38. Gradient-boosting model (M2)	66
39. Model comparison	68
40. Why Ridge won	69
41. Error analysis	70
PART VII — HYBRID PRICE RESPONSE	74
42. Why forecasting and price response were separated.....	74
43. Hybrid formulation	76
44. Native ML vs pooled vs shrunk.....	79
45. Out-of-time price-change validation	81
PART VIII — PRICE SIMULATION.....	84
46. Candidate price generation.....	84
47. Counterfactual feature construction.....	85
48. Cost treatment during simulation	86
PART IX — OPTIMIZATION.....	89
49. Revenue objective	89
50. Gross-profit objective.....	89
51. Why grid optimization was used	90
52. Analytical optimizer validation	90
53. Constraints	92
54. Constraint attribution.....	95
55. Guardrail ablation.....	99
56. Rule-only benchmark vs learned pricing.....	101
PART X — RISK AND DECISION POLICY	104
57. Recommendation risk layer.....	104
58. Decision states.....	106
59. High-risk gating.....	108
60. Reason codes.....	110

PART XI — BACKTESTING AND POLICY EVALUATION	113
61. Historical evaluation design	113
62. Demand forecast evaluation	113
63. Pricing policy evaluation.....	114
64. The circularity problem	115
65. What can actually be verified offline	117
PART XII — SYSTEM ARCHITECTURE	119
66. End-to-end architecture	119
67. Repository architecture.....	122
68. Training pipeline.....	124
69. Recommendation pipeline	125
PART XIII — FASTAPI.....	128
70. API architecture.....	128
71. Endpoints.....	130
72. Example API recommendation	132
PART XIV — STREAMLIT DASHBOARD.....	135
73. Dashboard design.....	135
74. Each dashboard page	136
75. Dashboard screenshots	140
PART XV — TESTING AND SOFTWARE QUALITY.....	146
76. Testing strategy	146
77. Important tests.....	146
78. Final test results	148
PART XVI — REPRODUCIBILITY AND MLOPS	152
79. Reproducible workflow	152
80. Configuration.....	153
81. Model versioning.....	155
82. Monitoring.....	156
83. Docker and CI	157
PART XVII — SCIENTIFIC AUDIT	159
84. Final scientific audit.....	159
85. What the model actually contributes.....	160
86. The weakest scientific component.....	162
87. The strongest scientific component.....	163
PART XVIII — LIMITATIONS	165
88. Data limitations	165
89. Modelling limitations	166

90. Econometric limitations	167
91. Optimization limitations.....	167
92. Business and deployment limitations	168
93. External validity.....	169
PART XIX — RESPONSIBLE PRICING.....	171
94. Ethics and pricing safety.....	171
95. Human approval.....	172
PART XX — PRODUCTION DEPLOYMENT	174
96. How this would become a real production system.....	174
97. Scaling.....	176
PART XXI — EXPERIMENTATION	178
98. Pricing A/B test.....	178
99. How real business uplift would be established	180
PART XXII — FUTURE WORK.....	182
100. Cross-price elasticity.....	182
101. Cannibalisation and multi-product optimization	183
102. Dynamic pricing, bandits and RL	184
103. Better causal identification	185
PART XXIII — BUSINESS CASE	187
104. Business interpretation	187
105. Example end-to-end pricing decision	188
106. Portfolio-level recommendation summary	190
PART XXIV — PROJECT EVOLUTION	192
107. How the project changed	192
PART XXV — INTERVIEW PREPARATION	197
108. The thirty most important questions	197
109. Five-minute project explanation	207
110. One-minute project explanation	209
111. Thirty-second version.....	209
PART XXVI — CLAIMS	210
112. Claims that are fully supported	210
113. Claims that require qualification	211
114. Claims that must not be made	212
PART XXVII — CV / GITHUB POSITIONING.....	214
115. Three CV bullets	214
116. GitHub repository description.....	215
117. Skills demonstrated	215

PART XXVIII — CONCLUSION	218
118. Final project assessment	218
119. Final lessons	219
PART XXIX — POST-FREEZE ENGINEERING.....	221
120. Phase O addendum (added Thursday, 2026-08-27).....	221
APPENDICES	221
Appendix A — Full dataset dictionary	221
Appendix B — Formula reference	224
Appendix C — Model hyperparameters (actual final values)	226
Appendix D — Configuration profiles.....	227
Appendix E — API schemas	228
Appendix F — Reason codes	230
Appendix G — Test inventory	231
Appendix H — Repository tree (actual, at report time)	233
Appendix I — Reproduction commands	234
Appendix J — Final consolidated metrics table.....	235
Appendix K — Final limitations register	244
Appendix L — Glossary.....	247

Report version	1.0
Report generated	2026-08-19
Repository state at report time	master, no commits yet (working tree only — see §81)
Pipeline artifacts dated	2026-08-18 (full make all run)
Verification re-run for this report	test suite, lint, <code>run_demo.py</code> , live FastAPI service, live Streamlit dashboard, all report figures — 2026-08-19
Environment	Windows 11 (10.0.26200), Python 3.13.0, pandas 2.2.3, numpy 2.0.2, scikit-learn 1.8.0, statsmodels 0.14.6
Post-freeze engineering addendum	Phase O (review/approval workflow, monitoring history + local alerting, batch scale harness) added Thursday, 2026-08-27 — no change to the scientific findings above; see §120 and <code>reports/22_POST_FREEZE_ENGINEERING.md</code>
Companion audit trail	<code>`reports/AI_PRICING_REVENUE_OPTIMIZATION_REPORT_SOURCES.md`</code>
Companion QA record	<code>`reports/REPORT_QA.md`</code>

How to read this report

Three rules are enforced throughout, and they are the difference between this document and a marketing deck.

1. **Every number comes from an artifact this repository produced.** Nothing is estimated by the author, rounded upward, or carried over from memory. The companion sources file maps each figure to the file, script and command that generated it.
2. **Four categories of number are never mixed.**

Category	Meaning	Example in this report
Observed historical	measured in the data as it happened	\$262,008,582 observed revenue over the panel
Predicted	model output at a price that <i>was</i> charged, scored against the realised outcome	test WAPE 0.4565
Model-internal counterfactual estimate	model output at a price that was never charged, scored by the same model that proposed it	+8.30% portfolio gross-profit uplift
Causal	the effect of <i>setting</i> a price	nothing in this project is in this category

3. **Where the repository's own prose disagrees with the current artifacts, the artifacts win**, and the discrepancy is recorded (§ "Documentation drift", and in full in the sources file).

Terminology is fixed for the whole document: *effective unit price*, *estimated AAC*, *decision-time unit cost*, *observed historical gross profit*, *predicted demand*, *candidate price*, *proposed candidate price*, *final recommended price*, *model-internal estimated counterfactual uplift*, *observational elasticity estimate*, **REVIEW_REQUIRED**.

PART I — EXECUTIVE AND BUSINESS CONTEXT

1. Executive summary

1.1 The business problem in one sentence

Given a product, a store, a week, the commercial context of that store-week and a feasible price range, **what unit price should a grocery retailer charge next week in order to maximise expected gross profit, without taking risks the evidence cannot support?**

1.2 Why this matters

Price is the only lever in retail that moves the top line and the bottom line at the same time, instantly, at zero marginal cost, and reversibly. To size the category - and only to size it - two arithmetic illustrations on the observed Cereals panel:

- a **1% relative** increase in the historical gross-profit total would correspond to roughly **\$0.4 million** over the eight years observed ($0.01 \times \$40,091,143 = \$400,911$);
- a **+1 percentage-point** increase in the realised gross-margin *rate*, applied to the same **\$262,008,582** of observed revenue, would correspond to roughly **\$2.6 million** ($0.01 \times \$262,008,582 = \$2,620,086$).

These two statements differ by a factor of 6.5, which is exactly why the units have to be stated. Both are arithmetic illustrations of the category's scale, **not** an estimated pricing uplift, **not** a realised gain, and **not** something this system has been shown to deliver: nothing in this report identifies a causal price effect (\$13, \$54, Appendix G). No promotional campaign, assortment change or supply-chain project has that combination of leverage and speed.

Price is also the fastest way to destroy value. Cereal is a heavily promoted, heavily substituted category: the controlled elasticity estimates in this project put the category around **-2.0**, meaning a naive 10% price rise would be expected to cost roughly 20% of unit volume. That is why the interesting engineering problem is not "predict demand" but "decide *when the evidence is strong enough to act*".

1.3 What was built

An end-to-end, auditable price-recommendation system:

```
official Kilts Center files
→ validated canonical panel (4,707,776 UPC × store × week rows)
→ pricing EDA + price-variation eligibility screen
→ observational elasticity analysis (naive → fixed effects → panel-robust inference)
→ demand models on a strict chronological split
→ native-ML price-response validation (does the model respond to price at all?)
→ separately estimated elasticity table (training weeks only, empirical-Bayes shrunk)
→ hybrid price response  $Q(p) = \hat{Q}(p_0) \cdot (p/p_0)^\epsilon$ 
→ vectorised counterfactual price simulation (unit cost held fixed)
→ constrained grid optimization with reason codes
→ heuristic risk layer + decision gate (RECOMMEND_CHANGE / KEEP_CURRENT / REVIEW_REQUIRED)
→ append-only recommendation audit log
→ FastAPI service (5 endpoints) + 9-page Streamlit dashboard
→ 519 tests, ruff clean, Dockerfile + CI workflow
→ a twelve-report scientific audit that measures how much of the answer the model actually supplies
```

1.4 Data

Source	Dominick's Finer Foods store-level scanner data, Cereals category, Kilts Center for Marketing, University of Chicago Booth
Files	upccer.csv (UPC metadata, 25,932 B) and wcer.zip →

	wcer.csv (movement, 458,400,475 B)
Raw movement rows	6,602,582
Canonical rows after documented exclusions	4,707,776 (71.30% retained)
Grain	one UPC × store × week observation, uniqueness asserted
Coverage	489 UPCs × 93 stores × 366 observed weeks , 1989-09-14 ... 1997-05-01
Observed historical revenue	\$262,008,582
Observed historical gross profit	\$40,091,143 (15.30% blended margin)
Structural + formula checks	15 / 15 pass

1.5 Modelling approach

- **Target:** move, weekly units sold. Revenue and gross profit are functions of the target and are therefore never inputs — money is derived *after* prediction, which is what makes the counterfactual coherent.
- **Split:** strictly chronological by Dominick's week index — train weeks **2–257**, validation **258–342**, test **343–399**. No shuffling anywhere.
- **Models compared:** four naive baselines, a ridge log-log regression and a HistGradientBoosting Poisson regressor.
- **Selected:** **M1 ridge log-log**, on a selection rule fixed in advance (lowest validation WAPE among *price-aware* models). Validation WAPE **0.4135**, test WAPE **0.4565** (MAE 8.503, RMSE 72.064, bias –3.883 units over 749,040 test rows).

1.6 Elasticity approach

The forecaster's *own* implied price response is far steeper than any controlled econometric estimate, so the two jobs were separated:

specification	elasticity
naive pooled log-log	–0.348
+ UPC fixed effects	–2.289
+ UPC × store fixed effects	–2.419
+ promotion, seasonality, trend	–1.909
pricing pooled, training weeks only (the number the engine uses)	–2.029 — two-way clustered (UPC, week) 95% CI [–2.236, –1.821]
pricing UPC × store FE, training weeks only	–2.222
shrunk per-UPC (median / p10 / p90)	–2.029 / –2.790 / –1.169
native ML implied (finite difference, 300 contexts)	–3.102

Per-product elasticities are estimated with panel-robust standard errors and shrunk toward the pooled estimate by empirical Bayes: $w_i = \tau^2 / (\tau^2 + se_i^2)$, with $\tau^2 = 0.766$ (REML) and a mean weight of **0.780** across the **239 of 372** products that survive the eligibility screen.

1.7 Optimization method

A **discrete grid** (\$0.05 candidate step, \$0.01 rounding) over the feasible interval produced by intersecting six guardrails: absolute bounds, cost floor, minimum gross margin, maximum change from the current price, an extrapolation guardrail keeping candidates inside the series' observed price support, and a materiality threshold. The optimizer is validated against closed-form optima for linear demand: profit $p^* = (a + bc)/(2b)$, revenue $p^* = a/(2b)$.

1.8 API, dashboard and audit

A FastAPI service (/health, /model/info, /predict-demand, /simulate-prices, /recommend-price) loads the pricing model once at startup and serves the most recent 8 weeks (112,763 decision contexts). A nine-page Streamlit dashboard renders only artifacts the pipeline produced. Every recommendation is appended to artifacts/recommendation_log.csv with its inputs, constraints, model version, risk level and reason codes.

1.9 Validation strategy

layer	what it establishes	verifiable against outcomes?
15 formula/structural checks	the canonical table obeys its own definitions	yes
12 leakage channels + poisoning tests	no future information reaches a decision	yes
chronological split, test scored once	forecasting skill at <i>observed</i> prices	yes
analytical optimizer fixtures	the optimizer finds the true optimum	yes (closed form)
decision-state invariants over 13,964 contexts	the policy layer cannot contradict itself	yes
out-of-time price-change episodes (154,899)	which price-response method best predicts what followed a <i>real</i> price move	partially — predictive, not causal
constraint attribution, guardrail and rule ablations	how much of the recommendation the model actually supplies	yes (decisions, not profits)
backtest policy economics	model-internal only — circular by construction	no

1.10 Key metrics

metric	value	category
Canonical rows	4,707,776	observed
Observed revenue / gross profit	\$262.0M / \$40.1M (15.30%)	observed
Selected model test WAPE	0.4565	predicted
Selected model validation WAPE	0.4135	predicted
Best naive baseline (rolling mean 4) test WAPE	0.7326	predicted
Pooled pricing elasticity	-2.029 [-2.236, -1.821]	observational estimate
Products with a usable own elasticity	239 / 372 (64.2% of decision contexts)	observed count
Out-of-time WAPE, non-promotion episodes: shrunk / ML / pooled / null	0.560 / 0.588 / 0.594 / 0.715	predicted
Batch recommendations (3,000 contexts, week 399)	58.9% actionable, 30.8% keep, 10.4% review	policy output
HIGH-risk contexts that auto-changed a price	0 of 1,039	policy output
Portfolio gross-profit uplift	+8.30%	model-internal counterfactual estimate
Share of final recommendations set by an interior model optimum	2.9%	audit result
A no-model rule matching the engine within one 5¢ step	82.5%	audit result
Tests / lint	519 passed, 0 failed / ruff clean	verified at Phase N, 2026-08-19

1.11 The most important findings

- This is a rule-bounded pricing system with a learned direction, not an ML pricing system.** Over all 13,964 decision contexts of week 399 under the default policy, only **2.9%** of final recommendations come from an

interior optimum of the estimated price response. **57.1%** sit on a guardrail corner, 25.3% are screened out before optimisation ever runs, 9.7% are risk-gated and 5.1% fail materiality. The constraint that binds first is the $\pm 10\%$ price-change cap, in **44.6%** of contexts.

5. **A rule that never looks at demand reproduces most of the engine's decisions.** "Always take the maximum allowed increase" lands within one 5-cent grid step of the engine's final price in **82.5%** of contexts and agrees on the decision state in **90.0%**.
6. **The demand forecast does not choose the price at all.** Under the hybrid response, $\hat{Q}(p_o)$ is a positive constant across candidate prices and therefore cancels out of $\text{argmax } (p - c) \cdot Q(p)$. The chosen price is a function of (p_o, c, ϵ) and the guardrails; the forecast sets the predicted *volume* and the dollar amounts. Verified numerically: scaling the base model by 10 $\times$ leaves every recommended price unchanged (tests/test_attribution.py).
7. **Naive elasticity is off by a factor of seven.** -0.348 pooled versus -2.419 with UPC $\times$ store fixed effects. That single comparison is the clearest demonstration in the project that association is not causation.
8. **Conventional standard errors were too small by a factor of 16.** Two-way clustering (UPC, week) widens the pooled elasticity standard error from 0.0065 to 0.1058, and the share of products significant at 5% falls from 90.1% to 75.2%. That correction alone moved the mean empirical-Bayes shrinkage weight from 0.954 to **0.780**.
9. **Category elasticity is stable across time; product-level elasticity is not.** The largest pairwise z-statistic between disjoint training windows is 1.36 (not significant), but the median rank correlation of *product* elasticities between disjoint windows is only **+0.19**.
10. **The elasticity assumption matters enormously in research mode and almost not at all in production mode.** The candidate profit-maximising price moves a median of **79.8%** across the four sensitivity elasticities when only a cost floor applies, **6.6%** once the historical-support band is added, and **0.0%** once the $\pm 10\%$ change cap is added.

1.12 The strongest limitation

No causal identification. Dominick's prices were set by the retailer in response to demand conditions, not randomised. The engine estimates $P(Q \mid \text{price}, X)$; the pricing decision needs $P(Q \mid \text{do}(\text{price}))$. Every price effect in this project is observational, every counterfactual uplift is a model-internal estimate produced by the same fitted response that proposed the price, and no number here may be described as a realised or causal business impact. The experiment that would settle it is designed in docs/PRICING_EXPERIMENT.md and was **not** run.

1.13 System diagram

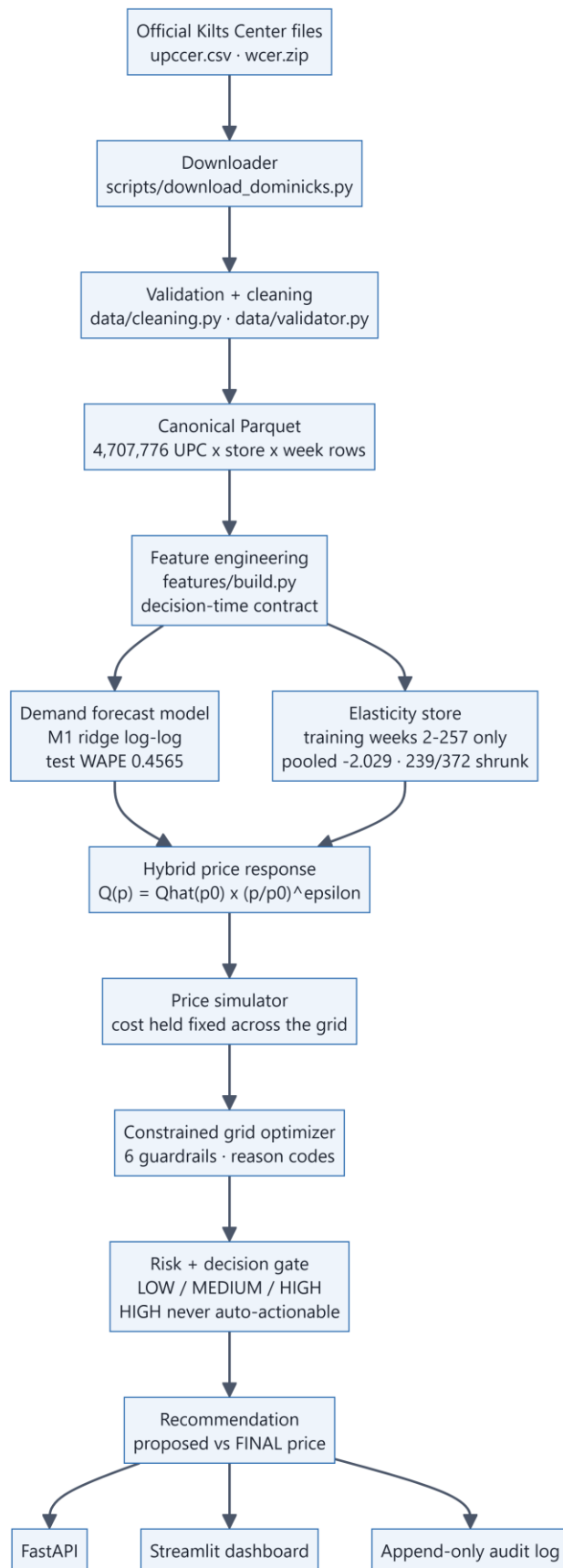


Diagram 1

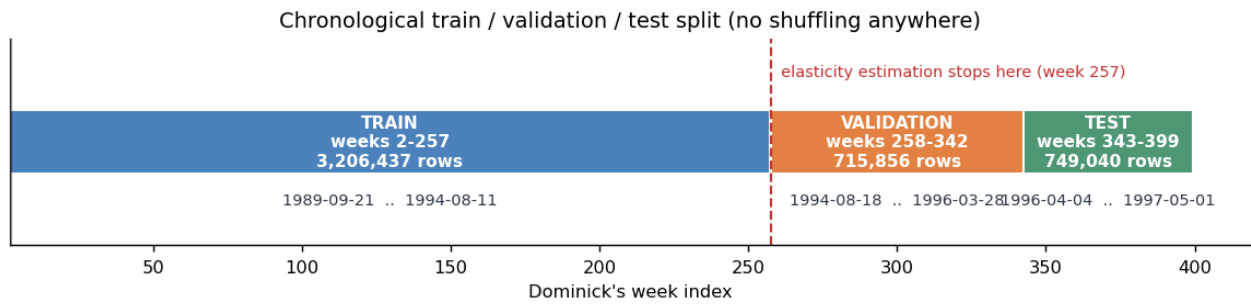


Figure 1 — Chronological train / validation / test split. Elasticity estimation is additionally restricted to weeks 2–257, so no elasticity used to price week 399 ever saw a validation or test outcome. Source: `artifacts/metrics/feature_build.json`.

2. Business problem

2.1 The retailer's decision

A category manager at a 93-store grocery chain sets, every week, a shelf price for each of ~489 cereal UPCs in each store. That is up to 45,477 price decisions per week. Each decision trades volume against margin, and each is made before the week's demand is observed.

Formally, let

- p — the effective unit price charged,
- X — the commercial context: product identity, store, week of year, planned promotion, recent demand history, recent price history, decision-time unit cost,
- $Q(p, X)$ — units sold,
- c — the unit cost the retailer pays.

Then:

$$Q = Q(p, X)$$

$$R(p) = p \cdot Q(p, X)$$

$$GP(p) = (p - c) \cdot Q(p, X)$$

2.2 Revenue and gross profit are maximised at different prices

This is the single most common confusion in pricing conversations, so it is worth deriving rather than asserting.

For **linear demand** $Q(p) = a - bp$ with $a, b > 0$:

$$R(p) = p(a - bp) = ap - bp^2 \Rightarrow dR/dp = a - 2bp = 0 \Rightarrow p_{R^*} = a/(2b)$$

$$GP(p) = (p - c)(a - bp) \Rightarrow dGP/dp = a + bc - 2bp = 0 \Rightarrow p_{GP^*} = (a + bc)/(2b)$$

so

$$p_{GP^*} - p_{R^*} = c/2 > 0 \quad \text{whenever } c > 0.$$

The profit-maximising price is always strictly above the revenue-maximising price when cost is positive, and the gap is exactly half the unit cost.

For **constant-elasticity demand** $Q(p) = k \cdot p^{-\epsilon}$ the same fact appears differently. Revenue $R(p) = k \cdot p^{1-\epsilon}$ is maximised where $\epsilon = -1$ (unit elastic). Gross profit has the closed form

$$p_{GP^*} = c \cdot \epsilon / (1 + \epsilon), \quad \epsilon < -1$$

and is **unbounded above** for $-1 < \epsilon < 0$: if demand is inelastic, raising the price always pays in this model, and the optimum is set entirely by constraints, not by the demand curve.

tests/test_optimizer.py::test_constant_elasticity_fixture_pushes_price_to_the_upper_bound encodes exactly that behaviour.

2.3 A worked numerical example from this project

Take the real decision context used throughout this report — **CAPN CRUNCH JUMBO CR** (UPC 3000006560) at **store 86**, decision week **399** (1997-05-01):

- current effective unit price $p_o = \$3.35$
- decision-time unit cost $c = \$2.5571$ (lagged, forward-filled AAC)
- applied elasticity $\varepsilon = -2.9401$ (shrunk product estimate)
- baseline predicted demand at p_o : $\hat{Q}(p_o) = 25.56$ units

Under the hybrid response $Q(p) = \hat{Q}(p_o) \cdot (p/p_o)^\varepsilon$:

candidate price	predicted units	expected revenue	expected gross profit
\$2.51	59.74	\$149.94	-\$2.81
\$2.91	38.68	\$112.54	\$13.65
\$3.35 (current)	25.56	\$85.64	\$20.27
\$3.66 (recommended)	19.71	\$72.13	\$21.74
\$3.88 (unconstrained optimum)	≈ 16.9	$\approx \$65.6$	$\approx \\$22.0$
\$4.11	14.01	\$57.60	\$21.76

Revenue falls monotonically across this whole range — because $|\varepsilon| > 1$, every price rise loses more volume proportionally than it gains in price. Gross profit rises to a peak at the closed-form optimum

$$p^* = c \cdot \varepsilon / (1 + \varepsilon) = 2.5571 \times (-2.9401) / (-1.9401) = \$3.875$$

and then falls. **Maximising revenue here would mean cutting the price to the floor and destroying most of the gross profit.** That is why this project optimises gross profit.

CAPN CRUNCH JUMBO CR (3000006560), store 86, week 399 (1997-05-01) - elasticity -2.940 (shrunk_product); shaded band = feasible interval 3.02 – 3.69. MODEL-INTERNAL estimates.

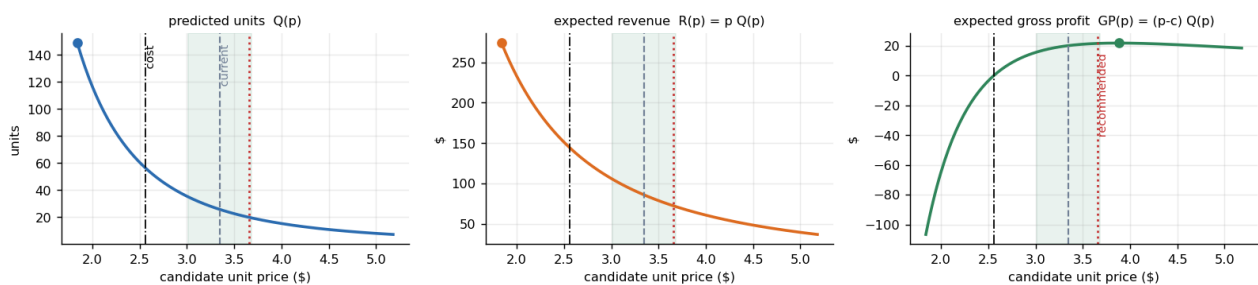


Figure 2 — The three objective curves for CAPN CRUNCH JUMBO CR at store 86, week 399. Black dash-dot = decision-time unit cost \$2.56; grey dashed = current price \$3.35; red dotted = final recommended price \$3.66; shaded band = the feasible interval \$3.02–\$3.69 the guardrails allow. The gross-profit optimum (\$3.88) lies **outside** the feasible band, so the recommendation is the upper guardrail corner — the single most important structural fact about this engine. All values are model-internal estimates. Source:

`scripts/make_report_figures.py`, live pipeline call.

2.4 What this project actually optimises

The shipped objective is **expected gross profit**:

$$\max_{(p \in \mathcal{F})} (p - c) \cdot \hat{Q}(p)$$

where

- $\mathcal{F}$ is the feasible set produced by the guardrails (\$53),

- c is a **decision-time** cost estimate, held **fixed** across the candidate grid (§48),
- $\hat{Q}(p)$ is the hybrid price response (§43).

Expected revenue is supported as a secondary objective (`optimization.objective: revenue`) because volume or share mandates are a real business reality, and both objectives use the *same* predicted demand so the only difference between them is the economics applied on top.

Two things the engine deliberately does **not** optimise: long-run customer equity (unobservable in scanner data) and category-level profit across substitutes (not modelled — see §101).

3. Why pricing is a difficult data-science problem

A model with excellent forecasting accuracy can still produce dangerous prices. This section explains why, with the evidence from this project attached to each mechanism.

3.1 The price–demand relationship is not observed; it is inferred

The retailer observes (p_t, Q_t) pairs. It never observes Q at a price it did not charge. Every counterfactual — the entire content of a price recommendation — is an extrapolation from a model fitted to prices that *were* charged. In this panel, the median UPC $\times$ store series has **8 distinct prices** across a median of 78 observations; there is genuine variation, but it is sparse, clustered, and not randomly assigned.

3.2 Elasticity is a slope, and slopes are fragile

Own-price elasticity $\epsilon = d \log Q / d \log p$ is a *derivative*, and derivatives are far harder to estimate than levels. In this project the same panel yields elasticities from **−0.348** to **−3.102** depending only on the specification. Conclusions that depend on the third decimal place of an elasticity should be treated as decoration.

3.3 Seasonality

Cereal demand has annual structure (back-to-school, holidays) and the panel spans eight years of secular drift. Without time controls, price and demand co-move for reasons that have nothing to do with the price. Adding `sin52`, `cos52` and a trend to the fixed-effects specification moves the estimate from **−2.419** to **−1.909**.

3.4 Promotion effects are bundled with price

A Bonus Buy is not a price cut. It is a price cut *plus* a feature ad, *plus* display space, *plus* an end-cap, *plus* a shopper-marketing budget — of which this data records only the price and a single-character code. Estimated separately on this panel, promotion weeks give **−2.662** and non-promotion weeks **−1.830** (`artifacts/metrics/elasticity.json`). The gap is the size of the bundling problem.

Its effect on the forecaster is equally visible: on the test window the selected model's WAPE is **0.374** on the 691,836 rows with no recorded promotion and **0.656** on the 57,204 rows with one, with a bias of **−37.2** units per row on promotion weeks against **−1.1** elsewhere.

3.5 Retailer anticipatory pricing (reverse causality)

Prices are cut *because* a demand surge is expected. Low price and high demand then coincide for a reason unrelated to the price, and the estimated elasticity is biased in a direction that cannot be signed a priori.

3.6 Endogeneity, stated formally

Write $\log Q_{it} = \alpha_i + \epsilon \cdot \log p_{it} + u_{it}$. OLS is consistent only if $E[u_{it} \cdot \log p_{it}] = 0$. That fails the moment the retailer conditions its price on anything in u — expected demand, competitor activity, inventory

position, trade-deal timing. It certainly does. The **factor-of-seven** move from -0.348 to -2.419 produced purely by adding fixed effects is direct evidence of exactly this contamination.

3.7 Confounding by unobserved marketing

Feature ads, display, competitor prices, local events, weather and shopper traffic are all unobserved here. Anything correlated with both price and demand loads onto the price coefficient.

3.8 Product heterogeneity

The per-UPC elasticity distribution in this panel runs from $p_{10} = -4.094$ to $p_{90} = -0.240$, and **5.1%** of UPCs have a *statistically significant positive* price coefficient. Positive coefficients are not bad data; they are evidence that for those series, price moves *with* unobserved demand. A single category elasticity is a useful prior and a poor description.

3.9 Store heterogeneity

Dominick's priced by zone. Store-level elasticities in this panel run from -0.947 to **$+0.297$** , and the median cross-store price spread for a given UPC is **13.6%** (p_{90} 27.5%). Cross-store price differences reflect local demand and competition, not exogenous variation — which is exactly why store fixed effects move the estimate.

3.10 Time effects

Eight years is long enough for category dynamics, competition and the price level itself to change. The monitoring artifact shows `series_reference_price` drifting from a reference-window mean of \$2.886 to \$3.247 in the test window (PSI 0.36, "large shift"). A model trained on 1989–1994 is scoring a materially different price regime in 1996–1997.

3.11 Cost uncertainty

The only cost available is an **Average Acquisition Cost** implied by an accounting margin (\$8). AAC lags replacement cost, moves with inventory ageing and forward buying, and is itself volatile: the median within-series AAC coefficient of variation is **0.088** (p_{90} 0.153). Since the gross-profit optimum is proportional to $\bar{c}$ under constant elasticity, a 10% error in $\bar{c}$ is a 10% error in the recommended price.

3.12 Extrapolation

A demand model asked for Q at a price the series has never carried is extrapolating, and log-linear models extrapolate confidently and wrongly. The project's response is the extrapolation guardrail (\$53.6), which keeps candidates inside the series' observed price support expanded by a configured tolerance. It is the *first binding constraint* in 13.4% of contexts and binds at the optimum in 2,103 of 13,964.

3.13 Price support

Only **54.08%** of the 36,443 UPC $\times$ store series pass the eligibility screen (≥ 40 observations, ≥ 5 distinct prices, price CV ≥ 0.05). The rest carry no usable information about price response no matter how good the model is, and always receive keep-current.

3.14 Substitution and cannibalisation

Cereal buyers substitute heavily between brands and sizes. This engine prices each UPC $\times$ store **independently**, so the category effect of re-pricing many products at once is **not** the sum of the per-product effects. This is arguably the largest missing mechanism in the project (\$101).

3.15 Stock-outs

Unavailability and demand collapse are indistinguishable in scanner data. A zero or suppressed sales week caused by an empty shelf is read by the model as weak demand. This project cannot separate the two.

3.16 Offline evaluation is fundamentally limited

The decisive point. Suppose the demand model is excellent. To evaluate a *pricing policy* offline you must score the prices it proposes — prices that were never charged — and the only thing available to score them with is the same model that proposed them. That is **circular**, and no amount of cross-validation fixes it. In this project the consequence is stated in the metric name itself: `model_internal_estimated_profit_uplift_pct`.

Therefore: good forecasting performance is necessary but nowhere near sufficient for safe pricing. A model can achieve any WAPE you like while having the price coefficient badly wrong, because at observed prices the lag and context features carry most of the signal. This project's own evidence: the selected model's implied elasticity is **-3.102**, materially steeper than every controlled econometric estimate (-1.909 to -2.419), and it was nonetheless the best forecaster in the comparison. That discrepancy is what motivated the entire hybrid architecture (§42).

PART II — DATA

4. Dataset selection

4.1 Why Dominick's Finer Foods

The dataset is the **Dominick's Finer Foods store-level scanner panel**, category **Cereals**, distributed by the **Kilts Center for Marketing, University of Chicago Booth School of Business** (<<https://www.chicagobooth.edu/research/kilts/research-data/dominicks>>).

Dominick's was a Chicago-area grocery chain; the Kilts Center release covers roughly nine years of weekly store-level movement across ~29 categories. It is one of the very few **publicly documented** retail scanner panels that carries all four things a pricing project needs simultaneously:

requirement	present in Dominick's?	why it matters here
store-level weekly price	yes (price, qty)	the decision variable
unit movement	yes (move)	the target
promotion coding	partially (sale: B/C/S)	lets promotion be controlled for <i>and</i> lets its incompleteness be measured
margin / cost	yes (profit, a gross-margin % on AAC)	without a cost there is no gross-profit objective at all

4.2 Academic research context

The panel underpins a large empirical-marketing and empirical-IO literature — work on promotion dynamics, category management, zone pricing and retail competition. That matters for a portfolio project in a specific way: the dataset's quirks (the `ok` flag, zero prices, bundle quantities, AAC-based margins) are *documented and discussed in public*, so decisions about them can be justified rather than invented.

4.3 Why the Cereals category

Cereals is a good stress test for a pricing engine:

- **Heavy promotion.** 7.35% of canonical rows carry a recorded promotion code, and promotion weeks average 54.5 units against 16.5 on non-promotion weeks — so the promotion-bundling problem is visible rather than hypothetical.
- **Genuine price variation.** Median 51 distinct prices per UPC, median within-series price CV 0.079, and 13.8% of consecutive-week pairs carry a price change.
- **High substitution.** Which makes the *absence* of cross-price modelling an honest, measurable limitation rather than a footnote.
- **Manageable scale.** 6.6M raw rows is large enough to be real and small enough to run end to end on a workstation.

4.4 Why this beats a generic sales dataset

A typical "retail sales" Kaggle-style dataset gives quantity and revenue with no cost, no promotion flag, no store dimension and often no genuine price variation. On such data you can build a demand forecaster, but you **cannot**:

- compute a gross-profit objective (no cost),
- control for promotion (no flag),
- exploit or even discuss zone pricing (no store),
- identify a price coefficient at all (no price movement),

- demonstrate the endogeneity problem (no controls to add).

4.5 Why the cost/margin field is the decisive feature

`profit` is a **gross-margin percentage**. Combined with the effective unit price it yields an implied unit cost, which is what turns the whole exercise from *revenue* optimization (a soluble but usually wrong objective) into *gross-profit* optimization (the business objective). Section 2.2 shows the two give different answers by exactly $c/2$ under linear demand — so without a cost field, the project could not have asked the right question.

5. Raw data acquisition

5.1 Official sources only

`scripts/download_dominicks.py` contacts **only** `chicagobooth.edu` URLs. There is no Kaggle fallback, no mirror, and no synthetic substitution. `tests/test_repo_and_downloader.py` asserts the URL host.

asset	file	recorded size	SHA-256 (first 16 hex)
UPC metadata	<code>upc_csv-files/upccer.csv</code>	25,932 B	<code>affa589f080ab18a...</code>
Movement archive	<code>movement_csv-files/wcer.zip</code>	42,402,094 B	<code>2d4a59f88e4b9725...</code>
Movement CSV (extracted)	<code>wcer.csv</code>	458,400,475 B	<code>a204106f6dff8a6c...</code>
Manual / codebook	<code>dominicks-manual-and-codebook_kiltscenter.pdf</code>	10,182,109 B	<code>d8dc133bfe402a21...</code>

Full hashes, source URLs, the usage restriction and the download timestamp (`2026-08-18T08:05:01Z`) are written to `data/raw/dominicks/cereals/SOURCE.json` and reproduced in `reports/01_DATA_AUDIT.md`. That file is present locally and git-ignored; the raw data are never redistributed.

5.2 Downloader behaviour

The script:

11. creates the target directories,
12. streams downloads with explicit timeouts,
13. validates the HTTP status **and** a non-zero content length,
14. writes to a temporary file and renames **atomically**, so a partial download can never be mistaken for a complete one,
15. extracts the archive with **zip-slip and absolute-path guards**,
16. normalises the extracted movement CSV to `wcer.csv` while recording the original archive member name,
17. writes `SOURCE.json` with URLs, sizes, SHA-256 hashes and timestamps.

5.3 Failure behaviour

If the network or the site blocks automated retrieval, the script prints exact manual download instructions and marks acquisition **BLOCKED**. It never fabricates or substitutes data. This actually happened during the original build: the first automated attempt failed mid-download with a connection reset, and a resumable retry completed it (`STATUS.md`, row 1).

5.4 Raw-data Git policy and attribution

The Kilts Center makes the Dominick's data available **for academic research purposes** and asks users to acknowledge the Kilts Center in publications. Enforced consequences:

- `data/raw/`, `data/interim/` and `data/processed/` are **git-ignored** and never committed.
`tests/test_repo_and_downloader.py` asserts the ignore rules.

- CI runs on small synthetic fixtures only and never downloads the licensed dataset on a public build service (`.github/workflows/ci.yml`).
- The Docker image contains **code only**; artifacts and processed data are mounted read-only at run time (`Dockerfile`, `docker-compose.yml`).

5.5 Reproducible download workflow

```
python scripts/download_dominicks.py      # official URLs only
python scripts/download_dominicks.py --force # re-download
make data                                  # download + build + audit
```

`bash`

6. Dominick's data model

6.1 Movement file fields

column	dtype loaded	meaning (per the Kilts manual)
upc	int64	Universal Product Code of the item
store	int32	Dominick's store number
week	int32	Dominick's week index; week 1 begins 1989-09-14
move	float64	number of individual units sold in that store-week
price	float64	price of the bundle actually scanned (may cover several units)
qty	float64	number of items in the bundle
sale	string	promotion code: B Bonus Buy, C Coupon, S simple price reduction (undocumented G and L also observed)
profit	float64	gross-margin percentage , computed on Average Acquisition Cost
ok	int8	1 = valid, 0 = suspect / "trash" per the manual

Two raw columns are deliberately **not** loaded: `PRICE_HEX` and `PROFIT_HEX`, which are redundant binary encodings of `price` and `profit`.

6.2 UPC metadata file (`upccer.csv`)

column	meaning
com_code	commodity code (Cereals)
descrip	product description, e.g. CAPN CRUNCH JUMBO CR
size	package size string, e.g. 12 OZ
case	units per case
nitem	internal item number

490 rows, **0 duplicate UPCs**, **0 nulls in any column** (`artifacts/metrics/raw_audit.json`).

6.3 The natural grain

UPC × Store × Week

This is not an arbitrary choice; it is what the data *is*, and it matters for three reasons.

18. **It is the decision grain.** Dominick's set prices by store zone, so the thing a category manager actually chooses is a price for one UPC in one store for one week. Modelling at UPC × week (pooling stores) would model a decision nobody makes.
19. **It is where the price variation lives.** The observed median cross-store price spread for a given UPC is **13.6%** — real, exploitable, and invisible at a coarser grain.
20. **It defines what a "series" is.** All lags, rolling windows, reference prices, eligibility screens, price support and risk assessments are computed *within* a UPC × store series. There are **36,443** such series.

Grain uniqueness is asserted, not assumed: `raw_duplicate_grain_rows = 0` in the raw file, and `grain_unique` is one of the 15 validation checks (0 duplicated UPC × store × week rows in the canonical table).

6.4 Time decoding

The manual anchors week 1 to **1989-09-14** (a Thursday); Dominick's store weeks run Thursday → Wednesday. So:

$$\text{week_start_date} = 1989-09-14 + (\text{week} - 1) \times 7 \text{ days}$$

This is implemented in `pricing_engine.data.cleaning.decode_week` and validated two ways: `week_date_mapping_unique` (each week index maps to exactly one calendar date) and `date_range_plausible` (the decoded range is **1989-09-14 ... 1997-05-01**, matching the documented coverage). Decoding also buys real calendar seasonality — `year`, `month`, `quarter`, `week_of_year` — which a bare integer week index cannot provide.

7. Price and quantity semantics

7.1 The problem

`price` is a **bundle** price, and `qty` is how many individual items that bundle contains. `move` counts **individual units**, not bundles. So `price` and `move` are denominated differently, and multiplying them is wrong whenever `qty > 1`.

7.2 The corrections

$$\text{effective_unit_price} = \text{price} / \text{qty}$$

$$\text{revenue} = \text{effective_unit_price} \times \text{move} = (\text{price} \times \text{move}) / \text{qty}$$

Implementation (`src/pricing_engine/data/cleaning.py`):

```
qty = out["qty"].where(out["qty"] > 0)          # never divide by zero silently
out["effective_unit_price"] = out["price"] / qty
out["revenue"] = out["effective_unit_price"] * out["move"]
```

python

The `.where(qty > 0)` produces `NaN` rather than `inf` for a non-positive quantity, and a dedicated validation check (`no_infinite_unit_price`) asserts that `effective_unit_price` is finite everywhere in the canonical table — i.e. that no silent divide-by-zero survived.

7.3 How much does this actually matter?

qty	rows	share
1	4,702,830	99.8949%
2	3,241	0.0688%
3	568	0.0121%
4	1,137	0.0242%
> 1 (total)	4,946	0.1051%

Only about one row in a thousand is a multi-unit bundle. But for those rows, using `price` directly would overstate the unit price by a factor of **2 to 4**, and a 4× price error propagates into the elasticity, the implied AAC, the gross-profit calculation, the feasible price interval and the recommendation. The correction costs one division and removes an entire class of silent error.

Its size is verified rather than assumed: `formula_effective_unit_price` and `formula_revenue` are validation checks with maximum absolute deviations of $0.000e+00$ and $1.421e-14$ respectively (floating-point noise).

7.4 Distributional consequence

The raw `price` field has mean 2.2412 (heavily dragged down by the 1,851,380 zero-price rows). After exclusion and unit conversion, the canonical `effective_unit_price` has **mean \$3.1157, median \$3.15, minimum \$0.05, maximum \$26.02**. The maximum is a genuine artefact worth naming: it belongs to `TONY THE TIGER T-SHIRT`, a non-cereal item that appears in the Cereals commodity file. Rows with a unit price above \$10 are 6.4×10^{-7} of the table — three rows.

8. Gross margin and AAC

8.1 What profit represents

`profit` is a **gross-margin percentage** computed by Dominick's accounting system on **Average Acquisition Cost (AAC)** — an inventory-accounting measure of what the retailer paid, on average, for the stock currently on hand.

8.2 The derivations

$$\text{gross_margin_rate} = \text{profit}/100$$

$$\text{estimated_unit_aac} = \text{effective_unit_price} \times (1 - \text{gross_margin_rate})$$

$$\text{gross_profit} = \text{revenue} \times \text{gross_margin_rate} = (p - \text{aac}) \times \text{move}$$

The second identity is the definition of a gross-margin rate rearranged: if $m = (p - c)/p$ then $c = p(1 - m)$.

The third follows immediately: $\text{revenue} \times m = p \cdot Q \cdot (p - c)/p = (p - c) \cdot Q$.

All three are validation checks with maximum absolute deviation $0.000e+00$.

8.3 Why the name `estimated_unit_aac` and never `true_unit_cost`

This naming decision (DECISIONS #4) is one of the more consequential small choices in the project. AAC is **not** necessarily the economically relevant cost at the moment of a pricing decision, for at least four reasons:

21. **Forward buying.** Retailers buy ahead of trade-deal expiry, so on-hand stock can be much cheaper than replacement stock.
22. **Inventory ageing.** AAC is a weighted average over stock of different vintages; it lags the current wholesale price by an unknown amount.
23. **Trade funds.** Off-invoice allowances, bill-backs and scan-downs may or may not be reflected in AAC depending on accounting treatment.
24. **The relevant cost for a marginal decision is replacement cost**, not average historical cost — a standard result in managerial economics.

Empirically, the AAC in this panel is not stable: the **median within-series AAC coefficient of variation is 0.088** with a p90 of 0.153 (`artifacts/metrics/eda_summary.json`). And it moves *with price*: on price cuts, the mean AAC change is **-9.16%** — consistent with promotions being funded by trade deals, and a direct warning against treating cost as independent of the pricing decision.

So the column is called `estimated_unit_aac` everywhere, and `docs/DATA_DICTIONARY.md` states the caveat inline. Renaming it to a generic "cost" would have made every downstream gross-profit number sound more authoritative than the evidence supports.

8.4 Consequence for the optimizer

Under constant elasticity the optimum is $p^* = c \cdot \epsilon / (1 + \epsilon)$, i.e. **linear in cost**. A 10% error in `c` produces a 10% error in the recommended price. This is the reason the project (a) uses a *lagged, decision-time* cost rather than the contemporaneous accounting margin (\$48), (b) holds cost **fixed** across the candidate grid, and (c) lists AAC accuracy as an open limitation rather than claiming the gross-profit figures are exact.

8.5 Sanity of the margin field

statistic	raw profit
mean	12.522
median	13.740
p01 / p99	0.00 / 46.21
min / max	-99.69 / 99.99
negative values	53,057
zero values	1,852,626

Negative margins are real (loss-leading, clearance, promotional deals funded after the fact) and are **not** deleted. Instead the canonical table carries a `margin_implausible_flag` for `|gross_margin_rate| ≥ 1` and an `aac_nonpositive_flag` for implied AAC ≤ 0. In the final canonical table both flags fire on **0.0%** of rows, because the rows carrying them were removed by the zero-price rule for independent reasons.

9. Data quality audit

9.1 Raw table

quantity	value
rows	6,602,582
columns	9
exact duplicate rows	0
duplicate UPC × store × week rows	0
distinct UPCs	490
distinct stores	93
distinct weeks	367 (index range 1 ... 399)
nulls in store,upc,week,move,qty,price,profit,ok	0
nulls in sale	6,242,568 (94.55% — no promotion code recorded)

That there are **zero** exact duplicates and **zero** grain collisions in a 6.6M-row raw file is itself a finding worth stating: the movement file is already at its natural grain, so no de-duplication judgement was needed.

9.2 The exclusion ledger

Every rule is applied in order to the frame that survived the previous rules, so the counts are "additional rows removed by this rule" and can be summed.

#	rule	rows removed	% of raw	rationale
1	<code>missing_key</code>	0	0.00%	UPC / store / week must be present to identify an observation

#	rule	rows removed	% of raw	rationale
2	missing_core_measure	0	0.00%	price, move, qty, profit are required for pricing economics
3	ok_flag_zero	141,285	2.1398%	the manual marks ok = 0 rows as suspect / trash
4	non_positive_price	1,753,521	26.5581%	a bundle price below \$0.01 cannot yield a unit price
5	non_positive_qty	0	0.00%	needed to compute a unit price
6	negative_move	0	0.00%	negative movement is not interpretable as demand
7	metadata_unmatched_kept	0	0.00%	unmatched rows would be kept and flagged, not dropped — none occurred
	total removed	1,894,806	28.6980%	
	canonical rows	4,707,776	71.3020%	

Note the interaction: 1,851,380 raw rows carry `price = 0`, but only 1,753,521 are removed by rule 4 — because 97,859 of them were already removed by rule 3 (`ok = 0`). The ledger is order-dependent by design and the order is recorded.

9.3 Metadata join

quantity	value
metadata rows	490
duplicate UPCs in metadata	0
UPCs present in movement	489
UPCs in metadata with no movement	1
UPCs in movement with no metadata	0
rows with no metadata after the join	0 (0.00%)

The join is executed with `validate="many_to_one"` and the row count is asserted before and after:

```
merged = movement.merge(meta[keep_cols], on="upc", how="left", validate="many_to_one")
if len(merged) != n_before:
    raise AssertionError(f"UPC metadata join changed the row count: {n_before} -> {len(merged)}.")
```

python

A silent fan-out on a join is one of the classic ways a pipeline gains rows it cannot account for. Here it is impossible by construction.

9.4 Canonical table dimensions

quantity	value
rows	4,707,776
UPCs	489
stores	93
weeks	366 (index 1 ... 399)
date range	1989-09-14 ... 1997-05-01
duplicated grain rows	0
rows with <code>ok ≠ 1</code>	0
observed total units	90,766,941
observed total revenue	\$262,008,582.295

quantity	value
observed total gross profit	\$40,091,143.487

366 observed weeks within an index range of 399 means **33 week indices carry no rows at all** after exclusion — real gaps in the panel, not an error.

9.5 The 15 validation checks

`scripts/validate_data.py` re-loads the persisted Parquet and re-derives every identity from scratch. All 15 pass.

#	check	result
1	non_empty	pass — 4,707,776 rows
2	grain_unique	pass — 0 duplicated UPC × store × week rows
3	only_ok_rows	pass — 0 rows with <code>ok ≠ 1</code>
4	formula_effective_unit_price	pass — max abs deviation 0.000e+00
5	formula_revenue	pass — max abs deviation 1.421e-14
6	formula_gross_margin_rate	pass — 0.000e+00
7	formula_estimated_unit_aac	pass — 0.000e+00
8	formula_gross_profit	pass — 0.000e+00
9	positive_price	pass — all bundle prices strictly positive
10	positive_qty	pass — all bundle quantities strictly positive
11	non_negative_move	pass — no negative unit movement
12	no_infinite_unit_price	pass — finite everywhere (no silent divide-by-zero)
13	week_date_mapping_unique	pass — each week index maps to exactly one date
14	date_range_plausible	pass — 1989-09-14 ... 1997-05-01
15	promotion_flag_binary	pass — recorded promotion share 0.0735

9.6 Suspicious distributions that were kept, flagged, not deleted

There is **no arbitrary outlier trimming anywhere in this project** (DECISIONS #7). Extremes are surfaced and bounded downstream by the extrapolation guardrail instead of being deleted to make plots tidy.

observation	magnitude	treatment
week-over-week price jumps > ±50%	10.07% of price-change events (77,152)	kept; real promotion behaviour mixed with possible recording artefacts; bounded by the extrapolation guardrail
maximum observed unit price	\$26.02 (TONY THE TIGER T-SHIRT)	kept; a non-cereal item in the Cereals commodity file
UPCs with negative total observed gross profit	6	kept; loss-leading is real
negative raw profit values	53,057	kept in raw; flagged via <code>margin_implausible_flag</code>
maximum weekly move	18,688 units	kept; the model applies a prediction cap instead (\$37.6)

9.7 Quality flags carried on every canonical row

flag	definition	share in canonical table
margin_implausible_flag	gross_margin_rate outside (-1, 1) or NaN	0.0%
aac_nonpositive_flag	implied unit AAC ≤ 0	0.0%
zero_move_flag	zero units sold that week	0.0%
bundle_flag	qty > 1	0.1051%
has_metadata_flag	UPC matched the metadata file	100%

zero_move_flag = 0.0% deserves a comment: after the zero-price exclusion, **no canonical row has zero sales** (n_upc_store_weeks_with_zero_sales = 0). That is a direct, mechanical consequence of the exclusion rule — zero-price and zero-move rows are essentially the same rows (\$10) — and it is the main reason the zero-price decision needed its own investigation.

10. Zero-price investigation

This exclusion removes **26.56% of the raw file**. It is by far the largest single judgement in the data pipeline, so it was given a dedicated audit (scripts/audit_zero_price.py → reports/09_ZERO_PRICE_AUDIT.md).

10.1 Scale

quantity	value
raw rows with price = 0	1,851,380
share of the raw file	28.0402%
of which move > 0 (recorded any sales)	677
of which move = 0	1,850,703
rows with price > 0 and move = 0	0
zero-price rows with non-zero profit	0
zero-price rows with qty > 1	0

The single most informative number here is **677**. Of 1.85 million zero-price rows, only 677 — **0.037%** — recorded any sales at all. A zero-price row in this file is overwhelmingly a week in which the item was **not offered**, not a week in which it was given away.

The second most informative number is the **0** in row 5: there is no such thing as a positively priced week with no sales. Price and movement are zero together. That is the signature of a "no offer this week" record, not of a demand event.

10.2 Interaction with the ok flag

	ok = 1	ok = 0
zero-price rows	1,753,521	97,859
all rows	6,461,297	141,285

69.3% of all ok = 0 rows are also zero-price rows. The two quality signals largely agree, which is mild independent evidence that the zero-price rows are indeed junk rather than information.

10.3 Distribution across time

statistic	value
weeks covered	367

statistic	value
minimum weekly zero share	19.98%
maximum weekly zero share	100.00%
mean weekly zero share	27.95%
standard deviation	6.90%
first week's zero share	42.34%
last week's zero share	28.21%
weeks above 50% zero	1

The phenomenon is **stationary**, not a data-collection failure confined to one era. One week is entirely unpriced.

10.4 Distribution across products

statistic	value
UPCs	490
median zero share per UPC	40.69%
p10 / p90	4.75% / 93.03%
UPCs above 80% zero	112
UPCs below 5% zero	57

The top-10 most affected UPCs are almost entirely seasonal, promotional or short-lived items — QUAKER CLUB PACK CAP (100.0% zero), ~KELL COMPLETE BRAN (99.98%), RICE KRISPIES HOLIDAY (98.57%), SPIDERMAN CEREAL (98.21%). These read as items that were carried briefly or in a few stores, exactly as a "not offered" interpretation predicts.

10.5 Distribution across stores

statistic	value
stores	93
median zero share	26.68%
min / max	8.09% / 43.01%
standard deviation	5.85%

Stores differ by a factor of five, consistent with genuinely different assortments — but no store is anywhere near 0% or 100%, so this is not a per-store recording failure.

10.6 Position within the series — the decisive evidence

Each zero-price row was classified by where it sits in its UPC × store series.

position	rows	share of zero-price rows
leading run, before the series' first priced week	291,284	15.73%
trailing run, after the series' last priced week	1,068,024	57.69%
interior gap between priced weeks	485,204	26.21%
series never priced at all	6,868 series	0.37% of rows
interior-gap rows that <i>did</i> record sales	660	—

73.4% of zero-price rows are leading or trailing runs — the item was not yet carried, or had been delisted. This is the interpretation the audit supports and it is not a guess: it is a positional fact about 1.36 million rows.

The remaining 26.2% are interior gaps — temporary de-listings, out-of-stocks or seasonal absences. Of those, only 660 recorded any sales.

10.7 What the manual says

Nothing. The official Kilts Center manual and codebook are silent on zero or missing prices (`KNOWN_LIMITATIONS.md` #32). The interpretation above is therefore **inferred from the data, not documented by the provider** — a distinction the project states explicitly rather than papering over.

10.8 Why the exclusion was retained

Four reasons, in order of weight:

25. **No unit price can be derived from a zero bundle price**, so the decision variable of the entire project is undefined for these rows.
26. **Imputing a price would fabricate the optimised variable.** Whatever value were imputed would flow into `effective_unit_price`, `estimated_unit_aac`, the elasticity regressions and the objective function. That is inventing data to feed a pricing model.
27. **99.96% of them carry no sales anyway**, so the information loss on the target is minimal.
28. **Keeping them with NaN price** would either propagate NaN through every downstream computation or force silent imputation at each step.

10.9 The selection bias this creates — measured, not asserted

Excluding these rows tilts the analysed sample toward continuously stocked, higher-volume series. The audit quantifies the tilt with a direct probe:

series group	n series	mean effective unit price	mean weekly units
zero-share > 50% ("intermittent")	12,529	\$2.717	3.05
zero-share ≤ 10% ("continuously stocked")	11,139	\$3.228	21.00

The continuously stocked series sell **6.9× more units per week** and carry **19% higher prices**. So:

- the elasticity estimates, the demand model and the recommendations are all fitted on and applied to the **high-volume, continuously stocked core** of the category;
- nothing in this project should be read as characterising intermittent, seasonal or short-lived items;
- because zero-move weeks are eliminated entirely, the demand model **never learns to predict a zero week**, and the whole intermittent-demand problem (Croston-style methods, zero-inflated models) is out of scope by construction.

That is the honest description of the sample this engine actually serves.

11. Promotion coding

11.1 The codes

code	meaning (manual)	raw rows
<i>(absent)</i>	no code recorded	6,242,568 (94.55%)
B	Bonus Buy	254,261
S	simple price reduction	91,259
G	undocumented	11,075
C	Coupon	3,418
L	undocumented	1

11.2 How the project encodes them

```
out["recorded_promotion_flag"] = code.isin(list(PROMOTION_CODES)).astype("int8")
out["recorded_promotion_type"] = code.map(PROMOTION_CODES).fillna("NONE_RECORDED")
unknown = code.notna() & ~code.isin(list(PROMOTION_CODES))
out.loc[unknown, "recorded_promotion_type"] = "UNKNOWN_CODE"
```

python

Three deliberate choices:

29. The flag is called `recorded_promotion_flag`, not `is_promo`. The value `0` means *no code was recorded*, not *no promotion happened*.
30. The type is `NONE_RECORDED`, not `NO_PROMOTION`.
31. The undocumented `G` and `L` codes are surfaced as `UNKNOWN_CODE`, never silently mapped to a documented meaning or to "no promotion". 11,076 rows carry a code nobody can interpret, and the pipeline says so.

In the canonical table, `recorded_promotion_flag = 1` on **7.35%** of rows — one of the 15 validation checks (`promotion_flag_binary`, recorded promotion share 0.0735).

11.3 Why absence of a code does not prove absence of a promotion

The Kilts documentation states the promotion coding is incomplete. Concretely:

- in-store **display** and **feature advertising** are not in this file at all, and both move volume without necessarily carrying a `sale` code;
- the codes are captured operationally, so coverage varies by store, era and operator;
- two codes (`G`, `L`) exist that the manual does not describe, so the mapping itself is incomplete;
- 94.55% of rows carry no code, which is implausible as a literal statement that 94.55% of store-weeks had no promotional activity of any kind in a category as promoted as cereal.

11.4 Consequences for causal interpretation

This is the mechanism that most directly undermines any causal reading of the price coefficient.

32. **The "controlled for promotion" specifications are only partially controlled.** `recorded_promotion_flag` captures the coded subset; the uncoded remainder loads onto the price coefficient.
33. **The "clean" non-promotion subsample is not clean.** Splitting the panel by the recorded flag gives -2.662 on promotion weeks and -1.830 on non-promotion weeks, but the second group certainly contains uncoded promotions, so -1.830 is itself contaminated toward the promotion value.
34. **Any estimated "price effect" partly measures the marketing bundle.** A Bonus Buy is a price cut *and* a feature *and* a display. The data can only see the price.
35. **The out-of-time validation inherits the problem.** 73.3% of the 154,899 observed price-change episodes carry a recorded promotion code in at least one of the two weeks (\$45), so even the naturalistic validation is dominated by bundled events. This is precisely why that report splits its results by promotion state.

11.5 The decision-time assumption

`recorded_promotion_flag` is used as a **model input for week t** , i.e. treated as known when the price is set. Strictly, the `sale` code is *observed* in the movement record — after the fact.

The assumption is: *a retailer setting next week's price already knows whether that item is scheduled for a Bonus Buy, coupon or price reduction that week*. Promotion calendars are agreed weeks in advance, so this is realistic — but it is an assumption, it is documented as leakage channel #13 in `docs/DATA_LEAKAGE_AUDIT.md`, and its impact is bounded and reported: it affects 7.35% of rows, and `reports/04_MODEL_COMPARISON.md` reports error separately by promotion state so the dependence is visible rather than hidden.

12. Canonical processed dataset

12.1 Storage and reproducibility

path	data/processed/dominicks_cereals.parquet
format	Apache Parquet (via pyarrow ≥ 14)
rows	4,707,776
columns	31
Parquet SHA-256	51f9148bd3955129a2d7e86953d487a4707296e7ec1b306d1ced5089799761a9
DataFrame fingerprint (content hash, storage-independent)	e9d26f2c0eda9f7e9d8dbb508053e56a932608668e16d8b5aeeec0032373722d0
built	2026-08-18T08:05:28Z
environment	Python 3.13.0, numpy 2.0.2, pandas 2.2.3, scikit-learn 1.8.0, Windows-11-10.0.26200

Two hashes are recorded because they answer different questions. The **Parquet SHA-256** answers "is this the same file?"; the **DataFrame fingerprint** answers "is this the same *data*?", surviving a re-write with different compression or row-group settings. The trained model artifact records the DataFrame fingerprint (`data_fingerprint` in `artifacts/models/demand_model_metadata.json`), so a model can always be tied to the exact dataset content it was fitted on.

12.2 Row grain and ordering

One row per (`upc`, `store`, `week`), sorted by `upc`, `store`, `week`, with grain uniqueness asserted. Column order is fixed by the `CANONICAL_COLUMNS` constant in `src/pricing_engine/data/schema.py` so the schema is stable across runs.

12.3 Schema

group	columns
keys	<code>upc</code> , <code>store</code> , <code>week</code>
calendar	<code>week_start_date</code> , <code>year</code> , <code>month</code> , <code>quarter</code> , <code>week_of_year</code>
raw measures	<code>move</code> , <code>price</code> , <code>qty</code> , <code>profit</code> , <code>sale</code> , <code>ok</code>
derived economics	<code>effective_unit_price</code> , <code>gross_margin_rate</code> , <code>estimated_unit_aac</code> , <code>revenue</code> , <code>gross_profit</code>
promotion	<code>recorded_promotion_flag</code> , <code>recorded_promotion_type</code>
product metadata	<code>com_code</code> , <code>descrip</code> , <code>size</code> , <code>case</code> , <code>nitem</code>
quality flags	<code>margin_implausible_flag</code> , <code>aac_nonpositive_flag</code> , <code>zero_move_flag</code> , <code>bundle_flag</code> , <code>has_metadata_flag</code>

The full data dictionary with per-column definitions is in **Appendix A** and in `docs/DATA_DICTIONARY.md`.

12.4 What is *not* in the canonical table

Modelling features (lags, rolling windows, decision-time cost, relative price features) are deliberately **not** here. They live in a separate table, `data/processed/dominicks_cereals_features.parquet`, built by `scripts/build_features.py`. The separation matters: the canonical table is a faithful, auditable representation of *what happened*, and the feature table is a modelling artefact governed by a decision-time availability contract (§20). Mixing the two is how outcome columns end up as predictors.

PART III — EXPLORATORY PRICING ANALYSIS

All numbers in this part come from `scripts/run_eda.py` → `artifacts/metrics/eda_summary.json` and `reports/02_PRICING_EDA.md`. Every figure named `eda_*.png` lives in `artifacts/figures/`.

13. Overall retail performance

13.1 Observed historical scale

metric	value	how it is computed
rows analysed	4,707,776	canonical table
UPCs × stores × weeks	489 × 93 × 366	distinct counts
date range	1989-09-14 ... 1997-05-01	decoded week start dates
observed total units	90,766,941	Σ move
observed total revenue	\$262,008,582	Σ (effective_unit_price × move)
observed total gross profit	\$40,091,143	Σ (revenue × gross_margin_rate)
blended gross margin	15.30%	Σ gross_profit / Σ revenue
mean gross-margin rate per row	17.42%	mean(gross_margin_rate)
mean effective unit price	\$3.1157	mean(effective_unit_price)
median effective unit price	\$3.15	median(effective_unit_price)
share of rows with a recorded promotion	7.35%	mean(recorded_promotion_flag)
share of bundle rows (qty > 1)	0.1051%	mean(bundle_flag)
share of zero-sales rows	0.00%	consequence of the zero-price exclusion (§10)

13.2 Why the two margin numbers differ

The **blended margin (15.30%)** is revenue-weighted: Σ gross_profit / Σ revenue. The **mean margin rate (17.42%)** is the unweighted average of the per-row rate. The blended number is lower, which says that **high-revenue store-weeks carry systematically lower margins** — exactly what you expect when volume is driven by promotions, and a useful sanity check that the derived economics behave the way retail economics should.

For anything portfolio-level, the blended figure is the correct one, and it is the number quoted throughout this report.

13.3 Weekly time series

`artifacts/figures/eda_weekly_revenue_profit.png` and `eda_weekly_units.png` (both visible in the dashboard screenshot, §75) show weekly totals across all stores. Two features are worth naming:

- **Gross profit is far noisier than revenue in relative terms**, and dips below zero in some weeks around 1993–1994 — deep promotional periods where the accounting margin went negative across enough of the category to swamp the rest.
- A visible **gap around 1995** in both series corresponds to the week indices with no surviving rows (§9.4).

14. Product concentration

metric	value
UPCs in the canonical table	489
UPCs generating 80% of observed revenue	121
top-10 UPCs' share of observed revenue	14.15%

metric	value
top-10 UPCs' share of observed gross profit	12.68%
UPCs with negative total observed gross profit	6

14.1 Reading the concentration curve

121 of 489 UPCs (**24.7%**) produce 80% of revenue. That is concentrated, but far less concentrated than the Pareto folklore would suggest — cereal is a long tail of brand × size × variant combinations, and the tail is not negligible.

The top-10 share of gross profit (12.68%) is **lower** than the top-10 share of revenue (14.15%), confirming at the product level what §13.2 showed at the row level: the biggest sellers are the most heavily promoted and carry thinner margins.

14.2 Implications actually taken in this project

36. **Business prioritisation.** A pricing team working this category would get most of the value from ~121 UPCs. Nothing in the engine hard-codes that, but it is the right way to stage a rollout.
37. **Modelling.** A single global model is defensible here precisely because revenue is *not* dominated by a handful of items — 121 products with real volume is enough for pooling to be informative without a few giants dominating the loss.
38. **The six negative-gross-profit UPCs are retained.** They are real loss-leaders (or accounting artefacts), and deleting them would remove the evidence that loss-leading exists in this data. The optimizer's cost floor ($p \geq c$ unless `allow_below_cost` is explicitly set) means the engine will never *propose* a loss-leader under any default profile.

15. Historical price variation

Price variation is the **identifying variation** for everything downstream. A series whose price never moves carries zero information about price response, however good the demand model is.

15.1 At the UPC level (pooling stores)

metric	value
median distinct prices per UPC	51
median price coefficient of variation	0.0807
median price range as % of mean price	60.61%
share of UPCs with price CV below 5%	28.02%

15.2 At the UPC × store series level — the decision grain

metric	value
number of series	36,443
median observations per series	78
median distinct prices per series	8
median price coefficient of variation	0.0791
median price-change rate (share of weeks with a change)	13.80%
series passing the eligibility screen	19,707
share eligible	54.08%

Eligibility criteria (`configs/config.yaml` → `eligibility`):

$$n_{\text{obs}} \geq 40 \quad \wedge \quad n_{\text{(distinct prices)}} \geq 5 \quad \wedge \quad CV_{\text{price}} \geq 0.05$$

15.3 The gap between the two levels is the whole story

A UPC has a median of **51** distinct prices; a UPC × store series has a median of **8**. The difference is cross-store price-level variation, not within-store price movement. If you estimate elasticity from pooled UPC-level data you are substantially identifying it from *where a store is*, not from *when a price moved* — and store location correlates with local demand and competition.

This is exactly why the elasticity specifications in §28 absorb store fixed effects, and why the estimate moves when they do.

15.4 Why the ineligible 45.92% matter

16,736 series fail the screen. They are not discarded from the data — they appear in the panel, the demand model scores them, and the API will answer for them. What happens instead is that the optimizer's eligibility screen fires first and returns `KEEP_CURRENT` with reason codes `INSUFFICIENT_HISTORY` and/or `INSUFFICIENT_PRICE_VARIATION`. In the full week-399 run those two codes account for 567 and 668 occurrences respectively in the 3,000-context batch, and **3,349 of 13,964** contexts in the full week are "screened out by the eligibility rules (never reaches the optimizer)".

An engine that returns "I don't know" for 25% of its inputs is behaving correctly. An engine that returned a confident price for all of them would be the problem.

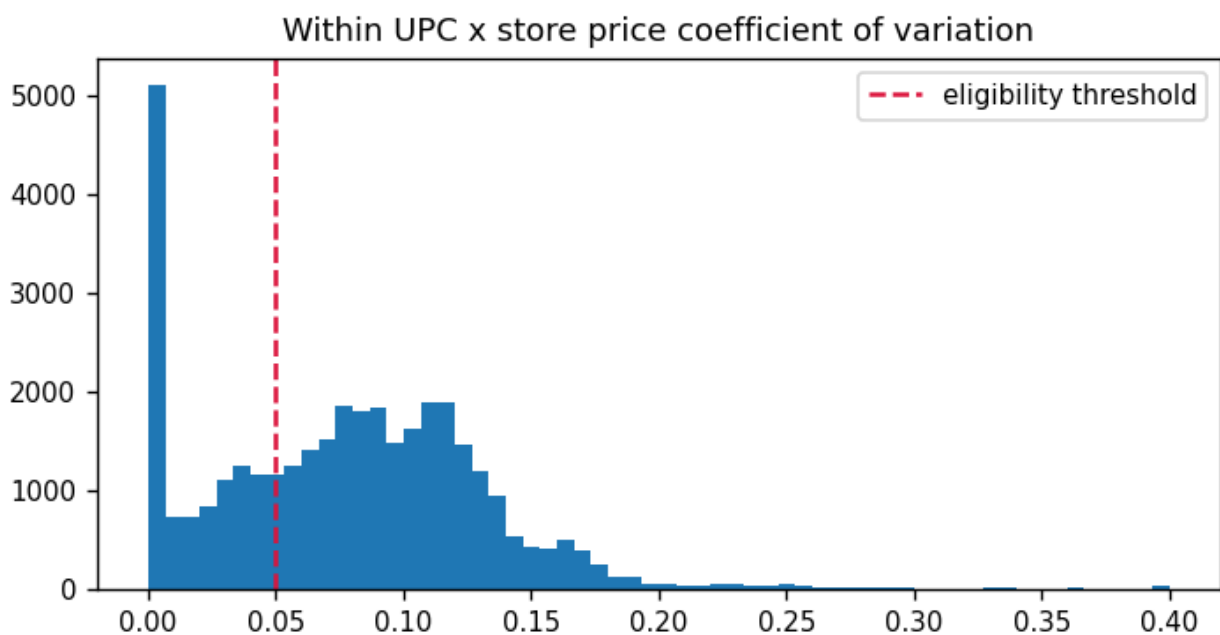


Figure 3 — Distribution of the within-series price coefficient of variation across all 36,443 UPC × store series, with the eligibility threshold ($CV \geq 0.05$) marked. Source: `scripts/run_edu.py`.

16. Cross-store pricing variation

metric	value
median cross-store price spread per UPC	13.63%
p90 cross-store price spread	27.45%
median cross-store coefficient of variation	0.0307

16.1 What this establishes

Dominick's genuinely priced by store zone. For a typical cereal UPC, the highest-priced store charges about **14% more** than the lowest-priced store in the same week, and for the top decile of products that gap reaches **27%**.

16.2 Why UPC × store is the right modelling grain

39. **It is the decision grain** — the chain set prices per store zone, so a recommendation must be per store.
40. **Ignoring it would bias the elasticity.** Cross-store price differences reflect local demand and competition, not exogenous variation. Pooling stores lets a "high-price store also has different demand" pattern masquerade as a price effect. Adding store fixed effects on top of UPC fixed effects moves the estimate from **-2.289** to **-2.419**.
41. **Price support is store-specific.** The extrapolation guardrail asks "has *this store* ever charged near this price for *this item*?" — a materially tighter and more defensible question than asking it of the chain.
42. **Store-level heterogeneity is large.** Store-level elasticity estimates in this panel run from **-0.947 to +0.297** — a range that includes the wrong sign, which is itself evidence that a store-level "elasticity" is mostly picking up composition.

16.3 The ethical flag this raises

Store identity is a proxy for place, and place correlates with demographics. `docs/RESPONSIBLE_PRICING.md` states plainly that zone pricing is precisely where "price by location" can shade into "charge more where a protected group shops", and lists what a real deployment would require: an explicit review of cross-store price differences against demographic data, a cap on cross-store dispersion for essential goods, and jurisdiction-by-jurisdiction legal sign-off. This project uses store as a decision dimension because the retailer did, and says so rather than treating it as a neutral feature.

17. Cost / AAC stability

metric	value
series with computable AAC	28,970
median within-series AAC coefficient of variation	0.0884
p90 within-series AAC CV	0.1528
share of rows with non-positive AAC	0.00%
share of rows with implausible margin	0.00%

17.1 Cost moves, and it moves *with price*

statistic	value
correlation between price and gross-margin rate	-0.0143
price-change events observed	766,384
share of price changes that are cuts	42.59%
mean margin-rate change on a price cut	-0.0811
mean AAC change on a price cut	-9.16%

Two facts here matter enormously for the optimizer.

Fact 1 — the margin rate is not constant across prices. On a price cut, the gross-margin *rate* falls by 8.1 percentage points on average. So you cannot hold the margin percentage fixed and re-price: doing so would mechanically manufacture profit at every candidate price. This is exactly the error the project's cost-handling rule exists to prevent (§48), and it is enforced by `tests/test_simulation.py::test_cost_is_held_fixed_across_price_grid`.

Fact 2 — AAC itself falls 9.2% on price cuts. Promotions in this era were substantially trade-funded: the retailer's acquisition cost dropped at the same time as the shelf price. That means the *contemporaneous* AAC of the week being priced is not merely an outcome in the temporal sense (§20) — it is an outcome that is **causally entangled with the pricing decision itself**.

17.2 Why contemporaneous AAC must not be used at recommendation time

Suppose the optimizer used week- t AAC when recommending week- t price. Then:

- it would be using information that does not exist at decision time (temporal leakage), **and**
- the cost it used would already reflect the promotion that the price change is supposed to represent — a circular dependency where a price cut looks cheap because the data already knows a trade deal accompanied it.

17.3 The leakage-safe design that follows

$\text{decision_time_unit_cost}(u, s, t) = \text{estimated_unit_aac}(u, s, t-1)$, forward-filled within the series

```
d["_lag_aac"] = g["estimated_unit_aac"].shift(1)
d["decision_time_unit_cost"] = d.groupby(SERIES_KEYS, observed=True, sort=False)["_lag_aac"].ffill()
```

python

The analyst knows what the item cost last week; they do not know what the accounting system will report for the week being priced. If no prior AAC exists, the optimizer **refuses** to optimise gross profit and returns the `COST_UNAVAILABLE` reason code rather than inventing a cost.

That this rule actually holds on the real data is proven, not assumed, by three independent checks in `reports/10_COST_LEAKAGE_AUDIT.md` (§23.4): 4,671,333 rows compared with **0 mismatches**; a future-cost poisoning test that changed nothing at or before the poisoned week and did change the following week; and 19 real recommendations of which **0** changed when future costs were poisoned.

A useful diagnostic from that audit: the decision-time cost happens to equal the same week's AAC in **55.44%** of rows. That is not leakage — it is just how often AAC did not move week over week. The remaining 44.56% is where the distinction bites, and it is why the poisoning test (which forces a difference) is the check with teeth.

18. Demand, time and promotions

18.1 Demand distribution

statistic	raw move
mean	14.08 units
median	8 units
p25 / p75	0 / 17
p99	103
max	18,688
standard deviation	50.59

Weekly unit demand is **extremely heavy-tailed**: the mean is 14 and the maximum is 18,688, a ratio of 1,327.

This shape drives several design decisions:

- **WAPE** rather than MAPE as the headline metric — MAPE is undefined at zero and explodes on small denominators (Appendix L).
- **log1p transforms** on the lagged-demand features for the linear model (`LOG1P_FEATURES` in `demand_model.py`), because a multiplicative demand specification implies working in logs anyway.

- **A prediction cap** at 5× the maximum training demand, because exponentiating a linear prediction can explode on extreme inputs (§37.6).
- **Poisson loss** for the boosted model, which is the right likelihood for weekly counts and guarantees non-negative predictions.

18.2 Temporal patterns

The weekly aggregate series (`eda_weekly_units.png`) shows annual seasonality plus a mild upward drift over eight years. The engine captures this through `week_of_year`, `month`, `quarter`, an explicit `time_index`, and — for the elasticity regressions and the ridge model — the trigonometric pair

$$\sin((2\pi w)/52), \quad \cos((2\pi w)/52)$$

which encodes annual periodicity in two smooth features rather than 52 dummies.

The monitoring artifact quantifies the drift between the training reference window (weeks 2–257) and the test window (343–399):

feature	PSI	band	reference mean → current mean
<code>time_index</code>	12.412	large shift	133.6 → 371.4
<code>series_age_weeks</code>	1.004	large shift	96.1 → 191.5
<code>price_vs_series_reference</code>	0.536	large shift	0.0486 → 0.0039
<code>series_reference_price</code>	0.363	large shift	\$2.886 → \$3.247
<code>roll_mean_price_4</code>	0.149	moderate	\$3.018 → \$3.231
<code>log_price</code>	0.107	moderate	1.068 → 1.145
<code>decision_time_unit_cost</code>	0.090	stable	\$2.545 → \$2.538
<code>recorded_promotion_flag</code>	0.000	stable	0.0699 → 0.0764

The two "large shift" features at the top are **expected by construction** — `time_index` and `series_age_weeks` are monotone in time, so any chronological split guarantees a large PSI. Reporting them anyway is deliberate: a drift monitor that suppresses its own trivially-true alarms is a monitor nobody audits. The interesting rows are the price ones: the **reference price level rose about 12.5%** between the two windows, which is real regime drift and is the single best argument for periodic retraining.

`time_index` being a large, unbounded, monotone drifter is also exactly why it is **excluded from the linear model** (`LINEAR_EXCLUDED_FEATURES`) — see §37.5.

18.3 Promotional periods

statistic	recorded promotion	no recorded promotion
mean weekly units	54.46	16.49
mean effective unit price	\$2.544	\$3.161

Promotion weeks sell **3.3× the units** at **80% of the price**. Note what this comparison is and is not: a naive elasticity computed from these two numbers would give roughly

$$(\ln(54.46/16.49))/(\ln(2.544/3.161)) = (1.195)/(-0.217) \approx -5.5$$

which is far steeper than any controlled estimate in this project. That gap is the promotion-bundling problem in a single line: the 3.3× volume lift is a price cut *plus* a feature ad *plus* display, and attributing all of it to price would overstate price sensitivity by roughly a factor of three.

This report makes no causal claim from the table above. It is an association between two subsamples that differ in many unobserved ways.

18.4 Store and product heterogeneity

dimension	median	p10 / min	p90 / max
per-UPC elasticity (355 UPCs, Phase D)	-2.358	-4.094	-0.240
per-store elasticity (93 stores)	-0.385	-0.947	+0.297
within-series price CV (36,443 series)	0.0791	—	—
within-series AAC CV (28,970 series)	0.0884	—	0.1528 (p90)
cross-store price spread per UPC	13.63%	—	27.45% (p90)

Product heterogeneity is large and meaningful. Store "elasticities" are centred near -0.39 and include positive values, which is a strong hint that a store-level own-price elasticity is largely composition: a store is a mix of products, and its aggregate price index moves with assortment as much as with pricing.

18.5 Major outliers, named rather than removed

observation	value	disposition
max weekly units	18,688	kept; prediction cap applies at $5\times$ max training demand
max effective unit price	\$26.02 (TONY THE TIGER T-SHIRT)	kept; a non-cereal item in the Cereals file
min effective unit price	\$0.05	kept
WoW price jumps $> \pm 50\%$	77,152 events (10.07%)	kept; bounded downstream by the extrapolation guardrail
UPCs with negative total gross profit	6	kept

The demo context used throughout this report contains a textbook example: in week 395, CAPN CRUNCH JUMBO CR at store 86 sold **17,824 units at \$1.50** — against 11 to 25 units in the surrounding weeks at \$3.35 — producing an observed gross profit of **-\$18,769**. That single store-week is a deep promotional event with negative accounting margin, it is retained in the data, and it is precisely the kind of observation that makes robust inference (\$29) and a materiality threshold (\$53.9) necessary rather than decorative.

18.6 No causal claims from this part

Everything in Part III is descriptive. The price-versus-demand scatter (`eda_price_vs_demand.png`) is an association between two jointly determined variables, and the dashboard labels it as such in-line: *"The scatter above is an association, not a demand curve: price and demand are jointly determined by promotions, seasonality and retailer decisions."*

PART IV — FEATURE ENGINEERING AND LEAKAGE

The rule enforced throughout `src/pricing_engine/features/build.py`:

A feature may only use information that a pricing analyst would already have when setting next week's price for a given UPC in a given store.

The decision being modelled is precise: **at the end of week $t-1$, choose the price for UPC u in store s for week t .**

19. Prediction target

$$y = \text{move} \quad (\text{units sold in week } t)$$

19.1 Why units and not money

Revenue and gross profit are **mechanical functions of the target**:

$$\text{revenue} = p \cdot \text{move}, \quad \text{gross_profit} = (p - c) \cdot \text{move}$$

Using either as a predictor is direct target leakage — a model given `revenue` would recover `move` exactly by dividing by the price it was also given. This is leakage channel #1 in `docs/DATA_LEAKAGE_AUDIT.md` and is enforced structurally: `FEATURE_COLUMNS` is a module-level constant that simply does not contain them, and `tests/test_features.py::test_no_outcome_columns_are_features` asserts it.

19.2 Why deriving money afterwards is what makes the counterfactual coherent

This is the deeper reason and it is worth stating carefully. The engine predicts *units* at a candidate price and then computes money from the candidate price and a fixed decision-time cost:

$$\hat{R}(p) = p \cdot \hat{Q}(p), \quad \hat{G}(p) = (p - c) \cdot \hat{Q}(p)$$

If instead the model predicted *revenue* directly at a candidate price, the price would appear on both sides of the relationship in a way the model could not disentangle, and the counterfactual would be incoherent: a "revenue model" asked what revenue would be at a 10% higher price has no principled way to separate the mechanical price effect from the behavioural volume effect.

Predicting units keeps exactly one behavioural quantity in the model and puts all the arithmetic outside it, where it is auditable.

19.3 The same logic applied to margin

The week- t accounting margin `gross_margin_rate` is also an outcome, and it is *mechanically tied to that week's price*. Using it would let the model recover the price and would also fix the margin percentage during simulation — manufacturing profit at any candidate price. It is excluded, and replaced at decision time by `decision_time_unit_cost` (\$17.3, \$48).

20. Feature availability framework

Every column is classified into one of four availability classes.

class	definition
KNOWN BEFORE DECISION	fully determined by weeks $\leq t-1$
KNOWN AT DECISION TIME	part of the decision itself, or planned in advance
KNOWN ONLY AFTER OUTCOME	realised during week t — never an input
REQUIRES ASSUMPTION	available only under a stated assumption

20.1 The classification (from docs/FEATURE_AVAILABILITY.md)

feature	class	justification
effective_unit_price	KNOWN AT DECISION TIME	it <i>is</i> the decision variable
log_price	KNOWN AT DECISION TIME	transform of the decision variable
price_vs_last_week	KNOWN AT DECISION TIME	candidate price vs lag_price_1
price_vs_series_reference	KNOWN AT DECISION TIME	candidate price vs expanding median of past prices
price_vs_recent_mean	KNOWN AT DECISION TIME	candidate price vs mean of previous 4 weeks
recorded_promotion_flag	REQUIRES ASSUMPTION	promotion calendars are agreed in advance; the <i>code</i> is observed after the fact
lag_move_1..4, roll_mean_move_4/8/13, roll_std_move_4	KNOWN BEFORE DECISION	shifted by ≥ 1 week by construction
lag_price_1, lag_price_2, roll_mean_price_4, series_reference_price	KNOWN BEFORE DECISION	past prices only
lag_promotion_1	KNOWN BEFORE DECISION	last week's promotion code
decision_time_unit_cost	REQUIRES ASSUMPTION	latest known implied AAC, lagged one week and forward-filled
store, upc, com_code, package_size_oz	KNOWN BEFORE DECISION	static identity / metadata
week_of_year, month, quarter, time_index	KNOWN BEFORE DECISION	calendar
series_age_weeks	KNOWN BEFORE DECISION	count of prior observations in the series

20.2 Explicitly excluded

column	class	why excluded
revenue, gross_profit	AFTER OUTCOME	mechanical functions of move
profit, gross_margin_rate (week t)	AFTER OUTCOME	realised accounting margin, also tied to that week's price
estimated_unit_aac (week t)	AFTER OUTCOME	derived from week t margin; replaced by decision_time_unit_cost
sale / recorded_promotion_type raw code of week t	partially AFTER OUTCOME	only the binary planned flag is used, under a documented assumption
ok	AFTER OUTCOME	a data-quality verdict; used for filtering only
qty, price (bundle form)	superseded	replaced by effective_unit_price = price / qty

20.3 Why the classification matters more here than in ordinary forecasting

In a pure forecasting project, leakage inflates a metric. In a *pricing* project it does something worse: it produces **confidently wrong prices**. A model that has already seen how the week's accounting margin turned out will happily recommend a price that "works" only because the answer was in the input. Since nobody ever observes the counterfactual, that error is invisible offline forever. The only defence is a contract enforced at build time and tested.

20.4 The two assumptions, stated as assumptions

43. **The promotion calendar is known in advance.** Realistic (calendars are agreed weeks ahead) but not derivable from this data. Impact bounded: 7.35% of rows, with error reported separately by promotion state.
44. **Last week's AAC is a usable proxy for this week's decision-time cost.** Temporally safe by construction and proven so (§23.4). Economically it is a proxy: AAC lags replacement cost (§8.3).

Neither is hidden; both are listed in `docs/DATA_LEAKAGE_AUDIT.md` and in the limitations register (Appendix K).

21. Temporal features

feature	construction	purpose
<code>week_of_year</code>	ISO week from <code>week_start_date</code>	annual seasonality (tree model)
<code>month, quarter</code>	from <code>week_start_date</code>	coarse seasonality
<code>sin52, cos52</code>	$\sin(2\pi w/52), \cos(2\pi w/52)$	smooth annual periodicity in two features instead of 52 dummies
<code>time_index</code>	the raw Dominick's week index as int32	secular trend (tree model only)
<code>series_age_weeks</code>	<code>groupby(upc, store).cumcount()</code>	how established this series is

21.1 Why the raw week index is never used bare

The integer `week` is decoded to a calendar date before anything else, so seasonality is real calendar seasonality rather than an arbitrary modulus. And `time_index` is **excluded from the linear model** entirely (`LINEAR_EXCLUDED_FEATURES = ("time_index",)`), because a standardised unbounded trend extrapolated into a future window explodes once the log-target prediction is exponentiated. That is not a hypothetical: the observed WAPE before the fix was **27.9** (DECISIONS #16). Trees saturate at the edge of their training range instead, so the boosted model keeps it.

21.2 Why `sin52/cos52` and not week-of-year dummies

52 dummies would cost 52 parameters, would treat weeks 52 and 1 as unrelated, and would be estimated from ~7 observations per week per series. The trigonometric pair costs two parameters, is continuous across the year boundary, and is the standard encoding for a single dominant annual harmonic. The same pair is used in the elasticity control set, so the seasonality control is identical in both places.

22. Lag and rolling features

22.1 The features

feature	formula	note
<code>lag_move_k</code> , $k \in \{1,2,3,4\}$	y_{t-k} within <code>(upc, store)</code>	recent demand level
<code>roll_mean_move_w</code> , $w \in \{4,8,13\}$	$1/w \sum_{j=1}^w y_{t-j}$, <code>min_periods=2</code>	smoothed demand level at three horizons
<code>roll_std_move_4</code>	$\text{sd}(y_{t-1}, \dots, y_{t-4})$, <code>min_periods=2</code>	recent demand volatility
<code>lag_price_1, lag_price_2</code>	p_{t-1}, p_{t-2}	recent price level
<code>roll_mean_price_4</code>	$1/4 \sum_{j=1}^4 p_{t-j}$, <code>min_periods=1</code>	recent price level, smoothed
<code>series_reference_price</code>	expanding median of $p_{t-1}, p_{t-2}, \dots$	the series' "normal" price
<code>lag_promotion_1</code>	promotion flag at $t-1$	promotion carry-over / stock-piling
<code>decision_time_unit_cost</code>	aac_{t-1} , forward-filled	\$17.3

Relative-price features, all computed **from the candidate price** and the fixed context:

feature	formula
price_vs_last_week	$p / p_{t-1} - 1$
price_vs_series_reference	$p / \text{series_reference_price} - 1$
price_vs_recent_mean	$p / \text{roll_mean_price_4} - 1$

22.2 Shifting: the correct and the leaking implementation

The classic error is to compute a rolling window on the raw target:

```
# WRONG – the window includes week t itself
d["roll_mean_move_4"] = g["move"].rolling(4).mean()
```

python

At week t that window covers $\{y_t, y_{t-1}, y_{t-2}, y_{t-3}\}$. It contains the answer. A model given this feature can achieve a spectacular WAPE and is useless.

The implementation used here shifts **first**, then rolls:

```
shifted_move = g[TARGET].shift(1)
d["_shifted_move"] = shifted_move
gs = d.groupby(SERIES_KEYS, observed=True, sort=False)["_shifted_move"]
for w in windows:
    d[f"roll_mean_move_{w}"] = gs.transform(lambda s, w=w: s.rolling(w, min_periods=2).mean())
d["roll_std_move_4"] = gs.transform(lambda s: s.rolling(4, min_periods=2).std())
```

python

At week t the window now covers $\{y_{t-1}, \dots, y_{t-4}\}$. Week t never sees its own value.

22.3 A worked example

Take a series with weekly units $[10, 20, 30, 40, 50]$ for weeks 1–5.

week	units	leaking <code>rolling(4).mean()</code>	correct <code>shift(1).rolling(4).mean()</code>
1	10	—	—
2	20	—	10.0
3	30	—	15.0
4	40	25.0 = mean(10,20,30,40)	20.0 = mean(10,20,30)
5	50	35.0 = mean(20,30,40,50)	25.0 = mean(10,20,30,40)

At week 5 the leaking version already contains 50 — the value being predicted.

`tests/test_features.py::test_rolling_windows_are_shifted` and

`test_lag_features_never_use_the_current_week` assert this numerically on synthetic fixtures where the answer is known in closed form.

22.4 The expanding reference price

`series_reference_price` is an **expanding median of shifted prices**:

```
d["_shifted_price"] = g["effective_unit_price"].shift(1)
gp = d.groupby(SERIES_KEYS, observed=True, sort=False)["_shifted_price"]
d["series_reference_price"] = gp.transform(lambda s: s.expanding(min_periods=1).median())
```

python

Two design points. **The shift** stops the candidate price from being embedded in its own reference (leakage channel #4). **The median** rather than the mean makes the reference robust to exactly the deep-promotion weeks documented in §18.5 — a single \$1.50 week in a \$3.35 series barely moves the median and would drag a mean noticeably.

22.5 Rows dropped for insufficient history

quantity	value
rows in the feature table	4,707,776
rows usable for training	4,671,333
rows dropped for no history	36,443
features built	29
build time	73.4 s

The 36,443 dropped rows are exactly the **first observation of each of the 36,443 UPC × store series** — the one row per series that has no `lag_move_1` and no `lag_price_1`. `training_frame()` requires both:

```
if require_lags:
    d = d[d["lag_move_1"].notna() & d["lag_price_1"].notna()]
```

python

Keeping them would mean imputing lag values, and the imputer's statistics are computed over the whole training set — which is a subtle, real leakage channel (#6). Dropping 0.77% of rows is the cheaper fix.

22.6 Missing-value structure in the surviving features

feature	missing share
lag_move_1, lag_price_1, price_vs_*, roll_mean_price_4, series_reference_price, decision_time_unit_cost, lag_promotion_1	0.774%
lag_move_2, roll_mean_move_4/8/13, roll_std_move_4, lag_price_2	1.536%
lag_move_3	2.282%
lag_move_4	3.017%
package_size_oz	0.705%

The pattern is exactly what a lag structure predicts — the deeper the lag, the more series-openings it misses. `package_size_oz` is missing where the metadata `size` string does not parse to a number.

The linear model imputes with the **median** and then standardises (`SimpleImputer(strategy="median")` → `StandardScaler`); the boosted model handles missing values natively. Neither imputer sees the target.

22.7 The single shared price-feature implementation

This is the most important engineering decision in the feature layer. `recompute_price_features(frame, candidate_price)` is the **only** place price-dependent features are computed, and it is called by:

- the training-table build,
- the counterfactual simulator (`simulate_price_grid`, `simulate_many`),
- the optimizer (before scoring the current price),
- the hybrid model (when evaluating the baseline at p_0),
- the API `/predict-demand` and `/simulate-prices` endpoints,
- the dashboard price simulator.

If the simulator had its own copy, a candidate price could be scored with a stale `log_price` or a stale `price_vs_last_week` — the model would see a new price in one feature and the old price in three others. That is leakage channel #7, and the single-implementation rule makes it structurally impossible.

The function also refuses bad input rather than coercing it:

```
if np.any(price <= 0):
    raise FeatureError("candidate prices must be strictly positive")
```

python

23. Leakage tests

Documentation asserting that a pipeline is leak-free is worth very little. Tests that would *fail* if it were not are worth a great deal. This project has both.

23.1 The twelve enumerated channels

#	channel	control	enforcing test
1	target-derived features (revenue, gross_profit)	never in FEATURE_COLUMNS (module constant)	test_no_outcome_columns_are_features
2	contemporaneous accounting margin / AAC	replaced by lagged, forward-filled decision_time_unit_cost	test_decision_time_cost_is_lagged_not_contemporaneous
3	unshifted rolling windows	all windows computed on the already-shifted series	test_rolling_windows_are_shifted, test_lag_features_never_use_the_current_week
4	expanding reference price including today	expanding median of shifted prices	test_series_reference_price_uses_only_past_prices
5	random train/test split on a time-indexed panel	chronological split by week index; no shuffling anywhere	test_temporal_split_is_chronological_and_disjoint
6	series with no history entering training	training_frame requires lag_move_1 and lag_price_1; 36,443 rows dropped	test_training_frame_drops_rows_without_history
7	price-feature staleness in simulation	one shared recompute_price_features	test_recompute_price_features_changes_only_price_features, test_price_features_recomputed_for_each_candidate
8	cost moving with the candidate price	cost held fixed across the grid	test_cost_is_held_fixed_across_price_grid
9	model selection on the test period	selection on validation only; test scored once	training log in artifacts/metrics/model_metrics.json records both
10	backtest using a full-sample model	backtest scores only weeks after the loaded model's training window	windows stated in reports/06_BACKTEST.md
11	elasticity fitted on evaluation outcomes	estimation restricted to training weeks 2–257; window stored in the table metadata; the script aborts if any input row exceeds it	test_build_elasticity_table_marks_sources_and_records_the_window
12	future cost information	lagged, forward-filled AAC, verified three ways on the real data	reports/10_COST_LEAKAGE_AUDIT.md, tests/test_cost_leakage.py (5 tests)

Channel 11 deserves emphasis. It is the leakage channel that a project which "separates forecasting from price response" is most likely to introduce accidentally: the elasticity is *also* a fitted quantity, and fitting it on the full panel would smuggle validation and test outcomes into every recommendation scored there. The guard is not just a convention — the script raises rather than run, and the approved window is written into the elasticity table's metadata (meta_training_weeks_low = 2, meta_training_weeks_high = 257).

23.2 Poisoning tests — why they are stronger than assertions

A test that checks "the lag column equals the previous row" verifies the implementation you wrote. A **poisoning** test verifies something better: it corrupts a *future* value and asserts that nothing at or before the decision point changes. If any hidden path reads the future, the test fails — including paths you did not think of.

The feature-level poisoning test lives in `tests/test_features.py`; the cost-level one is run on the **real 4.7M-row panel**, not on a fixture.

23.3 Cost-leakage audit — three independent checks on real data

`scripts/audit_cost_leakage.py` → `artifacts/metrics/cost_leakage_audit.json`,
`reports/10_COST_LEAKAGE_AUDIT.md`.

check	design	result
1. Identity	for every usable row, assert <code>decision_time_unit_cost</code> equals the series' most recent prior <code>estimated_unit_aac</code>	4,671,333 rows compared, 0 mismatches — pass
2. Future poisoning	overwrite the AAC of 25 real series from week 398 onward with corrupted values; rebuild features	decision-time cost unchanged at or before the poisoned week over 5,537 compared rows; did change the following week (so the test has teeth) — pass
3. Recommendation-level	regenerate real recommendations for week 397 for 19 series with poisoned future costs	0 of 19 recommendations changed — pass
		all_passed: true

Check 2's second half is the part that makes the audit credible. A poisoning test that finds no change *anywhere* has not proven safety — it has proven the poison never landed. This one confirms the poison landed in week 399 and did not propagate backwards.

23.4 What the audit does and does not prove

It proves **temporal availability**: no future information reaches a decision.

It does **not** prove **economic correctness**: the lagged AAC may still be a poor proxy for the true replacement cost at decision time (§8.3). The report states this distinction explicitly rather than letting "cost audit passed" be read as "the cost is right".

23.5 What no leakage control can buy

From `docs/DATA_LEAKAGE_AUDIT.md`, verbatim:

No leakage control turns an observational panel into an experiment. Even a perfectly leak-free model estimates $P(\text{units} \mid \text{price}, \text{context})$, not $P(\text{units} \mid \text{do}(\text{price}))$.

That sentence is the boundary between Part IV and Part V.

PART V — PRICE ELASTICITY

24. Economic definition

Own-price elasticity of demand is the ratio of proportional responses:

$$E = (\% \Delta Q) / (\% \Delta P) = (\partial \log Q) / (\partial \log P)$$

The logarithmic form is the useful one: it is unit-free, scale-free, and it is exactly the coefficient of a log-log regression.

regime	condition	a 1% price increase ...	revenue	gross profit
elastic	$ E > 1$	loses more than 1% of volume	falls	may still rise (depends on margin)
unit elastic	$ E = 1$	loses exactly 1% of volume	unchanged	rises whenever $c > 0$
inelastic	$ E < 1$	loses less than 1% of volume	rises	rises without bound in the CE model

24.1 Why the inelastic case is dangerous rather than convenient

For constant-elasticity demand $Q = k \cdot p^E$, gross profit is $(p - c) \cdot k \cdot p^E$. Setting the derivative to zero gives

$$p^* = c \cdot E / (1 + E)$$

which is only defined and positive for $E < -1$. For $-1 < E < 0$ the derivative never vanishes: **profit increases monotonically in price**, and the model's answer is "charge infinity". Any engine built on a constant-elasticity response must therefore have a constraint layer, not as a nicety but as the only thing standing between an inelastic estimate and a nonsensical recommendation.

This project encodes exactly that behaviour and tests it:

`tests/test_optimizer.py::test_constant_elasticity_fixture_pushes_price_to_the_upper_bound` asserts that an inelastic fixture recommends the feasible upper bound and raises the `PRICE_CHANGE_LIMIT` reason code.

It also matters empirically here: **p90 of the shrunk product elasticity distribution is -1.169**, and the raw per-product distribution reaches -0.331 at the least elastic end. Products near or above -1 exist in this catalogue, and for them the guardrails *are* the answer.

24.2 The clip band

`configs/config.yaml` → `pricing_response` clips $|\epsilon|$ into **[0.2, 6.0]** before use. In the shipped table **0 products were clipped**, so the band is currently a safety net rather than an active transformation — but it means a pathological estimate can never produce $p^* = c \cdot \epsilon / (1 + \epsilon)$ with ϵ arbitrarily close to -1 and an explosive optimum.

25. Arc elasticity

25.1 The midpoint formula

For two observations (P_1, Q_1) and (P_2, Q_2) :

$$E_{\text{arc}} = ((Q_2 - Q_1) / ((Q_1 + Q_2) / 2)) / ((P_2 - P_1) / ((P_1 + P_2) / 2))$$

The midpoint denominators make the estimate symmetric: going from \$3.00 to \$3.30 and back gives the same magnitude, which the naive percentage form does not.

25.2 Applied to consecutive weeks of this panel

statistic	value
consecutive week pairs available	4,509,599
usable pairs (both price and quantity move)	716,858
pairs where the formula is undefined	84.10%
median arc elasticity	-2.822
trimmed mean (1st–99th percentile)	-4.679
p25 / p75	-8.506 / +0.256
share negative	70.28%
share below -1 (elastic)	67.72%

25.3 What this tells us, and what it cannot

Useful: the median (-2.82) is negative and elastic, and 67.7% of usable pairs are elastic. The sign and rough magnitude are consistent with the controlled regressions.

Not useful for pricing, for four reasons:

45. **84.1% of pairs are undefined** — the price did not move, or quantity did not, so the ratio has a zero denominator or numerator. Restricting to weeks where the price moved is already a selected sample (§45).
46. **The interquartile range spans the sign.** p25 = -8.51 and p75 = **+0.26**. Any statistic with an IQR that wide is dominated by noise in the denominators.
47. **It controls for nothing.** No seasonality, no promotion, no trend, no product or store effects. Every confounder in Part III's list is loose.
48. **It is a two-point estimate.** A single week-to-week comparison has no ability to separate the price effect from whatever else changed that week.

The arc elasticity's real job in this project is as a **reference point**: it is the crudest possible estimator, it is reported honestly, and the fact that the controlled estimates land in the same neighbourhood (-1.9 to -2.4) is mild reassurance that the log-log machinery is not producing nonsense.

26. Log-log regression

26.1 Derivation

Assume a multiplicative demand system:

$$Q = A \cdot P^{\beta_1} \cdot \prod_k X_k^{\gamma_k} \cdot e^u$$

Take logs:

$$\log Q = \beta_0 + \beta_1 \log P + \sum_k \gamma_k \log X_k + u, \quad \beta_0 = \log A$$

Then, directly from the definition of elasticity:

$$(\partial \log Q) / (\partial \log P) = \beta_1$$

So **β_1 is the elasticity**, provided the functional form is right and the error is orthogonal to $\log P$.

26.2 The assumptions being bought

assumption	what breaks if it fails
constant elasticity over the price range	the estimate is a local average; extrapolation misleads
$E[u \cdot \log P] = 0$ (exogeneity)	the coefficient is not the price effect at all (§33)
additivity of the log-controls	omitted interactions load onto β_1

assumption	what breaks if it fails
$Q > 0$	$\log Q$ undefined at zero
independent errors	inference is wrong even if the point estimate is fine (§29)

The first is the reason the extrapolation guardrail exists. The second is the subject of §33. The fifth is the subject of §29.

26.3 Zero handling in this project

`prepare_loglog_frame(..., zero_handling="drop")` drops zero-sales rows. In this panel that removes **0 rows (0.00%)** — because the zero-price exclusion already removed every zero-sales row (§9.7). The handling is therefore inert here, but it is recorded in every fit's `zero_handling` field so the choice is visible: "drop (dropped 0 zero-sales rows (0.00%); fitted on a random subsample of 1,200,000 rows (seed=42))".

26.4 Subsampling

The pooled specifications are fitted on a **seeded random subsample of 1,200,000 rows** for tractability. This is recorded in every fit's metadata and listed as limitation #22. The one specification that was run on the **full 3,206,437-row training sample** (specification C in the inference audit) gives **−2.0443** against **−2.0289** on the subsample — a difference of 0.015, far inside any of the standard errors. The subsampling is not driving the result.

27. Naive elasticity result

27.1 The number

$$\log Q = \beta_0 + \beta_1 \log P + u, \quad \text{no controls}$$

estimate	−0.348
standard error	0.00321
95% CI	[−0.3543, −0.3417]
R ²	0.0126
n	1,200,000

27.2 Why this number is wrong, mechanism by mechanism

Taken literally, −0.348 says cereal demand is **inelastic** — a 10% price rise costs only 3.5% of volume — which would imply the retailer should raise prices essentially everywhere without bound. Every controlled specification says the opposite. Four mechanisms explain the gap.

(a) Composition across products (the dominant one). Different cereals sit at different price *levels* for reasons that have nothing to do with elasticity — brand equity, package size, ingredient cost. A premium \$4.50 granola sells fewer units than a \$2.50 store-brand corn flake. Pooling them, the regression sees "high price, low volume" and reads a negative slope — but it is reading *between-product* differences, not the response of any product to its own price change. Adding UPC fixed effects moves the estimate from **−0.348 to −2.289**, a factor of 6.6. That single step is the entire story.

(b) Composition across stores. High-price zones are systematically different places. Adding store fixed effects on top moves it further, to **−2.419**.

(c) Omitted promotion. Promotion weeks combine low prices with high volume for bundled reasons. Uncontrolled, this *steepens* the apparent response. Controlling for it (plus seasonality and trend) pulls the estimate back to **-1.909** — the opposite direction from (a) and (b), which is exactly what you expect: composition biases toward zero, promotion bundling biases away from it.

(d) Simultaneity. Price and quantity are jointly determined. Without an instrument, OLS estimates a mixture of the demand and supply relations.

27.3 The R^2 is the tell

$R^2 = 0.0126$. Log price alone explains **1.3%** of the variance in log demand. Adding UPC fixed effects takes it to 0.132; adding store effects to 0.197; adding controls to 0.227. Almost all the explanatory power in this panel is *who and where and when*, not *what price*. Any elasticity read off a model with an R^2 of 0.013 is a slope fitted through a cloud.

27.4 The general lesson

This is the cleanest demonstration in the whole project that **association is not causation**, and it is worth carrying into any interview: a seven-fold change in a headline coefficient, produced purely by adding controls that nobody could argue against, on real data, with no simulation and no storytelling.

28. Fixed-effects specifications

28.1 The ladder

`scripts/run_elasticity.py` → `artifacts/metrics/elasticity.json`. All on the same seeded 1.2M-row subsample.

#	specification	elasticity	SE	95% CI	R^2	Δ vs previous
M1	naive pooled	-0.348	0.0032	[-0.354, -0.342]	0.0126	—
M2	+ UPC fixed effects	-2.289	0.0072	[-2.304, -2.275]	0.1315	-1.941
M3	+ UPC × store fixed effects	-2.419	0.0711	[-2.558, -2.279]	0.1966	-0.129
M4	+ promotion, seasonality, trend	-1.909	0.0690	[-2.045, -1.774]	0.2268	+0.510

(Standard errors in this table are the clustered-by-UPC ones produced by `fit_LogLog`; §29 replaces them with a full covariance audit.)

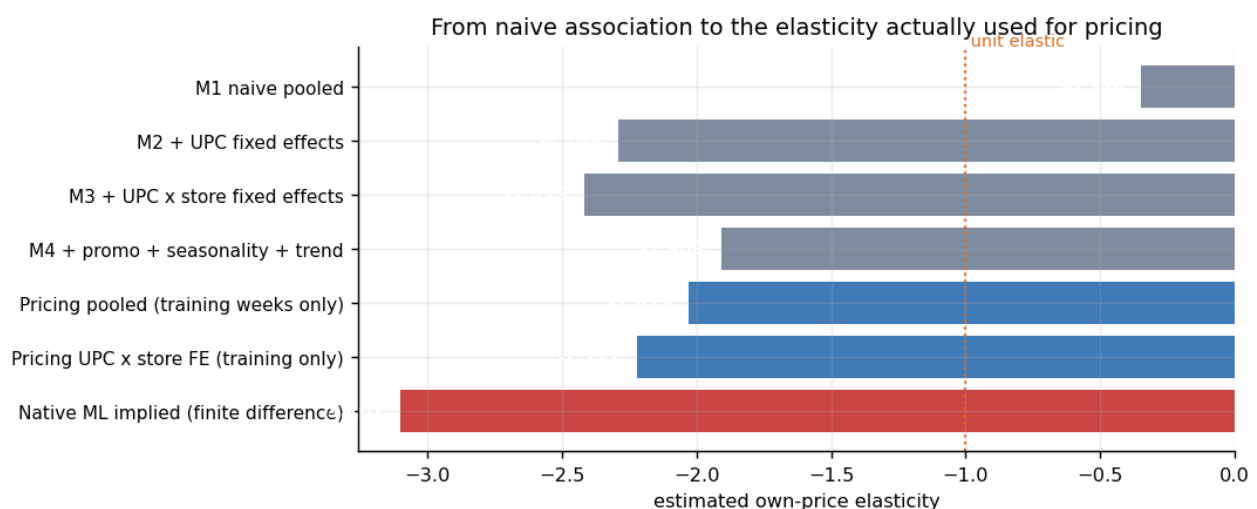


Figure 4 — Every elasticity estimate in this project on one axis. Grey = Phase D analysis specifications; blue = the pricing estimates fitted on training weeks only (the numbers the engine uses); red = the forecaster's own implied price response. Source: [artifacts/metrics/elasticity.json](#), [elasticity_estimation.json](#), [price_response.json](#).

28.2 What each step absorbs

UPC fixed effects (M2). A separate intercept per product absorbs every time-invariant product characteristic — brand, format, base demand level, base price level. The coefficient is now identified from *within-product* price movement.

Mechanically, the within transformation subtracts product means:

$$\log Q_{it} - \text{mean}(\log Q_i) = \beta_1(\log P_{it} - \text{mean}(\log P_i)) + u_{it}$$

UPC × store fixed effects (M3). Absorbs the store dimension as well, so the estimate uses only within-store, within-product variation. The move is small (−2.289 → −2.419) but the *standard error grows 10×* (0.0072 → 0.0711), which is the honest cost of throwing away between-store variation.

In the per-product estimator the same idea is implemented explicitly by demeaning within store codes before the regression:

```
store_codes = pd.factorize(sub["store"])[0]
if n_stores > 1:
    X = _demean(X, store_codes)
    y = _demean(y.reshape(-1, 1), store_codes).ravel()
```

python

Promotion + seasonality + trend (M4). Adds `recorded_promotion_flag`, `sin52`, `cos52` and a linear `trend = week/100`. This moves the estimate **back toward zero**, from −2.419 to −1.909.

28.3 Why M4 moves in the opposite direction, and why that is reassuring

Steps M2 and M3 remove *composition* bias, which pushed the naive estimate toward zero. Step M4 removes *promotion-bundling* bias, which pushed it away from zero. The two biases have opposite signs, and both corrections behave as theory predicts. If M4 had steepened the estimate further, something would have been wrong with the story.

28.4 The promotion split

Estimated separately with UPC × store fixed effects:

subsample	elasticity	SE	n	R ²
recorded promotion	−2.662	0.0074	345,945	0.2999
no recorded promotion	−1.830	0.0078	1,200,000	0.0987

The 0.83 gap is the bundling problem quantified. Note the caveat from §11.4: the "no recorded promotion" group certainly contains *uncoded* promotions, so −1.830 is itself biased toward the promotion value. The true clean-price elasticity is probably weaker than −1.830, not stronger.

28.5 The specification the engine actually uses

The Phase D ladder above is *analysis*. The number that drives recommendations is fitted separately, on **training weeks 2–257 only**, by `scripts/estimate_elasticity.py`:

specification	elasticity	n
pooled controlled (UPC × store FE + promo/season/trend)	−2.0289	1,200,000
UPC × store FE only	−2.2217	1,200,000

specification	elasticity	n
pooled controlled, full training sample	-2.0443	3,206,437

The pooled controlled figure **-2.029** is the project's headline price sensitivity, the prior for the empirical-Bayes shrinkage, and the fallback for every product without a usable own estimate.

Why it differs slightly from M4's -1.909: M4 is fitted on the *whole panel*, the pricing estimate on the *training window only*. The difference (0.12) is smaller than the two-way clustered standard error (0.106 → 95% half-width 0.21), so the two are statistically indistinguishable — but they are not the same number and this report never uses them interchangeably.

29. Robust and clustered inference

`scripts/audit_elasticity_inference.py` → `artifacts/metrics/elasticity_inference.json`,
`reports/15_ELASTICITY_INFERENCE_AUDIT.md`.

29.1 Why the conventional standard error is wrong here

OLS standard errors assume errors are independent and homoskedastic. In this panel neither holds, for structural reasons that are easy to name:

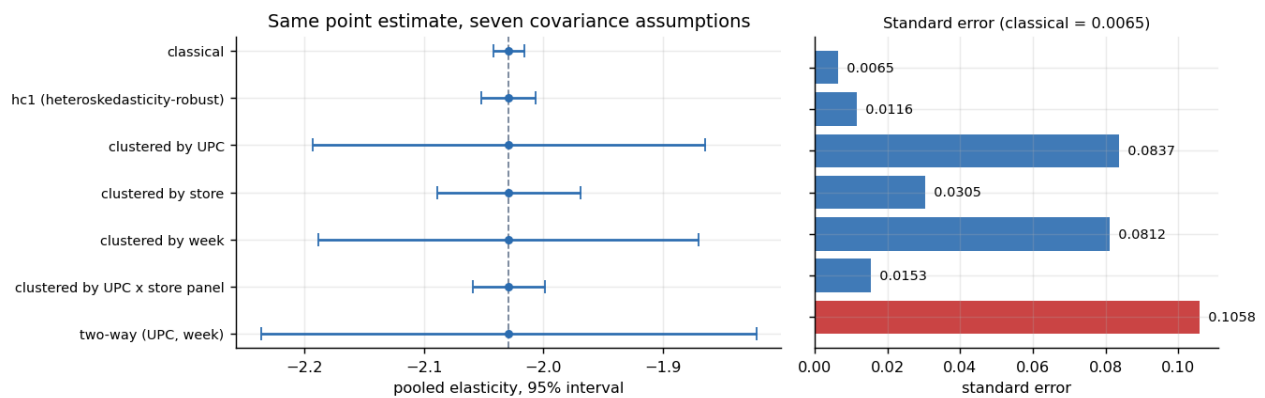
- **Within-store persistence.** A store's local demand shock (a new competitor, a road closure, a demographic shift) persists for months across all products.
- **Category-wide weekly shocks.** Holidays, weather, the chain's promotion calendar and national manufacturer advertising hit *every* product in the same week.
- **Within-product persistence.** A brand's national marketing cycle correlates its residuals across stores and weeks.

Every one of these means the effective number of independent observations is far smaller than the 1.2 million rows.

29.2 The full covariance audit

Specification A: pooled controlled (UPC × store FE + promo/season/trend), 1.2M subsample. Point estimate **-2.0289** in every row — only the precision changes.

covariance assumption	SE	ratio to classical	95% CI	clusters
classical (OLS)	0.00652	1.00×	[-2.042, -2.016]	—
HC1 (heteroskedasticity-robust)	0.01161	1.78×	[-2.052, -2.006]	—
clustered by UPC × store panel	0.01533	2.35×	[-2.059, -1.999]	26,413
clustered by store	0.03045	4.67×	[-2.089, -1.969]	86
clustered by week	0.08121	12.46×	[-2.188, -1.870]	255
clustered by UPC	0.08369	12.84×	[-2.193, -1.865]	369
two-way (UPC, week)	0.10585	16.24×	[-2.236, -1.821]	—



*Figure 5 — The pooled elasticity under classical, HC1 and five clustering schemes. The point estimate never moves; the interval widens by a factor of

49. Source: `artifacts/metrics/elasticity_inference.json`.*

29.3 Reading the ladder

The ordering is itself informative. Clustering at the **UPC × store panel** level (26,413 clusters) barely helps — 2.35× — because it only captures persistence within a single series. Clustering by **week** (12.46×) and by **UPC** (12.84×) each capture a genuinely large correlated dimension. Two-way (UPC, week) captures both and gives **16.24×**.

Concretely, the published 95% interval goes from **[−2.042, −2.016]** — a width of 0.026, which would let you claim you know the category elasticity to the second decimal — to **[−2.236, −1.821]**, a width of 0.415. The second interval is the honest one, and it comfortably contains M4's −1.909 and the full-sample −2.044, while **excluding** −2.42 and −3.10.

29.4 Per-product inference

statistic	value
products with per-UPC inference	262
median HC1 standard error	0.1071
median store-clustered SE	0.1445
median week-clustered SE	0.3832
median inflation ratio (robust vs HC1)	3.38×
p90 inflation ratio	6.83×
share significant at 5% under HC1	90.1%
share significant at 5% under robust SE	75.2%

One product in six that looked significant was not. Week clustering is where the damage is: category-wide weekly shocks are the dominant correlated dimension inside a single product's panel.

29.5 The estimator used in production

```
se = float(np.nanmax([se_store, se_week, se_hc1]))
```

python

The per-product standard error is the **maximum of the store-clustered, week-clustered and HC1** estimates — the conservative reading of the available one-way estimators. Two-way clustering is used at the **pooled** level, where there are enough clusters in both dimensions for the Cameron-Gelbach-Miller subtraction to be stable, and deliberately **not** inside a single product, where it is not (DECISIONS #44). `std_error_hc1` is retained in the output alongside it, and the ratio between them is what drives the shrinkage weights (§31).

29.6 Validation of the estimators themselves

The covariance estimators in `src/pricing_engine/economics/inference.py` were **validated against statsmodels to machine precision** (`reports/15_ELASTICITY_INFERENCE_AUDIT.md`). This matters: hand-rolled clustered covariance code is a classic source of silent factor-of-two errors, and "we checked it against a reference implementation" is a different claim from "we wrote it carefully".

29.7 Why this correction was the most consequential in the project

Understated standard errors do not just produce over-confident intervals. In this architecture they **silently disable the empirical-Bayes shrinkage**, because the weight $w_i = \tau^2 / (\tau^2 + se_i^2)$ is pushed toward 1 from both directions at once when `se_i` is too small (§31.7). Fixing the inference moved the mean shrinkage weight from **0.954 to 0.780**. An inference bug became a modelling bug.

30. Product-level elasticities

30.1 Why per-product estimates at all

The category elasticity is a good prior and a poor description. The per-UPC distribution runs from $p_{10} = -4.094$ to $p_{90} = -0.240$, and $\tau^2 = 0.766$ says there is genuine between-product variance in *true* elasticities well above sampling noise. Pricing every product at -2.029 would systematically under-price the elastic tail and over-price the inelastic tail.

30.2 Estimation design

For each UPC, `estimate_per_upc` fits

$$\log Q_{st} = \alpha_s + \beta \log P_{st} + \gamma' X_{st} + u_{st}$$

with **store fixed effects absorbed within the product** (so the coefficient comes from within-store price movement, not cross-store price levels), controls `recorded_promotion_flag`, `sin52`, `cos52`, `trend`, and panel-robust standard errors (§29.5). Columns that do not vary within a product are dropped before inversion (`keep_independent_columns`), so a degenerate control cannot make $X'X$ singular.

30.3 Eligibility thresholds

threshold	value	source
minimum observations per UPC	200	<code>elasticity.min_obs_per_upc</code>
minimum distinct prices per UPC	10	<code>elasticity.min_distinct_prices_per_upc</code>
maximum panel-robust SE to accept an estimate	1.5	<code>elasticity.max_se_for_product_estimate</code>
sign requirement	$\beta < 0$	hard-coded in <code>build_elasticity_table</code>

30.4 The eligibility funnel

`scripts/audit_eligibility_funnel.py` → `reports/18_ELASTICITY_ELIGIBILITY_FUNNEL.md`. Every transition is counted with a stated reason.

stage	UPCs	lost	reason
UPCs in the processed dataset	489	—	survived the raw data audit
... present in the modelling table	476	–13	needs ≥ 1 lagged price and ≥ 1 lagged demand observation
... observed in the training weeks	372	–104	estimation restricted to weeks 2–257 (no

stage	UPCs	lost	reason
			validation/test outcomes)
... with positive-sales training weeks	372	0	log demand undefined at zero (0 rows dropped)
... with ≥ 200 training observations	330	-42	a per-product regression on fewer rows is not worth reporting
... and ≥ 10 distinct prices	272	-58	the price coefficient is identified from price variation
... successfully estimated	272	0	regression converged with finite coefficient and SE
... after removing wrong-signed coefficients	248	-24	a positive price coefficient is evidence of endogeneity, not upward-sloping demand
... after removing imprecise coefficients	239	-9	panel-robust SE must be ≤ 1.5

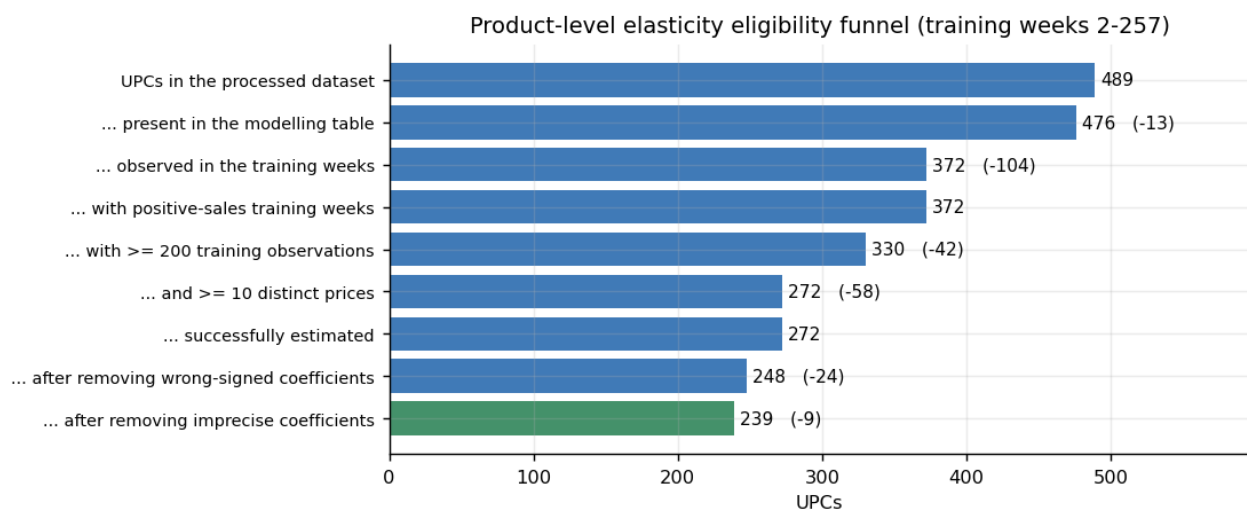


Figure 6 — 489 UPCs enter, 239 leave with a usable own elasticity. Source: `artifacts/metrics/elasticity_funnel.json`.

30.5 Rejection summary

reason	UPCs
insufficient observations or price variation	100
wrong sign	24
standard error too large	9
total rejected of 372 in the training window	133

30.6 Why wrong-signed coefficients are rejected rather than shrunk

A positive price coefficient does not mean demand slopes upward. It means that, for that product, price moved *with* unobserved demand strongly enough to flip the sign — the endogeneity problem winning outright. Shrinking such an estimate toward the pooled value would blend contamination into the prior-weighted answer instead of discarding it. Rejecting it and falling back to the pooled estimate is the conservative choice (DECISIONS #35).

Note the difference from the *Phase D analysis*, where **5.1% of UPCs had a significant positive coefficient**. Here 24 of 272 (**8.8%**) are wrong-signed at all, significant or not — a slightly different population (training weeks only,

different thresholds) measured a slightly different way. Both numbers are reported and neither is used as the other.

30.7 Coverage at decision time

At week 399 there are **13,964** UPC × store decision contexts:

elasticity source	contexts	share
shrunk_product	8,962	64.18%
pooled_fallback	5,002	35.82%

More than a third of decision contexts are priced with the category-level number. Those recommendations carry the `POOLED_ELASTICITY_FALLBACK` reason code and can **never be rated LOW risk** — the risk layer promotes them to at least MEDIUM (§57.4). That is the mechanism by which an estimation limitation propagates into a policy consequence instead of disappearing.

30.8 Distribution of the shipped product elasticities

Over the 239 products with their own shrunk estimate (`artifacts/models/elasticity_table.csv`):

statistic	value
mean	-1.954
median	-1.898
standard deviation	0.769
min / max	-4.058 / -0.331
p25 / p75	-2.464 / -1.396
clipped to the [0.2, 6.0] band	0

Including the pooled fallback across all 372 products, the final distribution has median **-2.029**, p10 **-2.790**, p90 **-1.169**.

31. Empirical-Bayes shrinkage

This is the most statistically substantial component of the project, and the one that was found to be broken and then fixed.

31.1 The problem it solves

You have 239 noisy per-product estimates $\hat{\epsilon}_i$ with standard errors se_i , and one precise pooled estimate $\mu = -2.029$. Using $\hat{\epsilon}_i$ directly over-fits the noise; using μ everywhere throws away real heterogeneity. Empirical Bayes is the principled compromise.

31.2 The hierarchical model

$$\hat{\epsilon}_i \mid \theta_i \sim N(\theta_i, se_i^2) \quad (\text{sampling})$$

$$\theta_i \sim N(\mu, \tau^2) \quad (\text{between products})$$

The posterior mean of θ_i is a precision-weighted average:

$$\llbracket \hat{\theta}_i = w_i \hat{\epsilon}_i + (1-w_i)\mu, \quad w_i = \tau^2 / (\tau^2 + se_i^2) \rrbracket$$

31.3 Why the weight has this form, intuitively

τ^2 is how much true elasticities genuinely differ between products. se_i^2 is how much noise sits on product i 's estimate. The weight is the share of the *observed* variance in $\hat{\epsilon}_i$ that is real signal:

- $se_i \rightarrow 0$ (a perfectly precise estimate) $\Rightarrow w_i \rightarrow 1 \Rightarrow$ keep your own number.
- $se_i \rightarrow \infty$ (pure noise) $\Rightarrow w_i \rightarrow 0 \Rightarrow$ fall back to the category.
- $\tau^2 \rightarrow 0$ (all products identical) $\Rightarrow w_i \rightarrow 0$ for everyone $\Rightarrow$ pool completely.
- $\tau^2 \rightarrow \infty$ (products wildly different) $\Rightarrow w_i \rightarrow 1 \Rightarrow$ trust each product.

w_i is monotonically decreasing in se_i and lies in $[0, 1]$ by construction. `tests/test_shrinkage.py` (12 tests) pins all four limiting cases, the monotonicity and the bounds.

31.4 Estimating τ^2

Two estimators are implemented.

Method of moments — transparent, and what Phase L used:

$$\tau_{\text{mom}}^2 = \max(\text{mean}((\varepsilon_i - \mu)^2) - \text{mean}(se_i^2), 0)$$

i.e. observed dispersion minus average sampling variance, floored at zero.

REML — the shipped default. The standard restricted-likelihood fixed point from random-effects meta-analysis:

$$w_i = 1/(\tau^2 + se_i^2),$$

$$\tau^2 \leftarrow (\sum_i w_i^2 [(\varepsilon_i - \mu)^2 - se_i^2 + 1/(\sum_j w_j)]) / (\sum_i w_i^2)$$

The $+ 1/\sum w$ term is the restricted correction for having estimated μ ; dropping it gives the ML estimator, which is biased downward. Floored at zero, because a negative variance is not a variance.

REML is the default because the moment estimator is noisier and truncates at zero more often (DECISIONS #45).

31.5 The prior mean

Two choices are implemented; the shipped one is **prior_mean: pooled** — shrink toward the pooled controlled elasticity, which is *also* the fallback used for products with no usable estimate. This makes the two internally consistent: a product with $w_i = 0$ and a product with no estimate at all receive exactly the same number.

The alternative (estimate the prior mean freely from the product coefficients) gives **-1.933** against the pooled **-2.029** — close enough that this choice mattered far less than the standard-error fix.

Note also that τ^2 is measured **around the prior mean the shrinkage actually targets**, not around the sample mean. Measuring dispersion around one centre while shrinking toward another is internally inconsistent, and Phase L did exactly that.

31.6 The shipped numbers

quantity	value
τ^2 (REML, around the pooled prior)	0.7661
τ	0.8753
τ^2 (method of moments, same inputs)	0.6731
prior mean μ	-2.0289 (pooled)
prior mean if estimated freely	-1.9334
products used	239
median per-product robust SE	0.3904
mean shrinkage weight	0.7797
median weight	0.8341
min / max weight	0.2693 / 0.9834
converged	true
weights monotone in SE	true (asserted)

Weight distribution:

band	products
< 0.25	0
0.25 – 0.50	25
0.50 – 0.75	48
0.75 – 0.90	96
0.90 – 0.99	70
> 0.99	0

That **no** product sits above 0.99 or below 0.25 is the signature of a shrinkage estimator that is actually doing work. Under the Phase L inputs the median weight was 0.989 and the distribution piled against the ceiling.

31.7 The audit and the correction — how the estimator was found to be broken

`scripts/audit_shrinkage.py` → `reports/14_SHRINKAGE_AUDIT.md` re-derived the estimator from first principles and ran seven variants on the same data.

variant	τ^2	median SE	mean weight	median shrunk ϵ
Phase L: moment τ^2 , sample-mean prior, HC1 SE, filtered	0.906	0.101	0.955	–1.842
moment τ^2 , pooled prior, HC1 SE, filtered	0.913	0.101	0.955	–1.846
REML τ^2 , pooled prior, HC1 SE, filtered	0.901	0.101	0.955	–1.846
moment τ^2 , pooled prior, ROBUST SE, filtered	0.673	0.390	0.761	–1.904
Phase M (shipped): REML τ^2, pooled prior, ROBUST SE, filtered	0.766	0.390	0.780	–1.898
REML τ^2 , pooled prior, ROBUST SE, unfiltered (272 products)	1.246	0.404	0.813	–1.816
REML τ^2 , free prior, ROBUST SE, filtered	0.759	0.390	0.778	–1.866

The finding: the formula was correct; the inputs were wrong. Rows 1–3 show that changing the τ^2 estimator or the prior mean, holding HC1 standard errors fixed, moves the mean weight by less than 0.001. Rows 4–5 show that changing the standard errors — and nothing else — moves it from 0.955 to 0.76–0.78.

The mechanism is that understated standard errors inflate $w = \tau^2 / (\tau^2 + se^2)$ **from both directions at once**: the denominator's se^2 term shrinks directly, *and* τ^2 is estimated as observed dispersion minus average sampling variance, so under-stating se^2 inflates τ^2 too. Both moves push w toward 1.

The median per-product SE inflation from HC1 to panel-robust is **3.79×**, so se^2 grows by roughly 14×. That is why a purely inferential bug silently switched the shrinkage off.

`tests/test_shrinkage.py` now includes a regression test for exactly this failure mode: that understated standard errors silently disable the shrinkage.

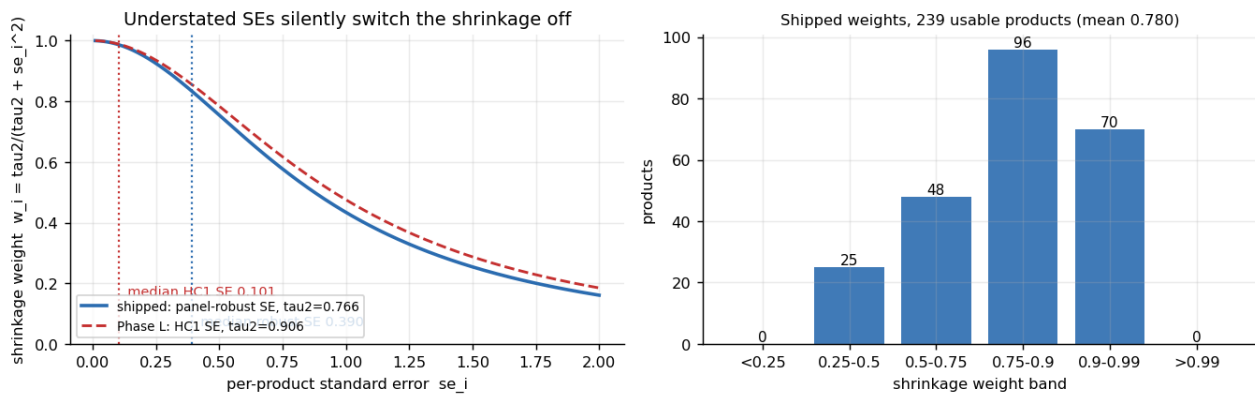


Figure 7 — Left: $w = \tau^2 / (\tau^2 + se^2)$ under the shipped panel-robust inputs (blue) and the Phase L HC1 inputs (red), with each regime's median standard error marked. At the HC1 median SE of 0.101 the weight is 0.99 — effectively no shrinkage. At the robust median of 0.390 it is 0.83. Right: the shipped weight distribution across the 239 usable products. Source: [artifacts/metrics/shrinkage_audit.json](#).

31.8 A worked example from the actual table

UPC **3000006560** (CAPN CRUNCH JUMBO CR), the context used throughout this report:

quantity	value
training observations	20,125
distinct prices	215
raw estimate $\hat{\epsilon}_i$	-3.0178
HC1 standard error	0.0398
panel-robust standard error se_i	0.2557
SE inflation	6.43×
τ^2	0.7661
weight $w_i = 0.7661 / (0.7661 + 0.2557^2)$	0.9214
pooled prior μ	-2.0289
shrunk $0.9214 \times (-3.0178) + 0.0786 \times (-2.0289)$	-2.9401
clipped?	no
source	shrunk_product

Check the arithmetic: $0.2557^2 = 0.06538$; $0.7661 / (0.7661 + 0.06538) = 0.9214$; $0.9214 \times (-3.0178) = -2.7807$; $0.0786 \times (-2.0289) = -0.1595$; sum = **-2.9401**. ✓

This is a well-identified product — 20,125 observations, 215 distinct prices — so it keeps 92% of its own estimate. Had the HC1 standard error of 0.0398 been used, the weight would have been $0.7661 / (0.7661 + 0.00158) = 0.9979$ and the shrunk value **-3.0157** — essentially the raw estimate, with the shrinkage doing nothing at all.

A documentation-drift note: README.md §3 quotes this context's elasticity as "-3.016 (shrunk_product)". That is the *raw* estimate (-3.0178) carrying the *shrunk* label. The value actually applied by the current artifacts is **-2.9401**, confirmed by a live `run_demo.py` execution and a live `/recommend-price` API call on 2026-08-19. This report uses -2.9401 throughout.

31.9 Fallback behaviour

```
def epsilon_for(self, upcs):
    lookup = self.products.set_index("upc")["elasticity_final"]
    return pd.Series(np.asarray(upcs)).map(lookup).fillna(self.pooled).to_numpy("float64")
```

python

A UPC not in the table, or one whose estimate was rejected, gets the pooled value and the source string `pooled_fallback`. There is no silent NaN path and no interpolation.

31.10 What shrinkage does and does not buy

Buys: stability. Noisy per-product coefficients are pulled toward a well-identified category value in proportion to how noisy they are, and the weight is recorded per product so any recommendation can be traced to how much of its elasticity was its own.

Does not buy: identification. If the pooled elasticity is biased by promotion contamination — and §11.4 argues it is — then **every shrunk product estimate inherits a share $(1 - w_i)$ of that bias**, and the products with the *least* own information inherit the most. Shrinkage fixes variance, not bias. This is limitation #34 in `KNOWN_LIMITATIONS.md` and it is stated in exactly those terms.

32. Elasticity stability over time

`scripts/audit_elasticity_stability.py` → `artifacts/metrics/elasticity_stability.json`,
`reports/17_ELASTICITY_STABILITY.md`.

32.1 Design

The entire elasticity stack — pooled controlled, UPC × store FE, per-UPC with panel-robust SEs, REML empirical-Bayes shrinkage — is re-estimated on five historical windows. **Every window lies strictly inside the approved training period (weeks 2–257)**, and the script raises rather than run if a window would cross into validation or test. W1/W2/W3 are **disjoint thirds**, so per-product comparisons between them use non-overlapping data.

32.2 Category-level results

window	weeks	rows	pooled	SE (two-way)	FE	τ^2	mean w	usable products	median shrunk	p10 / p90
W1 early	2–87	979,559	–1.903	0.223	–2.279	1.016	0.663	85 / 259	–1.90	[–3.12, –0.98]
W2 middle	88–129	545,034	–2.323	0.213	–3.280	1.227	0.620	87 / 221	–2.22	[–3.35, –1.37]
W3 late	130–257	1,681,844	–2.240	0.125	–2.486	1.070	0.837	218 / 297	–2.02	[–3.17, –0.75]
W4 expanding	2–129	1,524,593	–1.917	0.166	–2.304	1.045	0.682	146 / 285	–1.99	[–3.02, –1.09]
W5 full window	2–257	3,206,437	–2.029	0.106	–2.222	0.766	0.780	239 / 372	–1.90	[–3.05, –0.95]

The pooled estimate ranges over **0.420** between the most extreme windows.

32.3 Is that drift or noise? A formal test

Using each window's own two-way clustered standard error on the three **disjoint** windows:

pair	pooled A	pooled B	difference	SE of difference	z	significant at 5%?
W1 vs W2	–1.903	–2.323	+0.420	0.309	+1.36	no
W1 vs W3	–1.903	–2.240	+0.337	0.256	+1.32	no
W2 vs W3	–2.323	–2.240	–0.083	0.247	–0.33	no

Largest absolute z between disjoint windows: 1.36. No pair differs by more than sampling noise once the panel clustering is admitted. **The category elasticity is stable within the precision this data supports.**

Note how much the inference work in §29 is doing here. Under classical standard errors these differences would be enormously significant, and the project would have concluded that elasticity drifts violently over time. The correct covariance assumption changes the scientific conclusion, not just the interval.

32.4 Product-level stability

pair	common products	Spearman rank corr	Pearson	sign stability (raw)	median abs difference
W1 vs W2	34	-0.054	-0.054	100.0%	0.952
W1 vs W3	34	+0.350	+0.294	100.0%	0.753
W2 vs W3	34	+0.187	+0.175	100.0%	0.695

Median pairwise rank correlation: +0.187. Median sign stability: 100%.

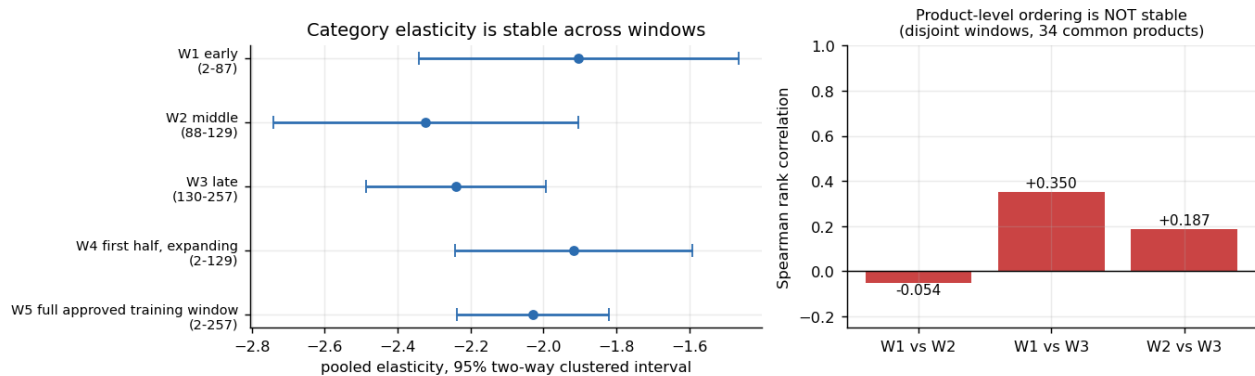


Figure 8 — Left: pooled elasticity with 95% two-way clustered intervals across all five windows; every interval overlaps every other. Right: Spearman rank correlation of shrunk product elasticities between disjoint windows. Source: [artifacts/metrics/elasticity_stability.json](#).

32.5 What this means for pricing policy

The two findings pull in opposite directions and both must be stated.

Defensible: "cereal demand in this chain is elastic, around -2 , and that is stable across the training period." The category number reproduces.

Not defensible: "product X is more price-sensitive than product Y, and that ordering is a durable asset." A rank correlation of $+0.19$ between disjoint windows means the *ordering* barely reproduces. The magnitudes differ by a median of 0.70 – 0.95 in absolute elasticity between windows — which is roughly the size of τ (0.875) itself.

Sign stability of 100% is the one genuinely encouraging product-level result: whatever else moves, a product that looked elastic in one window looked elastic in the others.

Consequences the project actually draws:

50. Product-level elasticity must not be described as a durable asset (`STATUS.md`, open issue 5).
51. The shrinkage is doing more work than it appears — pulling unstable product estimates toward a stable category number is exactly the right response to this pattern.
52. A long-lived pricing policy built on product-specific elasticities would decay. Re-estimation cadence is a real operational requirement, not a nice-to-have.
53. Only 34 products are common to all three disjoint windows, so the product-level conclusion itself rests on a thin sample and is reported with that caveat.

33. Causal limitations

This chapter is the boundary condition on everything else in this report.

33.1 The formal statement

The engine estimates the **observational conditional distribution**

$$P(Q \mid \text{Price} = p, X)$$

The pricing decision requires the **interventional distribution**

$$P(Q \mid \text{do}(\text{Price} = p), X)$$

These are equal only if price is assigned independently of everything else that affects demand — conditional on the observed covariates. In this data it demonstrably is not.

$$P(Q \mid \text{Price}, X) \neq P(Q \mid \text{do}(\text{Price}), X)$$

33.2 The project's own evidence that they differ

A **seven-fold** change in the headline coefficient ($-0.348 \rightarrow -2.419$) produced purely by adding controls nobody could argue against. If the naive association were the causal effect, controls would not move it.

And nothing guarantees the last specification is causal either — it is only the *least contaminated* of the four.

33.3 The specific threats, with evidence

1. Promotion bundling. Bonus Buys arrive with feature ads and display that the data does not record. Estimated separately: -2.662 on promotion weeks versus -1.830 on non-promotion weeks.

2. Incomplete promotion coding. 94.55% of rows carry no code, and two undocumented codes (G, L) exist. Every promotion control is partial by construction, so the "clean" subsample is not clean.

3. Retailer anticipation / reverse causality. Prices are cut *because* a demand surge is expected. Low price and high demand then coincide for reasons unrelated to the price. The direction of the resulting bias cannot be signed a priori.

4. Seasonality and trend. Category demand and price levels both drift over eight years; the monitoring artifact shows the reference price level rising $\sim 12.5\%$ between the training and test windows.

5. Store heterogeneity and zone pricing. Zones were tied to local competition and demographics. Cross-store price differences reflect local demand, not exogenous variation — which is why store fixed effects move the estimate.

6. Product heterogeneity. $p_{10} -4.094$ to $p_{90} -0.240$, with 5.1% of UPCs showing a *significant positive* coefficient. Those are not bad data; they are series where price moves with unobserved demand.

7. Stock-outs. Unavailability and demand collapse are indistinguishable in scanner data.

8. Competitive response. Rival stores' prices are entirely unobserved.

33.4 About the Dominick's experiments

The Dominick's research programme at Chicago Booth **did** include in-store pricing experiments in several categories. **This project claims none of its estimates are experimental.** The Cereals movement file used here carries no experiment assignment, no treatment window and no randomisation indicator that could be verified, so nothing here is claimed as causal identification.

33.5 What genuine causal pricing claims would require

In rough order of strength:

approach	what it needs	feasible on this data?
Randomised price experiment	prices assigned at random within stores/weeks; the design in <code>docs/PRICING_EXPERIMENT.md</code>	no — no randomisation exists in this extract; requires a live retailer
Instrumental variables	an instrument that shifts price without	no — no such instrument is present in

approach	what it needs	feasible on this data?
	affecting demand directly — e.g. wholesale-cost shocks, distance-to-warehouse fuel costs, upstream trade-deal calendars	this file
Natural experiment / regression discontinuity	a documented policy change, zone reassignment or discrete pricing rule	no — none identified in the Cereals extract
Difference-in-differences	a treated and a control group with credible parallel pre-trends	no — no exogenous treatment to define
Causal ML (double ML, causal forests)	still requires unconfoundedness given observables — it improves <i>estimation</i> , not <i>identification</i>	would not fix it — the confounders here are unobserved

The last row is worth emphasising because it is a common interview trap: double machine learning and causal forests are **estimation** technologies. They relax functional-form assumptions; they do not manufacture identification. Applying them here would produce a more flexible, equally confounded estimate.

33.6 What this project does instead

It does not pretend. Specifically:

- elasticity results are labelled **observational elasticity estimates** throughout;
- the counterfactual uplift metric is named `model_internal_estimated_profit_uplift_pct` — the circularity is in the field name, where it cannot be dropped from a slide;
- the out-of-time validation report opens with a boxed warning that it is predictive, not causal (\$45);
- `scripts/audit_claims.py --strict` scans the whole repository for ten watched phrases and fails the build if an UNSUPPORTED claim reappears — currently **0 UNSUPPORTED** across 49 files and 363 occurrences — re-run with this report itself included;
- the experiment that *would* settle it is fully designed in `docs/PRICING_EXPERIMENT.md` and explicitly **not run**.

PART VI — DEMAND FORECASTING

34. Modelling objective

$$\hat{Q} = f(P, X)$$

where X is the decision-time context of §20. The model is fitted by minimising a loss on **observed** (P, X, Q) triples — that is, it learns the conditional expectation of demand at prices that were actually charged.

34.1 Prediction and price response are different jobs

	forecasting	price response
question	what will demand be next week?	what would demand be at a price we did not charge?
evaluated on	realised outcomes at observed prices	nothing directly observable
dominant signal	lags, seasonality, promotion, series level	the price coefficient alone
a good answer needs	conditional-mean accuracy	correct <i>slope</i> , identified
this project's metric	WAPE 0.4565 on 749,040 held-out rows	no direct offline metric exists

A model can be excellent at the left column and wrong in the right one, because at observed prices the lag and context features carry most of the signal and the price coefficient can absorb whatever is left. That is not a hypothetical here: the selected model is the best forecaster in the comparison *and* has an implied elasticity (−3.102) materially steeper than every controlled econometric estimate (§42).

Part VI is entirely about the left column. Part VII is about the right one.

35. Temporal train / validation / test split

35.1 The split

`temporal_split()` orders the distinct week indices and cuts at fixed fractions (70 / 15 / 15 from `configs/config.yaml` → `modeling`).

split	weeks	dates	rows	share
train	2 – 257	1989-09-21 ... 1994-08-11	3,206,437	68.6%
validation	258 – 342	1994-08-18 ... 1996-03-28	715,856	15.3%
test	343 – 399	1996-04-04 ... 1997-05-01	749,040	16.0%
	365 distinct weeks		4,671,333	100%

The split is on **weeks**, not rows, which is why the row shares are not exactly 70/15/15 — later weeks carry more active series.

Week 1 does not appear: it is entirely consumed by the "no lagged history" drop (§22.5).

35.2 Why a random split would be indefensible

Three distinct failures, all of which a random split commits at once:

(a) Within-series future leakage. A random split puts week 300 of a series in train and week 299 in test. But `lag_move_1` for week 300 is week 299's demand. The model would be given the test row's target as a training feature.

(b) Autocorrelation. Weekly demand is strongly autocorrelated. Neighbouring weeks are near-duplicates, so a random split makes train and test near-identical and the metric measures interpolation, not forecasting.

(c) Regime leakage. The panel spans eight years of price-level drift (§18.2). A random split lets the model see the 1996 price regime while "predicting" it.

The combined effect is a metric that is optimistic by a wide and unknowable margin.

`tests/test_features.py::test_temporal_split_is_chronological_and_disjoint` asserts the split is chronological and the three sets are disjoint.

35.3 How the split is used

- **Train (2–257):** model fitting, the UPC demand prior, the prediction cap, and — separately — the entire elasticity table.
- **Validation (258–342):** model *selection*. The rule was fixed in advance: lowest validation WAPE among **price-aware** models.
- **Test (343–399):** scored **once**, after selection, for the final report. No tuning decision was made on it.

That last point is leakage channel #9 and the guard is procedural rather than mechanical — but the training log in `artifacts/metrics/model_metrics.json` records both windows for every model, so the sequence is auditable: only M1 and M2 (the two price-aware models) carry validation metrics, and only the selected M1 carries a `test` block plus segment breakdowns.

35.4 The elasticity window is *stricter* than the model window

The elasticity table is fitted on weeks **2–257 only** — the same as the model training window. This is deliberate and it is leakage channel #11: an elasticity fitted on the full panel would embed validation and test outcomes into every recommendation scored in those weeks. The window is recorded in the elasticity table metadata and the estimation script aborts if any input row exceeds it.

36. Baselines

Four naive baselines were implemented in `src/pricing_engine/models/baselines.py`. None is price-aware — which is the point: they establish how much of weekly demand is predictable from *series level and season alone*, with no reference to price.

model	formula	intuition
M0a last-week naive	$\hat{Q}_t = Q_{t-1}$	random walk; the classic hard-to-beat forecast baseline
M0b rolling-mean(4)	$\hat{Q}_t = 1/4 \sum_{j=1}^4 Q_{t-j}$	smooths week-to-week noise
M0c series historical mean	$\hat{Q}_t = \bar{Q}_{(\text{series}, \text{train})}$	the series' long-run level
M0d seasonal naive (52w)	$\hat{Q}_t = Q_{t-52}$	same week last year

36.1 Results

model	valid WAPE	valid MAE	valid RMSE	valid bias	test WAPE	test MAE	test RMSE	test bias
M0a last-week naive	0.8007	15.251	63.889	+0.075	0.7735	14.407	88.014	+0.141
M0b rolling-mean(4)	0.7471	14.230	51.926	+0.219	0.7326	13.644	70.211	+0.227
M0c series historical mean	0.7607	14.489	47.196	+1.971	0.7441	13.860	62.329	+1.443
M0d seasonal naive (52w)	0.7920	15.086	56.413	+0.794	0.8207	15.287	76.395	+1.041

36.2 Reading the baselines

M0b is the best naive model on both WAPE and MAE — smoothing beats the pure random walk, which tells you weekly demand here is noisy around a slowly moving level rather than a true random walk.

M0c has the lowest RMSE but a worse WAPE than M0b. That is not a contradiction: RMSE punishes large errors quadratically, and a long-run mean never chases a spike, so it avoids the catastrophic misses that a lagged forecast makes on the week *after* a promotion. It pays for that with a large positive bias (+1.97 units) — it systematically over-forecasts, because the mean of a right-skewed series sits above its median.

M0d is the worst on the test window (0.8207, and worse than on validation). Year-ago demand is a poor guide in a category where promotion timing shifts between years — which is itself useful evidence that the seasonality here is *not* a rigid annual pattern.

Every baseline has positive bias; the selected model has negative bias. Naive methods over-forecast because they chase past spikes; the log-target ridge under-forecasts because $E[\exp(x)] > \exp(E[x])$ and the model is fitted on the log scale (§41.4).

36.3 Their real purpose

WAPE 0.73–0.82 is the price of admission. Any price-aware model that cannot beat 0.73 on the test window is not adding information about price — it is just a more expensive way of computing a moving average. The selected model reaches **0.4565**, a **37.7% relative improvement** over the best baseline. That gap is the evidence that the feature set and the price signal carry real information.

37. Ridge log-log model (M1 — selected)

37.1 Specification

$$\log(1 + Q) = \beta_0 + \beta_p \log P + \beta_n' Z_n + \beta_c' D_c + \varepsilon$$

fitted by ridge regression with L2 penalty $\alpha = 1.0$, then inverted with `expm1` and clipped:

$$\hat{Q} = \min(\max(e^{\hat{y}} - 1, 0), \text{cap})$$

37.2 Why the log target

- 54. **Weekly demand is heavy-tailed** (mean 14, max 18,688). On the raw scale a handful of promotional store-weeks would dominate the squared-error loss.
- 55. **$\log(1+Q)$ makes the model multiplicative**, which is what demand economics implies: a promotion multiplies volume, a price change scales it.
- 56. **It makes β_p directly interpretable** as a partial elasticity — although see §37.7 for why that reading is misleading here.
- 57. **$\log1p$ rather than $\log$ handles $Q = 0$ gracefully** (moot in this dataset, but correct).

37.3 Feature preprocessing

```
numeric = Pipeline([("impute", SimpleImputer(strategy="median")),
                    ("scale", StandardScaler())])
categorical = OneHotEncoder(handle_unknown="ignore", min_frequency=50)
pre = ColumnTransformer([("num", numeric, list(self.numeric_features)),
                        ("cat", categorical, list(CATEGORICAL_FEATURES))])
Pipeline([("pre", pre), ("ridge", Ridge(alpha=alpha, random_state=self.seed))])
```

python

choice	reason
median imputation	robust to the heavy tail; means would be dragged by spikes

choice	reason
standardisation	ridge penalises coefficients on their own scale, so unscaled features get arbitrarily different shrinkage
<code>handle_unknown="ignore"</code>	an unseen store at inference time produces an all-zero block, not a crash
<code>min_frequency=50</code>	rare categories are pooled instead of getting a coefficient fitted on a handful of rows
<code>log1p</code> on demand-level features	<code>LOG1P_FEATURES</code> — the lagged/rolling demand columns are as heavy-tailed as the target

37.4 The 26 feature columns

Numeric (24, after exclusions): `log_price`, `price_vs_last_week`, `price_vs_series_reference`, `price_vs_recent_mean`, `lag_move_1..4`, `roll_mean_move_4/8/13`, `roll_std_move_4`, `lag_price_1`, `lag_price_2`, `roll_mean_price_4`, `series_reference_price`, `decision_time_unit_cost`, `lag_promotion_1`, `recorded_promotion_flag`, `series_age_weeks`, `package_size_oz`, `upc_demand_prior`, `sin52`, `cos52`.

Categorical (2, one-hot): `store`, `com_code`.

37.5 Two features that are *not* there, and why

`time_index` is excluded from the linear model only. A standardised, unbounded, monotone trend extrapolated into a future window explodes once the log prediction is exponentiated. Observed WAPE before the fix: **27.9** (DECISIONS #16). Trees saturate at the edge of their training range, so the boosted model keeps it. The monitoring artifact confirms the diagnosis independently: `time_index` has the largest PSI of any feature (12.41) between the training and test windows.

`upc` is not used as a raw categorical. 489 one-hot columns would be unwieldy, and `HistGradientBoostingRegressor` caps native categorical cardinality at 255 anyway. Instead the model learns a **smoothed UPC demand prior** on training rows only:

$$\text{prior}_u = (\sum_{i \in u} \log(1+Q_i) + \lambda \bar{y}) / (n_u + \lambda), \quad \lambda = 50$$

```
self.upc_prior_ = (stats["sum"] + UPC_PRIOR_SMOOTHING * self.global_prior_) / (
    stats["count"] + UPC_PRIOR_SMOOTHING)
```

python

This is empirical-Bayes shrinkage again, in a different guise: a product with few training rows is pulled toward the global mean log-demand. Fitted on training rows only, so it cannot leak; unseen UPCs at inference get the global prior via `.fillna(self.global_prior_)`.

37.6 The prediction cap

```
PREDICTION_CAP_MULTIPLE = 5.0
self.prediction_cap_ = float(np.nanmax(y) * PREDICTION_CAP_MULTIPLE)
```

python

Exponentiating a linear prediction can explode on extreme inputs. Predictions are clipped to 5× the largest weekly demand ever observed in training. The cap is stored **in the artifact**, so it travels with the model and is visible to anyone auditing it (limitation #13). It is a documented safety bound, not a silent clip.

37.7 Interpreting the price coefficient — a trap the code warns about

The raw ridge coefficient on `log_price` is **-0.0135**. That is *not* the model's price response, and the code says so in a docstring:

```
def raw_price_coefficient(self) -> float | None:
    """Ridge coefficient on ``log_price``, back on the raw scale.

    This is NOT the model's price response: ``log_price`` is collinear with
    the relative-price features and with lagged prices, so the response is
    spread across several coefficients. Use :meth:`implied_elasticity` for
    the number that actually matters.
    """
```

python

`log_price` sits alongside `price_vs_last_week`, `price_vs_series_reference`, `price_vs_recent_mean`, `lag_price_1`, `lag_price_2`, `roll_mean_price_4` and `series_reference_price` — all of which move when the price moves. Ridge spreads the response across the whole collinear block. Reading -0.0135 as "the elasticity is -0.01 " would be badly wrong.

37.8 The correct way to measure the model's price response

`implied_elasticity()` measures it by **central finite difference**, recomputing every price-dependent feature at $(1 \pm \text{rel}) \cdot p$:

```
q_up = self.predict(recompute_price_features(d, base * (1.0 + rel)))
q_dn = self.predict(recompute_price_features(d, base * (1.0 - rel)))
elasticity = (np.log(q_up + eps) - np.log(q_dn + eps)) / (np.log(1+rel) - np.log(1-rel))
```

python

This is model-family agnostic — it works identically for the linear and the boosted model — and it measures **the elasticity the optimizer will actually exploit**, which is the only one that matters.

Results with `rel = 0.02` on a 50,000-row sample:

window	median	mean	p10	p90	share negative
validation (at training time)	-3.322	-3.345	-3.862	-2.980	99.996%
test (at evaluation time)	-3.188	-3.259	-3.747	-2.833	99.988%
300 decision contexts, $\pm 30\%$ sweep (Phase F)	-3.102	—	-3.482	-2.667	100%

All three are steeper than every controlled econometric estimate. \$42 is about what was done in response.

37.9 Training cost

31.0 seconds to fit on 3,206,437 rows — against 181.9 s for the boosted model.

38. Gradient-boosting model (M2)

38.1 Specification

`HistGradientBoostingRegressor` with **Poisson loss**, on the raw (untransformed) target.

hyperparameter	value	reason
loss	poisson	the right likelihood for weekly counts; guarantees $\hat{Q} \geq 0$ by construction
learning_rate	0.06	conservative, paired with many iterations
max_iter	500	
max_leaf_nodes	63	moderately deep interactions
min_samples_leaf	40	regularisation on a 3.2M-row panel

hyperparameter	value	reason
l2_regularization	1.0	
early_stopping	False	the internal split would be random, breaking the chronological discipline
max_bins	255	default
categorical_features	"from_dtype"	native categorical handling for store, com_code
random_state	42	

The `early_stopping=False` choice is worth naming: `HistGradientBoosting`'s internal early-stopping split is *random*, which would leak future weeks into the stopping decision. A fixed iteration budget with strong per-leaf regularisation is the disciplined alternative.

38.2 Results

	valid WAPE	valid MAE	valid RMSE	valid bias	valid sMAPE	fit time
M1 ridge log-log	0.4135	7.875	39.597	-3.706	0.3807	31.0 s
M2 HGB Poisson	0.4226	8.050	38.378	+0.363	0.3809	181.9 s

M2 has a **lower RMSE** (38.378 vs 39.597) and a **far better bias** (+0.363 vs -3.706) — the Poisson loss on the raw scale is unbiased where the log-target model is not. But it loses on WAPE, MAE and sMAPE, and the selection rule was fixed in advance on WAPE.

M2 was **not scored on the test window**, correctly: only the selected model is.

38.3 The decisive problem with M2 — its price response

	median	mean	p10	p90	share negative
M1 implied elasticity	-3.322	-3.345	-3.862	-2.980	99.996%
M2 implied elasticity	-4.512	-5.069	-10.109	-0.389	93.02%

Read the last column first. **For 7% of sampled contexts the boosted model implies that raising the price *increases* demand.** That is not a subtle calibration issue — it is an upward-sloping demand curve, and an optimizer handed one will happily recommend a price increase for the wrong reason entirely.

Read the spread second. M2's implied elasticity runs from **-10.1 at p10 to -0.39 at p90** — a 26-fold range. M1's runs from -3.86 to -2.98, a 1.3-fold range. The tree model's price response is wildly unstable across contexts.

The mechanism is easy to state: gradient-boosted trees fit **step functions**. Near a split threshold the local price derivative is enormous; between thresholds it is exactly zero; and nothing in the objective constrains the response to be monotone. Sign, magnitude and smoothness of the price response are all unconstrained.

`monotonic_cst` (monotonic constraints) is available in scikit-learn and would address the sign problem. It was not used here — a legitimate criticism of this project and an obvious extension (§102).

38.4 Was M2 wasted?

No. It establishes three things that only a second model family can establish:

58. **The linear model is not leaving obvious accuracy on the table.** A strong boosted model with 500 iterations and native categorical scored *worse* on the selection metric, so the log-linear structure is not a crude approximation here.

59. **Price-response instability is a model-family property, not a data property.** Two models on identical features give -3.32 and -4.51 with completely different dispersions.
60. **It is the counter-example that justifies the price-aware selection restriction** (§39.2).

39. Model comparison

39.1 The complete table

Chronological split: train weeks 2–257, validation 258–342, test 343–399. Validation $n = 715,856$; test $n = 749,040$.

model	price-aware	valid WAPE	valid MAE	valid RMSE	valid bias	test WAPE	test MAE	test RMSE	test bias	interpretability	price-response suitability
M0a last-week naive	no	0.8007	15.251	63.889	+0.075	0.7735	14.407	88.014	+0.141	trivial	none — no price term
M0b rolling-mean(4)	no	0.7471	14.230	51.926	+0.219	0.7326	13.644	70.211	+0.227	trivial	none
M0c series historical mean	no	0.7607	14.489	47.196	+1.971	0.7441	13.860	62.329	+1.443	trivial	none
M0d seasonal naive (52w)	no	0.7920	15.086	56.413	+0.794	0.8207	15.287	76.395	+1.041	trivial	none
M1 ridge log-log ☒	yes	0.4135	7.875	39.597	-3.706	0.4565	8.503	72.064	-3.883	high — linear, inspectable	usable but too steep (implied -3.10)
M2 HGB Poisson	yes	0.4226	8.050	38.378	+0.363	not scored	—	—	—	low	unsafe — 7% wrong-signed, p10 -10.1

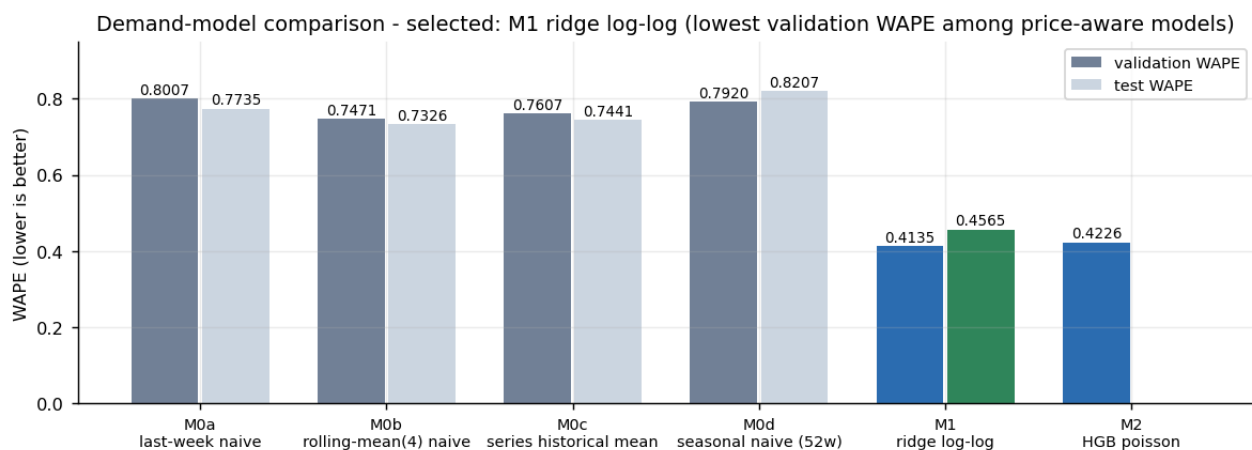


Figure 9 — Validation and test WAPE for all six models. Grey = not price-aware. Only the selected model was scored on the test window.
Source: `artifacts/metrics/model_metrics.json`.

39.2 Why selection was restricted to price-aware models

This restriction is still in force and it is not arbitrary. The system's purpose is to answer "what happens if we change the price?". A model with no price term answers that question with "nothing" — and M0b would have been a perfectly respectable *forecaster* at WAPE 0.7326.

Selecting on forecast accuracy alone, across a candidate set that includes price-blind models, would eventually select a model that cannot do the job at all. The restriction encodes the actual requirement in the selection rule.

39.3 Test-set discipline

Only M1 has a `test` block in `model_metrics.json`. Selection used the validation window; the test window was scored **once**, after the decision, and never revisited. Reloading the persisted artifact and re-scoring gives a **reload WAPE difference of exactly 0.00e+00** (`scripts/evaluate.py` → `artifacts/metrics/evaluation.json`), so the reported test number is a property of the saved model, not of the training session's memory.

40. Why Ridge won

"Because validation WAPE was lower" is the decision rule, not the explanation. Here are the mechanisms, with the evidence for each.

40.1 The gap is small, and that is itself informative

0.4135 vs 0.4226 — a **2.2% relative** difference. The honest first statement is that a well-specified linear model and a 500-iteration gradient booster are **roughly equally good** at this task. Anyone claiming a decisive victory for either has over-read a 0.009 WAPE gap.

40.2 Strong log-linear structure

Retail demand really is close to multiplicative: promotion multiplies, price scales, seasonality modulates. $\log(1+Q) \sim \log(P) + \log\text{-lags}$ is not a convenient approximation here — it is close to the correct functional form. When the true structure is (log-)linear, a linear model with the right transforms wins, because the tree has to *rediscover* a smooth surface by stacking axis-aligned steps.

Evidence: the `LOG1P_FEATURES` transform is applied to exactly the demand-level features, and the resulting model beats a booster given the same raw features and full freedom to find interactions.

40.3 Well-designed features leave little for interactions to find

The feature set already encodes the interactions that matter:

- `price_vs_last_week`, `price_vs_series_reference`, `price_vs_recent_mean` are *ratios* — they encode price × history interactions explicitly;
- `upc_demand_prior` encodes the product-level demand scale, which is the main thing a tree would otherwise have to learn by splitting;
- `sin52/cos52` encode annual seasonality smoothly.

A booster's advantage is discovering interactions you did not encode. When you have encoded them, the advantage shrinks to noise. This is the clearest practical lesson in the modelling section: **feature engineering transferred the boosted model's edge into the linear model**.

40.4 Regularisation on a very noisy panel

Weekly UPC × store demand is noisy: median 8 units, p99 103, max 18,688. Ridge with $\alpha = 1.0$ on standardised features applies uniform shrinkage across the collinear price block, which is close to the right prior when several correlated features carry the same signal. The booster, with 500 iterations, 63 leaves and no early stopping, has far more capacity to fit noise — and its `min_samples_leaf = 40` is the only thing holding it back.

40.5 Limited incremental non-linear signal in this category

Cereal is a stable, mature category. There is no evidence in these results of a strong threshold effect ("demand collapses above \$4.49") that a tree would capture and a linear model would miss. The booster's lower RMSE (38.378 vs 39.597) suggests it *does* fit the extreme promotional weeks somewhat better — which is exactly

where trees should help — but that gain is outweighed on WAPE, which is volume-weighted rather than squared-error weighted.

40.6 The tie-breaker that actually mattered: interpretability and safety

Even had the numbers been reversed, M2 would have been a problem:

property	M1	M2
implied elasticity, median	-3.322	-4.512
implied elasticity, p10 / p90	-3.862 / -2.980	-10.109 / -0.389
share of contexts with a negative (correct-signed) response	99.996%	93.02%
demand curve monotone decreasing across a ±30% sweep	100% of 300 contexts	not validated
price coefficient inspectable	yes	no
fit time	31 s	182 s

A pricing engine built on a response that points the wrong way 7% of the time would produce a confidently wrong price increase roughly once every fourteen recommendations.

40.7 The honest caveat

M1 won **and its price response is still too steep** (-3.10 vs a controlled -1.9 to -2.4). Winning the forecasting comparison did not make it right about price. That is the subject of Part VII, and it is the finding that reshaped the whole architecture.

41. Error analysis

All figures below are the **selected model on the test window** (weeks 343–399, $n = 749,040$).

41.1 Headline test metrics

metric	value
WAPE	0.4565
MAE	8.503 units
RMSE	72.064 units
bias (mean signed error)	-3.883 units
sMAPE	0.4000

41.2 What WAPE 0.4565 means in business terms

$$\text{WAPE} = (\sum_i |Q_i - \hat{Q}_i|) / (\sum_i Q_i)$$

WAPE is **volume-weighted**, so it is dominated by the store-weeks that matter commercially, and it is well defined when individual weeks are zero.

0.4565 means: across all test store-weeks, the total absolute forecast error equals 45.65% of total actual units. Equivalently, for a typical store-week selling 19 units, the model is off by about 8.5 units.

Is that good? Three reference points:

61. **Against the best naive baseline (0.7326), it is a 37.7% relative improvement.**
62. **For weekly UPC × store retail demand, 0.40–0.55 is a normal range.** At this grain, most of the variance is genuinely irreducible: shopper arrival is a stochastic process, competitor activity is unobserved, and a single basket can double a slow item's week.

63. **It is an honest number.** It was measured once, on a chronologically held out window, after selection, and it is *worse* than the validation number (0.4135) — as an honest temporal split usually is.

The business consequence is the important part: **any counterfactual profit difference smaller than this error deserves scepticism.** That is the entire rationale for the materiality threshold (§53.9). A 1% predicted uplift computed from a model with 46% error on a single store-week is noise; the threshold turns that observation into a decision rule.

41.3 Error by promotion state — the dominant split

segment	n	units	MAE	RMSE	WAPE	bias
no recorded promotion	691,836	9,868,496	5.336	56.089	0.3741	-1.125
recorded promotion	57,204	4,082,965	46.800	173.070	0.6557	-37.238

This is the single most important error pattern in the project.

- **7.6% of rows carry 29.3% of the units.** Promotion weeks are where the volume is.
- **MAE is 8.8× higher on promotion weeks** (46.8 vs 5.3 units).
- **The bias on promotion weeks is -37.2 units per row.** The model systematically and massively **under-forecasts promotional lift.**

Why: the model knows *whether* a promotion code is present, but not its depth, its display support, its feature-ad placement or its co-op funding. A Bonus Buy that triples volume and one that adds 20% look identical in the feature set. The model learns the average lift and misses both tails.

Consequence for pricing: the engine is least reliable exactly where price moves most. This is why the risk layer promotes contexts with `promotion_share > 0.5` out of LOW risk — "more than half of this series' weeks carry a promotion code: the price effect is entangled with promotion activity" (§57.4).

41.4 Why the model under-forecasts overall (bias -3.883)

Two mechanisms compound.

(a) Jensen's inequality on the log target. The model minimises squared error on $\log(1+Q)$ and is inverted with `expm1`. For a convex transform, $E[\exp(x)] > \exp(E[x])$, so back-transforming a conditional *mean of logs* yields something closer to a conditional *median* of levels. On a right-skewed distribution the median is below the mean. A smearing correction (Duan's estimator) would address this and was not applied — a fair criticism.

(b) Promotional under-prediction, per §41.3, which contributes $57,204 \times (-37.24) / 749,040 \approx -2.84$ of the total -3.883. **Roughly 73% of the overall bias comes from 7.6% of the rows.**

Note that the boosted model has bias **+0.363** — Poisson loss on the raw scale is unbiased in levels by construction. The selected model traded bias for WAPE.

41.5 Error by within-series price variation

price CV bucket	n	units	MAE	RMSE	WAPE	bias
CV ≤ 0.05 (stable prices)	73,077	1,066,404	5.840	18.149	0.4002	-1.955
0.05 – 0.10	243,380	3,356,144	4.952	28.601	0.3591	-1.437
0.10 – 0.15	351,047	7,368,028	9.903	67.390	0.4718	-4.834
> 0.15 (volatile prices)	81,536	2,160,885	15.460	159.434	0.5833	-8.819

The pattern is monotone above CV 0.05: **the more a series' price moves, the worse the forecast.** WAPE runs 0.359 → 0.472 → 0.583.

Two readings, both true:

- Volatile-price series are the promoted series, so this is largely \$41.3 seen through a different lens.
- **It is a genuine tension in the architecture.** Price variation is what makes a series *eligible* for optimization (§15.2) — and it is also where the demand model is least accurate. The engine is most confident exactly where it forecasts worst.

The eligibility screen and the risk layer are the response: eligibility asks "can this be modelled at all", risk asks "how much do we trust acting on it", and the risk bands are deliberately stricter than the eligibility thresholds.

41.6 Error by store — the worst ten

store	n	units	MAE	RMSE	WAPE	bias
98	9,491	165,046	12.592	316.710	0.7241	+1.116
86	9,365	417,882	30.827	377.157	0.6908	-26.144
89	8,741	97,786	6.778	39.875	0.6059	-3.543
92	3,416	28,776	4.785	24.578	0.5681	-1.201
111	8,522	121,666	8.097	49.631	0.5672	-4.556
90	8,541	83,239	5.524	31.681	0.5668	-2.582
18	9,196	174,135	10.661	229.781	0.5630	-1.384
106	8,945	85,182	5.305	23.934	0.5571	-2.076
95	9,100	130,567	7.983	45.518	0.5564	-4.471
40	9,050	107,128	6.577	35.630	0.5556	-2.317

The spread is wide — 0.556 to 0.724 against a chain average of 0.457. **Store 86** is the stand-out: 417,882 units (by far the largest volume in this list) with a bias of -26.1 units per row. It is a very high-volume, very heavily promoted store, and it under-forecasts hardest. It is also, coincidentally, the store in the demo context used throughout this report — which is a useful reminder that the worked example is not a cherry-picked easy case.

Store 92's small row count (3,416 against ~9,000 for the others) suggests partial coverage in the test window.

41.7 The largest individual errors

The biggest misses are, structurally, promotional spikes. The demo series provides a clean instance: week 395 at store 86 sold **17,824 units at \$1.50** against 11–25 units at \$3.35 in surrounding weeks — a 700× movement. No model built on these features predicts that; the feature set contains a binary promotion flag, and in that week the flag was in fact NONE_RECORDED.

This also explains the RMSE/MAE ratio of **8.5** (72.06 / 8.50). A handful of enormous misses dominate the squared error while barely moving the absolute error. Reporting both, plus WAPE, plus bias, plus segment breakdowns is what makes the error structure legible instead of hiding behind one number.

41.8 Stability through time

From the monitoring artifact, test-window accuracy by calendar quarter:

period	n	MAE	RMSE	WAPE	bias
1996 Q2	174,937	7.566	42.336	0.4091	-2.928
1996 Q3	171,403	8.870	53.044	0.4577	-4.140
1996 Q4	150,001	7.771	121.335	0.4645	-2.775
1997 Q1	182,298	8.653	41.732	0.4634	-4.592
1997 Q2	70,401	11.108	89.848	0.5278	-6.156

Accuracy degrades monotonically the further the model gets from its training window — 0.409 → 0.528 over four quarters. That is model decay, visible and measured. The last quarter is also the shortest (70,401 rows) so it is the noisiest estimate, but the direction is consistent across all five periods.

The operational conclusion: **a deployed version of this model would need periodic retraining**, and the monitoring script is the instrument that would trigger it.

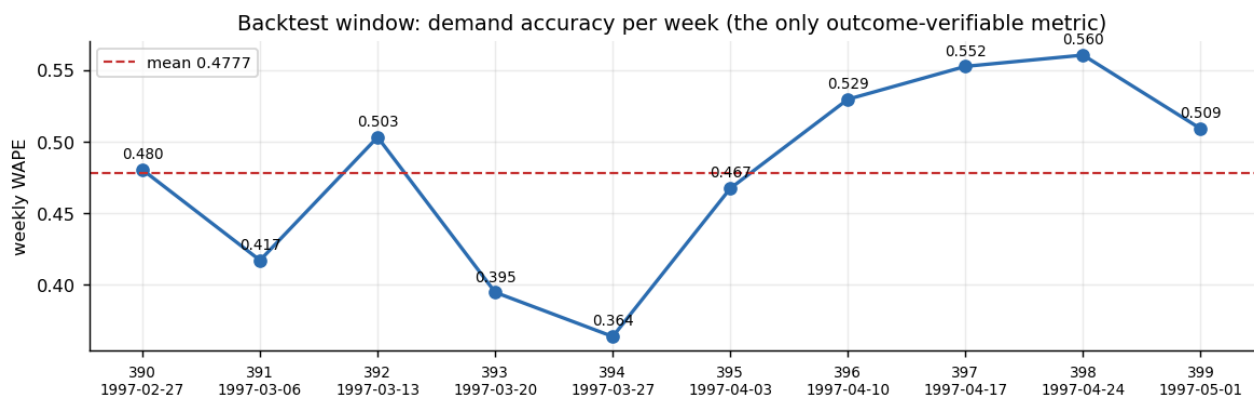


Figure 10 — Weekly WAPE across the ten backtest weeks 390–399 (the last ten weeks of the test window). Mean 0.4777, range 0.364–0.560. This is the only outcome-verifiable metric in the backtest. Source: [artifacts/metrics/backtest.json](#).

PART VII — HYBRID PRICE RESPONSE

42. Why forecasting and price response were separated

42.1 The discovery

`scripts/price_response.py` (Phase F) exists to answer a question that no accuracy metric can: **does the fitted model actually respond to price, and does it respond sensibly?** It takes 300 real decision contexts from week 399, sweeps the price $\pm 30\%$ around the current level, recomputes every price-dependent feature at each candidate, and measures the resulting demand curve.

diagnostic	result
contexts tested	300
share of demand curves monotone decreasing	100.0%
share with any increasing segment	0.0%
share flat (relative demand span < 1%)	0.0%
share producing negative predicted units	0.0%
median relative demand span across the sweep	1.832
median local implied elasticity	-3.102
p10 / p90 local implied elasticity	-3.482 / -2.667
share of contexts where the profit optimum is below the current price	26.67%
share of contexts where the revenue optimum is below the current price	100.00%
share of contexts where the profit optimum sits at a grid edge	7.67%

The first five rows are a clean pass. The model responds to price, always in the right direction, never flat, never negative. Nothing here is broken.

Row seven is the problem. A median implied elasticity of **-3.102** against controlled econometric estimates of **-1.909 to -2.419** — and against a training-window pooled estimate of **-2.029** whose two-way clustered 95% interval is **[-2.236, -1.821]**. The forecaster's price response is not merely at the edge of that interval; it is far outside it.

Note also row 12: the revenue optimum is below the current price in **100%** of contexts, while the profit optimum is below it in only 27%. Same demand curve, two objectives, two completely different answers — the §2.2 result appearing in real data.

42.2 Why the discrepancy matters so much

Suppose the truth is -2.0 and the model believes -3.1. Then at a candidate price 10% above current, the model predicts

$$(1.10)^{-3.1} = 0.735 \quad \text{versus the truth} \quad (1.10)^{-2.0} = 0.826$$

— it expects to lose **26.5%** of volume where the truth is **17.4%**. The model is systematically pessimistic about price increases, which biases every recommendation downward. The error compounds with the size of the move, and it is invisible to any accuracy metric because the counterfactual is never observed.

42.3 Why the forecaster gets the slope wrong

Three mechanisms, in rough order of importance.

(a) It is not identified on price. The model's job is conditional-mean accuracy at *observed* prices. Nothing in the loss function requires the price coefficient to be the causal slope; it only has to help predict. The lag, seasonality and promotion features carry most of the signal, and `log_price` plus seven correlated relative-price features absorb whatever residual structure remains — including the endogeneity that fixed effects remove in the econometric specifications.

(b) It has no fixed effects. The econometric estimates get to -2.4 by absorbing UPC and store effects. The ridge model has `store` and `com_code` one-hots and a smoothed `upc_demand_prior`, but no $\text{UPC} \times \text{store}$ fixed effect and no within transformation. Much of the composition bias that §27.2 diagnoses is therefore still live in its price block.

(c) Promotion contamination is worse in the forecaster. The econometric specifications control for the promotion flag *and* seasonality *and* trend simultaneously. The forecaster has the flag, but its price features and its promotion feature are fitted jointly on data where the two move together, so the price block absorbs part of the promotional lift.

42.4 The principle this generalises to

The best forecaster is not necessarily the best intervention model.

A predictive model is optimised for $P(Y | X)$ on the distribution it was trained on. An intervention model needs the *response surface* to be right in a direction the training distribution barely explores. These are different objectives, they are optimised by different estimators, and a model can be excellent at one and unreliable at the other.

This is the single most portable lesson in the project.

42.5 The architectural response

Rather than discard the forecaster (its contextual skill is real and measured) or trust its price coefficient (it is not defensible), Phase L **split the two jobs**:

job	owner	evidence for it
where is demand <i>now</i> , given all context?	the ML forecaster	test WAPE 0.4565 on 749,040 held-out rows
how does demand move <i>with price</i> ?	a separately estimated elasticity	pooled -2.029 , two-way clustered CI $[-2.236, -1.821]$, 239 shrunk product estimates

The native ML response was **not deleted**. It is retained as `PriceResponseMethod.ML` and is the benchmark in `reports/08_PRICE_RESPONSE_COMPARISON.md` and `reports/16_OUT_OF_TIME_PRICE_RESPONSE.md`. Keeping the thing you rejected, and measuring against it, is what makes the decision auditable rather than assertive (DECISIONS #33).

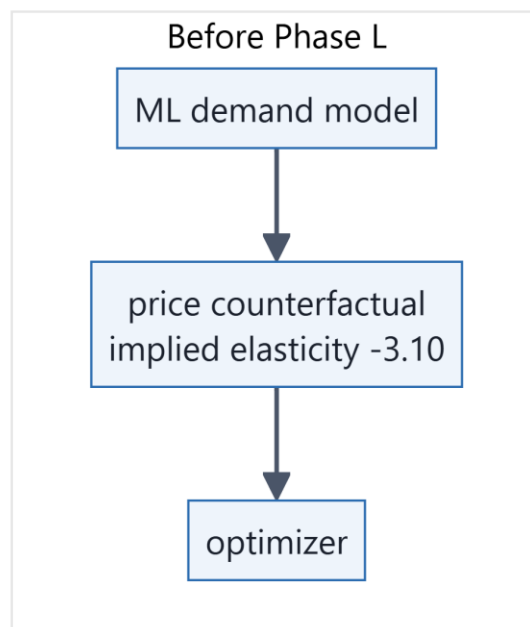
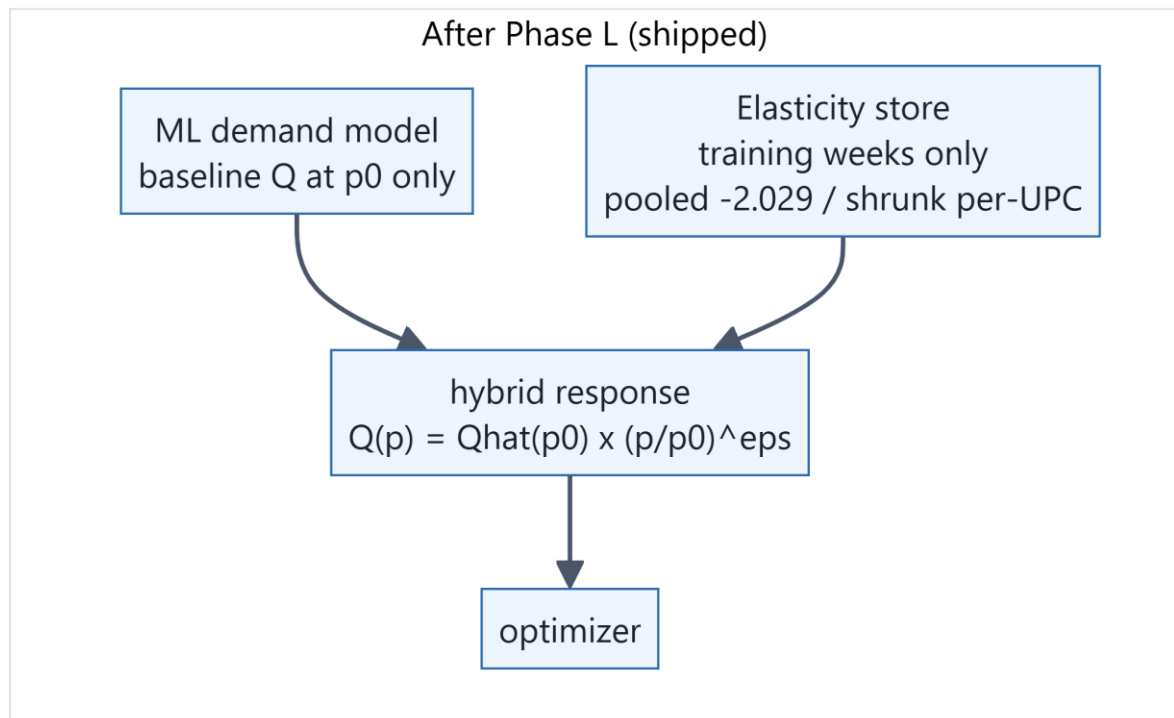


Diagram 2

43. Hybrid formulation

43.1 The equation

$$[Q(p) = \hat{Q}(p_0) \cdot (p/p_0)^\epsilon]$$

43.2 Every term defined

term	definition	provenance
------	------------	------------

term	definition	provenance
p	the candidate price being scored	the decision variable
p_0	the reference price — the observed current price of the decision context	stamped from the observed row <i>before</i> any candidate is substituted
$\hat{Q}(p_0)$	the ML forecaster's prediction evaluated at the reference price , with all price-dependent features recomputed at p_0	M1 ridge log-log
ϵ	the price elasticity applied to this row	pooled -2.029 , or the shrunk per-UPC estimate; clipped to $ \epsilon \in [0.2, 6.0]$
$Q(p)$	the counterfactual demand used by the simulator and optimizer	—

43.3 Implementation

```

p = frame["effective_unit_price"].to_numpy("float64")
p0 = frame[REFERENCE_PRICE_COLUMN].to_numpy("float64")
if np.any(~np.isfinite(p0)) or np.any(p0 <= 0):
    raise HybridModelError("reference_price must be positive and finite for every row")

# Baseline demand is evaluated AT the reference price, so every price-dependent
# feature is consistent with p0 and the price effect comes only from epsilon.
baseline_frame = recompute_price_features(frame, p0)
q0 = np.asarray(self.base_model.predict(baseline_frame), dtype="float64")

epsilon = self.elasticity_for(frame)
ratio = np.power(p / p0, epsilon)
units = np.clip(q0 * ratio, 0.0, None)
cap = getattr(self.base_model, "prediction_cap_", None)
return np.clip(units, 0.0, cap) if cap else units

```

python

Three details worth naming:

64. **recompute_price_features(frame, p0)** is called **explicitly**. Even though the frame arrives with the candidate price in `effective_unit_price`, the baseline is re-evaluated with *every* price-dependent feature reset to p_0 . Without this, `price_vs_last_week` and friends would still carry the candidate, and the baseline would not be a baseline.
65. **A missing reference price raises**. During candidate scoring the row's price *is* the candidate, so a silent fallback to "the row's current price" would make $p/p_0 = 1$, the elasticity term would vanish, and the engine would quietly return baseline demand for every candidate — a silent no-op that would look like a working system (DECISIONS #41).
66. **The base model's prediction cap is inherited**, so the hybrid cannot exceed the forecaster's own safety bound.

43.4 The anchoring property, stated precisely

At $p = p_0$:

$$Q(p_0) = \hat{Q}(p_0) \cdot (p_0/p_0)^\epsilon = \hat{Q}(p_0) \cdot 1^\epsilon = \hat{Q}(p_0)$$

The hybrid reproduces the base model's prediction **exactly at the reference price**, bit for bit. This is pinned by 15 parametrised regression tests in `tests/test_phase_1_pricing.py` spanning 5 elasticities $\times$ 3 price levels.

43.5 What this does NOT mean — the claim that was removed

An earlier version of this repository said "*no forecast skill is lost*." That claim was **removed repository-wide** in Phase M (DECISIONS #51), because it is a statement about **all** prices and the equality only holds at **one**.

The precise statement, from the module docstring:

Preserving the prediction at p_o says nothing about accuracy at any other price. Away from the reference price the hybrid's demand curve is the elasticity model's, not the forecaster's, and its counterfactual accuracy is a separate empirical question — measured out of time in `reports/16_OUT_OF_TIME_PRICE_RESPONSE.md`, not assumed here.

This report uses the precise statement everywhere: **the hybrid preserves the forecaster's baseline prediction at the reference price, not necessarily its accuracy at counterfactual prices.**

43.6 The property with the largest downstream consequence

Because $\hat{Q}(p_o)$ does not depend on p , it is a **positive constant across the candidate grid**. Therefore, for the gross-profit objective:

$$\begin{aligned} p (p - c) \cdot \hat{Q}(p_o) (p/p_o)^\epsilon \\ = p (p - c) (p/p_o)^\epsilon \end{aligned}$$

$\hat{Q}(p_o)$ **cancels out of the arg-max**. The demand *forecast* sets the predicted volume and every dollar amount reported, but it **does not choose the price**. The chosen price is a function of (p_o, c, ϵ) and the guardrails only.

This is verified numerically, not just algebraically: `tests/test_attribution.py` scales the base model by 10x and asserts that **every recommended price is unchanged**.

It also yields a closed form for the unconstrained optimum:

$$p^* = c \cdot \epsilon / (1 + \epsilon) \quad (\epsilon < -1), \quad p^* \rightarrow \infty \quad (-1 < \epsilon < 0)$$

which is the basis of the constraint-attribution audit (§54).

43.7 The three available methods

method	ϵ used	role
m1	none — the base model answers price counterfactuals directly	benchmark, retained
pooled	one category elasticity, -2.029 , for every row	simple, transparent
shrunk ☒	per-UPC empirical-Bayes estimate; pooled fallback	shipped default

Selected on out-of-time evidence, not inheritance (§45, DECISIONS #50).

43.8 What the hybrid buys and what it costs

Buys:

- the price counterfactual is governed by an estimate whose provenance, standard error, clustering, shrinkage weight and estimation window are all known and recorded;
- the forecaster's contextual skill (seasonality, promotion state, series level, lag structure) is retained in the baseline;
- the two components can be audited, criticised and replaced independently.

Costs:

- the response is now constant-elasticity by assumption.** The functional form is imposed, not learned. A real demand curve with kinks, reference-price effects or threshold behaviour is not representable;

- **no price × context interaction.** The elasticity is per-UPC, so it cannot differ between a promotion week and a normal week for the same product — even though \$28.4 shows those elasticities differ by 0.83;
- **the elasticity is still observational.** The hybrid replaces one observational estimate with a better-identified observational estimate. It does not manufacture causal identification.

44. Native ML vs pooled vs shrunk

scripts/compare_price_response.py --n-contexts 300 → artifacts/metrics/price_response_comparison.json, reports/08_PRICE_RESPONSE_COMPARISON.md.
Week 399, standard policy profile, **identical contexts and identical constraints** for all three methods — the only thing that varies is the price response.

44.1 Recommendation behaviour by method

metric	ml	pooled	shrunk
share actionable (RECOMMEND_CHANGE)	77.33%	70.00%	74.67%
share KEEP_CURRENT	10.33%	16.67%	11.67%
share REVIEW_REQUIRED	12.33%	13.33%	13.67%
median absolute price change	4.84%	8.36%	7.54%
share of changes that are increases	73.28%	94.76%	91.96%
median model-internal estimated uplift	8.28%	17.57%	13.91%
mean recommended price	\$3.181	\$3.268	\$3.268

44.2 Reading the differences

The direction of the recommendation is the biggest divergence. The native ML response proposes an *increase* 73.3% of the time; the pooled response 94.8% of the time. The mechanism is exactly the closed form: with $\epsilon = -3.10$, $p^* = c \cdot \epsilon / (1 + \epsilon) = 1.476 \cdot c$; with $\epsilon = -2.029$, $p^* = 1.972 \cdot c$. A steeper elasticity puts the optimum **closer to cost**, so the ML response proposes price cuts far more often.

The magnitude difference follows. Median absolute change 4.84% (ml) versus 8.36% (pooled): the pooled response's optimum is further above the current price, so it runs into the $\pm 10\%$ cap more often.

The internal uplift claim more than doubles between methods — 8.28% to 17.57%. Two methods, the same model, the same constraints, the same contexts, and one claims twice the benefit of the other. **This is why no single uplift number in this project is quoted without naming the price-response method that produced it.**

shrunk sits between the two, closer to pooled. That is the expected consequence of shrinking toward the pooled prior with a mean weight of 0.78 and using the pooled fallback for 35.8% of contexts.

44.3 Pairwise disagreement on identical contexts

pair	mean abs price gap	median	p90	mean gap as % of price	identical price	same decision state	opposite direction	correlation
ml vs pooled	\$0.087	\$0.00	\$0.300	2.74%	63.0%	79.7%	4.0%	0.981
ml vs shrunk	\$0.090	\$0.00	\$0.302	2.87%	64.7%	86.7%	7.7%	0.975
pooled vs shrunk	\$0.032	\$0.00	\$0.100	0.95%	86.7%	91.7%	2.0%	0.989

Three observations:

67. **The median gap is \$0.00 in all three pairs.** Most of the time the constraints bind and all three methods land on the same guardrail corner. The disagreement lives in the tail: the p90 gap between ml and pooled is **\$0.30**, which on a ~\$3.18 price is a 9.4% difference.
68. **ml and pooled agree on the decision state only 79.7% of the time.** One context in five gets a different *decision* — not merely a different price — purely from the price-response assumption.
69. **They point in opposite directions in 4.0–7.7% of contexts.** One method says raise, the other says cut.
- Correlations of 0.975–0.989 look reassuring and are somewhat misleading: they are dominated by the fact that all three methods start from the same current price and are bounded by the same $\pm 10\%$ cap.

A documentation-drift note: README.md, KNOWN_LIMITATIONS.md #28 and reports/VALIDATION_SUMMARY.md row 15 used to quote "ml vs shrunk 87.3%" and "pooled vs shrunk 89.0%". The current artifact (artifacts/metrics/price_response_comparison.json) gives **86.7%** and **91.7%**. The ml-vs-pooled figure (79.7%) matches. This report uses the artifact values.

44.4 The elasticity sensitivity probe

The same 20 real contexts were re-optimised under four **assumed** elasticities — -1.5 , -1.9 , -2.4 , -3.1 — under two guardrail regimes.

regime	median spread of the candidate profit-maximising price across the four scenarios
wide what-if guardrails	21.51%
production guardrails (standard)	0.00%

Under research conditions, the elasticity assumption moves the answer by more than a fifth of the price. Under production conditions it moves it **not at all**, because the $\pm 10\%$ change cap binds first in almost every context.

A worked row from the sensitivity table — HONEY NUT CHEERIOS (1600068290), store 116, current price \$4.79, unit cost \$3.628:

assumed ϵ	analytic CE optimum $c \cdot \epsilon / (1 + \epsilon)$	candidate price, wide	candidate price, constrained	decision
-1.5	\$10.885	\$7.63	\$5.11	REVIEW_REQUIRED
-1.9	\$7.660	\$7.63	\$5.11	REVIEW_REQUIRED
-2.4	\$6.220	\$6.23	\$5.11	REVIEW_REQUIRED

The unconstrained optimum ranges over **\$6.22 to \$10.89** — a 75% spread — and the constrained candidate is **\$5.11 in every case**. The guardrails have absorbed the entire elasticity uncertainty.

44.5 What these differences teach

70. **Recommendations depend materially on the price-response assumption**, and that dependence is measurable. This is the strongest argument for why the elasticity had to be estimated properly rather than inherited from the forecaster.
71. **The production guardrails largely neutralise the dependence** — which is simultaneously reassuring (the system is robust to being wrong about elasticity) and deflating (the system is therefore not very sensitive to being *right* about it either). §55 pursues this honestly.
72. **Any uplift figure is meaningless without its method label**, since the two plausible methods differ by 2× on the same contexts.

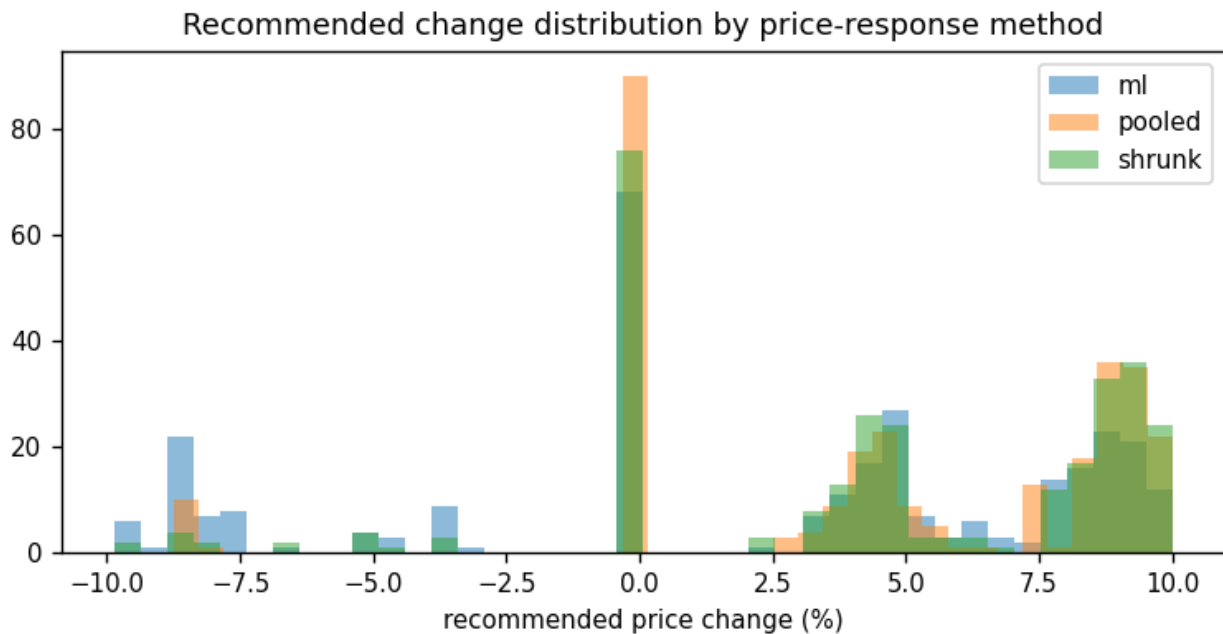


Figure 11 — Distribution of recommended price changes under the three price-response methods on identical contexts. Source: `scripts/compare_price_response.py`.

45. Out-of-time price-change validation

`scripts/audit_out_of_time_response.py` → `artifacts/metrics/out_of_time_price_response.json`,
`reports/16_OUT_OF_TIME_PRICE_RESPONSE.md`.

This is predictive / naturalistic validation, not causal identification.

Prices in the Dominick's data were set by the retailer, not randomised. An observed demand move after a price move confounds the price effect with whatever caused the retailer to change the price. Nothing in this section identifies a causal price effect, and no number here may be described as one.

45.1 The question it does answer

Given a price change that **actually happened** in weeks the elasticity estimator never saw, which price-response method best predicts the demand that followed?

This is the closest thing to an out-of-sample test of a counterfactual model that observational data allows.

45.2 Design

- Elasticities fitted on training weeks **2–257** only. Every episode is from weeks **258–399** (validation + test).
- An **episode** is a UPC × store series observed in two consecutive weeks, both with positive price and positive units, where the price moved by **≥ 5% and ≤ 60%**.
- **154,899 episodes** qualify. Three further episodes were dropped because the contextual baseline predicted zero units, making every ratio metric undefined.
- The contextual baseline uses **week t's context at the previous price**, so seasonality, trend, promotion coding and the lagged demand level are all absorbed and the elasticity only has to explain the deviation the price move caused:

$$\text{baseline}_t = Q_ML(p_{t-1} \mid \text{context of week } t),$$

$$\hat{Q}_t = \text{baseline}_t \cdot (p_t/p_{t-1})^\epsilon$$

- $\epsilon = 0$ (baseline only) is included as the null model. A price-response method that cannot beat it is adding nothing.
- 73.3% of episodes carry a recorded promotion code in at least one of the two weeks, so promotion and non-promotion splits are reported separately.

45.3 Results — all episodes (n = 154,899)

method	WAPE	MAE (units)	bias	sign accuracy	median abs error in log demand change	rank corr
null ($\epsilon = 0$)	0.8086	45.43	-37.26	n/a	0.697	n/a
pooled elasticity	0.6249	35.11	-29.35	81.7%	0.436	+0.796
shrunk product elasticity	0.5853	32.89	-24.99	81.7%	0.437	+0.784
native ML price response	0.6042	33.95	-26.07	81.7%	0.459	+0.786

45.4 Results — no recorded promotion (n = 41,310)

This is the cleanest split available and the one that should carry the most weight, because it removes the episodes where price moved as part of a bundle.

method	WAPE	MAE	bias	sign accuracy	median abs log error	rank corr
null ($\epsilon = 0$)	0.7152	24.05	-18.33	n/a	0.440	n/a
pooled elasticity	0.5944	19.99	-14.77	68.3%	0.392	+0.553
shrunk product elasticity	0.5596	18.82	-12.76	68.3%	0.396	+0.543
native ML price response	0.5877	19.76	-13.29	68.3%	0.436	+0.543

45.5 Results — recorded promotion in either week (n = 113,589)

method	WAPE	MAE	bias	sign accuracy	median abs log error	rank corr
null ($\epsilon = 0$)	0.8263	53.21	-44.15	n/a	0.819	n/a
pooled elasticity	0.6307	40.61	-34.65	86.6%	0.454	+0.818
shrunk product elasticity	0.5902	38.00	-29.44	86.6%	0.454	+0.810
native ML price response	0.6073	39.10	-30.72	86.6%	0.467	+0.812

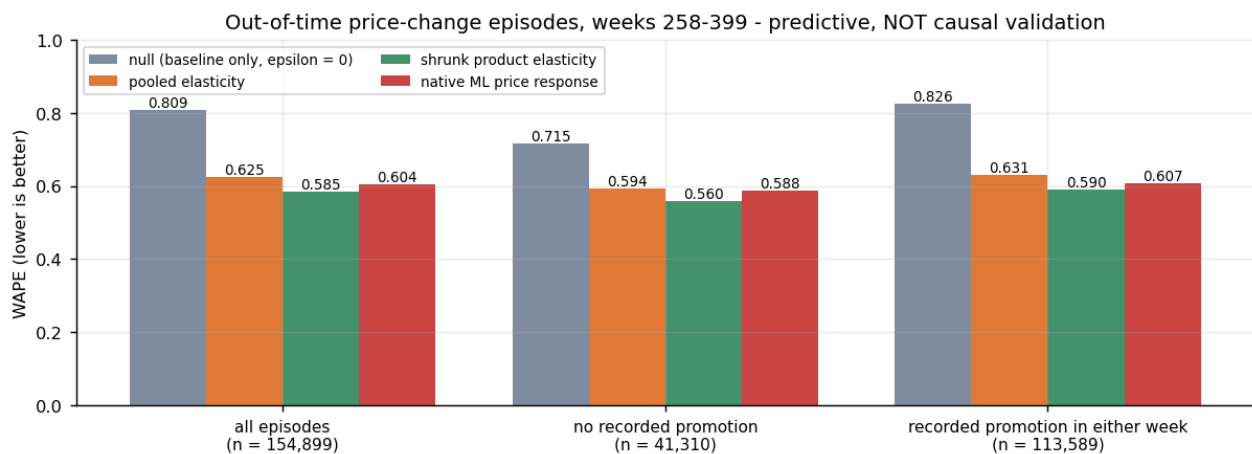


Figure 12 — WAPE by method across the three episode splits. *shrunk* wins in every split. All four methods beat the no-price-response null.
Source: `artifacts/metrics/out_of_time_price_response.json`.

45.6 What can be concluded

Every price-response method beats the null in every split. Modelling a price response is worth doing: on non-promotion episodes, WAPE falls from 0.715 to 0.560 — a **21.7% relative improvement** — purely from applying an elasticity to the contextual baseline.

shrunk has the lowest WAPE in every split, and by a consistent margin: 0.5853 / 0.5596 / 0.5902 against native ML's 0.6042 / 0.5877 / 0.6073 and pooled's 0.6249 / 0.5944 / 0.6307. This is the evidence on which **shrunk** was retained as the default (DECISIONS #50) — **on measurement, not inheritance**.

shrunk also has the smallest bias, materially: -12.76 units on non-promotion episodes against -18.33 for the null. All methods still under-predict, consistent with \$41.4.

Sign accuracy is identical across methods (81.7% / 68.3% / 86.6%) to three decimal places. That is expected and worth understanding: sign accuracy asks "did demand move in the direction the price move implies?", and since every method uses a negative elasticity, they all predict the same *direction*. The metric therefore discriminates between having a price response and not having one, not between methods.

The observed implied elasticity of these episodes is a useful independent anchor: median **-2.452** across all episodes, **-1.862** on non-promotion episodes. The non-promotion figure sits very close to the training-window pooled estimate of -2.029 and its interval [-2.236, -1.821], estimated on completely disjoint weeks. That is meaningful corroboration — and it is further evidence against the forecaster's -3.10.

Interesting anomaly: pooled has the *best* median-absolute-log-error (0.392 vs shrunk's 0.396) and the best rank correlation (+0.553 vs +0.543) on non-promotion episodes, while losing decisively on WAPE and MAE. The interpretation is that the pooled estimate is marginally better on *typical* episodes, and the shrunk estimate is materially better on the **high-volume** episodes that WAPE weights. Since the business cares about volume-weighted error, WAPE is the right tie-breaker — but the disagreement is real and is reported rather than smoothed away.

45.7 What cannot be concluded — the four limits

73. **Weeks in which the retailer changed price are not a random sample of weeks.** Selection on the treatment is exactly the condition under which observational comparisons fail.
74. **73.3% of episodes carry a promotion code**, so even the "all episodes" result is dominated by bundled events. This is why the non-promotion split is reported separately — and even that split contains *uncoded* promotions (§11.4).
75. **The baseline uses week t 's context**, including week t 's promotion flag. That is the right design for isolating the price effect, but it means the comparison is conditional on knowing the promotion state.
76. **This validates a price-response predictor, not a causal effect.** A method could win here by correctly predicting the confounded association and still be wrong about the intervention.

45.8 Why this is nonetheless the strongest empirical result in the project

Look at what it is not. It is not a self-scored simulation (the backtest is — §64). It is not a claim about a price nobody charged. It is a comparison of four candidate models against **154,899 realised outcomes**, on weeks the estimator never saw, where the losing methods include the forecaster's own native response and a no-price-response null. The winner was then made the default.

That is a defensible model-selection procedure for a counterfactual component, executed on real data, with the causal limitation stated in a box at the top of its own report.

PART VIII — PRICE SIMULATION

46. Candidate price generation

46.1 The grid

src/pricing_engine/simulation/price_grid.py:

$$G = \{\text{round}(\ell + k \cdot s, r) : k = 0, 1, \dots, \lfloor (h - \ell) / s \rfloor\} \cup \{\text{round}(p_0, r)\}$$

then filtered to $[\ell, h]$, de-duplicated, and required to be strictly positive.

parameter	config key	value
step s	optimization.price_step	\$0.05
rounding r	optimization.price_rounding	\$0.01
charm pricing (snap to x.x9)	optimization.charm_pricing	false
lower bound ℓ , upper bound h	from build_bounds (\$53)	per context

46.2 The current price is always a candidate

```
if include_current:
    prices = np.append(prices, current_price)
```

python

"Keep the current price" must be a **scoreable option**, not a special case handled by a separate branch. This matters for correctness: the materiality check compares the best candidate's objective value against the objective value *at the current price on the same grid*, and both must be produced by the same scoring path.

46.3 Why \$0.05 and \$0.01

\$0.05 is the granularity at which grocery shelf prices actually move; \$0.01 is the granularity at which they are *displayed*. Scoring at 5-cent resolution and rounding to the cent matches the physical decision.

The cost of the grid is measured rather than assumed: the grid optimum sits a **median of \$0.025** (p90 **\$0.043**) from the continuous optimum inside the same feasible interval — about 1% of a typical \$3.18 price. \$54.4 uses that number.

46.4 Failure modes are loud

```
if high < low:
    raise PriceGridError(
        f"empty feasible price range [{low:.4f}, {high:.4f}]: the constraints "
        "leave no candidate price.")
if prices.size == 0:
    raise PriceGridError(
        f"no candidate prices survived rounding in [{low:.4f}, {high:.4f}] with step {step}")
```

python

An empty grid raises rather than returning an empty array that would silently produce `argmax` on nothing. The optimizer catches the upstream case (`bounds.is_empty`) and returns `KEEP_CURRENT` with `NO_FEASIBLE_PRICE`.

46.5 Grid size in practice

For the worked context (current \$3.35, feasible \$3.015–\$3.685): 14 grid points at \$0.05 spacing plus the current price. Across all 13,964 week-399 contexts, the $\pm 10\%$ change cap alone removes a median of **293** candidates from the wide research grid, and the extrapolation guardrail a median of **264**.

47. Counterfactual feature construction

47.1 The two invariants

Stated in the module docstring of `simulation/counterfactual.py` and enforced by tests:

1. **Only price-dependent features change.** All context features (lags, rolling means, calendar, promotion, cost) are held fixed.
2. **Cost is held fixed.** The historical accounting margin is not reused at a new price.

47.2 What changes when the candidate price changes

Exactly the five features in `PRICE_DEPENDENT_FEATURES`:

feature	recomputation
<code>effective_unit_price</code>	<code>= p</code>
<code>log_price</code>	<code>= log(p)</code>
<code>price_vs_last_week</code>	<code>= p / lag_price_1 - 1</code>
<code>price_vs_series_reference</code>	<code>= p / series_reference_price - 1</code>
<code>price_vs_recent_mean</code>	<code>= p / roll_mean_price_4 - 1</code>

The denominators — `lag_price_1`, `series_reference_price`, `roll_mean_price_4` — are **historical context and do not move**. That is the whole point: the relative-price features must express *this candidate against the actual history*, not against itself.

47.3 What stays fixed

Everything in `CONTEXT_FEATURES` (24 columns): `store`, `upc`, `com_code`, `week_of_year`, `month`, `quarter`, `time_index`, `recorded_promotion_flag`, `lag_move_1..4`, `roll_mean_move_4/8/13`, `roll_std_move_4`, `lag_price_1`, `lag_price_2`, `roll_mean_price_4`, `series_reference_price`, `decision_time_unit_cost`, `lag_promotion_1`, `series_age_weeks`, `package_size_oz`.

47.4 Why target-derived context cannot be changed consistently

Suppose you wanted the simulation to be "more realistic" by letting the demand history respond to the candidate price. You immediately face an impossible regress:

- `lag_move_1` is last week's **realised** demand. Changing this week's price cannot change it.
- To make a multi-week counterfactual coherent you would need a **dynamic** model: choose `pt`, predict `Qt`, feed `Qt` into `lag_move_1` for `t+1`, and so on — accumulating prediction error at every step and requiring a *policy* over the whole horizon rather than a single price.
- Changing *some* context (say, letting `roll_mean_move_4` respond) while leaving the rest fixed produces an internally inconsistent frame: the model would see a demand history that never happened alongside a price history that did.

The engine therefore answers a precise, single-period question: **holding this week's context fixed at what actually obtained, what would demand have been at price `p`?** That is a well-posed question. "What would the world look like after eight weeks of a different pricing policy?" is not answerable with this architecture, and the project does not pretend otherwise.

47.5 Batched scoring

```
frame = pd.concat([row] * grid.size, ignore_index=True)
frame = recompute_price_features(frame, grid)
units = np.asarray(model.predict(frame), dtype="float64")
```

python

The whole candidate grid is scored in **one** model call, not a Python loop. `simulate_many` generalises this to `(n_contexts × n_candidates)` rows in a single call — the path that would make batch pricing over all 36,443 series tractable. (The batch script does not currently use it; limitation #18.)

47.6 Output validation

```
if np.any(units < 0):  
    raise SimulationError("demand model returned negative predicted units")
```

python

Negative predicted demand is a hard error, not something to clip and carry on with. In the Phase F validation over 300 contexts and a $\pm 30\%$ sweep, **0%** of predictions were negative.

47.7 Curve diagnostics

`demand_curve_diagnostics()` returns, for every simulated curve:

diagnostic	why it matters
share_segments_increasing	fraction of adjacent grid steps where demand <i>rises</i> with price
monotone_decreasing	boolean — does the curve slope the right way everywhere?
relative_demand_span	$(\max - \min)/\text{mean}$ — does the curve move at all?
median_local_elasticity	arc elasticity between adjacent grid points
flat_response	<code>relative_demand_span < 0.01</code>

These are the diagnostics that make an *accurate* model *dangerous* visible: a model can have excellent WAPE and produce an upward-sloping or flat demand curve, and only this check would catch it. Under the hybrid response, monotonicity is guaranteed by construction for $\epsilon < 0$; the diagnostics remain because the `m1` method is still selectable and because guarantees that are also checked are worth more than guarantees that are only asserted.

48. Cost treatment during simulation

This section is short and is one of the two or three most important in the report.

48.1 The error this prevents

The canonical table contains `gross_margin_rate` for every historical observation. The tempting shortcut is:

```
# WRONG  
expected_gross_profit = candidate_price * predicted_units * historical_margin_rate
```

python

This is catastrophic, and it is catastrophic in a way that *looks like a result*. Holding the margin **percentage** fixed means the implied unit cost scales with the candidate price:

$$c(p) = p(1 - m) \implies GP(p) = pmQ(p)$$

Since `m` is a constant, gross profit becomes proportional to **revenue**. The gross-profit objective silently collapses into a revenue objective — and worse, because raising the price raises the implied cost, the model never sees the margin expansion that a real price increase produces. Every recommendation would be wrong, and the numbers would look entirely plausible.

48.2 The correct method

$$GP(p) = (p - c) \cdot \hat{Q}(p)$$

where `c` is a decision-time cost estimate held fixed across the entire candidate grid.

```
if unit_cost is not None and np.isfinite(unit_cost):
    out["unit_cost"] = float(unit_cost)
    out["expected_gross_profit"] = (grid - float(unit_cost)) * units
    out["expected_margin_rate"] = np.where(grid > 0, (grid - float(unit_cost)) / grid, np.nan)
```

python

Note that `expected_margin_rate` is now an **output**, computed from the candidate price and the fixed cost — not an input assumed constant. In the worked example (§2.3) it ranges from −1.9% at \$2.51 to +37.8% at \$4.11, which is exactly the margin expansion a price increase produces and exactly what the wrong method would have hidden.

`tests/test_simulation.py::test_cost_is_held_fixed_across_price_grid` asserts that the `unit_cost` column is constant across every candidate.

48.3 Why holding cost fixed is the right economics, not just the safe choice

Within a single week's pricing decision, the retailer's acquisition cost is already determined: the stock is bought, the trade deal is agreed, the invoice is written. The shelf price does not change what the retailer paid. So `c` is genuinely a constant of the decision.

The *empirical* complication from §17.1 — that AAC falls 9.2% on price cuts — does not contradict this. It says that price cuts and cost reductions **co-occur**, because promotions are trade-funded. It does not say that choosing a lower shelf price *causes* the cost to fall. Treating cost as responding to the candidate price would import that correlation into the counterfactual as if it were a mechanism, which is a second, subtler version of the same error.

48.4 Which cost, exactly

`c = decision_time_unit_cost(u, s, t) = estimated_unit_aac(u, s, t-1)`, forward-filled within the series

Three properties:

- **temporally available** — the week-`t` accounting margin is an outcome (§20);
- **causally clean** — it predates the pricing decision, so it cannot embed the trade deal that accompanied it;
- **provably so** — 4,671,333 rows compared with 0 mismatches, plus a future poisoning test with teeth and a recommendation-level check (§23.3).

48.5 When the cost is unavailable

```
if objective == "gross_profit" and unit_cost is None:
    reasons.append(ReasonCode.COST_UNAVAILABLE.value)
```

python

and the optimizer returns `KEEP_CURRENT`. The objective function refuses too:

```
if unit_cost is None or not np.isfinite(unit_cost):
    raise ObjectiveError(
        "Gross-profit optimization requires an available unit cost (estimated AAC) "
        "for this UPC and store. Use objective='revenue' or supply a cost.")
```

python

No cost is ever invented. The alternatives — impute the category median, use the current price times a default margin — would put a fabricated number directly into the objective the engine maximises. Returning "I cannot answer this" is the correct behaviour and it is tested.

48.6 The full chain of custody for cost

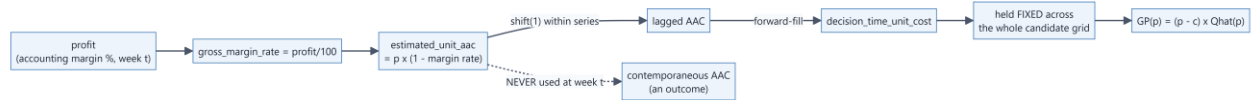


Diagram 3

PART IX — OPTIMIZATION

49. Revenue objective

$$R(p) = p \cdot \hat{Q}(p)$$

```
if objective == "revenue":  
    return p * q
```

python

Revenue optimization needs no cost, so it is the fallback when `decision_time_unit_cost` is unavailable, and it is a legitimate primary objective when the business mandate is volume or category share rather than profit.

Under the hybrid response $Q(p) = \hat{Q}(p_0)(p/p_0)^\epsilon$, revenue is $\hat{Q}(p_0) \cdot p_0^{-\epsilon} \cdot p^{1+\epsilon}$, which is:

- **monotonically decreasing** in p for $\epsilon < -1$ (elastic),
- **monotonically increasing** for $-1 < \epsilon < 0$ (inelastic),
- flat for $\epsilon = -1$.

There is no interior revenue optimum under constant elasticity — the revenue optimum is always a **boundary** of the feasible interval. Since 80.8% of UPCs in this panel are elastic, the revenue optimum is almost always the **lower** bound. Phase F confirms this empirically: **100% of 300 contexts have their revenue optimum below the current price**, against 26.7% for profit.

That is precisely the §2.2 divergence, measured.

50. Gross-profit objective

$$GP(p) = (p - c) \cdot \hat{Q}(p)$$

```
if unit_cost is None or not np.isfinite(unit_cost):  
    raise ObjectiveError("Gross-profit optimization requires an available unit cost ...")  
return (p - float(unit_cost)) * q
```

python

This is the primary objective (`optimization.objective: gross_profit`).

50.1 Why

77. It is the business objective. A retailer that maximises revenue on an elastic category prices toward cost and loses money.
78. It has a genuine **interior** optimum under constant elasticity ($p^* = c \cdot \epsilon / (1 + \epsilon)$ for $\epsilon < -1$), so the answer comes from the demand curve rather than from a boundary — at least in principle (§54 shows how often the guardrails intervene in practice).
79. It forces the cost question to be confronted honestly (§8, §48). A revenue engine can pretend cost does not exist.
80. Both objectives use the **same** predicted demand, so the only difference between them is the economics applied on top — which makes the comparison between them meaningful rather than a comparison of two different models.

50.2 What it is not

It is expected **gross** profit at the item level: no labour, no shelf space, no handling, no shrink, no basket effects, no cross-elasticity, no customer lifetime value. It optimises $(p - c) \cdot Q$ for one UPC in one store in one week.

51. Why grid optimization was used

51.1 The choice

A discrete grid search over `g` rather than a continuous solver (`scipy.optimize`). DECISIONS #19.

51.2 The six reasons

- 1. Retail prices are genuinely discrete.** Shelf prices move in 5-cent steps and are displayed in cents. A continuous optimum of \$3.6473 must be rounded anyway, and rounding *after* optimising can push the answer outside the feasible interval or across a constraint boundary. Optimising *on* the grid the price will actually take avoids that entirely.
- 2. The demand model may be a step function.** The architecture supports `hgb_poisson`, whose prediction surface is piecewise constant. Gradients are zero almost everywhere and undefined on the boundaries; every gradient-based solver fails silently. A grid is indifferent to the model family — and the `m1` price-response method is still selectable.
- 3. Business constraints are not smooth.** Materiality is a threshold. The risk gate is a discrete state machine. Rounding is a step function. Charm pricing is a floor-plus-offset. Encoding these as smooth penalties would be an exercise in making a solver converge, not in making a better decision.
- 4. Auditability.** Every candidate is scored and the whole curve is available. The engine can answer "why not \$3.71?" with "it was outside the feasible interval" or "its expected gross profit was \$21.68 against \$21.74 at \$3.66" — from stored data, not from a re-run. That is what makes reason codes possible.
- 5. Low dimensionality.** One decision variable over a bounded interval. The typical feasible interval here contains 10–30 candidates; the wide research grid used by the attribution audit contains a few hundred. Exhaustive search is trivially affordable, and it is **globally optimal on the grid by construction** — no local minima, no initialisation, no convergence criterion.
- 6. Vectorisation.** The entire grid is one batched model call (\$47.5). A continuous solver would make dozens of sequential single-row predictions — slower *and* less amenable to batching across contexts.

51.3 The cost, measured

The grid quantises the answer. Measured over all 13,964 week-399 contexts, the grid optimum sits a **median of \$0.025** (p90 \$0.043) from the continuous optimum *inside the same feasible interval* — about **1%** of a typical price (limitation #41). Given a demand model with 46% WAPE, a 1% quantisation error is not the binding constraint on accuracy.

51.4 When a continuous solver would be the right choice

If the decision variable were multi-dimensional — pricing a whole category jointly, with cross-elasticities and a category-level margin constraint — a grid would be exponentially expensive and a constrained continuous solver (or a mathematical program) would be the correct tool. That is \$101's territory.

52. Analytical optimizer validation

The optimizer is validated against problems whose answers are known in closed form. This is the strongest kind of test available for an optimizer: **deterministic, closed-form ground truth, no reference to the trained model.**

52.1 Linear demand — profit optimum

$$Q(p) = a - bp, \quad \Pi(p) = (p - c)(a - bp)$$

$$d\Pi/dp = a - bp - b(p - c) = a + bc - 2bp = 0 \Rightarrow \llbracket p^* = (a + bc)/(2b) \rrbracket$$

Second-order condition: $d^2\Pi/dp^2 = -2b < 0$ for $b > 0$, so it is a maximum.

Fixture: $Q = 100 - 10p$, $c = \$2.50$.

$$p^* = (100 + 10 \times 2.50)/(2 \times 10) = 125/20 = \$6.25$$

Result: recovered **within one grid step (\$0.05)**.

```
def analytical_linear_optimum(a: float, b: float, c: float) -> float:
    """Profit-maximising price for the textbook linear demand Q(p) = a - b p."""
    if b <= 0:
        raise ObjectiveError("linear demand slope b must be positive")
    return (a + b * c) / (2.0 * b)
```

python

52.2 Linear demand — revenue optimum

$$R(p) = p(a - bp) \Rightarrow dR/dp = a - 2bp = 0 \Rightarrow \llbracket p_{R^*} = a/(2b) \rrbracket$$

Fixture: same demand. $p_{R^*} = 100/20 = \$5.00$. **Result:** recovered within one grid step.

And note the relationship the two fixtures demonstrate together: $\$6.25 - \$5.00 = \$1.25 = c/2$. The optimizer reproduces the theoretical gap between the two objectives exactly.

52.3 The full analytical test suite

tests/test_optimizer.py (23 tests) plus tests/test_attribution.py (69):

fixture	closed-form ground truth	result
linear demand profit optimum, $Q = 100 - 10p$, $c = 2.50$	$p^* = (a+bc)/(2b) = 6.25$	recovered within one grid step (0.05)
linear demand revenue optimum	$p^* = a/(2b) = 5.00$	recovered within one grid step
constant elasticity, inelastic ($\epsilon = -0.5$)	optimum at the upper guardrail	recommended price equals the feasible upper bound; PRICE_CHANGE_LIMIT raised
starting at the optimum	no change	KEEP_CURRENT, KEEP_CURRENT_OPTIMAL
gain below materiality (0.45% vs 1% threshold)	no change	KEEP_CURRENT, NON_MATERIAL_UPLIFT
hybrid with fixed ϵ	$Q(p) = Q_0(p/p_0)^{\epsilon}$ exactly	reproduced to floating-point tolerance
empirical-Bayes shrinkage	$w = \tau^2/(\tau^2 + se^2)$	weights and blend match the closed form
unconstrained CE optimum	$p^* = c \cdot \epsilon / (1 + \epsilon)$	matched by the attribution replica
base model scaled 10×	recommended price unchanged	0 changes across the test set

52.4 The inelastic fixture is the important one

test_constant_elasticity_fixture_pushes_price_to_the_upper_bound encodes the §24.1 result: for $|\epsilon| < 1$ the profit function has no interior maximum, so the optimizer *must* return the feasible upper bound and *must* say so via PRICE_CHANGE_LIMIT. A test that only checked interior optima would miss the case where the constraint layer is the only thing between the model and an absurd recommendation.

52.5 What the analytical tests do and do not prove

Prove: the search, the objective evaluation, the constraint intersection, the materiality check, the rounding and the reason-code emission are all correct. Given a demand curve, the engine finds its optimum subject to its constraints.

Do not prove: that the demand curve is right. That is Part V, Part VI and Part VII's problem, and no optimizer test can substitute for it.

53. Constraints

All guardrails are intersected into **one feasible interval** $[\ell, h]$ by `build_bounds()`, and the constraints that are **actually active at the edges** of the final interval are recorded as `binding`. Each gets its own subsection below with business rationale, mathematical form, implementation and a worked example from the demo context (current price \$3.35, unit cost \$2.5571, observed support \$1.50–\$3.79).

53.1 Absolute minimum price

- **Business rationale.** A hard floor below which no price is ever legal or sensible, independent of cost or history.
- **Form.** $p \geq p_{\min}$, default \$0.01.
- **Implementation.** `lower["ABSOLUTE_PRICE_FLOOR"] = 0.01 if abs_min_price is None else float(abs_min_price)`
- **Example.** `ABSOLUTE_PRICE_FLOOR = $0.01` — never binds here.
- **Empirical.** Present in 13,964 contexts, **removes candidates in 0**, binds at the optimum in **0**. It is a backstop, and it correctly never fires.

53.2 Absolute maximum price

- **Business rationale.** A category-level ceiling (a gouging cap, a legal cap, a brand-positioning cap).
- **Form.** $p \leq p_{\max}$.
- **Implementation.** Optional; `upper["ABSOLUTE_PRICE_CEILING"]` is only added when `abs_max_price` is supplied.
- **Example.** Not configured in any shipped profile.
- **Empirical. Present in 0 contexts.** This is a genuine gap: a production system for essential goods would need one (\$94).

53.3 Cost floor

- **Business rationale.** Never sell below cost unless loss-leading is an explicit, deliberate policy.
- **Form.** $p \geq c$ unless `allow_below_cost` is true.
- **Implementation.**

```
if not allow_below_cost:
    lower["COST_FLOOR"] = float(unit_cost)
```

python

- **Example.** `COST_FLOOR = $2.5571`.
- **Empirical.** Present in 13,964 contexts, **removes a median of 36 candidates** from the wide research grid, binds at the optimum in **0**, changes the final price in **0**. Interpretation: it is always doing work (removing sub-cost candidates) and it is never the *binding* constraint, because the minimum-margin floor sits strictly above it whenever `min_gross_margin_rate > 0`.
- `allow_below_cost` is **false in all three shipped profiles**, including the DEMO `aggressive` one.

53.4 Minimum gross margin

- **Business rationale.** A category margin floor that survives a bad cost estimate. Even if `c` is wrong, a 5% floor keeps the recommendation off the cliff.
- **Form.** Require $(p - c)/p \geq m$, i.e.

$$p \geq c/(1 - m)$$

- **Implementation.**

```
if min_margin >= 1.0:
    raise ConstraintError("min_gross_margin_rate must be < 1")
margin_floor = float(unit_cost) / (1.0 - min_margin)
lower["MIN_MARGIN"] = margin_floor
```

python

- **Example.** $m = 0.05$ (standard) $\Rightarrow \text{MIN_MARGIN} = 2.5571/0.95 = \2.6916 .
- **Empirical.** Present in 13,964, **removes a median of 39 candidates, binds at the optimum in 982**, changes the decision state in **78**, changes the final price in **802**.
- **Profile sensitivity.** conservative $m = 0.10$, standard $m = 0.05$, aggressive $m = 0.00$.

53.5 Maximum price change

- **Business rationale.** Operational and reputational. Large weekly price swings confuse shoppers, break shelf-tag processes, invite competitor response and erode trust. It also bounds the size of any single mistake.
- **Form.** $p \in [p_0(1-\delta), p_0(1+\delta)]$.
- **Implementation.**

```
max_change = float(policy.get("max_price_change_pct", 0.10))
lower["PRICE_CHANGE_LIMIT"] = current_price * (1.0 - max_change)
upper["PRICE_CHANGE_LIMIT"] = current_price * (1.0 + max_change)
```

python

- **Example.** $\delta = 0.10 \Rightarrow$ window $[\$3.015, \$3.685]$. **Both edges of the final interval come from this constraint** in the worked context.
- **Empirical.** Present in 13,964, **removes a median of 293 candidates, binds at the optimum in 10,329 (74.0%)**, changes the decision state in **166**, changes the final price in **7,874 (56.4%)**.
- **This is the dominant constraint in the system.** It is the *first* binding constraint in **44.6%** of contexts.
- **Risk interaction.** A MEDIUM-risk context has its cap tightened further:

```
policy["max_price_change_pct"] = min(float(policy.get("max_price_change_pct", 0.10)),
                                     float(cap))
```

python

standard 10% $\rightarrow$ 5%, conservative 5% $\rightarrow$ 3%.

53.6 Extrapolation guardrail

- **Business rationale.** A demand model asked about a price the series has never carried is extrapolating, and log-linear models extrapolate confidently and wrongly. Do not let the optimizer wander outside observed evidence.
- **Form.** With observed support $[p_{\min}^{\text{hist}}, p_{\max}^{\text{hist}}]$ for that UPC $\times$ store series and tolerance t :

$$p \in [p_{\min}^{\text{hist}}(1-t), p_{\max}^{\text{hist}}(1+t)]$$

- **Implementation.**

```
tol = float(policy.get("extrapolation_tolerance_pct", 0.05))
lower["OUTSIDE_EXTRAPOLATION_RANGE"] = float(hist_price_min) * (1.0 - tol)
upper["OUTSIDE_EXTRAPOLATION_RANGE"] = float(hist_price_max) * (1.0 + tol)
```

python

- **Example.** Support \$1.50–\$3.79, $t = 0.05 \Rightarrow [\$1.425, \$3.9795]$. Wider than the change window here, so it does not bind in this context.
- **Empirical.** Present in 13,964, **removes a median of 264 candidates, binds at the optimum in 2,103, changes the decision state in 1,166**, changes the final price in **1,165**.

- **Why series-specific rather than a universal $\pm X\%$.** Support is a property of the series. A UPC $\times$ store that has ranged \$1.50–\$3.79 over 359 weeks supports a much wider recommendation than one that has only ever been \$3.29 or \$3.39. A single magic number would be arbitrary in both directions (DECISIONS #22).

53.7 Historical support (as a risk input)

The extrapolation guardrail and the historical-support screen are related but distinct, and the attribution audit tracks them separately.

- **EXTRAPOLATION** is the hard interval above.
- **HISTORICAL_SUPPORT** is the *risk* pathway: `extrapolation_distance()` measures how far a candidate sits outside the observed support, and the risk bands use it directly (`max_extrapolation_pct`: 0.02 for LOW, 0.05 for MEDIUM). A candidate outside support cannot be LOW risk.
- **Empirical.** **HISTORICAL_SUPPORT binds at the optimum in 3,373 contexts (24.2%), changes the decision state in 2,668**, but changes the final price in only **24**. Read that carefully: it almost never moves the *price*; it very often changes the *decision* — by pushing a context out of LOW risk and into the gate. It is the **second most frequent first-binding constraint** (24.2%).

53.8 Rounding

- **Business rationale.** Prices are displayed in cents; shelf tags are printed at a fixed granularity.
- **Form.** $p \leftarrow r \cdot \text{round}(p / r)$, $r = 0.01$. Optional charm pricing snaps to $x.x9$: `np.floor(values * 10.0) / 10.0 + 0.09` (disabled by default).
- **Empirical.** Removes no candidates, **binds at the optimum in 6,978 (50.0%)**, changes the decision state in **0**, changes the final price in **6,978**. So in half of all contexts the reported price differs from the unrounded grid optimum — by a median of \$0.025 — and in **no** context does that change the decision.

53.9 Materiality

- **Business rationale.** With weekly WAPE around 0.46, a sub-1% estimated gain is indistinguishable from noise. Acting on it burns operational effort and customer goodwill for nothing. It also produces the **KEEP_CURRENT** outcome that any credible engine must have — a system that always finds a better price is not a system anyone should trust.
- **Form.** Let $V(p)$ be the objective. Recommend a change only if

$$(V(p^*) - V(p_0)) / (|V(p_0)|) \geq \theta$$

- **Implementation.**

```
materiality = float(policy.get("materiality_threshold_pct", 0.01))
if abs(price_change_pct) < 1e-9:
    return keep(*reasons, KEEP_CURRENT_OPTIMAL, ...)
if not np.isfinite(uplift) or uplift < materiality:
    return keep(*reasons, NON_MATERIAL_UPLIFT, KEEP_CURRENT_OPTIMAL, ...)
```

python

- **Example.** $\theta = 0.01$ (standard). The demo context's estimated uplift is **+7.23%**, comfortably material.
- **Empirical.** Binds at the optimum in **716**, **changes the decision state in 716**, changes the final price in **700**. It is the *first* binding constraint in **4.0%** of contexts. In the 3,000-context batch, **NON_MATERIAL_UPLIFT** fires **136** times.
- **Honest caveat.** 1% is **asserted, not fitted**. Calibrating it would mean fitting realised out-of-sample error against the threshold, which was not done (limitation #40).

53.10 Risk gating

- **Business rationale.** Constraints bound the *size* of a mistake; they do not bound its *likelihood*. A thin series with almost no price variation can produce a perfectly feasible recommendation that nobody should act on automatically.
- **Form.** A discrete state machine, not an interval. See §57–§59.
- **Implementation.**

```
if final_risk.level is RiskLevel.HIGH and high_risk_action != "recommend":
    if high_risk_action == "keep_current":
        return keep([*reasons, HIGH_RISK_KEEP_CURRENT, LOW_CONFIDENCE, KEEP_CURRENT_OPTIMAL], ...)
    decision = DecisionState.REVIEW_REQUIRED
    reasons.extend([HIGH_RISK_REVIEW_REQUIRED, LOW_CONFIDENCE])
```

python

- **Empirical.** Binds at the optimum in **3,370**, **changes the decision state in 1,354**, changes the final price in **2,988**. It is the *first* binding constraint in only **0.02%** of contexts (3) — because by the time it fires, something else has usually already bound.

53.11 The intersection, and a Phase M correction worth naming

```
low = max(lower.values())
high = min(upper.values()) if upper else float("inf")
binding = [name for name, value in lower.items() if abs(value - low) <= 1e-9 * max(1.0, abs(low))]
        + [name for name, value in upper.items() if abs(value - high) <= 1e-9 * max(1.0, abs(high))]
```

python

`binding` lists the constraints **actually active at the edges of the final interval**, not every constraint that tightened a running bound along the way.

An earlier version appended `PRICE_CHANGE_LIMIT` whenever the change window was narrower than the absolute floor, **even when the extrapolation guardrail later superseded it** — which over-reported how often the change cap binds. The bug was found while validating the attribution replica against the real optimizer (DECISIONS #48). It is a good example of a measurement instrument catching a defect in the thing it was built to measure.

53.12 The full stack on the worked example

bound	source	value
lower	ABSOLUTE_PRICE_FLOOR	\$0.01
lower	OUTSIDE_EXTRAPOLATION_RANGE	\$1.4250
lower	COST_FLOOR	\$2.5571
lower	MIN_MARGIN	\$2.6916
lower	PRICE_CHANGE_LIMIT ← binds	\$3.0150
upper	OUTSIDE_EXTRAPOLATION_RANGE	\$3.9795
upper	PRICE_CHANGE_LIMIT ← binds	\$3.6850
	feasible interval	[\$3.015, \$3.685]

Both edges come from the $\pm 10\%$ change cap. The unconstrained gross-profit optimum is $c \cdot \epsilon / (1 + \epsilon) = \3.875 , which sits **above** the upper bound — so the recommendation is the guardrail corner, \$3.66 after rounding to the grid.

54. Constraint attribution

`scripts/audit_constraints.py` → `artifacts/metrics/constraint_attribution.json`,
`reports/11_CONSTRAINT_ATTRIBUTION.md`. **This is the most important chapter in the report.**

54.1 The question

When this engine recommends a price, is the price chosen by the estimated demand response, or is it simply the nearest point the guardrails allow?

54.2 Method

Every one of the **13,964** UPC × store decision contexts of week 399 is run through the full pipeline **eleven times**: once unconstrained on a wide research grid (0.2× to 5× the current price), once under the **standard** policy, and once per constraint with **that constraint and only that constraint relaxed** (leave-one-out). Price response: **shrunk**.

The audit uses `pricing_engine.optimization.attribution.evaluate`, a fast replica of `optimize_price` (~450× faster, which is what makes 11 × 13,964 runs feasible). It is trusted **only** because `tests/test_attribution.py` asserts it reproduces the real optimizer's decision state, final price and reason codes exactly across **60 parameter combinations**, and it was additionally checked against **400 real week-399 contexts with zero mismatches** (DECISIONS #49).

54.3 The closed form that makes the audit possible

Under the hybrid response, the unconstrained gross-profit optimum is

$$p^* = c \cdot \varepsilon / (1 + \varepsilon) \quad (\varepsilon < -1), \quad p^* \rightarrow \infty \quad (-1 < \varepsilon < 0)$$

because $\hat{Q}(p_0)$ cancels (\$43.6). So the "what would the model want, with no guardrails?" counterfactual has an analytic answer.

54.4 Three progressively stricter readings of "the model decided this"

reading	contexts	share
The unconstrained optimum is inside the feasible interval at all	968	6.9%
The optimizer's proposal lands on it (within one 5¢ grid step)	638	4.6%
The final price lands on it (also survived materiality and the risk gate)	635	4.5%

The gap between rows 1 and 2 is **rounding**: the 5-cent grid is anchored on the lower bound, so the grid optimum sits a median of \$0.025 (p90 \$0.043) from the continuous optimum inside the same interval. The gap between rows 2 and 3 is the **non-actionable states**: a REVIEW_REQUIRED or KEEP_CURRENT context ships the current price whatever the optimizer proposed.

54.5 Where the unconstrained optimum actually wants to go

statistic	value
median unconstrained price change	+49.0%
p10 / p90	-4.7% / +174.6%
share of contexts whose unconstrained optimum is above the current price	86.7%
median FINAL price change actually recommended	+3.7%

The model, left alone, wants to raise prices by half. The system recommends +3.7%. **The guardrails are absorbing an order of magnitude.**

Why does the model want +49%? Because $p^* = c \cdot \epsilon / (1 + \epsilon)$ and, at the median shrunk elasticity of -2.03 , $p^* = 1.97 \cdot c$ — nearly twice cost. Historical prices in this panel sit at a blended 15.3% margin, i.e. around $1.18 \cdot c$. A constant-elasticity model with $\epsilon \approx -2$ and no competitive dynamics simply believes the category is under-priced. Whether that is a real finding about 1990s grocery pricing or an artefact of an unidentified elasticity is exactly the question this project cannot answer (§33) — and it is the strongest possible argument for the guardrails.

54.6 Per-constraint attribution

Counts out of 13,964 contexts.

constraint	present	removes candidates	median removed when active	binding at optimum	changed decision state	changed final price
MAX_PRICE_CHANGE	13,964	13,964	293	10,329	166	7,874
MIN_PRICE	13,964	0	0	0	0	0
MAX_PRICE	0	0	0	0	0	0
MIN_MARGIN	13,964	13,964	39	982	78	802
COST_FLOOR	13,964	13,964	36	0	0	0
EXTRAPOLATION	13,964	13,964	264	2,103	1,166	1,165
ROUNDING	13,964	0	0	6,978	0	6,978
MATERIALITY	13,964	0	0	716	716	700
RISK_GATE	13,964	0	0	3,370	1,354	2,988
HISTORICAL_SUPPORT	13,964	0	0	3,373	2,668	24

Reading the columns:

- **Removes candidates** — the constraint excludes at least one price from the wide research grid.
- **Binding at optimum** — its own bound is the active edge at the chosen price (or, for the non-interval guardrails, it is the rule that fired).
- **Changed decision state / final price** — leave-one-out: relaxing *only* this constraint changes the outcome.

Three rows deserve attention:

COST_FLOOR: removes candidates in 100% of contexts, binds in 0. It is always working and never decisive, because **MIN_MARGIN** sits strictly above it whenever the margin floor is positive.

HISTORICAL_SUPPORT: changes the decision in 2,668 contexts but the price in 24. It operates almost entirely through the *risk* pathway, not the price pathway.

RISK_GATE: changes the price in 2,988 contexts. Because a gated context ships the current price instead of the proposal.

54.7 First binding constraint

constraint	contexts	share
MAX_PRICE_CHANGE	6,222	44.6%
HISTORICAL_SUPPORT	3,373	24.2%
NONE	1,924	13.8%
EXTRAPOLATION	1,877	13.4%
MATERIALITY	565	4.0%
RISK_GATE	3	0.02%

The $\pm 10\%$ price-change cap is the first thing that bites in 44.6% of contexts. In only 13.8% does nothing bind at all.

54.8 The headline: what determines the final recommendation

determinant	contexts	share
guardrail corner (optimum outside the feasible interval)	7,967	57.1%
screened out before optimisation (eligibility / cost)	3,528	25.3%
risk gate (REVIEW_REQUIRED, not actionable)	1,353	9.7%
materiality threshold (KEEP_CURRENT)	716	5.1%
learned signal (interior optimum, actionable)	400	2.9%

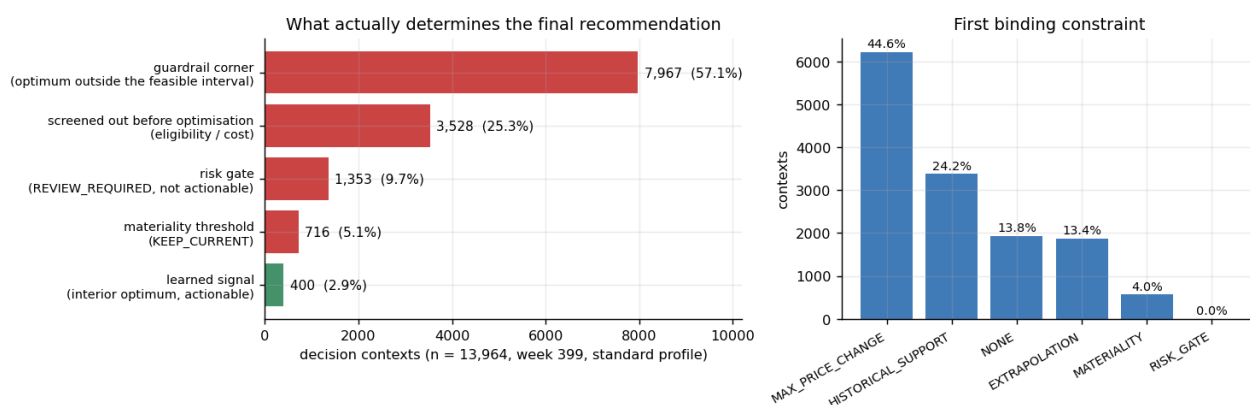


Figure 13 — Left: the final determinant of each of the 13,964 week-399 recommendations. Green is the only category in which the estimated price response chose the price. Right: the first binding constraint. Source: [artifacts/metrics/constraint_attribution.json](#).

54.9 The answer, stated plainly

How much of the final recommendation comes from the model versus policy rules?

2.9% of final recommendations are set by an interior optimum of the estimated price response. **57.1%** are guardrail corners. **25.3%** never reach the optimizer at all. **9.7%** are risk-gated. **5.1%** fail materiality.

This is a **rule-bounded pricing system with a learned direction**, not an ML pricing system.

54.10 Why this was published as the headline rather than buried

DECISIONS #43 records the choice explicitly. The alternative — present the project as "ML pricing" and put the attribution in an appendix — was rejected on the grounds that hiding it would make every other claim suspect, whereas leading with it makes the project a demonstration of **auditing**, which is the more valuable and rarer skill to show.

The finding is also, on reflection, **the correct architecture**. Given:

- an elasticity that is observational and not causally identified (§33),
- a demand model with 46% WAPE (§41),
- product-level elasticities that do not reproduce across time windows (§32),
- and no ability to measure realised outcomes at unobserved prices (§64),

...a system that let the model move prices by 49% would be reckless. The guardrails are not covering for a weak model; they are the appropriate response to a well-characterised uncertainty. The model's job in this architecture is to supply **direction and eligibility** — *should this price move, and which way?* — and the rules supply **magnitude**.

§85 returns to the question of whether that job is worth doing.

55. Guardrail ablation

scripts/audit_guardrails.py → artifacts/metrics/guardrail_ablation.json,
reports/13_GUARDRAIL_ABLATION.md.

55.1 The question

At which policy layer does the pricing signal stop determining the recommendation?

55.2 Method

All 13,964 week-399 contexts are re-optimised under five progressively stricter guardrail regimes, each under **four assumed elasticities** (−1.5, −1.9, −2.4, −3.1). The diagnostic is: how much does the recommended price move when the elasticity assumption changes?

55.3 Results

layer	disabled constraints	median price change	p10 / p90	RECOMMEND_CHANGE	KEEP_CURRENT	REVIEW_REQUIRED	share where the elasticity changes the price	median price variation across scenarios	p90 variation	share where a constraint moves the optimum
A research (cost floor only)	EXTRAP, SUPPORT, MATERIALITY, MAX_CHANGE, MIN_MARGIN, RISK_GATE	+49.1%	−4.7% / +174.4%	99.2%	0.8%	0.0%	99.99%	79.8%	80.4%	0.0%
B extrapolation only	SUPPORT, MATERIALITY, MAX_CHANGE, MIN_MARGIN, RISK_GATE	+10.8%	−4.7% / +34.2%	98.3%	1.7%	0.0%	61.0%	6.6%	38.5%	74.6%
C + price-change cap	SUPPORT, MATERIALITY, MIN_MARGIN, RISK_GATE	+7.5%	−3.8% / +9.6%	98.5%	1.5%	0.0%	30.6%	0.0%	16.9%	90.3%
D standard (shipped)	—	+3.7%	0.0% / +9.3%	59.2%	31.1%	9.7%	21.3%	0.0%	14.5%	95.9%
E conservative	—	0.0%	0.0% / +4.4%	45.5%	54.5%	0.0%	20.2%	0.0%	4.1%	98.5%

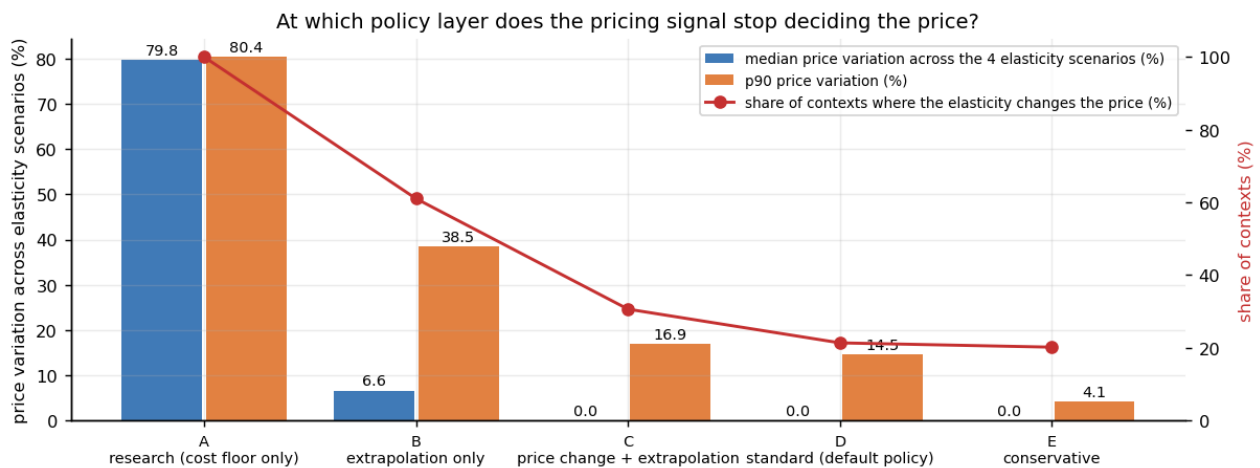


Figure 14 — Median and p90 price variation across the four elasticity scenarios (bars, left axis) and the share of contexts where the elasticity changes the price at all (line, right axis), by guardrail layer. Source: `artifacts/metrics/guardrail_ablation.json`.

55.4 The finding

The elasticity assumption matters enormously in research mode and essentially not at all in production mode.

- **Layer A (cost floor only):** the recommended price moves a median of **79.8%** across the four elasticities, and the elasticity changes the price in **99.99%** of contexts. In pure research conditions the pricing signal is everything.
- **Layer B (add the historical-support band):** median variation collapses to **6.6%** — a factor of twelve — and a constraint moves the optimum in **74.6%** of contexts. The support band alone does most of the work.
- **Layer C (add the $\pm 10\%$ change cap):** median variation reaches **0.0%**. The p90 is still **16.9%**, so the elasticity still matters in the tail, but for the median context the guardrails now decide.
- **Layers D and E:** median variation stays at **0.0%**; the p90 falls further (**14.5% $\rightarrow$ 4.1%**).

The pricing signal stops determining the median recommendation at layer C — the moment the price-change cap is added.

55.5 The honest interpretation

There are two readings and both are true.

Reading 1 — the system is robust. If the elasticity is wrong by ± 0.8 (the full -1.5 to -2.4 span, which is wider than the two-way clustered confidence interval), the shipped recommendation for the median context does not change at all. That is a real safety property, and it is why an unidentified observational elasticity can be deployed responsibly here.

Reading 2 — the system is therefore not very sensitive to being right either. You cannot claim robustness to being wrong without accepting insensitivity to being right. If the elasticity moved from -2.0 to the true value, the median recommendation would not move either.

Both are stated in the repository (`KNOWN_LIMITATIONS.md` #29, #35; `STATUS.md` open issue 1), and neither is presented without the other.

55.6 Where the elasticity does still matter

The median is **0.0%** under production guardrails, but three things remain:

81. **The tail.** p90 variation is **14.5%** under the standard profile — one context in ten still has its price materially determined by the elasticity.
82. **The direction.** The elasticity decides whether the optimum is above or below the current price. §44.1 shows this is where the methods diverge most: `m1` proposes increases **73.3%** of the time, `pooled` **94.8%**.

83. **Eligibility and risk.** A product with no usable own estimate gets POOLED_ELASTICITY_FALLBACK and can never be LOW risk (\$57.4). The elasticity's *provenance* changes the decision state even when its *value* does not change the price.

56. Rule-only benchmark vs learned pricing

scripts/audit_model_value.py → artifacts/metrics/model_value_ablation.json,
reports/12_MODEL_VALUE_ABLATION.md.

56.1 The question — and, crucially, the non-question

The question. Does the learned demand/elasticity layer produce materially different *decisions* from reasonable rules?

Not the question. Whether the model earns more money. Scoring a rule with the model's own demand curve and declaring the model better is **circular** — the same fitted curve would be both proposer and judge. **No profit comparison is made in this audit.** That restraint is the single best methodological decision in the whole Phase M suite.

56.2 Setup

All 13,964 week-399 contexts, standard profile, shrunk response. Every rule is pushed through **the same policy layer** as the model — eligibility screen, risk gate, feasible interval, rounding, materiality, decision state — so the comparison isolates the **pricing signal**, not the guardrails. Rule inputs (median margin, modal price) are computed from **training weeks only**.

56.3 The rules

rule	definition
R0 keep current price	never move a price; the only policy whose outcome was actually observed
R1 hold historical margin	$p = c / (1 - m)$, m = that series' median historical gross-margin rate
R2 cost-plus 25% margin	$p = c / (1 - 0.25)$, one fixed category margin target
R3 nearest historical modal price	snap to the most frequent historical price of that series
R4 always take the max allowed increase	$p = p_o \cdot (1 + \text{effective change cap})$ — no demand model at all

56.4 Decision agreement with the engine

rule	final price differs at all	differs by > 1 grid step (5¢)	mean abs diff	median abs diff	median abs diff %	same decision state	model keeps / rule changes	model changes / rule keeps
R0 keep current	59.7%	59.1%	\$0.141	\$0.130	4.3%	30.7%	0.5%	59.1%
R1 hold historical margin	61.4%	51.5%	\$0.215	\$0.120	4.2%	90.0%	5.7%	4.0%
R2 cost-plus 25%	59.7%	34.9%	\$0.107	\$0.030	1.0%	88.8%	5.6%	3.8%
R3 nearest modal price	62.2%	49.5%	\$0.179	\$0.050	2.5%	78.9%	4.6%	7.6%
R4 max allowed increase	59.2%	17.5%	\$0.060	\$0.019	0.6%	90.0%	5.6%	4.2%

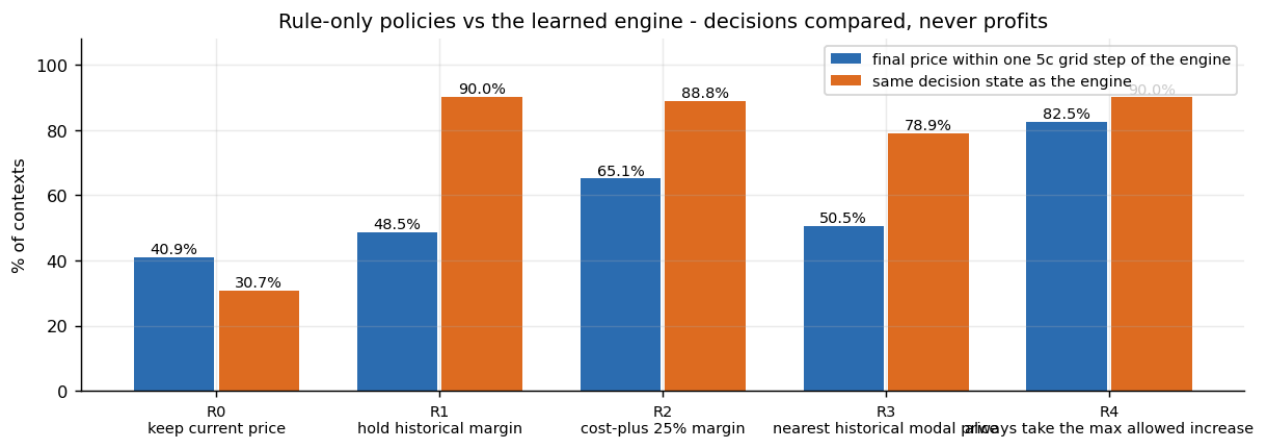


Figure 15 — For each rule: share of contexts where its final price lands within one 5-cent grid step of the engine's, and share where it reaches the same decision state. Decisions are compared; profits are not. Source: [artifacts/metrics/model_value_ablation.json](#).

56.5 The headline

R4 — "always take the maximum allowed increase", a rule that never looks at demand, cost-response, elasticity or any model output — lands within one 5-cent grid step of the engine's final price in 82.5% of contexts (100% – 17.5%), agrees on the decision state in 90.0%, and differs by a median of \$0.019 (0.6% of price).

The exact-match rate is lower (40.8%) only because the 5-cent candidate grid rarely contains the cap itself.

This is \$54 seen from the other direction: the model's unconstrained optimum sits above the change cap in 86.7% of contexts, so the constrained answer is "go to the cap" — which is exactly what R4 does by construction.

56.6 Where the model and R4 genuinely differ

The learned layer is not decorative. It differs from R4 in the contexts where:

84. **the estimated elasticity implies an optimum *below* the current price** — a price cut, which R4 can never propose;
85. **the optimum is interior**, so the recommended move is smaller than the cap (the 2.9% from \$54.8, plus the near-interior cases);
86. **the estimated uplift is immaterial** and the model keeps the price while R4 would still move it;
87. **the context is ineligible or the elasticity source is a fallback**, which changes the risk level and therefore the decision state.

Quantitatively: the model and R4 disagree on the decision state in **10.0%** of contexts, and the model's median actionable change is **+8.36%** against R4's **+10.0%** (the cap itself).

56.7 The other rules are informative too

R0 (never move) agrees on the decision state only 30.7% of the time and the engine proposes a change where R0 keeps in **59.1%** of contexts. So the engine is emphatically *not* equivalent to doing nothing — it is very active. It is just that its activity is largely "go to the cap".

R2 (flat cost-plus 25%) has the smallest median price gap of the margin rules (\$0.030, 1.0%) — notable because the observed blended margin in the panel is 15.3% and R2 targets 25%. That the engine's answers cluster near a 25% cost-plus rule says the model is pushing prices well above their historical margin, consistent with \$54.5.

R3 (snap to the modal historical price) has the worst decision agreement of the price rules (78.9%) and the highest "model changes / rule keeps" rate (7.6%). Anchoring to history is the rule most different in kind from what the engine does.

56.8 What this ablation does and does not establish

Establishes: the *magnitude* of most recommendations is reproducible by a trivial rule. Any claim that "the ML model sets the price" is false for this system as configured.

Does not establish: that the rule is as *good*. Nobody knows which policy earns more, because that would require observing outcomes at both sets of prices, which is precisely what offline data cannot provide (§64). The audit deliberately refuses to answer it rather than answering it circularly.

The honest summary: *under these guardrails, on this category, at this decision cadence*, the learned layer contributes direction, eligibility and risk — not magnitude. Whether that is worth the engineering is a business judgement, and §85 gives the evidence-based answer.

PART X — RISK AND DECISION POLICY

57. Recommendation risk layer

57.1 What it is — stated first, because the naming matters

It is a heuristic risk score. It is not a calibrated statistical probability, and it is never described as one.

From the module docstring of `optimization/risk.py`:

*This is **not** a calibrated statistical confidence interval, and it is never described as one. It is a transparent rule-based score over the factors that actually make a price recommendation unreliable in this dataset.*

The direction of the scale is also fixed and stated:

level	meaning	consequence
LOW	plenty of history and price variation; the recommendation sits inside the observed price support	may be actionable
MEDIUM	usable but thinner evidence	actionable under a tighter price-change cap
HIGH	thin evidence and/or extrapolation	never an automatic price change under the default profiles

57.2 The bands

`configs/config.yaml` → `risk`:

band	min observations	min distinct prices	max extrapolation distance
LOW	150	15	2%
MEDIUM	80	8	5%
HIGH	<i>(anything that meets neither)</i>		

Compare with the **eligibility** screen (40 obs, 5 distinct prices, $CV \geq 0.05$). The risk bands are deliberately **stricter**, and the config says why:

*Risk bands are deliberately **STRICTER** than the eligibility screen: eligibility answers "can this series be modelled at all", risk answers "how much do we trust acting on it". A series can therefore be eligible and still be **HIGH** risk, which is what routes it to **REVIEW_REQUIRED** instead of an automatic price change.*

That two-tier design is the mechanism that produces a **human queue** rather than a binary act/don't-act split.

57.3 The extrapolation distance

```
def extrapolation_distance(price, hist_min, hist_max) -> float:
    if hist_min is None or hist_max is None:
        return float("nan")
    if price < hist_min: return (hist_min - price) / hist_min
    if price > hist_max: return (price - hist_max) / hist_max
    return 0.0
```

python

Zero inside the observed support, otherwise the relative distance outside it, **NaN** when no support exists at all. NaN forces HIGH:

```
no_support = dist != dist # NaN
if no_support:
    level = RiskLevel.HIGH
```

python

A series with no observed price support cannot be assessed, so it is treated as maximally risky rather than defaulting to permissive. This is the right direction for the failure mode.

57.4 Two promotions out of LOW risk

Beyond the bands, two conditions demote a LOW assessment to MEDIUM:

```
if promotion_share is not None and promotion_share > 0.5:
    notes.append("more than half of this series' weeks carry a promotion code: "
                "the price effect is entangled with promotion activity")
    if level is RiskLevel.LOW: level = RiskLevel.MEDIUM

if elasticity_source == "pooled_fallback":
    notes.append("no usable product-specific elasticity: the category-level "
                "(pooled) estimate is being applied to this product")
    if level is RiskLevel.LOW: level = RiskLevel.MEDIUM
```

python

The second is how an **estimation** limitation becomes a **policy** consequence. 35.8% of week-399 contexts have no usable own elasticity; none of them can be LOW risk; every one carries the `POOLED_ELASTICITY_FALLBACK` reason code. That is limitation propagation done properly — the shortfall shows up in the decision, not only in a footnote.

57.5 Risk is assessed twice

```
preliminary_risk = risk_at(current_price) # before any price is proposed
...
final_risk = risk_at(best_price) # at the proposed price
```

python

The **preliminary** assessment can gate a context before optimisation ever runs (under the `conservative` profile) and can tighten the change cap for MEDIUM risk. The **final** assessment evaluates the extrapolation distance of *the proposal*, which the preliminary one cannot know. Both are needed: a series can be well-evidenced at its current price and the proposal can still extrapolate.

57.6 What is recorded

Every assessment returns the factors behind it, and they are logged:

```
factors = {"n_obs", "n_distinct_prices", "price_cv", "extrapolation_distance",
           "abs_price_change_pct", "promotion_share", "elasticity_source"}
```

python

plus free-text `notes`. So "why is this HIGH risk?" is always answerable from the audit log.

57.7 The honest limitation

The thresholds — 150/15/2% and 80/8/5% — are **asserted, not fitted**. Calibrating them would mean regressing realised out-of-sample error against the band boundaries and choosing cut-points that achieve a target error rate. That was **not done** (limitation #40, `STATUS.md` open issue 7).

Calling an uncalibrated score "confidence" would have been dishonest, so the project calls it risk, labels it heuristic in the config, the code, the API response, the dashboard and every report, and lists it as an open issue (DECISIONS #24).

57.8 The Phase L naming inversion — a failure worth recording

The scale used to read the other way round: "HIGH" meant *high confidence* and sat on actionable recommendations. That is a genuinely dangerous naming bug — a reviewer scanning a queue would read "HIGH" as "safe to auto-apply" when it meant the opposite of the current semantics.

Phase L inverted it so that **HIGH means risky**, and added the gate (DECISIONS #36). The fact that this had to be fixed is reported rather than hidden, and `tests/test_decision_states.py` (246 tests) now pins the semantics across every combination of risk level, decision state, actionability, proposal and final price, for all three profiles.

58. Decision states

58.1 The three states

```
class DecisionState(StrEnum):
    RECOMMEND_CHANGE = "RECOMMEND_CHANGE"
    KEEP_CURRENT      = "KEEP_CURRENT"
    REVIEW_REQUIRED   = "REVIEW_REQUIRED"
```

python

state	meaning	actionable	final price
RECOMMEND_CHANGE	an actionable price change	true	the proposal
KEEP_CURRENT	leave the price alone	false	the current price
REVIEW_REQUIRED	a proposal exists but a human must approve it	false	the current price

58.2 Why three and not two

A binary change/keep split throws information away. A HIGH-risk context can still deserve a human's attention: the model has found something, but the evidence is too thin to act on automatically. Collapsing that into `KEEP_CURRENT` silently discards a real signal; collapsing it into `RECOMMEND_CHANGE` acts on evidence that does not support it.

`REVIEW_REQUIRED` is the honest third option, and it is what makes the system a **decision-support** tool rather than an autonomous pricer (DECISIONS #37).

58.3 The proposal / final-price split

Phase M split what used to be one field into two (DECISIONS #47):

field	meaning
proposed_candidate_price	what the optimizer proposed. Not a business price on its own.
final_recommended_price	what would actually be charged: the proposal only when actionable , otherwise the current price
proposed_price_change_pct	change of the proposal
price_change_pct	change of the final price — 0.0 whenever not actionable
model_internal_estimated_profit_uplift_pct	the proposal's estimated uplift
realisable_profit_uplift_pct	forced to 0.0 whenever not actionable

```

actionable = decision is DecisionState.RECOMMEND_CHANGE
final_price = proposed_candidate_price if actionable else current_price
...
realisable_profit_uplift_pct = (profit_uplift if (actionable and profit_uplift is not None) else 0.0)

```

python

Why this matters. With a single `recommended_price` field, any consumer that forgot to read the `actionable` flag would display a price the policy had refused to authorise — and, worse, a dashboard summing uplift across a batch would include proposals that will never be applied. Forcing `realisable_profit_uplift_pct` to zero makes that class of error impossible by construction rather than by convention.

58.4 The invariants

Six invariants are asserted over **every** decision:

#	invariant
1	HIGH risk is never actionable (under default profiles)
2	<code>actionable iff decision == RECOMMEND_CHANGE</code>
3	<code>REVIEW_REQUIRED</code> keeps the current price
4	<code>KEEP_CURRENT</code> keeps the current price
5	<code>KEEP_CURRENT</code> proposes nothing (proposal == current price)
6	<code>RECOMMEND_CHANGE</code> actually changes the price

58.5 Invariant verification over every real context

`scripts/audit_decision_states.py` → `artifacts/metrics/decision_state_audit.json`,
`reports/19_DECISION_STATE_AUDIT.md`. All **13,964** week-399 contexts × three profiles.

profile	inv. 1	inv. 2	inv. 3	inv. 4	inv. 5	inv. 6
standard	0	0	0	0	0	0
conservative	0	0	0	0	0	0
aggressive (DEMO)	4,717	0	0	0	0	0

Zero violations under both default profiles. The 4,717 under `aggressive` are **by design** — that profile sets `high_risk_action: recommend` and is explicitly labelled **DEMO ONLY** in the configuration:

```

aggressive:
  # DEMO ONLY: explicitly allows acting on HIGH-risk contexts. Never the default.
  high_risk_action: recommend

```

yaml

The audit reporting them as violations rather than suppressing them is the correct behaviour: the invariant is stated absolutely, and the one profile that breaks it is named.

58.6 Decision-state distribution by profile

All 13,964 week-399 contexts:

profile	RECOMMEND_CHANGE	KEEP_CURRENT	REVIEW_REQUIRED	LOW risk	MEDIUM	HIGH	HIGH & actionable
standard	8,265 (59.2%)	4,346 (31.1%)	1,353 (9.7%)	5,511	3,693	4,760	0
conservative	6,354 (45.5%)	7,610 (54.5%)	0	7,163	2,041	4,760	0
aggressive (DEMO)	9,885 (70.8%)	4,079 (29.2%)	0	4,032	1,830	8,102	4,717

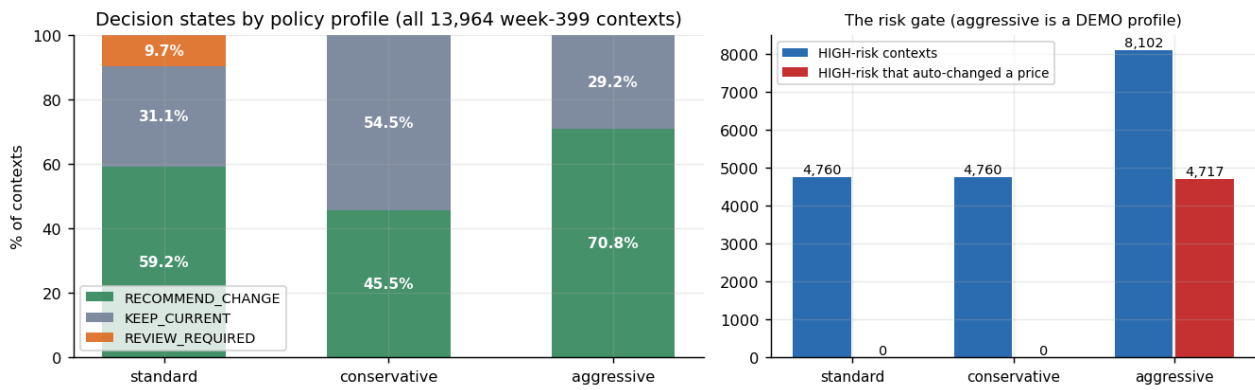


Figure 16 — Left: decision-state mix by policy profile over all 13,964 contexts. Right: HIGH-risk contexts and how many of them auto-changed a price. Zero under both default profiles. Source: `artifacts/metrics/decision_state_audit.json`.

Two things worth reading off this table:

- **conservative produces no REVIEW_REQUIRED at all**, because its `high_risk_action` is `keep_current` — HIGH-risk contexts are silently kept rather than queued. That is a deliberate difference in operating model, not an inconsistency.
- **aggressive has more HIGH-risk contexts (8,102 vs 4,760)**, which is initially surprising. The mechanism: its wider $\pm 20\%$ change cap and 10% extrapolation tolerance let the optimizer propose prices further outside the observed support, and the **final** risk assessment is made *at the proposal*. Looser guardrails manufacture risk.

58.7 The 246 property tests

`tests/test_decision_states.py` contains **246** tests that walk every combination of risk level $\times$ decision state $\times$ actionability $\times$ proposal $\times$ final price across all three profiles, plus audit-log schema rotation. It is the single largest test file in the project, and appropriately so: the decision state is the contract every downstream consumer relies on.

59. High-risk gating

59.1 The batch statistics

3,000-context batch (`reports/07_RECOMMENDATION_SUMMARY.md`, week 399, standard profile, `shrunk` response):

quantity	value
contexts scored	3,000
RECOMMEND_CHANGE	1,766 (58.87%)
KEEP_CURRENT	923 (30.77%)
REVIEW_REQUIRED	311 (10.37%)
LOW risk	1,175
MEDIUM risk	786
HIGH risk	1,039
HIGH-risk contexts that auto-changed a price	0 of 1,039

Full week (13,964 contexts, `reports/19`): HIGH risk 4,760, **HIGH-risk actionable 0**.

59.2 What happens to HIGH-risk contexts

Under the `standard` profile, 4,760 HIGH-risk contexts split as:

outcome	contexts
KEEP_CURRENT	3,407
REVIEW_REQUIRED	1,353
RECOMMEND_CHANGE	0

And the KEEP_CURRENT ones break down by cause (causes overlap):

cause	contexts
screened out by the eligibility rules (never reaches the optimizer)	3,349
the optimum is the current price	3,391
no feasible price inside the guardrails	41
estimated uplift below the materiality threshold	16

Most HIGH-risk contexts never reach the risk gate at all: the eligibility screen catches them first, which is the cheaper and earlier defence.

59.3 Why HIGH-risk recommendations are not auto-actioned

Constraints bound the size of a mistake; they do not bound its likelihood.

The guardrails guarantee that no single recommendation moves a price more than 10%, stays above cost, keeps a 5% margin and stays near observed support. They say nothing about whether the *evidence* supports moving at all. A series with 45 observations and 6 distinct prices can produce a perfectly feasible +9.3% recommendation with a large estimated uplift — computed from an elasticity fitted on almost no price variation, or falling back to the category number entirely.

That is a different kind of error from "the move is too big", and it needs a different control.

59.4 Human-in-the-loop pricing

The intended lifecycle, implemented in `src/pricing_engine/audit.py`:

GENERATED → REVIEWED → APPROVED / REJECTED → PUBLISHED

Every recommendation is appended to `artifacts/recommendation_log.csv` (34 columns) with its inputs, constraints, model version, elasticity used and source, risk level, decision state, both prices, both uplift figures and the reason codes. The current log holds **3,002 rows**.

REVIEW_REQUIRED is where the human enters. The engine's contribution is to make the queue **small and prioritised**: 9.7% of contexts under the standard profile — 1,353 of 13,964.

59.5 The limitation the project states about its own queue

REVIEW_REQUIRED is a queue, not a solution. In a real deployment somebody has to work that queue; this demo only produces it.
(KNOWN_LIMITATIONS.md #33)

1,353 items per week is a real staffing question. A category manager with 45,477 weekly price decisions cannot review 1,353 of them by hand every week without a prioritisation layer — sorting by estimated value at risk, or by product importance. That layer does not exist here, and saying so is more useful than pretending the queue is free.

59.6 The aggressive profile and why it exists

`aggressive` sets `high_risk_action`: `recommend` and would auto-change 4,717 HIGH-risk prices. It exists for exactly one reason: to make the **sensitivity of the output to the policy** visible and measurable. Turning the gate off and showing that 4,717 prices move is a far more convincing demonstration that the gate does something than asserting it.

It is labelled DEMO ONLY in the config comment, in the dashboard selector ("Policy profile (DEMO setting)"), in `DECISIONS.md` #25 and #38, and in the decision-state audit, which reports its 4,717 invariant violations rather than excluding it.

60. Reason codes

Every recommendation carries a sorted, de-duplicated list of reason codes.

```
reason_codes=sorted(set(reasons)).
```

60.1 The full catalogue

code	meaning	when it fires
<code>KEEP_CURRENT_OPTIMAL</code>	the engine recommends leaving the price alone	attached to every <code>KEEP_CURRENT</code> outcome
<code>LOW_CONFIDENCE</code>	the evidence does not support automatic action	with HIGH-risk gating
<code>INSUFFICIENT_HISTORY</code>	fewer observations than <code>eligibility.min_observations</code> (40)	eligibility screen
<code>INSUFFICIENT_PRICE_VARIATION</code>	fewer than 5 distinct prices, or price CV < 0.05	eligibility screen
<code>OUTSIDE_EXTRAPOLATION_RANGE</code>	the extrapolation guardrail is an active edge of the feasible interval	constraint binding
<code>COST_UNAVAILABLE</code>	no decision-time cost for a gross-profit objective	eligibility screen
<code>MARGIN_CONSTRAINT</code>	the cost floor or minimum-margin floor is an active edge	constraint binding
<code>PRICE_CHANGE_LIMIT</code>	the maximum-change window is an active edge	constraint binding
<code>PROFIT_UPLIFT_POSITIVE</code>	the proposal beats the current price on estimated gross profit by more than materiality	objective = <code>gross_profit</code>
<code>REVENUE_UPLIFT_POSITIVE</code>	as above, revenue objective	objective = <code>revenue</code>
<code>NO_FEASIBLE_PRICE</code>	the constraint intersection is empty, or no candidate survives	constraints
<code>NON_MATERIAL_UPLIFT</code>	the estimated gain is below the materiality threshold	materiality check
<code>DEMAND_CURVE_NOT DECREASING</code>	the simulated demand curve is not monotone decreasing	curve diagnostic
<code>HIGH_RISK_REVIEW_REQUIRED</code>	HIGH risk under a profile whose <code>high_risk_action</code> is <code>review_required</code>	risk gate
<code>HIGH_RISK_KEEP_CURRENT</code>	HIGH risk under a profile whose <code>high_risk_action</code> is <code>keep_current</code>	risk gate
<code>MEDIUM_RISK_CONSERVATIVE</code>	MEDIUM risk; the price-change cap has been tightened	risk adjustment
<code>POOLED_ELASTICITY_FALLBACK</code>	no usable product-specific elasticity; the category estimate is applied	elasticity provenance

60.2 Observed frequencies (3,000-context batch, week 399, standard)

code	count	share of contexts
PRICE_CHANGE_LIMIT	2,216	73.9%
PROFIT_UPLIFT_POSITIVE	2,077	69.2%
KEEP_CURRENT_OPTIMAL	923	30.8%
INSUFFICIENT_PRICE_VARIATION	668	22.3%
INSUFFICIENT_HISTORY	567	18.9%
POOLED_ELASTICITY_FALLBACK	523	17.4%
OUTSIDE_EXTRAPOLATION_RANGE	465	15.5%
MEDIUM_RISK_CONSERVATIVE	416	13.9%
LOW_CONFIDENCE	311	10.4%
HIGH_RISK_REVIEW_REQUIRED	311	10.4%
MARGIN_CONSTRAINT	208	6.9%
NON_MATERIAL_UPLIFT	136	4.5%
NO_FEASIBLE_PRICE	37	1.2%

Sanity checks that the codes are internally consistent:

- LOW_CONFIDENCE (311) and HIGH_RISK_REVIEW_REQUIRED (311) match exactly, and both equal the REVIEW_REQUIRED count (311). ✓
- KEEP_CURRENT_OPTIMAL (923) equals the KEEP_CURRENT count (923). ✓
- PRICE_CHANGE_LIMIT is the most frequent code, consistent with the attribution finding that it is the first binding constraint in 44.6% of contexts. ✓
- POOLED_ELASTICITY_FALLBACK (523/3,000 = 17.4%) is lower than the full-week fallback share (35.8%) because the 3,000-context batch is a sample and because the code is only emitted for contexts that pass the eligibility screen — a fallback context that is screened out never reaches that line. ✓
- DEMAND_CURVE_NOT DECREASING **never fires**, consistent with the hybrid response being monotone by construction for $\epsilon < 0$. ✓

60.3 Worked examples

Example A — actionable, at the guardrail corner (the demo context):

```
CAPN CRUNCH JUMBO CR (3000006560), store 86, week 399
$3.35 → $3.66 (+9.25%) RECOMMEND_CHANGE, actionable, LOW risk
reason codes: PRICE_CHANGE_LIMIT, PROFIT_UPLIFT_POSITIVE
```

Read as: *the proposal beats the current price on estimated gross profit (PROFIT_UPLIFT_POSITIVE), and it sits at the edge of the $\pm 10\%$ change window (PRICE_CHANGE_LIMIT) — the model wanted to go further.*

Example B — screened out before optimisation:

```
KEEP_CURRENT, not actionable
reason codes: INSUFFICIENT_HISTORY, INSUFFICIENT_PRICE_VARIATION,
              KEEP_CURRENT_OPTIMAL
```

Read as: *this series has too few observations and too little price variation to model, so no price was proposed at all.*

Example C — a proposal a human must approve:

```
REVIEW_REQUIRED, not actionable, HIGH risk
reason codes: HIGH_RISK_REVIEW_REQUIRED, LOW_CONFIDENCE,
              POOLED_ELASTICITY_FALLBACK, PRICE_CHANGE_LIMIT,
              PROFIT_UPLIFT_POSITIVE
```

Read as: *a proposal exists and looks profitable, but this product has no usable own elasticity and the evidence is thin — final price stays where it is until a human approves.*

60.4 Why reason codes rather than feature importances

A SHAP plot explains a *prediction*. A reason code explains a **decision** — which is what a category manager, an auditor or a regulator actually asks about. "Why this price?" is answered with `PRICE_CHANGE_LIMIT`, `PROFIT_UPLIFT_POSITIVE` and a feasible interval, not with a bar chart of feature attributions.

Reason codes are also **testable**: `tests/test_optimizer.py` and `tests/test_attribution.py` assert exact code sets for specific fixtures, and the attribution replica must reproduce them exactly to be trusted. A SHAP explanation cannot be unit-tested in that way.

PART XI — BACKTESTING AND POLICY EVALUATION

61. Historical evaluation design

`scripts/backtest.py --weeks 10 --contexts-per-week 400 → artifacts/metrics/backtest.json, reports/06_BACKTEST.md.`

parameter	value
weeks evaluated	390 – 399 (1997-02-27 ... 1997-05-01)
contexts per week	400 (sampled)
model	ridge_loglog-20260818-132847, trained on weeks 2–257
policy profile	standard
price response	shrunk
objective	gross profit

The backtest window sits **entirely inside the test window (343–399)** and therefore **entirely after** the model's training window and the elasticity estimation window (both ending at week 257). Leakage channel #10 is closed by construction.

61.1 The two halves, which must not be confused

The backtest produces two categorically different kinds of number:

half	what it measures	verifiable against outcomes?
demand accuracy per week	how well the forecaster predicted realised demand at the prices actually charged	YES
policy economics	what four pricing policies would have earned, scored by the model	NO — circular

The report keeps them separate, and so does this section. §62 is the first; §63 is the second; §64 explains exactly why the second cannot be trusted.

61.2 Which model scores which half

A design detail worth naming (DECISIONS #42): **accuracy is scored with the baseline forecaster; policies are scored with the pricing model.** WAPE measures forecasting at observed prices, which is the base model's job; the price-response layer only matters for counterfactuals. Mixing them would make the accuracy number depend on the elasticity, which it should not.

62. Demand forecast evaluation

62.1 Weekly accuracy

week	week start	n	MAE	RMSE	WAPE	bias	sMAPE
390	1997-02-27	14,053	8.092	33.862	0.4802	–2.954	0.4104
391	1997-03-06	14,036	6.871	56.248	0.4170	–2.299	0.3767
392	1997-03-13	14,010	9.531	69.951	0.5030	–5.013	0.3755
393	1997-03-20	14,097	6.323	20.311	0.3947	–2.144	0.3786
394	1997-03-27	14,255	6.396	23.998	0.3638	–3.434	0.3135
395	1997-04-03	14,166	8.225	168.758	0.4673	–2.859	0.3922
396	1997-04-10	14,151	11.094	57.358	0.5294	–7.213	0.4011
397	1997-04-17	14,074	13.363	64.906	0.5524	–9.169	0.4042
398	1997-04-24	14,046	13.570	57.362	0.5602	–9.691	0.5039

week	week start	n	MAE	RMSE	WAPE	bias	sMAPE
399	1997-05-01	13,964	9.298	30.309	0.5089	-1.836	0.4450

anchor statistic	value
mean weekly WAPE	0.4777
min / max weekly WAPE	0.3638 / 0.5602
mean weekly bias	-4.661 units

62.2 Reading it

The mean weekly WAPE of **0.4777** is consistent with the test-window figure of **0.4565** — slightly worse, as expected, because these are the *last* ten weeks of the test window and accuracy degrades with distance from training (§41.8).

The range **0.364–0.560** is wide, and that spread is the operationally important number. It says the *same model on the same category* can be 36% or 56% wrong depending on the week. It is the direct justification for the materiality threshold: a counterfactual difference smaller than the week-to-week variation in the model's own error is not a signal.

Week 395 has an RMSE of **168.8** against a MAE of **8.2** — a ratio of 20.5. That is one or more enormous individual misses. It is the week containing the 17,824-unit promotional spike in the demo series (§18.5), and it is a clean illustration of why RMSE alone would be a misleading headline metric here.

Bias is negative in every single week, ranging -1.8 to -9.7. The systematic under-forecast (§41.4) is stable, not episodic. A production system would either apply a smearing correction or calibrate the bias out; this one reports it.

62.3 This is the only outcome-verifiable metric in the backtest

Every number in §62 is checkable against what actually happened: prices were charged, units were sold, the model predicted, the errors are real. Nothing in §63 has that property.

63. Pricing policy evaluation

63.1 The four policies

Implemented in `src/pricing_engine/optimization/policies.py`, all pushed through the same constraint layer:

policy	definition
HistoricalPricePolicy	charge what was actually charged — the only policy whose outcome was observed
SimpleMarginPolicy	cost-plus to a target margin
ElasticityBaselinePolicy	the closed-form CE optimum $p^* = c \cdot \epsilon / (1 + \epsilon)$, clipped to the feasible interval
MLPricingPolicy	the full engine: hybrid response, constrained optimizer, risk gate

63.2 Results — model-internal estimates only

Totals across 10 weeks × ~400 contexts.

policy	expected revenue	expected gross profit	predicted units	mean price	share unchanged	median abs change	share outside support	share constrained	GP vs historical	rev vs historical
--------	------------------	-----------------------	-----------------	------------	-----------------	-------------------	-----------------------	-------------------	------------------	-------------------

policy	expected revenue	expected gross profit	predicted units	mean price	share unchanged	median abs change	share outside support	share constrained	GP vs historical	rev vs historical
HistoricalPricePolicy	\$170,709	\$37,679	53,834	\$3.181	100%	0.0%	0.0%	0%	0.00%	0.00%
SimpleMarginPolicy	\$173,228	\$38,546	55,352	\$3.134	0%	4.37%	25.1%	100%	+2.30%	+1.48%
ElasticityBaselinePolicy	\$161,290	\$42,834	48,526	\$3.336	6.4%	9.99%	37.9%	100%	+13.68%	-5.52%
MLPricingPolicy	\$163,340	\$41,617	49,620	\$3.320	36.9%	4.40%	21.4%	78.4%	+10.45%	-4.32%

63.3 Reading the table — carefully

Every "vs historical" figure is a model-internal estimate. The same fitted price response both proposes and scores the prices. See §64 before drawing any conclusion.

That said, the *relative* patterns are informative about the policies' behaviour, which is a legitimate reading:

All profit-seeking policies trade revenue for profit. Elasticity baseline: +13.7% gross profit, -5.5% revenue. ML: +10.5% profit, -4.3% revenue. That is \$2.2 in action — with $|\varepsilon| > 1$, raising price loses revenue and can still gain profit.

SimpleMarginPolicy is the outlier: it *raises* revenue (+1.5%) and barely raises profit (+2.3%), because a fixed margin target on a 15.3%-blended-margin category mostly *lowers* prices (mean \$3.134 vs \$3.181 historical).

ElasticityBaselinePolicy "wins" on estimated profit and is the least deployable. It moves 93.6% of prices, by a median of 10.0% (the cap), and puts **37.9%** of them outside the observed price support. It is the unconstrained model optimum with the guardrails only partially applied — the research policy of §55 layer A, essentially.

MLPricingPolicy is deliberately more conservative: 36.9% of prices unchanged, median change 4.4%, only 21.4% outside support, and 78.4% of its recommendations touched by a binding constraint. It gives up 3.2 percentage points of *estimated* profit for a materially safer profile. That trade is the whole point of the risk and materiality layers, and it is visible in the table.

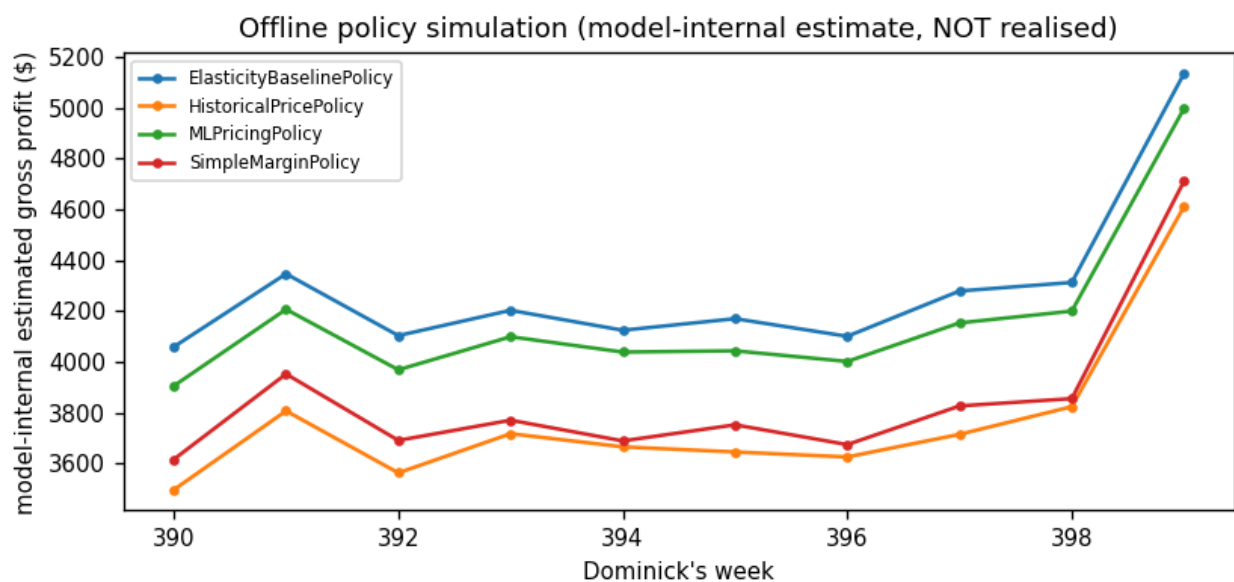


Figure 17 — Model-internal estimated gross profit by policy across weeks 390–399. Only the HistoricalPricePolicy bar corresponds to an outcome that was actually observed. Source: `scripts/backtest.py`.

64. The circularity problem

This section exists because it is the single most important caveat attached to §63, and because a report that buried it would be misleading.

64.1 The problem, stated

To evaluate a pricing policy you must know what would have happened at the prices it proposes. Those prices were **never charged**. The only instrument available to estimate the outcome is the **same fitted price-response model that proposed them**.

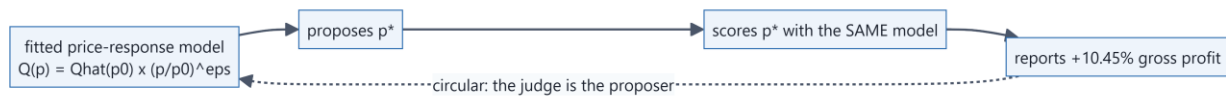


Diagram 4

64.2 What the +10.45% actually means

"If the fitted price-response model is correct, then applying the ML policy's prices instead of the historical ones would have produced 10.45% more gross profit — as computed by that same model."

The conditional clause is not decoration. It is the entire content of the statement.

64.3 Why this is not fixable with better offline methodology

Standard fixes do not apply.

- **Cross-validation** validates predictions at *observed* prices. It says nothing about unobserved ones.
- **A held-out test set** contains the retailer's prices, not the policy's.
- **Off-policy evaluation** (importance sampling, doubly-robust estimators) requires a **known behaviour policy with positive propensity** over the actions being evaluated. The retailer's pricing rule is unknown, deterministic in effect, and assigns zero probability to most candidate prices. The importance weights are undefined.
- **A better demand model** reduces the variance of a circular estimate. It does not make it non-circular.

The only escape is **intervention** — assigning prices exogenously (§98).

64.4 A concrete illustration of how the bias arises

Suppose the true elasticity is -2.0 and the model believes -2.4 .

The model computes its optimum as $p^* = c \cdot 2.4 / 1.4 = 1.714 \cdot c$, whereas the true optimum is $c \cdot 2.0 / 1.0 = 2.0 \cdot c$. It then *scores* $1.714 \cdot c$ using its own $\epsilon = -2.4$ curve, which — relative to the truth — **understates** how much volume survives a price rise, so it under-states the profit at its own chosen price. In this particular direction the bias happens to be conservative.

Reverse the error (model believes -1.6 , truth -2.0) and the model both over-shoots the price *and* over-states the profit there. **The direction of the bias depends on the direction of the model's error, which is unknown.** That is why the number cannot be signed, let alone trusted.

64.5 How the project handles it

mechanism	where
the metric is named <code>model_internal_estimated_profit_uplift_pct</code>	code, API schema, dashboard, audit log, all reports
<code>realisable_profit_uplift_pct</code> is forced to 0 for non-actionable rows	<code>optimizer.py</code>
every dashboard page carries the disclaimer	<code>dashboard/app.py</code> <code>DISCLAIMER</code> constant
the API <code>ModelInfoResponse</code> and <code>RecommendPriceResponse</code> carry <code>disclaimer</code> fields	<code>api/schemas.py</code>

mechanism	where
the backtest report is labelled <code>VALIDATED (LIMITED)</code> in <code>STATUS.md</code>	<code>STATUS.md</code>
<code>scripts/audit_claims.py --strict</code> fails the build if an unsupported claim reappears	CI-enforceable
the model-value ablation refuses to compare profits at all (§56.1)	<code>audit_model_value.py</code>

64.6 The one thing that partly escapes circularity

§45's out-of-time validation compares four candidate price-response methods against **154,899 realised outcomes** following **real** price changes. That is not self-scoring — the ground truth is observed demand. It is still not causal (§45.7), but it is genuinely out-of-sample, and it is why `shrunk` was chosen on evidence rather than assertion.

65. What can actually be verified offline

65.1 Strongly verifiable

These are checkable against outcomes or against closed-form ground truth, and this report presents them as facts.

claim	evidence	value
demand-model accuracy on held-out time	<code>artifacts/metrics/evaluation.json</code> , 749,040 rows scored once	test WAPE 0.4565 , MAE 8.503, RMSE 72.064, bias -3.883
accuracy vs naive baselines	<code>model_metrics.json</code>	37.7% relative WAPE improvement over the best baseline
accuracy through time	<code>backtest.json</code> , <code>monitoring.json</code>	weekly WAPE 0.364–0.560; quarterly 0.409 → 0.528
model persistence is lossless	<code>evaluate.py</code>	reload WAPE difference 0.00e+00
optimizer correctness	<code>tests/test_optimizer.py</code>	linear-demand profit optimum $(a+bc)/(2b) = 6.25$ and revenue optimum $a/(2b) = 5.00$ recovered within one grid step
constraint behaviour	<code>constraint_attribution.json</code> , 13,964 contexts × 11 runs	every constraint's present / removes / binds / changes counts
decision-state invariants	<code>decision_state_audit.json</code>	0 violations across 13,964 contexts × 2 default profiles
no cost leakage	<code>cost_leakage_audit.json</code>	4,671,333 rows, 0 mismatches; poisoning test with teeth; 0 of 19 recommendations changed
no feature leakage	<code>tests/test_features.py</code>	12 enumerated channels, each with a failing-if-broken test
elasticity estimated only on training weeks	elasticity table metadata	window 2–257 recorded; script aborts otherwise
out-of-time price-response ranking	<code>out_of_time_price_response.json</code>	154,899 realised episodes; <code>shrunk</code> lowest WAPE in every split
elasticity inference	<code>elasticity_inference.json</code>	estimators validated against statsmodels; two-way clustered SE 16.24× classical
elasticity stability (category)	<code>elasticity_stability.json</code>	max pairwise z between disjoint windows 1.36 — not significant
rule-vs-model decision agreement	<code>model_value_ablation.json</code>	R4 within one grid step 82.5%, same decision state 90.0%
test suite and lint	live re-run at Phase N, 2026-08-19	519 passed, 0 failed ; ruff clean

65.2 Not directly verifiable

These are estimates, and this report labels every one of them as such.

claim	why it cannot be verified here
true counterfactual demand at any unobserved price	never observed; the model is the only estimator and it is also the proposer
the +8.30% portfolio uplift	model-internal; circular (§64)
the +10.45% backtest policy uplift	model-internal; circular
the causal effect of a price change	requires intervention (§33, §98)
whether the ML policy beats the rule-only policy	would require a profit comparison, which would be circular (§56.1)
whether the risk bands are correctly calibrated	never fitted against realised out-of-sample error
whether the AAC proxy is economically correct	only its temporal availability was proven
whether product-level elasticity is a durable asset	rank correlation +0.19 across disjoint windows says probably not

65.3 The dividing line

Everything about the *machinery* can be verified. Almost nothing about the *economics* can.

The engine's data pipeline, feature contract, model persistence, optimizer, constraint layer, decision logic and audit trail are all testable and tested. The one thing it exists to claim — that these prices would earn more money — cannot be established offline by any method, and requires the experiment in §98.

A project that understands that distinction, states it, and then builds the verification apparatus for everything on the left side of the line is doing the job correctly. A project that reports "+10% profit uplift" as a result is not.

PART XII — SYSTEM ARCHITECTURE

66. End-to-end architecture

66.1 The full pipeline

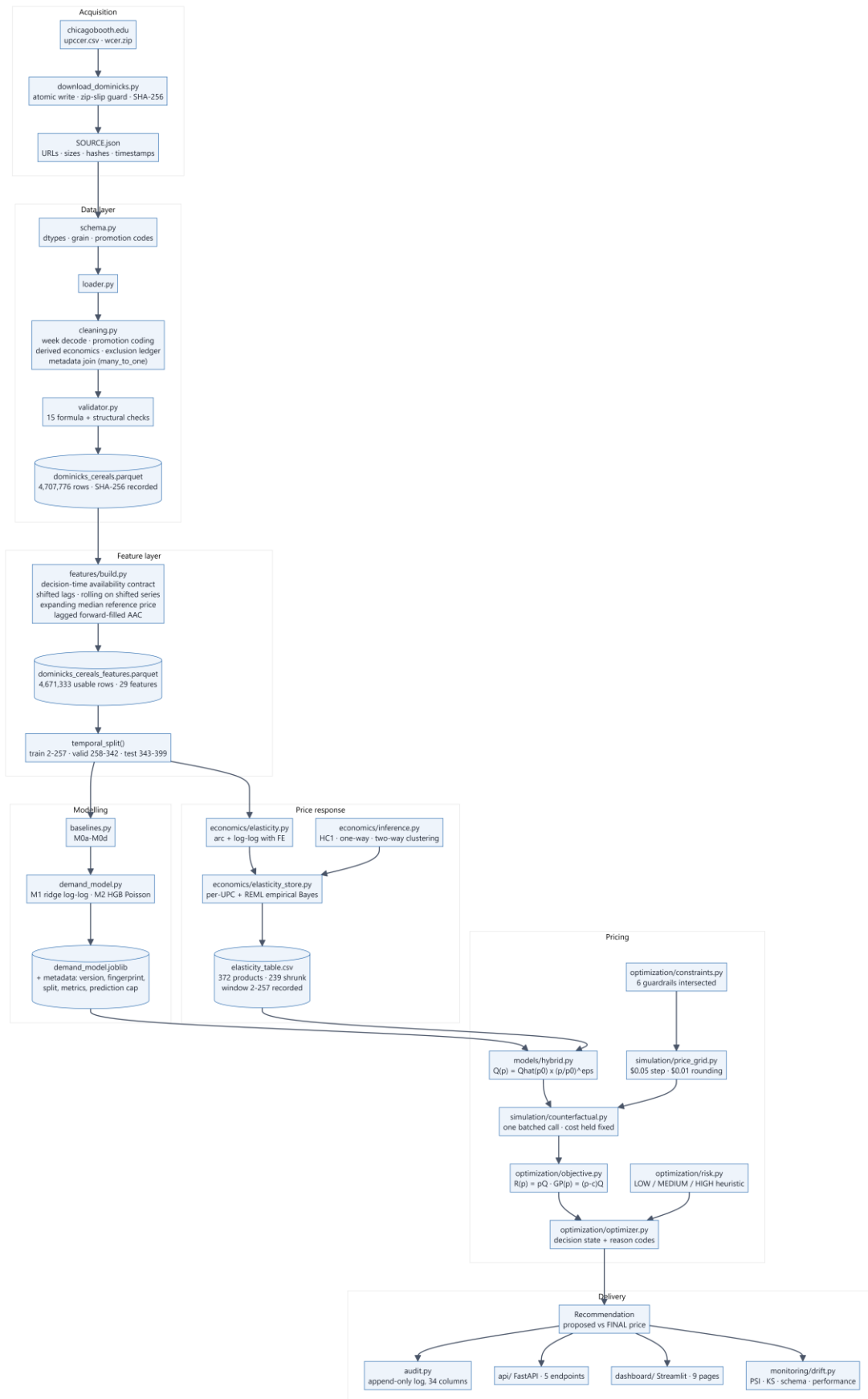


Diagram 5

66.2 The decision path for one recommendation

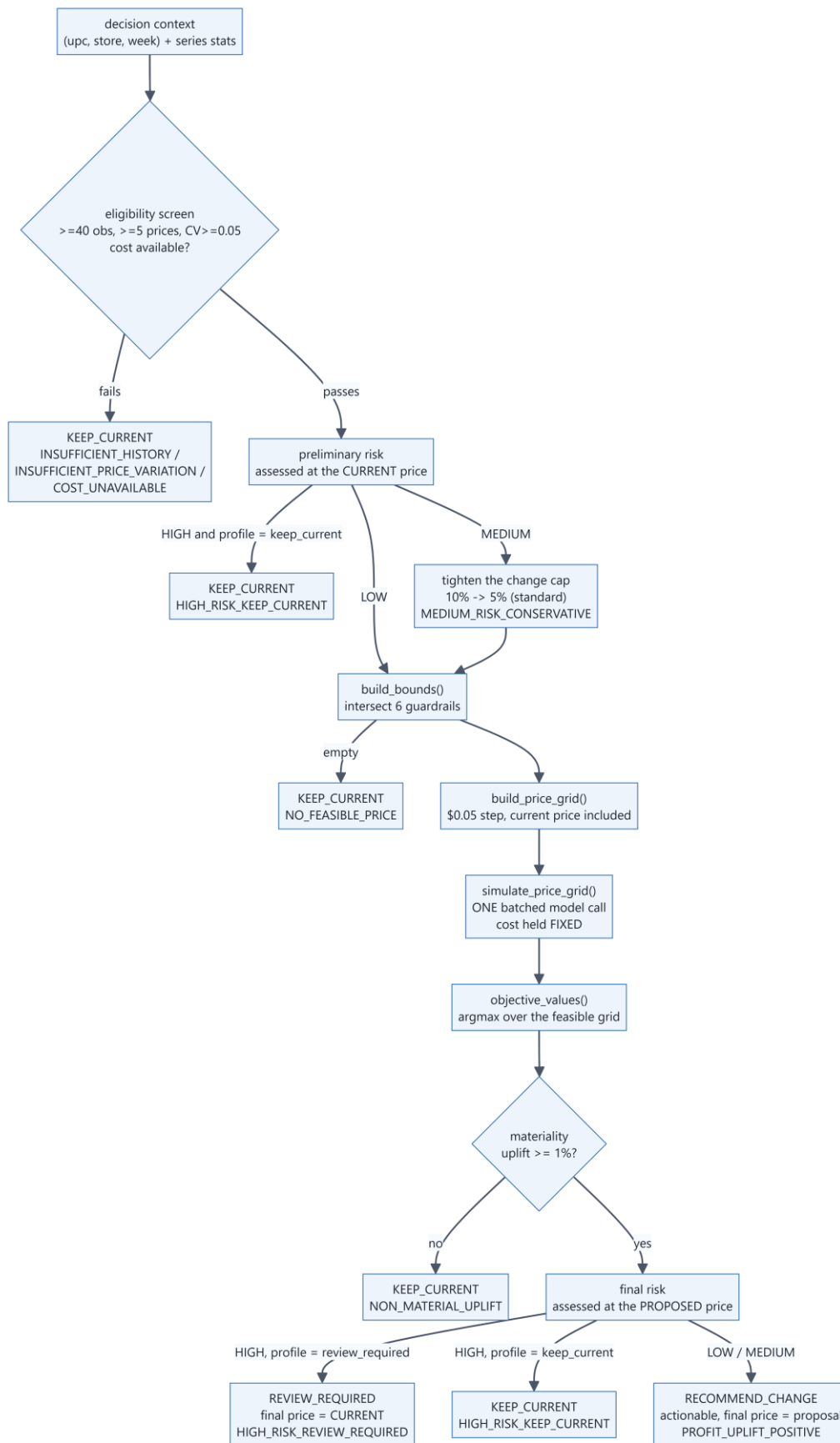


Diagram 6

66.3 The layering principle

Each layer answers exactly one question, and the boundaries are where the scientific claims change:

layer	question	claim strength
data	what happened?	observed fact
features	what was knowable at decision time?	contract, tested
forecast	how much will sell at this context?	predicted , measured on held-out time
elasticity	how does demand move with price?	observational estimate , with intervals
hybrid	what would demand be at price p ?	model-internal counterfactual
optimizer	which feasible price maximises the objective?	exact , given the curve
constraints	which prices are allowed?	policy
risk	should we act automatically?	heuristic
audit	what did we decide, and why?	recorded

67. Repository architecture

67.1 Package layout and responsibilities

module	responsibility	why it is separate
config.py	loads configs/config.yaml; get / require / path / policy accessors	one source of truth for every threshold; code never hardcodes a magic number, and PRICING_ENGINE_CONFIG allows overriding the whole tree
data/schema.py	canonical column list, grain, dtypes, promotion-code map	the schema is a constant, so column order and grain are stable across runs
data/loader.py	reads the raw CSVs with explicit dtypes	prevents pandas type inference from silently changing behaviour between runs
data/cleaning.py	week decoding, promotion coding, derived economics, the exclusion ledger, the metadata join	every exclusion is counted; the join asserts its own cardinality
data/validator.py	the 15 formula and structural checks	re-derives identities from the persisted file, independently of the code that wrote it
economics/metrics.py	revenue, gross_profit, gross_margin_rate, price_change_pct, price_variation_summary, eligible_series	one definition of revenue project-wide — EDA, simulator, optimizer, API and dashboard all call the same functions
economics/elasticity.py	arc and log-log elasticity with fixed effects	the analysis-layer estimators
economics/inference.py	OLS, HC1, one-way and two-way clustered covariance, demeaning	validated against statsmodels; used by both the analysis and the pricing estimators
economics/elasticity_store.py	per-UPC estimation, REML empirical Bayes, the servable ElasticityTable	the pricing elasticity is a first-class artifact with its own provenance, not a number in a script
features/build.py	the decision-time feature contract; recompute_price_features; temporal_split	the single implementation of price-dependent features , shared by training, simulation, optimization, API and dashboard
models/baselines.py	four naive forecasts	the price of admission
models/demand_model.py	DemandModel — one interface over ridge and HGB; UPC prior; prediction cap; implied_elasticity; persistence	downstream code never branches on model family
models/hybrid.py	HybridPricingModel — the elasticity price response	exposes the same predict(frame) → units contract , so the simulator, optimizer, API and dashboard need no

module	responsibility	why it is separate
		special case
models/metrics.py	WAPE, MAE, RMSE, bias, sMAPE	one definition, used everywhere
simulation/price_grid.py	candidate grid construction, rounding, charm pricing	grid failures raise instead of returning empty
simulation/counterfactual.py	simulate_price_grid, simulate_many, demand_curve_diagnostics	enforces both simulation invariants
optimization/objective.py	the two objectives plus the analytical optima used as test ground truth	closed-form answers live next to the code they validate
optimization/constraints.py	build_bounds, apply_constraints, PriceBounds	one intersection, with provenance
optimization/risk.py	the heuristic risk layer	isolated so it can be replaced by a calibrated model without touching the optimizer
optimization/optimizer.py	optimize_price → Recommendation	the orchestrator; the only place decision states are assigned
optimization/policies.py	the four backtest policies	so alternative policies run through the <i>same</i> constraint layer
optimization/attribution.py	the fast optimizer replica used by the Phase M audits	~450× faster; only trusted because tests pin it against the real optimizer
monitoring/drift.py	PSI, KS, schema checks, performance windows	offline monitoring
audit.py	append-only recommendation log with lifecycle states	every recommendation is reconstructible after the fact
utils/io.py	atomic writes, ensure_dir, utc_now, fingerprinting	so a crashed run cannot leave a half-written artifact

67.2 The interaction that matters most

```

features.build.recompute_price_features
      ↑      ↑      ↑      ↑      ↑
    training simulator optimizer hybrid model API

```

One function, five callers. This is what makes the guarantee "a candidate price is scored exactly as a training row would be" structural rather than aspirational. Any divergence would be a leakage bug (channel #7) and it is tested from both directions: `test_recompute_price_features_changes_only_price_features` and `test_price_features_recomputed_for_each_candidate`.

The second most important interaction is the **model interface**: `DemandModel.predict(frame) → units` and `HybridPricingModel.predict(frame) → units` have identical contracts, so `simulate_price_grid`, `optimize_price`, the API and the dashboard are all model-agnostic. Switching from `m1` to `pooled` to `shrunk` is a config change, not a code change.

67.3 Scripts — one CLI per pipeline stage

29 pipeline scripts plus this report's figure generator. Grouped:

group	scripts
acquisition & data	download_dominicks.py, build_dataset.py, validate_data.py, build_features.py
analysis	run_eda.py, run_elasticity.py
modelling	train.py, evaluate.py, price_response.py
pricing	estimate_elasticity.py, compare_price_response.py, optimize.py, backtest.py, run_demo.py
monitoring & smoke	monitor.py, smoke_dashboard.py
Phase I audits	audit_zero_price.py, audit_cost_leakage.py

group	scripts
Phase M audits	audit_constraints.py, audit_model_value.py, audit_guardrails.py, audit_shrinkage.py, audit_elasticity_inference.py, audit_out_of_time_response.py, audit_elasticity_stability.py, audit_eligibility_funnel.py, audit_decision_states.py, audit_claims.py
reporting	make_report_figures.py (added for this report)

Every script writes both a **machine-readable JSON** to `artifacts/metrics/` and a **human-readable Markdown report** to `reports/`. That pairing is what makes the reports auditable: the prose is generated from the same dictionary the JSON serialises.

67.4 Src-layout and why

`pyproject.toml` uses `[tool.setuptools.packages.find] where = ["src"]` and the package is installed editable. So `import pricing_engine` resolves identically for scripts, tests, the API and the dashboard — no `sys.path` manipulation in library code, and no chance of importing a stale copy from the working directory (DECISIONS #30).

68. Training pipeline

68.1 The command

```
python scripts/train.py # or: make train
```

bash

68.2 What it does, in order

88. **Load** `data/processed/dominicks_cereals_features.parquet`.
89. **Restrict to usable rows** via `training_frame()` — requires `move`, `lag_move_1` and `lag_price_1` (drops 36,443 series-opening rows).
90. **Chronological split** via `temporal_split()` — train 2–257 (3,206,437), validation 258–342 (715,856), test 343–399 (749,040). Prints the windows.
91. **Fit and score the four naive baselines** on validation *and* test.
92. **Fit M1 ridge log-log** — 31.0 s. Fit the smoothed UPC demand prior on training rows only; record the prediction cap at 5× max training demand.
93. **Fit M2 HGB Poisson** — 181.9 s.
94. **Measure each price-aware model's implied elasticity** by central finite difference on a 50,000-row sample.
95. **Select** the price-aware model with the lowest **validation WAPE** → **M1 ridge log-log** (0.4135 vs 0.4226).
96. **Score the selected model once on the test window** → WAPE 0.4565.
97. **Compute segment breakdowns** for the selected model: by promotion state, by within-series price-variation bucket, worst 10 stores by WAPE.
98. **Persist** `artifacts/models/demand_model.joblib` with `compress=3`, plus `demand_model_metadata.json`.
99. **Write** `artifacts/metrics/model_metrics.json` and `reports/04_MODEL_COMPARISON.md`.

Total runtime: **243 s**.

68.3 What is stored in the artifact

```
joblib.dump({
    "kind": self.kind, "name": self.name, "params": self.params,
    "estimator": self.estimator,          # the whole sklearn Pipeline
    "upc_prior": self.upc_prior_,          # fitted on training rows only
    "global_prior": self.global_prior_,
```

```
"prediction_cap": self.prediction_cap_,
"metadata": self.metadata.as_dict(),
}, p, compress=3)
```

python

The **preprocessing travels with the model**, so a candidate price scored at inference goes through exactly the same imputation, scaling and encoding as a training row.

68.4 The separate elasticity fit

```
python scripts/estimate_elasticity.py # or: make estimate-elasticity
```

bash

Restricted to training weeks 2–257 (the script aborts otherwise), 35.0 s. Produces `artifacts/models/elasticity_table.csv` — 372 product rows with `elasticity_raw`, `std_error`, `std_error_hc1`, `std_error_cluster_store`, `std_error_cluster_week`, `se_inflation_vs_hc1`, `t_stat`, `CI bounds`, `r_squared`, `n_obs`, `n_distinct_prices`, `usable`, `reject_reason`, `shrinkage_weight`, `elasticity_shrunk`, `elasticity_final`, `elasticity_source`, `clipped` — plus 23 `meta_*` columns carrying the pooled estimate, its several standard errors, τ^2 , the prior mean, the estimation window and the clip band.

The elasticity table is a versioned model artifact, not a number in a script.

68.5 The full pipeline

```
make all
```

bash

which runs, in dependency order: `data` → `validate` → `features` → `eda` → `elasticity` → `train` → `evaluate` → `price-response` → `estimate-elasticity` → `compare-response` → `optimize` → `backtest` → `monitor` → `audit-zero-price` → `audit-cost` → `audit` (reports 11–20).

69. Recommendation pipeline

69.1 A real recommendation, end to end

Traced from a **live API call** made on 2026-08-19 (§72 shows the raw JSON).

Step 0 — the request

```
{"upc": 3000006560, "store": 86, "objective": "gross_profit", "policy_profile": "standard"}
```

json

Step 1 — context lookup. `AppState.context(3000006560, 86, None)` filters the in-memory serving slice (the most recent 8 weeks, 112,763 contexts) and takes the latest row: **week 399, 1997-05-01**.

Step 2 — the observed context

field	value
product	CAPN CRUNCH JUMBO CR
current effective unit price p_0	\$3.35
decision-time unit cost c (lagged, forward-filled AAC)	\$2.5571
observed price support	\$1.50 – \$3.79 over 359 weeks, 56 distinct prices
recent weeks	392–394: 12–18 units @ \$3.35 · 395: 17,824 units @ \$1.50 · 396: 465 @ \$3.19 (\$) · 397–399: 11–25 @ \$3.35

Step 3 — reference price stamped. `attach_reference_price(row)` records `reference_price = $3.35` from the **observed row**, before any candidate is substituted.

Step 4 — elasticity lookup. UPC 3000006560 has a usable own estimate: raw -3.0178 , panel-robust SE 0.2557 , weight $0.9214 \Rightarrow \epsilon = -2.9401$, source `shrunk_product`.

Step 5 — baseline demand. `recompute_price_features(frame, $3.35)` then `M1.predict(...)` $\Rightarrow \hat{Q}(p_o) = 25.5647$ units. (Actual observed that week: 12 — a 113% over-forecast on this single store-week, which is entirely typical at WAPE 0.46.)

Step 6 — eligibility. 359 observations ≥ 40 ✓, 56 distinct prices ≥ 5 ✓, price CV ≥ 0.05 ✓, cost available ✓.
Passes.

Step 7 — preliminary risk at \$3.35. 359 obs ≥ 150 ✓, 56 prices ≥ 15 ✓, extrapolation distance $0.0 \leq 0.02$ ✓, elasticity source is `shrunk_product` (not a fallback), promotion share $\leq 0.5 \Rightarrow$ **LOW**. No cap tightening.

Step 8 — constraints.

bound	source	value
lower	ABSOLUTE_PRICE_FLOOR	\$0.0100
lower	OUTSIDE_EXTRAPOLATION_RANGE (\$1.50 $\times$ 0.95)	\$1.4250
lower	COST_FLOOR	\$2.5571
lower	MIN_MARGIN (\$2.5571 / 0.95)	\$2.6916
lower	PRICE_CHANGE_LIMIT (\$3.35 $\times$ 0.90) ←	\$3.0150
upper	OUTSIDE_EXTRAPOLATION_RANGE (\$3.79 $\times$ 1.05)	\$3.9795
upper	PRICE_CHANGE_LIMIT (\$3.35 $\times$ 1.10) ←	\$3.6850

Feasible interval: [\$3.015, \$3.685]. Both edges are the change cap.

Step 9 — candidate grid. \$0.05 step from \$3.015 to \$3.685, rounded to \$0.01, plus the current price: **{ \$3.02, \$3.07, ..., \$3.66, \$3.35 }** — 15 candidates.

Step 10 — simulate. One batched call. Every candidate gets `log_price`, `price_vs_last_week`, `price_vs_series_reference`, `price_vs_recent_mean` recomputed; every context feature and the cost stay fixed.

Step 11 — objective. $GP(p) = (p - \$2.5571) \cdot Q(p)$. The unconstrained CE optimum is $\$2.5571 \times 2.9401/1.9401 = \3.875 , **outside** the feasible interval, so `argmax` lands on the **upper edge: \$3.66**.

	current \$3.35	proposed \$3.66
predicted units	25.5647	19.7077
expected revenue	\$85.64	\$72.13
expected gross profit	\$20.2714	\$21.7365
expected margin rate	23.7%	30.1%

Step 12 — materiality. $(21.7365 - 20.2714)/20.2714 = 0.0723 = +7.23\% \geq 1\% \Rightarrow$ material.
`PROFIT_UPLIFT_POSITIVE` raised.

Step 13 — final risk at \$3.66. \$3.66 is inside the observed support (\$1.50–\$3.79), so extrapolation distance is $0.0 \Rightarrow$ still **LOW**.

Step 14 — decision. LOW risk $\Rightarrow$ **RECOMMEND_CHANGE**, `actionable = true`, `final_recommended_price = $3.66`.

Step 15 — reason codes. `PRICE_CHANGE_LIMIT` (the change window is an active edge — the model wanted \$3.875), `PROFIT_UPLIFT_POSITIVE`.

Step 16 — audit. Appended to `artifacts/recommendation_log.csv` with all 34 columns, lifecycle state `GENERATED`.

69.2 What this trace demonstrates

- The engine **wanted +15.7%** (to \$3.875) and **recommended +9.25%** (to \$3.66). The guardrail absorbed the difference — this is one of the 7,967 "guardrail corner" contexts (\$54.8).
- The elasticity applied was **92% the product's own** and 8% the category prior.
- The demand forecast over-predicted this particular week by 113% and **the recommended price would have been identical had it predicted 12 units**, because $\hat{Q}(p_o)$ cancels from the arg-max (\$43.6). The forecast set the dollar amounts, not the price.
- Every threshold used came from `configs/config.yaml`; nothing was hardcoded.

PART XIII — FASTAPI

70. API architecture

70.1 Startup and model loading

```
@asynccontextmanager
async def lifespan(app: FastAPI):
    try:
        app.state.engine = build_state()
    except FileNotFoundError as exc:      # keep /health informative instead of crashing
        app.state.engine = None
        app.state.startup_error = str(exc)
    yield
    app.state.engine = None
```

python

The model artifact, the elasticity table and the serving context slice are loaded **once** during the FastAPI lifespan, not per request. Reloading a model per request is the classic way to turn a 5 ms endpoint into a 2 s one.

If artifacts are missing the app still starts, and `/health` reports `model_loaded: false` rather than the process dying. That is the right failure mode for a service behind a load balancer.

70.2 What is served

```
model = load_pricing_model(base_model, cfg=cfg)  # the elasticity hybrid, not the raw forecaster
feats = pd.read_parquet(features_path)
contexts = feats[feats["week"] > max_week - SERVING_WEEKS]      # SERVING_WEEKS = 8
contexts = contexts[contexts["lag_price_1"].notna()]
stats = pd.read_csv(cfg.path("metrics_dir") / "price_variation_upc_store.csv")
```

python

Verified live on 2026-08-19:

```
{"status": "ok", "model_loaded": true, "contexts_loaded": 112763,
 "decision_weeks": [392, 393, 394, 395, 396, 397, 398, 399]}
```

json

The service deliberately serves the **configured pricing model** (`shrunk` by default), not the raw forecaster — so the API and the batch pipeline cannot disagree about what a recommendation means.

70.3 Validation strategy

From the schemas module docstring:

Validation is deliberately strict: a pricing endpoint that silently accepts a negative price or an impossible range is a liability.

rule	mechanism
<code>store ≥ 1, week ≥ 1</code>	<code>Pydantic Field(..., ge=1)</code>
<code>price > 0, unit_cost > 0</code>	<code>Field(..., gt=0)</code>
<code>min_price > 0, max_price > 0, 0 < step ≤ 5.0</code>	Field bounds
<code>max_price ≥ min_price</code>	<code>@model_validator(mode="after")</code>
<code>grid ≤ 500 candidates</code>	<code>@model_validator</code> — <code>(max-min)/step > 500</code> rejected
<code>objective ∈ {gross_profit, revenue}</code>	Literal
<code>policy_profile ∈ {conservative, standard, aggressive}</code>	Literal
unknown context	<code>ContextNotFound</code> → 404 with the served week range
model not loaded	503

rule	mechanism
optimizer refusal (e.g. no cost)	OptimizerError → 422

The 500-candidate cap is a small but real piece of production thinking: without it, `{"min_price": 0.01, "max_price": 100, "step": 0.001}` would allocate a 100,000-row frame and score it.

70.4 Disclaimers as schema fields

`ModelInfoResponse` and `RecommendPriceResponse` both carry a `disclaimer` field with a default value. That means the caveat is **in the response payload** — it cannot be lost when a consumer copies a number into a spreadsheet:

*"Predictions are observational model estimates, not causal effects.
Counterfactual economics at unobserved prices are MODEL-INTERNAL estimates:
the same fitted price-response model both proposes and scores candidate
prices, so they are an internal simulation, not an unbiased policy value."*

*"REVIEW_REQUIRED and KEEP_CURRENT are not actionable: for those states
final_recommended_price equals current_price and only
proposed_candidate_price carries the optimizer's proposal. Uplift figures are
model-internal estimates, never realised or causal impact."*

70.5 What the API does not have

No authentication, no rate limiting, no request logging, no tracing, no per-tenant isolation. **Demo scope**, and `STATUS.md` says so.

71. Endpoints

AI Pricing & Revenue Optimization Engine 0.1.0 OAS 3.1

/openapi.json

Price recommendation service built on the Dominick's Finer Foods Cereals scanner panel (Kilts Center, University of Chicago Booth).

Scientific note. All counterfactual economics returned by this API are *model-internal estimates*: demand at prices that were never charged was never observed, and the same fitted price-response model both proposes and scores the candidate prices. Nothing here is a causal or realised uplift.

Decision states. RECOMMEND_CHANGE is actionable; KEEP_CURRENT leaves the price alone; REVIEW_REQUIRED means a proposal exists but a human must approve it - HIGH-risk contexts never auto-change a price under the default policy.

pricing ^	
POST	/predict-demand Predict Demand v
POST	/simulate-prices Simulate Prices v
POST	/recommend-price Recommend Price v
meta ^	
GET	/health Health v
GET	/model/info Model Info v
Schemas ^	
HTTPValidationError > Expand all object	
HealthResponse > Expand all object	
ModelInfoResponse > Expand all object	
PredictDemandRequest > Expand all object	
PredictDemandResponse > Expand all object	
RecommendPriceRequest > Expand all object	
RecommendPriceResponse > Expand all object	
SimulatePricesRequest > Expand all object	

Figure 18 — The live `/docs` page, captured from a running instance on 2026-08-19. Source: headless Chrome against `http://127.0.0.1:8077/docs`.

71.1 GET /health

purpose	liveness plus enough state to diagnose a bad deploy
request	none
response	status, model_loaded, contexts_loaded, decision_weeks
live example	{"status":"ok", "model_loaded":true, "contexts_loaded":112763, "decision_weeks":[392,...,399]}
failure modes	never fails; reports model_loaded: false if artifacts are missing

71.2 GET /model/info

purpose	model provenance and headline metrics, for auditability
request	none
response	name, kind, version, price_response_method, pooled_elasticity, elasticity_training_weeks, trained_at_utc, target, n_features, feature_columns, train/valid/test_weeks, metrics, data_fingerprint, disclaimer
failure modes	503 if the model is not loaded; returns nulls if metadata is

	absent

Live response (abridged), 2026-08-19:

```
{
  "name": "M1 ridge log-log",
  "kind": "ridge_loglog",
  "version": "ridge_loglog-20260818-132847",
  "price_response_method": "shrunk",
  "pooled_elasticity": -2.028859937952365,
  "elasticity_training_weeks": [2, 257],
  "trained_at_utc": "2026-08-18T09:28:47+00:00",
  "target": "move",
  "n_features": 26,
  "train_weeks": [2, 257],
  "valid_weeks": [258, 342],
  "test_weeks": [343, 399],
  "metrics": {
    "valid": {
      "wape": 0.41345926697946545,
      "mae": 7.875275193954714,
      ...
    },
    "test": {
      "wape": 0.45649432400768286,
      "mae": 8.502566963198962,
      ...
    }
  },
  "data_fingerprint": "e9d26f2c0eda9f7e9d8dbb508053e56a932608668e16d8b5aeeec0032373722d0"
}
```

json

Note that the response carries **the elasticity estimation window** and the **dataset fingerprint** — so a consumer can verify which data the served model was fitted on without access to the training environment.

71.3 POST /predict-demand

purpose	predicted units for one context, optionally at an override price
request	upc, store, week?, price? (gt=0)
response	upc, store, week, week_start_date, product_description, price_used, observed_price, predicted_units, model_version
validation	store ≥ 1, week ≥ 1, price > 0
failure modes	404 unknown context (message names the served window); 422 invalid payload; 503 model not loaded

The reference price is stamped **before** the candidate is substituted:

```
frame = recompute_price_features(attach_reference_price(row), np.array([price]))
```

python

Without that, the hybrid would compute $p/p_0 = 1$ and silently return baseline demand for every price.

71.4 POST /simulate-prices

purpose	the full demand / revenue / gross-profit curve over a requested price range
request	upc, store, week?, min_price, max_price, step (default 0.05), unit_cost?
response	unit_cost_used, model_version, price_response_method, elasticity_used, elasticity_source, n_candidates, curve[], diagnostics
curve point	candidate_price, predicted_units, expected_revenue, expected_gross_profit, expected_margin_rate
diagnostics	n_candidates, share_segments_increasing, monotone_decreasing, relative_demand_span, median_local_elasticity, min/max_predicted_units, flat_response
validation	max_price ≥ min_price; ≤ 500 candidates; step ∈ (0, 5]
failure modes	422 empty or oversized grid; 404 unknown context

`expected_gross_profit` and `expected_margin_rate` are `null` — not zero — when no cost is available. That distinction is preserved through the Pydantic model (`float | None`), so a consumer cannot mistake "unknown" for "zero".

71.5 POST /recommend-price

purpose	the full constrained recommendation with reason codes
request	<code>upc</code> , <code>store</code> , <code>week?</code> , <code>objective</code> (default <code>gross_profit</code>), <code>policy_profile</code> (default <code>standard</code>), <code>unit_cost?</code>
response	30 fields — see §72
validation	Literal on objective and profile; <code>unit_cost > 0</code>
failure modes	422 on <code>OptimizerError</code> (e.g. non-positive current price); 404 unknown context; 503 model not loaded

`diagnostics` is deliberately **popped** from the response (`d.pop("diagnostics", None)`) — it holds internal state (grid bounds, objective values, series stats) that belongs in the audit log rather than in a public contract.

72. Example API recommendation

A **live** call, executed 2026-08-19 against a running instance. (The capture instance ran on port 8077 to avoid colliding with anything on the default port; `make api` serves on 8000.)

```
curl -X POST localhost:8000/recommend-price \  
-H "content-type: application/json" \  
-d '{"upc": 3000006560, "store": 86, "objective": "gross_profit", "policy_profile": "standard"}'
```

bash

```
{  
  "upc": 3000006560,  
  "store": 86,  
  "decision_week": 399,  
  "decision_week_start_date": "1997-05-01",  
  "product_description": "CAPN CRUNCH JUMBO CR",  
  "objective": "gross_profit",  
  "policy_profile": "standard",  
  "model_version": "ridge_loglog-20260818-132847",  
  "price_response_method": "shrunk",  
  "elasticity_used": -2.9400591555147004,  
  "elasticity_source": "shrunk_product",  
  "current_price": 3.35,  
  "proposed_candidate_price": 3.66,  
  "final_recommended_price": 3.66,  
  "proposed_price_change_pct": 0.09253731343283578,  
  "price_change_pct": 0.09253731343283578,  
  "decision": "RECOMMEND_CHANGE",  
  "actionable": true,  
  "predicted_units_current": 25.564729608414478,  
  "predicted_units_recommended": 19.707714927805956,  
  "expected_revenue_current": 85.6418441881885,  
  "expected_revenue_recommended": 72.1302366357698,  
  "expected_gross_profit_current": 20.271424609064187,  
  "expected_gross_profit_recommended": 21.73652571021359,  
  "model_internal_estimated_profit_uplift_pct": 0.0722742051633755,  
  "model_internal_estimated_revenue_uplift_pct": -0.15776875989181677,  
  "realisable_profit_uplift_pct": 0.0722742051633755,  
  "unit_cost_used": 2.5570549964904785,  
  "risk_level": "LOW",  
  "reason_codes": ["PRICE_CHANGE_LIMIT", "PROFIT_UPLIFT_POSITIVE"],  
  "risk_notes": [],  
  "constraints": {  
    "bounds": [3.015, 3.6850000000000005],  
    "max_price_change_pct": 0.1,  
    "price_change_window": [3.015, 3.6850000000000005],  
  },  
}
```

```

    "extrapolation_tolerance_pct": 0.05,
    "support_low": 1.424999999999998,
    "support_high": 3.9795000000000003,
    "unit_cost": 2.5570549964904785,
    "margin_floor": 2.6916368384110303,
    "lower_bounds": {"ABSOLUTE_PRICE_FLOOR": 0.01, "PRICE_CHANGE_LIMIT": 3.015,
                     "OUTSIDE_EXTRAPOLATION_RANGE": 1.424999999999998,
                     "COST_FLOOR": 2.5570549964904785, "MIN_MARGIN": 2.6916368384110303},
    "upper_bounds": {"PRICE_CHANGE_LIMIT": 3.6850000000000005,
                     "OUTSIDE_EXTRAPOLATION_RANGE": 3.9795000000000003}
  },
  "disclaimer": "REVIEW_REQUIRED and KEEP_CURRENT are not actionable: ..."
}

```

json

72.1 Every field explained

field	value	meaning
upc, store	3000006560, 86	the decision key
decision_week, decision_week_start_date	399, 1997-05-01	the week being priced
product_description	CAPN CRUNCH JUMBO CR	from the UPC metadata join
objective	gross_profit	which objective was maximised
policy_profile	standard	which guardrail set applied
model_version	ridge_loglog-20260818-132847	ties the answer to a specific artifact
price_response_method	shrunk	which counterfactual assumption produced this
elasticity_used	-2.9401	the ϵ actually applied
elasticity_source	shrunk_product	its provenance — not the pooled fallback
current_price	\$3.35	observed p_0
proposed_candidate_price	\$3.66	the optimizer's proposal — not a business price on its own
final_recommended_price	\$3.66	what would actually be charged (equal here only because the decision is actionable)
proposed_price_change_pct	+9.254%	change of the proposal
price_change_pct	+9.254%	change of the final price — would be 0.0 if not actionable
decision	RECOMMEND_CHANGE	the decision state
actionable	true	the single boolean a consumer must respect
predicted_units_current → _recommended	25.56 → 19.71	model-internal demand at both prices
expected_revenue_current → _recommended	\$85.64 → \$72.13	revenue falls — $ \epsilon > 1$
expected_gross_profit_current → _recommended	\$20.27 → \$21.74	profit rises
model_internal_estimated_profit_uplift_pct	+7.227%	the same model proposed and scored this — not realised, not causal
model_internal_estimated_revenue_uplift_pct	-15.777%	the revenue/profit trade-off, quantified
realisable_profit_uplift_pct	+7.227%	equals the estimate only because actionable ; forced to 0.0 otherwise
unit_cost_used	\$2.5571	the lagged, forward-filled AAC held fixed across the grid
risk_level	LOW	heuristic, not calibrated confidence
reason_codes	PRICE_CHANGE_LIMIT, PROFIT_UPLIFT_POSITIVE	the change cap is an active edge; the proposal beats current on

field	value	meaning
		estimated profit
risk_notes	[]	no promotion-entanglement or fallback caveat applies
constraints.bounds	[\$3.015, \$3.685]	the feasible interval
constraints.lower_bounds / upper_bounds	5 lower, 2 upper	every guardrail's value, not just the binding one — so "why not \$3.90?" is answerable from the payload
disclaimer	(text)	travels with the response

72.2 The two most important fields

price_response_method + elasticity_source. Without them, +7.227% is uninterpretable — \$44.1 showed the same engine reports 8.28% or 17.57% median uplift depending only on the method. The payload always says which one produced this number.

proposed_candidate_price vs final_recommended_price. They are equal here only because **actionable** is true. For a **REVIEW_REQUIRED** context, the proposal would be \$3.66 and the final price \$3.35, and any consumer reading only one field would be wrong in one direction or the other.

PART XIV — STREAMLIT DASHBOARD

73. Dashboard design

73.1 Why it exists

A JSON API answers "what price?". It does not let a **pricing manager** ask "why?", "what if?", and "how much do I trust this?" — which are the three questions that decide whether a recommendation gets applied.

73.2 Target user

A **category / pricing manager** who owns the price decision, plus a **data scientist or analyst** auditing the engine's behaviour. Neither reads Parquet files. Both need to interrogate a single decision *and* see the portfolio.

73.3 Design principles, enforced in code

1. No number is typed into the app. From the module docstring:

Nine pages, all driven by artifacts produced by the pipeline — no numbers are typed into this file. Anything the pipeline has not produced yet is reported as missing rather than faked.

Every metric comes from `artifacts/metrics/*.json`, `artifacts/figures/*.png`, the feature Parquet or the model artifact.

2. Missing artifacts are reported, never faked.

```
def missing(what: str, command: str) -> None:
    st.warning(f"{what} is not available yet. Run `{command}` first.")

def figure(name: str):
    path = CFG.path("figures_dir") / name
    if path.exists(): st.image(str(path), use_container_width=True)
    else: st.info(f"Figure not generated yet: {name}")
```

python

3. The disclaimer is unavoidable. A module-level `DISCLAIMER` constant is rendered on every page that shows counterfactual economics, and a permanent sidebar warning appears on **all nine pages**:

"Counterfactual economics shown anywhere in this app are MODEL-INTERNAL estimates - never realised or causal uplift. HIGH risk means risky: those contexts are gated to REVIEW_REQUIRED by default."

4. Caching is explicit. `@st.cache_data` for artifacts and the panel, `@st.cache_resource` for the model. The panel is ~4.7M rows and is loaded with an explicit column list (`PANEL_COLUMNS`, 38 columns) rather than the whole table.

5. The dashboard serves the pricing model, not the forecaster.

```
base = load_model(path, cfg=CFG)
try:    return load_pricing_model(base, cfg=CFG)
except Exception: return load_pricing_model(base, cfg=CFG, method="ml")
```

python

So the dashboard, the API and the batch pipeline cannot disagree about what a recommendation means. The `ml` fallback exists so the app still runs when the elasticity table is missing — and it changes the displayed `price_response_method`, so the substitution is visible.

6. Plotly, deliberately. `st.bar_chart/st.line_chart` pull in Altair, which does not import on Python 3.13 in this environment. The choice is commented in the code and recorded in DECISIONS #29.

74. Each dashboard page

Nine pages, verified rendering by `scripts/smoke_dashboard.py` and captured live on 2026-08-19 (\$75).

74.1 Page 1 — Executive overview

aspect	detail
purpose	the whole project on one screen for a non-technical stakeholder
filters	none — this is the portfolio view
KPIs (row 1)	historical observed revenue \$262.0M ; historical observed gross profit \$40.1M ; products × stores 489 × 93 ; weeks covered 366
KPIs (row 2)	selected model M1 ridge log-log ; test WAPE 0.4565 ; series eligible for pricing 19,707 / 36,443 ; model-internal estimated portfolio uplift +8.30% (with a hover: <i>"Internal simulation only. Not realised, not causal, not an unbiased policy value."</i>)
charts	weekly observed revenue and gross profit; weekly observed units; reason-code frequency bar chart
batch panel	contexts scored 3,000; actionable 58.9%; review required 10.4%; keep current 30.8%; plus the caption <i>"HIGH-risk contexts that still auto-changed a price: 0 of 1,039 (must be zero under the default policy profiles)"</i> *
business interpretation	scale, model quality, coverage and the current recommendation mix, in one view
limitations shown	the uplift KPI is labelled model-internal in its own tooltip; the page-level disclaimer is directly under the title

74.2 Page 2 — Product / store explorer

aspect	detail
purpose	inspect the raw history of one UPC × store series before trusting any recommendation for it
filters	product (sorted by total units), store
KPIs	observations; observed revenue; observed gross profit; mean gross margin
charts	dual-axis time series — effective unit price and estimated unit AAC (dotted) on the left axis, units sold as bars on the right, with triangles marking recorded promotion weeks
table	the raw series: week, date, price, units, revenue, gross profit, margin rate, estimated AAC, promotion type
business interpretation	"has this series ever been priced near where the engine wants to put it?" and "does its cost move with its price?"
limitations shown	in-line caption: <i>"Triangles mark weeks with a recorded promotion code. Absence of a marker does NOT prove there was no promotion — the coding is incomplete."</i>

74.3 Page 3 — Pricing and demand

aspect	detail
purpose	category-level pricing structure
KPIs	mean effective unit price \$3.12 ; rows with a promotion code

aspect	detail
	7.35% ; median cross-store price spread 13.6%
charts	price distribution; price-versus-demand scatter; within-series price-CV histogram with the eligibility threshold (0.05) marked by a dashed line
table	top 200 series by units: n_obs, n_distinct_prices, price_cv, price_min/max, n_price_changes, eligible
business interpretation	how much price variation exists, and which series can be optimised at all
limitations shown	in-line caption: <i>"The scatter above is an association, not a demand curve: price and demand are jointly determined by promotions, seasonality and retailer decisions."</i>

74.4 Page 4 — Elasticity analysis

aspect	detail
purpose	show the naive-to-controlled elasticity ladder and the product-level spread
table	the four log-log specifications with elasticity, SE, CI, R ² , n, controls
callout	computed live: <i>"The naive pooled coefficient and the fixed-effects coefficient differ by a factor of 7.0. That gap is the clearest evidence that the raw association is not a causal effect."</i>
KPIs	median per-UPC elasticity -2.36 ; p10 / p90 -4.09 / -0.24 ; UPCs with a wrong-signed significant coefficient 5.1%
charts	per-UPC elasticity histogram; arc-elasticity histogram
table	per-UPC estimates with SE, t, n, R ²
business interpretation	how price-sensitive the category is, and how much products differ
limitations shown	the wrong-sign KPI is displayed as a headline metric, not buried

74.5 Page 5 — Price simulator

aspect	detail
purpose	free-form what-if on one context, with no policy layer
filters	product, store (latest week only)
KPIs	current price; decision-time unit cost (lagged AAC); observed price support; elasticity applied with a tooltip naming the source and method
input	a candidate-price slider, $\pm 40\%$ around the current price, \$0.05 step
outputs	predicted units; expected revenue; expected gross profit; price change %
chart	dual-axis — predicted units on the left, expected revenue and expected gross profit on the right, with vertical markers for the current price and the candidate
guards	red error if the candidate is outside the observed price support : <i>"Extrapolation warning: \$X is outside this series' observed price support (\$A – \$B). The model has no data there."</i> — and a second if the candidate is below cost
business interpretation	build intuition about the trade-off before looking at what the engine recommends
limitations	this page has no guardrails — it will happily simulate a price

aspect	detail
	the policy layer would refuse. That is the point, and the two red warnings are how it stays honest.

74.6 Page 6 — Recommendation engine

aspect	detail	
purpose	the product: a full recommendation with its reasoning	
filters	product, store, policy profile (labelled "Policy profile (DEMO setting)"), objective (gross profit / revenue)	
KPIs	current price; proposed candidate price (with tooltip: "The optimizer's proposal. It only becomes a business price when the decision state is RECOMMEND_CHANGE."); FINAL recommended price (tooltip: "What would actually be charged under this policy: the current price whenever the decision is not actionable."); decision state	
status banner	green for RECOMMEND_CHANGE, blue for KEEP_CURRENT, amber warning for REVIEW_REQUIRED with the explicit text: "this context is HIGH risk, so the proposal is NOT applied automatically. A human must approve it. The final price stays at \$X."	
provenance line	`price response: shrunk`	elasticity -3.248 (shrunk_product)`
economics	predicted units, expected revenue and expected gross profit, each shown as current → recommended, with the uplift labelled "(model-internal estimate)"	
reason codes	rendered as inline code chips	
risk notes	rendered as italic bullets when present	
feasible range	printed explicitly	
expander	full constraints and diagnostics JSON	
chart	demand / revenue / gross-profit curves with the current price, the proposed candidate, a shaded feasible band , and — when the decision is not actionable — a red "not actionable" annotation on the proposal marker	
audit log	expander showing the last 200 rows of artifacts/recommendation_log.csv	
business interpretation	everything a manager needs to accept or reject a single price	
limitations shown	the proposed/final split is explicit in the KPI row; the risk caption states the score is heuristic and not calibrated confidence	

74.7 Page 7 — Model performance

aspect	detail
--------	--------

aspect	detail
purpose	the evidence behind the model, for a technical reviewer
split panel	train / validation / test weeks and dates, with the sentence <i>"No random splitting is used anywhere."</i>
table	all six models: price-aware flag, validation WAPE/MAE, test WAPE/MAE, implied elasticity
callout	<i>"Selected model: M1 ridge log-log (lowest validation WAPE among price-aware models)"</i>
charts	actual vs predicted weekly; prediction scatter; residuals
segment tables	error by recorded promotion state; error by within-series price variation
price-response panel	curves monotone decreasing 100.0% ; median implied elasticity -3.10 ; flat curves 0.00% ; demand-curve figure
method-comparison panel	the per-method table and the pairwise-disagreement table, with the caption <i>"The same baseline demand model and constraints produce different prices depending only on the price-response assumption."</i>
elasticity-estimation panel	pooled controlled -2.029 ; UPC × store FE -2.222 ; products usable 239/372 ; mean shrinkage weight 0.78
backtest panel	the four-policy table and weekly-WAPE figure, under the heading <i>"Offline policy comparison (model-internal estimate)"</i>
limitations shown	the implied-elasticity column sits directly beside the WAPE column, so the -3.10 vs -2.03 tension is visible in one table

74.8 Page 8 — Data quality

aspect	detail
purpose	prove the data pipeline rather than assert it
KPIs	raw rows 6,602,582 ; canonical rows 4,707,776 ; rows removed 1,894,806 (28.7%) ; duplicate grain rows 0
table	the complete exclusion ledger — rule, rows removed, % of raw, reason
JSON	metadata-join quality (duplicate metadata UPCs 0, movement UPCs 489, metadata UPCs 490, rows without metadata 0)
tables	ok flag counts; recorded sale-code counts, with the caption <i>"B = Bonus Buy, C = Coupon, S = simple price reduction. G and L are undocumented codes found in the Cereals file."</i>
checks	the 15 formula/structural checks with a green <i>"15 / 15 checks pass"</i>
zero-price panel	rows with price = 0 1,851,380 (28.0% of raw) ; of which recorded any sales 677 ; leading/trailing runs 73%
cost-leakage panel	the three audit results as JSON, all true
reproducibility panel	Parquet SHA-256, DataFrame fingerprint, full environment
business interpretation	"can I trust the inputs?" answered with counts, not adjectives
limitations shown	the 26.6% zero-price exclusion is displayed as a headline KPI rather than hidden in the ledger

74.9 Page 9 — Methodology and limitations

aspect	detail
purpose	the caveats, in full, inside the product
banner	a red error box: "This engine does not measure causal price effects. Dominick's prices were set by the retailer, not randomised. Every counterfactual number here is a model

aspect	detail
	estimate."*
tabs	Limitations (KNOWN_LIMITATIONS.md in full) · Causal (docs/CAUSAL_LIMITATIONS.md) · Pricing science (docs/PRICING_SCIENCE.md) · Data source (docs/DATA_SOURCE.md) · Model card (docs/MODEL_CARD.md) · Reports (a selector over 12 generated reports)
business interpretation	a manager can read every caveat without leaving the app
limitations shown	this page is the limitations

One gap worth naming: the Reports tab selector lists 12 reports (01–10, MONITORING_DESIGN, VALIDATION_SUMMARY) and does **not** include the ten Phase M audits (11–20). Those are the reports containing the project's most important findings — the constraint attribution, the rule ablation, the shrinkage correction. A user browsing only the dashboard would not reach them. This is a real, small defect and it is recorded in the limitations register (Appendix K, #L14).

75. Dashboard screenshots

These are real screenshots of the running application, captured on 2026-08-19 by driving the live Streamlit server (port 8511) with headless Chrome via Selenium and saving full-page captures. No image in this report is mocked, drawn or edited.

All nine are stored in `artifacts/report_figures/`:

#	page	file
1	Executive overview	dashboard_1_executive_overview.png
2	Product / store explorer	dashboard_2_product_store_explorer.png
3	Pricing & demand	dashboard_3_pricing_and_demand.png
4	Elasticity analysis	dashboard_4_elasticity_analysis.png
5	Price simulator	dashboard_5_price_simulator.png
6	Recommendation engine	dashboard_6_recommendation_engine.png
7	Model performance	dashboard_7_model_performance.png
8	Data quality	dashboard_8_data_quality.png
9	Methodology & limitations	dashboard_9_methodology_limitations.png

Plus the API documentation: `api_openapi_docs.png` (§71).

75.1 Executive overview

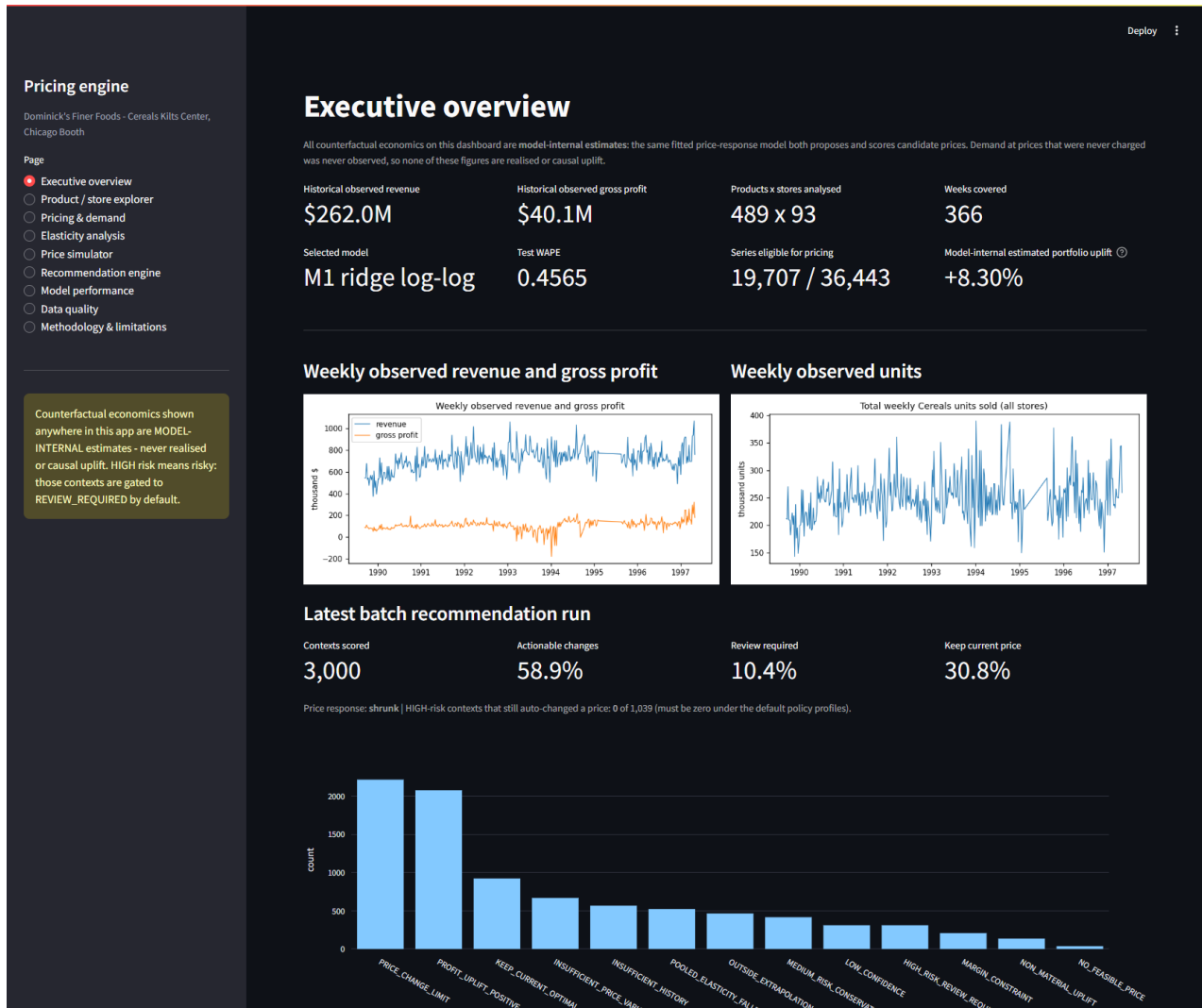


Figure 19 — Every KPI on this page is read from a pipeline artifact. Note the +8.30% uplift tile, which carries a tooltip stating it is an internal simulation, and the caption confirming 0 of 1,039 HIGH-risk contexts auto-changed a price.

75.2 Price simulator

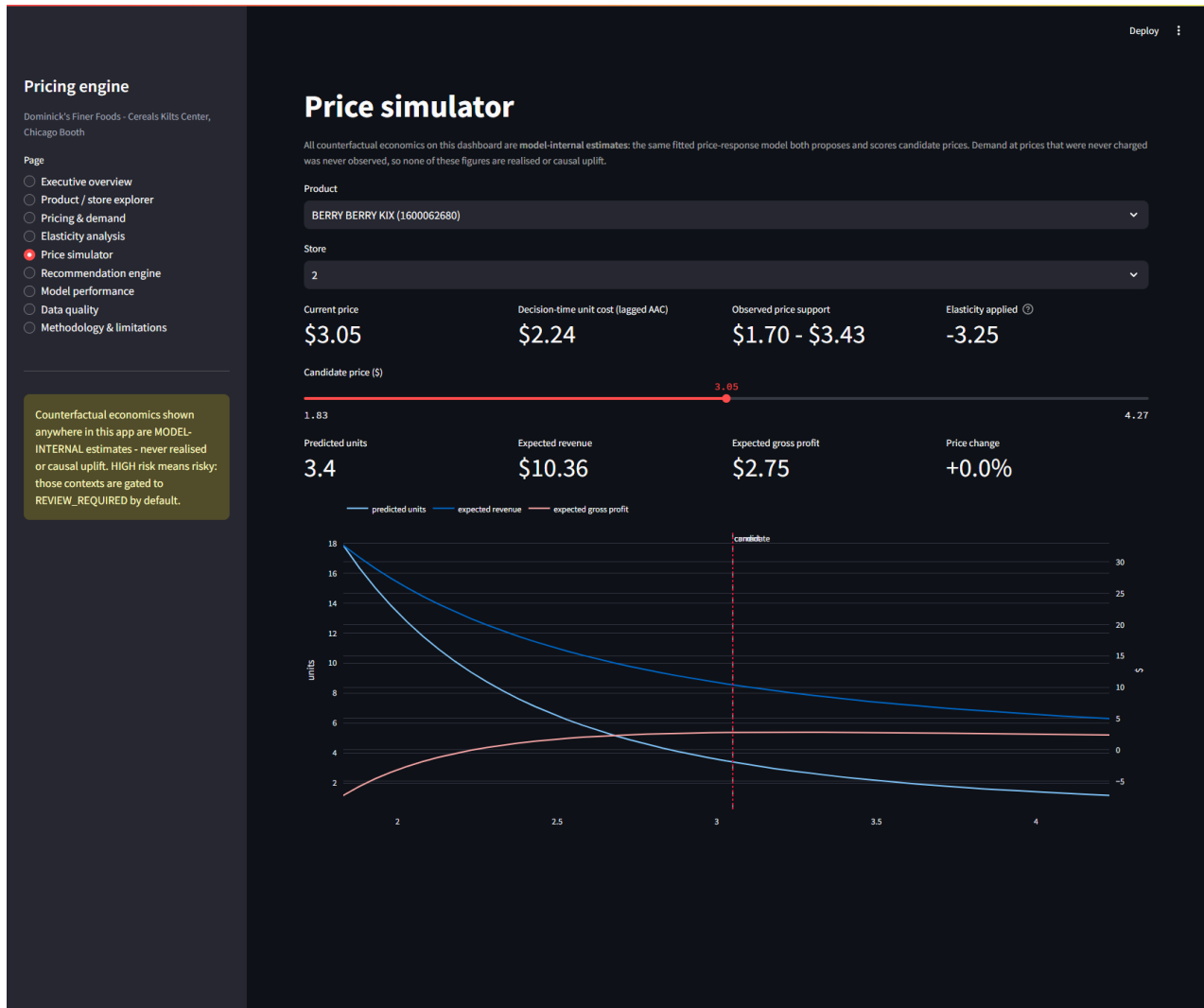


Figure 20 — BERRY BERRY KIX (1600062680) at store 2. Current price \$3.05, decision-time unit cost \$2.24, observed support \$1.70–\$3.43, applied elasticity –3.25. The gross-profit curve (red) peaks well to the right of the current price; the revenue curve (blue) falls monotonically — the \$2.2 divergence, visible in the product.

75.3 Recommendation engine

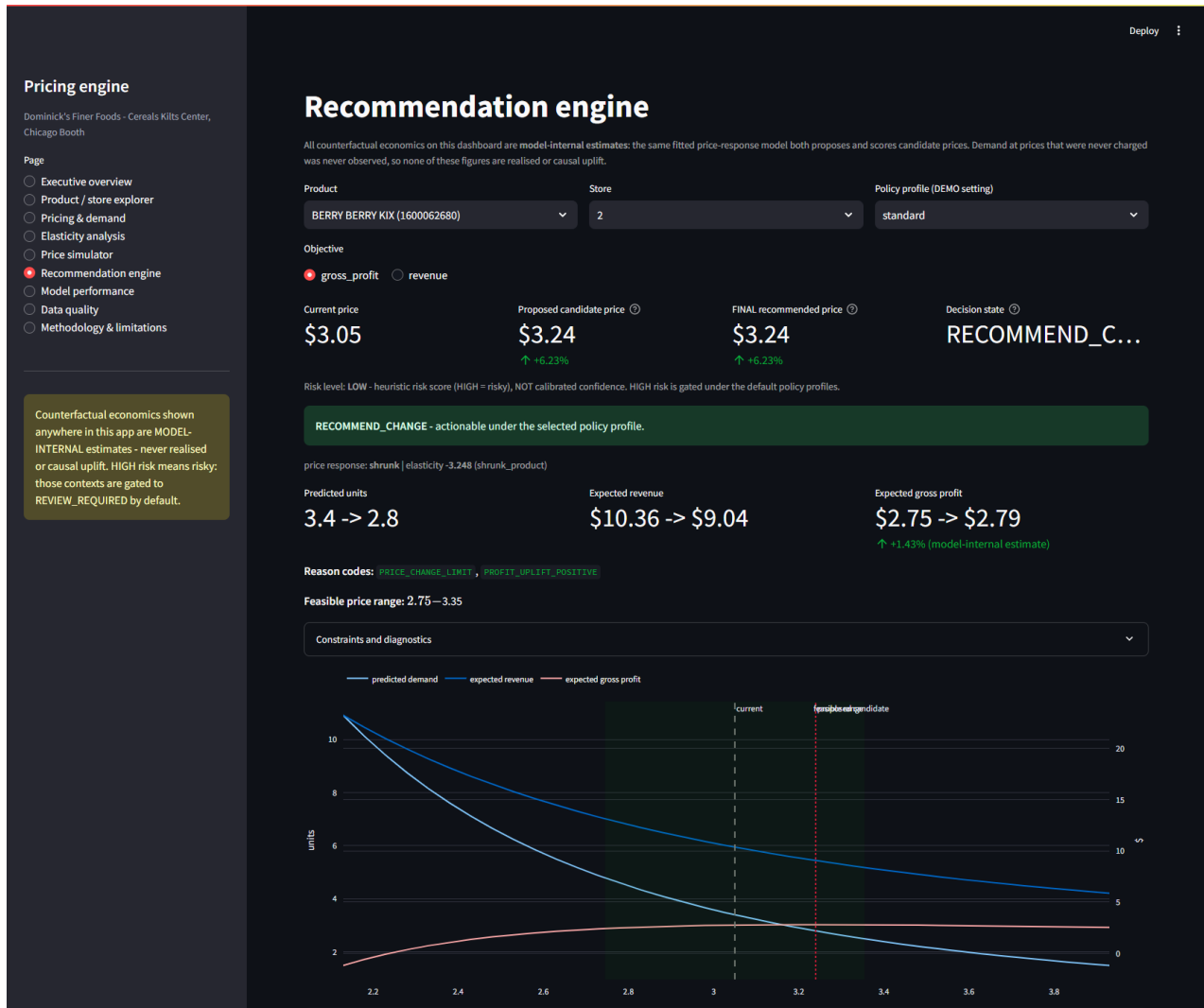


Figure 21 — The same context under the *standard* profile: \$3.05 → \$3.24 (+6.23%), *RECOMMEND_CHANGE*, *LOW* risk, elasticity -3.248 (*shrunk_product*), model-internal estimated uplift +1.43%, reason codes *PRICE_CHANGE_LIMIT* and *PROFIT_UPLIFT_POSITIVE*, feasible range \$2.75–\$3.35 shaded on the chart. Note that **proposed** and **FINAL** are shown as separate tiles.

75.4 Data quality

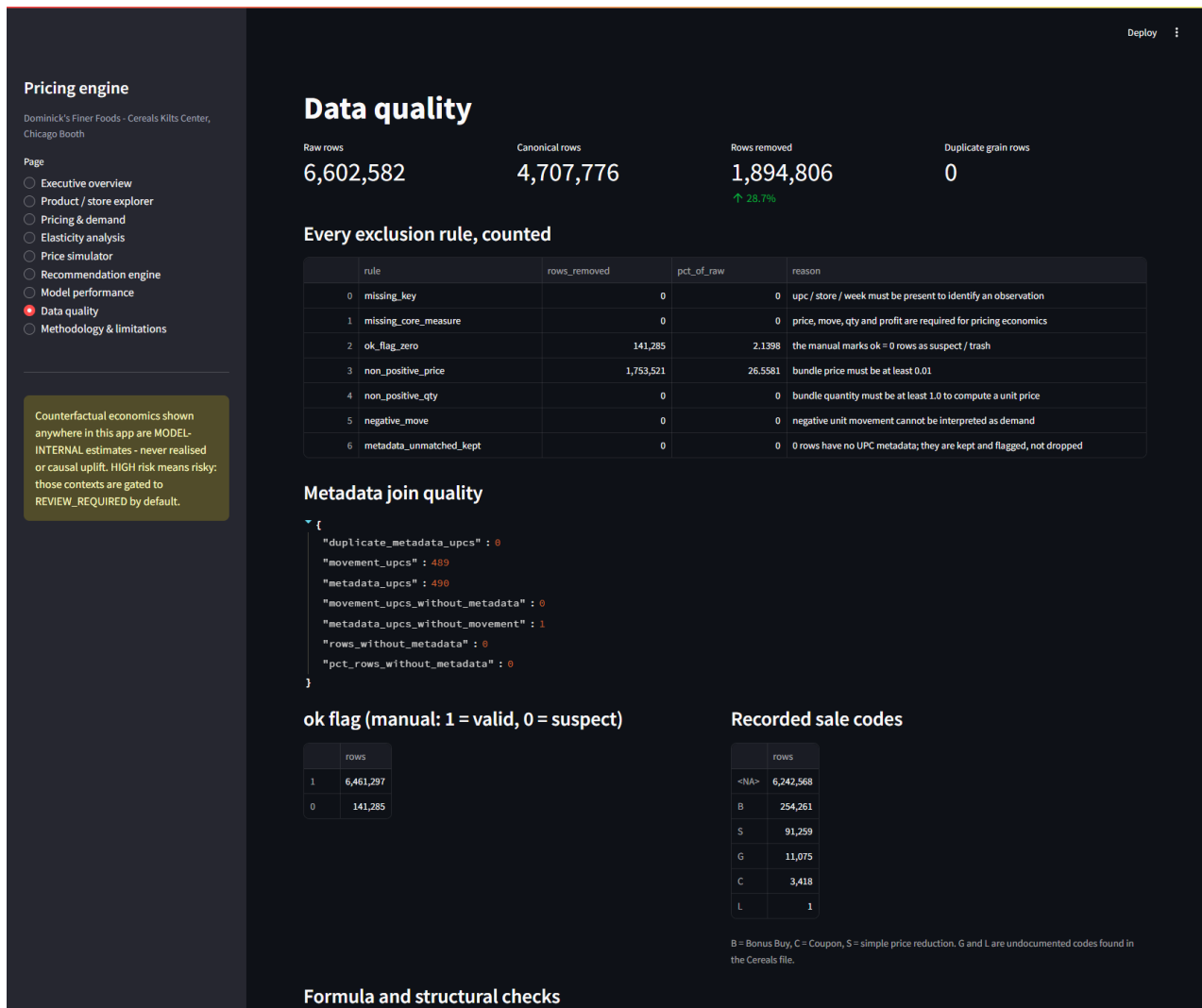


Figure 22 — The exclusion ledger, the metadata-join report, the `ok` flag counts and the recorded sale-code counts including the undocumented `G` (11,075) and `L` (1) codes, all rendered directly from `build_audit.json` and `raw_audit.json`.

75.5 The remaining five pages

Captured and stored; referenced here rather than embedded to keep the report readable:

- `dashboard_2_product_store_explorer.png` — the dual-axis price / cost / units series with promotion markers.
- `dashboard_3_pricing_and_demand.png` — price distribution, price-vs-demand scatter with its "association, not a demand curve" caption, and the price-CV histogram with the eligibility threshold marked.
- `dashboard_4_elasticity_analysis.png` — the four-specification table, the live factor-of-7.0 callout, and the per-UPC elasticity distribution.
- `dashboard_7_model_performance.png` — the six-model comparison table, the segment breakdowns, the price-response validation panel and the method-comparison tables.
- `dashboard_9_methodology_limitations.png` — the red causal banner and the six documentation tabs.

75.6 Capture method, for reproducibility

```
# terminal 1
python -m streamlit run dashboard/app.py --server.port 8511 --server.headless true
# terminal 2 — headless Chrome via Selenium, click each sidebar radio label,
# grow the viewport to the full document height, save a full-page PNG
```

bash

The first page load takes ~40 s because the app materialises the ~4.7M-row feature panel and loads the model; subsequent page switches take 25–45 s depending on how much the page computes. Those load times are themselves a finding, and they are limitation #19: *"The dashboard loads the full feature panel (~4.7 M rows) into memory with caching. Fine on a workstation, not a deployment pattern."*

PART XV — TESTING AND SOFTWARE QUALITY

76. Testing strategy

76.1 The principle

Decision-critical paths are prioritised over coverage percentage. There is no coverage-percentage target anywhere in the project. Instead, every place where a wrong answer would be *invisible* — leakage, cost handling, constraint logic, decision states, the analytical optimum — has a test that fails if it breaks.

76.2 The seven kinds of test present

kind	what it establishes	example
unit	a pure function computes what it claims	revenue, gross_margin_rate, price_change_pct identities
property / invariant	a relationship holds across every combination	HIGH risk is never actionable, across 3 profiles × every state
analytical	the implementation recovers a closed-form answer	$p^* = (a+bc)/(2b) = 6.25$
leakage / poisoning	corrupting the future changes nothing at or before the decision	test_decision_time_cost_is_lagged_not_contemporaneous, the 25-series cost poisoning
integration	the whole chain runs on synthetic fixtures	raw → processed → features → model → simulation → optimization → audit log
contract / API	the service honours its schema and rejects bad input	15 API tests
replica-equivalence	a fast audit implementation matches the real one exactly	69 attribution tests over 60 parameter combinations

76.3 The synthetic-fixture discipline

CI never downloads the licensed Dominick's data. Tests that need the real artifacts **skip themselves** via the `realdata` pytest marker; everything else runs on deterministic synthetic fixtures where the correct answer is known in closed form. That is what makes the analytical tests possible: on a fixture with $Q = 100 - 10p$, the optimum is not "whatever the model says", it is exactly \$6.25.

76.4 The two tests-of-the-tests

Two places in the suite verify that a test would actually catch a failure:

100. **The cost-poisoning test has teeth.** It asserts not only that the poisoned future left the decision-time cost unchanged at or before the poisoned week, but that it **did change** the following week. Without that second assertion the test would pass if the poison never landed.
101. **The shrinkage regression test.** `tests/test_shrinkage.py` asserts that understated standard errors *silently disable the shrinkage* — encoding the exact failure mode Phase M discovered, so it cannot recur unnoticed.

77. Important tests

77.1 Pricing formulas — `tests/test_data_pipeline.py`, `test_economics.py`

Assert the five derived identities to floating-point tolerance, plus edge cases: zero-qty safety (NaN, not `inf`), zero-price handling in `gross_margin_rate` (NaN, not a silent 0 or `inf`), and `price_change_pct` with a zero

reference. The corresponding production checks on the **real 4.7M-row table** are the 15 validation checks (§9.5), which report maximum absolute deviations of `0.000e+00` and `1.421e-14`.

77.2 Merge cardinality — `tests/test_data_pipeline.py`

The UPC metadata join must not multiply rows. Enforced in production code by `validate="many_to_one"` and an explicit row-count assertion that raises `AssertionError`, and covered by a test with a deliberately duplicated metadata UPC.

77.3 Lag leakage — `tests/test_features.py` (13 tests)

- `test_lag_features_never_use_the_current_week`
- `test_rolling_windows_are_shifted`
- `test_series_reference_price_uses_only_past_prices`
- `test_temporal_split_is_chronological_and_disjoint`
- `test_training_frame_drops_rows_without_history`
- `test_no_outcome_columns_are_features`
- `test_recompute_price_features_changes_only_price_features`

The last is a two-directional check: the five price-dependent features **must** change and the 24 context features **must not**.

77.4 Cost poisoning — `tests/test_cost_leakage.py` (5 tests)

Plus the real-data audit (§23.3): 4,671,333 rows compared with 0 mismatches, a 25-series future poisoning from week 398 that changed nothing at or before the poison and did change the next week, and 19 real recommendations of which 0 changed.

Also tested: the optimizer **refuses** rather than inventing a cost when none is available (`COST_UNAVAILABLE / ObjectiveError`).

77.5 Hybrid anchor — `tests/test_phase_1_pricing.py` (40 tests)

The equality $Q_{\text{hybrid}}(p_0) = Q_{\text{ML}}(p_0)$ is pinned by **15 parametrised tests** spanning **5 elasticities** × **3 price levels**. Also covered: the hybrid reproduces $Q(p) = Q_0(p/p_0)^\epsilon$ exactly for a fixed ϵ ; a missing `reference_price` **raises** rather than silently returning baseline demand; the elasticity table records its estimation window and marks each row's source; and risk gating behaves correctly across all three profiles.

77.6 Analytical optimizer optima — `tests/test_optimizer.py` (23 tests)

The full fixture table is in §52.3. The two headline cases:

fixture	ground truth	result
$Q = 100 - 10p$, $c = 2.50$, profit	$(a+bc)/(2b) = 6.25$	recovered within one grid step
same, revenue	$a/(2b) = 5.00$	recovered within one grid step

Plus every constraint individually, the inelastic-pushes-to-the-bound case, starting-at-the-optimum, sub-materiality, and objective errors.

77.7 High-risk policy invariants — `tests/test_decision_states.py` (246 tests)

The largest file in the suite. Walks every combination of risk level × decision state × actionability × proposal × final price for all three profiles, plus audit-log schema rotation. The production counterpart is the real-data audit: **0 invariant violations** across 13,964 contexts × 2 default profiles (§58.5).

77.8 Attribution replica equivalence — tests/test_attribution.py (69 tests)

The Phase M audits use a fast replica of `optimize_price` (~450× faster). These 69 tests assert it reproduces the real optimizer's **decision state, final price and reason codes exactly** across **60 parameter combinations**; it was additionally checked against **400 real week-399 contexts with zero mismatches**.

They also contain the numerical proof of §43.6: **scaling the base demand model by 10× leaves every recommended price unchanged**, and the closed-form unconstrained optimum $p^* = c \cdot \varepsilon / (1 + \varepsilon)$ is matched.

77.9 API validation — tests/test_api.py (15 tests)

Health, model info, **model loaded once** (not per request), the prediction contract, validation rejections (negative price, inverted range, oversized grid), decision states, and all three policy profiles.

77.10 Shrinkage mathematics — tests/test_shrinkage.py (12 tests)

The four limiting cases (imprecise $\rightarrow w \rightarrow 0$; precise $\rightarrow w \rightarrow 1$; zero heterogeneity $\rightarrow$ strong pooling; large heterogeneity $\rightarrow$ weak pooling), monotonicity of w in se , the $[0,1]$ bounds, prior-mean handling, and the regression test for the Phase M failure mode.

77.11 Repository hygiene — tests/test_repo_and_downloader.py (10 tests)

Licensed data is git-ignored; the downloader contacts **official URLs only** (host asserted); zip-slip and absolute-path archive members are rejected; archive-member selection is correct.

77.12 End-to-end integration — tests/test_integration.py (2 tests)

The complete chain on synthetic fixtures: raw $\rightarrow$ processed $\rightarrow$ features $\rightarrow$ model $\rightarrow$ simulation $\rightarrow$ optimization $\rightarrow$ audit log.

78. Final test results

78.1 Test suite — verified 2026-08-19

```
$ python -m pytest
519 passed in 7.05s
```

metric	value
tests passed	493
tests failed	0
tests skipped	0
tests xfailed / xpassed	0
runtime	7.05 s

78.2 Composition by file (verified by `pytest --collect-only`)

file	tests	covers
<code>test_decision_states.py</code>	246	risk $\times$ decision $\times$ actionable $\times$ proposal $\times$ final-price invariants over every combination and all three profiles; audit-log schema rotation
<code>test_attribution.py</code>	69	the audit replica reproduces <code>optimize_price</code> exactly across 60 parameter combinations; the baseline demand level cannot change the recommended price; closed-form

file	tests	covers
		unconstrained optimum; constraint intervals
test_phase_1_pricing.py	40	hybrid mechanics, shrinkage maths, per-UPC estimator, risk gating across all three profiles, uplift naming, elasticity provenance, $Q_{\text{hybrid}}(p_o) == Q_{\text{ML}}(p_o)$ across 5 elasticities $\times$ 3 price levels
test_optimizer.py	23	analytical optima, every constraint, keep-current paths, objective errors, risk levels
test_data_pipeline.py	17	derived formulas, zero-qty safety, promotion coding, week decoding, exclusion rules, metadata join, grain uniqueness
test_economics.py	16	revenue/margin identities, price variation, arc-elasticity edge cases, log-log recovery, fixed-effects confounding
test_simulation.py	16	price grid, only price features move, cost held fixed, single batched call, curve diagnostics
test_api.py	15	health, model info, model loaded once, prediction contract, validation rejections, decision states, policy profiles
test_features.py	13	lag alignment, shifted rolling windows, leakage poisoning, decision-time cost, price-feature recomputation, temporal split
test_shrinkage.py	12	four limiting cases, monotonicity, bounds, prior-mean handling, and that understated SEs silently disable the shrinkage
test_repo_and_downloader.py	10	licensed data git-ignored, official URLs only, zip-slip and absolute-path rejection, archive-member selection
test_monitoring.py	9	PSI, KS, schema checks, drift ranking, performance windows
test_cost_leakage.py	5	decision-cost identity, future poisoning (and that the test has teeth), recommendation invariance, missing-cost refusal
test_integration.py	2	raw $\rightarrow$ processed $\rightarrow$ features $\rightarrow$ model $\rightarrow$ simulation $\rightarrow$ optimization $\rightarrow$ audit log
total	493	

78.3 Lint — verified 2026-08-19

```
$ ruff check .
All checks passed!
```

Configuration: line-length = 100, target-version = "py311", select = ["E", "F", "I", "UP", "B", "W"] (pycodestyle errors, pyflakes, isort, pyupgrade, flake8-bugbear, warnings), ignore = ["E501"], notebooks excluded.

B (bugbear) is the notable inclusion — it catches mutable default arguments, bare `except`, `zip()` without `strict=`, and loop-variable binding bugs, which are exactly the classes of defect that survive review.

78.4 Dashboard smoke test

```
$ python scripts/smoke_dashboard.py
```

PASS — all 9 pages render without exceptions. Independently confirmed for this report by driving the live application with headless Chrome and capturing all nine pages (§75).

78.5 Docker — NOT TESTED

Dockerfile and docker-compose.yml are written and are included in the CI workflow. The image was **never built in this environment**: the Docker CLI (29.7.2) is present but the daemon is not running (failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine).

STATUS.md records this as **IMPLEMENTED (BLOCKED locally)** and reports/VALIDATION_SUMMARY.md row 36 records it as **NOT TESTED — not PASS**. This report repeats that verdict without softening it.

78.6 CI — implemented, not executed

.github/workflows/ci.yml runs on push to main/master and on pull requests, across Python 3.11 and 3.12:

102. checkout, setup-python with pip cache
103. pip install -e ".[api,dev]"
104. ruff check .
105. smoke imports (pricing_engine, the optimizer, api.main)
106. pytest -q
107. docker build -t pricing-engine:ci . (3.11 only)

Never executed, because there is no GitHub remote in this environment and the repository has no commits (§81.4). Status: **IMPLEMENTED**, not **VALIDATED**.

78.7 Claim audit — the meta-check

scripts/audit_claims.py --strict scans the repository for ten watched phrases (validated, guarantee, proven, causal, production ready, actual uplift, ...), classifies every occurrence, recognises disclaimers as disclaimers, and **fails the build** if an **UNSUPPORTED** claim reappears.

verdict	before this report (43 files)	including this report (49 files)
UNSUPPORTED	0	0
NEEDS QUALIFICATION	11	38
SAFE	151	325
total occurrences	162	363

The scanner was re-run **with this report included**, and still reports **0 UNSUPPORTED** — a useful independent check on the document you are reading. It caught exactly one violation on the first pass: the phrase "*realised margin*", used here to mean the accounting margin a week actually produced, matched the watched pattern for a claimed business outcome. It was reworded. That is the scanner behaving correctly on a genuinely ambiguous phrase.

Of the 38 "needs qualification" hits, 27 are in this report and 11 are pre-existing. All are uses of *validated*, *guarantees* or *profit uplift* either as pipeline-stage labels (**Status:** **VALIDATED**, make features # modelling table with decision-time guarantees) or carrying the required qualifier (§113). They are surfaced rather than whitelisted, which is the right default for a scanner of this kind.

78.8 Summary status table

check	status	evidence
-------	--------	----------

check	status	evidence
pytest	519 passed, 0 failed, 0 skipped	live run at Phase N, 2026-08-19
ruff	All checks passed	live run 2026-08-19
dashboard smoke	PASS — 9/9 pages	smoke_dashboard.py; independently re-verified by live capture
API live	PASS	/health, /model/info, /recommend-price called live 2026-08-19
demo pipeline	PASS	run_demo.py executed live 2026-08-19
report figures	PASS — 14 generated	make_report_figures.py, live pipeline calls
Docker build	NOT TESTED	daemon unavailable
CI workflow	IMPLEMENTED, not executed	no remote; no commits
claim audit	0 UNSUPPORTED	audit_claims.py --strict

PART XVI — REPRODUCIBILITY AND MLOPS

79. Reproducible workflow

79.1 From an empty environment to a running application

```
# 0. Python 3.11+ (this build: 3.13.0)
python -m pip install -e ".[api,dashboard,dev]"

# 1. Data — official Kilts Center URLs only; never mirrored, never synthetic
make data          # download_dominicks.py + build_dataset.py + reports/01
make validate      # 15 formula / structural checks on the persisted Parquet

# 2. Features — the decision-time availability contract
make features      # 4,671,333 usable rows, 29 features, ~73 s

# 3. Analysis
make eda           # reports/02 + 6 figures
make elasticity     # reports/03 — naive → fixed effects ladder

# 4. Models
make train         # 6 models compared, selection on validation WAPE, ~243 s
make evaluate      # re-score the SAVED artifact on the test window
make price-response # reports/05 — does the model respond to price?

# 5. Pricing layer
make estimate-elasticity # training weeks 2-257 ONLY, ~35 s
make compare-response   # reports/08 — ml vs pooled vs shrunk + sensitivity
make optimize           # 3,000 recommendations + audit log, ~105 s
make backtest           # reports/06 — weeks 390-399
make monitor            # drift / schema / performance

# 6. The scientific audit
make audit-zero-price   # reports/09
make audit-cost         # reports/10
make audit              # reports/11-20 (ten Phase M audits)

# 7. Demo, quality gates and applications
make demo               # one real UPC × store, end to end
make test lint          # 519 tests at Phase N, ruff
make api                # http://127.0.0.1:8000/docs
make dashboard          # Streamlit, 9 pages

# or everything at once
make all
```

bash

79.2 What makes a re-run reproducible

mechanism	detail
single seed	<code>project.random_seed: 42</code> , threaded through <code>Config.seed</code> into the ridge, the HGB, the elasticity subsampling and every <code>.sample()</code> call
chronological split by rule	the split is derived from the week index and fixed fractions — no stored index list to drift
dataset fingerprints	Parquet SHA-256 and a storage-independent DataFrame fingerprint; the feature table has its own fingerprint
environment capture	Python, platform, numpy, pandas, scikit-learn versions and a UTC timestamp written into <code>dataset_fingerprint.json</code> and the model metadata
provenance for the raw data	<code>SOURCE.json</code> with URLs, byte sizes, SHA-256 hashes, timestamps
atomic writes	downloads and artifacts are written to a temp file and renamed, so a crash cannot leave a half-written artifact that

mechanism	detail
	looks complete
config-only thresholds	every threshold is in <code>configs/config.yaml</code> ; changing a guardrail never requires a code edit
JSON + Markdown pairing	each script emits a machine-readable JSON and a human-readable report generated from the same dictionary

79.3 What is not reproducible from this repository alone

The data. Raw and processed data are git-ignored and never redistributed (academic-use licence). A reader must run `scripts/download_dominicks.py` themselves. The downloader records exactly what it fetched so the inputs can be verified byte for byte.

79.4 Runtime budget for a full run

stage	time
dataset build	(dominated by reading the 458 MB CSV)
feature build	73.4 s
model training (6 models)	243 s
elasticity estimation	35.0 s
price-response comparison (300 contexts)	38.5 s
batch optimize (3,000 contexts)	105.3 s
elasticity inference audit	22.3 s
elasticity stability audit	51.3 s
guardrail ablation	28.8 s
constraint attribution	15.8 s
cost-leakage audit	12.8 s
eligibility funnel	10.1 s
model-value ablation	9.6 s
decision-state audit	6.4 s
out-of-time price response	5.7 s
shrinkage audit	< 1 s
claim audit	0.2 s
test suite	7.05 s

The Phase M audit suite — twelve scripts producing ten reports over 13,964 contexts × 11 pipeline runs — completes in under four minutes, which is what the 450× attribution replica bought.

80. Configuration

80.1 One file, one source of truth

```
"""Everything tunable (paths, thresholds, policy profiles, model
hyper-parameters) lives in ``configs/config.yaml``. Code should never hardcode
a magic number; it should read it from here."""
```

python

`Config` exposes `get(dotted, default)`, `require(dotted)` (raises when absent), `path(name)`, `policy(profile)` and `seed`. The whole tree can be overridden with the `PRICING_ENGINE_CONFIG` environment variable.

80.2 The configuration blocks

block	contents
project	name, random_seed: 42, category
paths	raw / interim / processed dirs, the two Parquet tables, artifacts, models, metrics, figures, reports, the recommendation log
data	week1_start_date: 1989-09-14, keep_only_ok_rows: true, min_price: 0.01, min_qty: 1
modeling	target, split fractions 0.70/0.15/0.15, min_weeks_per_series: 20, lags [1,2,3,4], rolling windows [4,8,13], HGB hyperparameters
eligibility	min_observations: 40, min_distinct_prices: 5, min_price_cv: 0.05
optimization	objective: gross_profit, price_step: 0.05, price_rounding: 0.01, charm_pricing: false
policy_profiles	conservative / standard / aggressive (below)
risk	LOW and MEDIUM band thresholds
pricing_response	method: shrunk, elasticity-table path, min/max_abs_elasticity: 0.2 / 6.0, fallback: pooled
elasticity	min_obs_per_upc: 200, min_distinct_prices_per_upc: 10, shrinkage: empirical_bayes, shrinkage_tau2_estimator: reml, shrinkage_prior_mean: pooled, max_se_for_product_estimate: 1.5, sensitivity_scenarios: [-1.5, -1.9, -2.4, -3.1]

80.3 The three policy profiles

These are DEMO settings, labelled as such in the file itself:

```
# DEMO POLICY SETTINGS - illustrative guardrails for a portfolio demo, NOT
# universal retail best-practice thresholds.
```

parameter	conservative	standard ☒	aggressive (DEMO ONLY)
max_price_change_pct	5%	10%	20%
extrapolation_tolerance_pct	2%	5%	10%
min_gross_margin_rate	10%	5%	0%
materiality_threshold_pct	2%	1%	0.5%
allow_below_cost	false	false	false
high_risk_action	keep_current	review_required	recommend ⚠
medium_risk_max_price_change_pct	3%	5%	20%

Behavioural consequences over all 13,964 week-399 contexts:

profile	RECOMMEND_CHANGE	KEEP_CURRENT	REVIEW_REQUIRED	median change	HIGH-risk auto-changed
conservative	45.5%	54.5%	0%	0.0%	0
standard	59.2%	31.1%	9.7%	+3.7%	0
aggressive	70.8%	29.2%	0%	—	4,717 ⚠

Having three profiles is a design decision in itself (DECISIONS #25): it makes the **sensitivity of the output to the policy** explicit and measurable, rather than leaving one arbitrary threshold set looking like a law of nature.

80.4 The risk configuration

```
risk:
  low_risk: {min_observations: 150, min_distinct_prices: 15, max_extrapolation_pct: 0.02}
  medium_risk: {min_observations: 80, min_distinct_prices: 8, max_extrapolation_pct: 0.05}
```

yaml

with the comment explaining both the direction of the scale and why the bands are stricter than the eligibility screen (§57.2).

81. Model versioning

81.1 Model metadata

artifacts/models/demand_model_metadata.json — 20 fields:

field	value in the shipped artifact
name / kind	M1 ridge log-log / ridge_loglog
version	ridge_loglog-20260818-132847
trained_at_utc	2026-08-18T09:28:47+00:00
seed	42
feature_columns	the 26 columns, in order
categorical_features	store, com_code
target	move
params	the complete sklearn Pipeline parameter dump (37 entries)
train_weeks / valid_weeks / test_weeks	[2,257] / [258,342] / [343,399]
train_dates / valid_dates / test_dates	the calendar equivalents
n_train_rows	3,206,437
metrics	validation and test blocks
data_fingerprint	e9d26f2c0eda9f7e9d8dbb508053e56a932608668e16d8b5aeec0032373722d0
environment	timestamp, Python 3.13.0, platform, numpy 2.0.2, pandas 2.2.3, scikit-learn 1.8.0, git_commit

81.2 The fingerprint chain

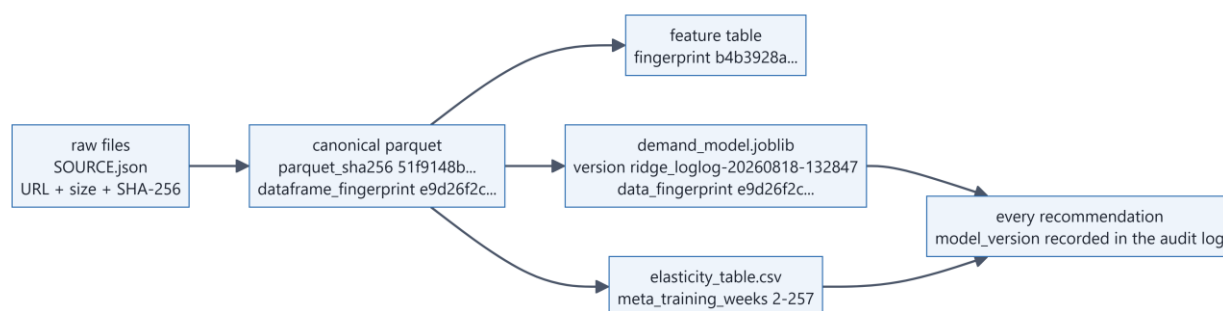


Diagram 7

A recommendation in `artifacts/recommendation_log.csv` carries `model_version`, `price_response_method`, `elasticity_used` and `elasticity_source`. From those four fields you can reach the model artifact, its `data_fingerprint`, the canonical Parquet hash and the raw file hashes. **Full lineage, from a shelf price back to a downloaded byte.**

81.3 Persistence is verified, not assumed

`scripts/evaluate.py` reloads the persisted artifact and re-scores the test window:

```
"reload_wape_difference": 0.0
```

`json`

Exactly zero. So the reported test WAPE is a property of the saved model, not of the training session's in-memory state.

81.4 The version-control gap — stated plainly

The repository has no commits. `git log` returns *"your current branch 'master' does not have any commits yet"*, and every file is untracked. Consequently:

- `git_commit` in the model metadata reads the literal string `"HEAD"` rather than a SHA;
- the CI workflow has never run, because there is nothing to push;
- the model artifact cannot be tied to a code revision.

This is a genuine reproducibility gap and it is the one thing in this section that a reviewer should fix first: `git add -A && git commit` would immediately populate `git_commit` on the next training run and make the whole lineage chain complete. It is recorded as limitation **#L15** in Appendix K.

81.5 What a real model registry would add

capability	present here	a real registry
immutable versioned artifacts	filename + metadata	content-addressed store
lineage to code revision	missing (no commits)	commit SHA + build ID
lineage to data	yes (fingerprints)	yes
stage transitions (staging → production)	none	explicit promotion
rollback	manual file copy	one command
approval gates	none	required reviewers
A/B / shadow serving	none	traffic splitting

82. Monitoring

Only what is actually implemented is described here.

82.1 What exists

`src/pricing_engine/monitoring/drift.py` + `scripts/monitor.py` → `artifacts/metrics/monitoring.json`, `reports/MONITORING_DESIGN.md`. Four checks, run **offline**, comparing the reference window (training weeks 2–257) against the current window (test weeks 343–399, 749,040 rows).

1. Schema check — column presence, dtypes, null shares. Result: **passed**, 0 missing columns, 0 unexpected dtypes.

2. Feature drift — Population Stability Index and Kolmogorov-Smirnov statistic per feature, banded *stable* / *moderate shift* / *large shift*, and ranked. Results are in §18.2; the headline is that `time_index` (PSI 12.41) and `series_age_weeks` (1.00) are trivially large by construction on a chronological split, while the genuinely informative drift is in the price block: `series_reference_price` PSI 0.363 with the mean rising **\$2.886 → \$3.247**.

3. Prediction drift — PSI **0.0556** (*stable*), KS 0.0950, mean prediction 16.69 → 15.00, p95 40.50 → 35.26. **The model's output distribution is stable even though its inputs have drifted**, which is a useful and non-obvious result.

4. Performance by period — WAPE, MAE, RMSE, bias and sMAPE by calendar quarter. Results in §41.8: **0.409 → 0.528 over five quarters**, monotone degradation with distance from training.

82.2 What the monitoring establishes

Drift alone is not the alarm; drift *plus* degrading performance is. This artifact shows both, and they point the same way: the price regime moved, and accuracy decayed. A production system would use the quarterly WAPE trend as the retraining trigger and the feature PSI as the diagnostic explaining it.

82.3 What is not implemented

capability	status
a metric store (Prometheus, a time-series DB)	absent
alerting (thresholds, pager, email)	absent
scheduled execution (cron, Airflow, Dagster)	absent
serving-latency / error-rate monitoring	absent
data-quality monitoring on incoming data	absent
a monitoring dashboard	absent (the Streamlit app shows the artifact, it does not monitor)
automatic retraining triggers	absent
recommendation-outcome monitoring	absent — and impossible without a live deployment

The last is the important one. Even a fully instrumented version of this system could not monitor whether its recommendations *worked* without applying them and observing the result — which is §98's territory.

STATUS.md records monitoring as IMPLEMENTED, not VALIDATED, with the remaining issue "Offline only; no metric store, no alerting." That is accurate.

83. Docker and CI

83.1 Dockerfile

python:3.11-slim, with these deliberate properties:

property	detail
code only	<code>COPY src, api, configs, scripts</code> — no data, no artifacts
licensed data excluded	the Dominick's data are academic-use only and are never baked into an image
runtime mounts	<code>./artifacts</code> and <code>./data/processed</code> mounted read-only
non-root user	<code>useradd --create-home appuser, USER appuser</code>
layer ordering	<code>pyproject.toml</code> + <code>src</code> copied and installed <i>before</i> <code>api/configs/scripts</code> , so dependency layers cache across code changes
healthcheck	<code>HEALTHCHECK</code> polling <code>/health</code> every 30 s
env hygiene	<code>PYTHONDONTWRITEBYTECODE=1</code> , <code>PYTHONUNBUFFERED=1</code> , <code>PIP_NO_CACHE_DIR=1</code>
entrypoint	<code>uvicorn api.main:app --host 0.0.0.0 --port 8000</code>

`docker-compose.yml` adds port mapping, the two read-only mounts, a `PRICING_ENGINE_POLICY_PROFILE=standard` environment variable and `restart: unless-stopped`.

83.2 Docker status: NOT TESTED

The image was **never built**. The Docker CLI (29.7.2) is present but the daemon is not running in this environment:

```
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine
```

reports/VALIDATION_SUMMARY.md row 36 records this as **NOT TESTED — not PASS**, and STATUS.md as IMPLEMENTED (BLOCKED locally). This report does not soften either verdict. A Dockerfile that has never been built is a plausible Dockerfile, not a working one.

83.3 CI workflow

.github/workflows/ci.yml, on push to main/master and on pull requests, matrix over Python 3.11 and 3.12:

108. checkout + setup-python with pip cache
109. pip install -e ".[api,dev]"
110. lint — ruff check .
111. smoke imports — pricing_engine, optimize_price, api.main
112. tests — pytest -q
113. Docker build — docker build -t pricing-engine:ci . (3.11 only)

With the explicit comment:

```
# CI never downloads the licensed Dominick's dataset. Tests that need  
# the real artifacts skip themselves; everything else runs on the  
# deterministic synthetic fixtures.
```

yaml

83.4 CI status: implemented, never executed

There is no GitHub remote in this environment and **the repository has no commits**, so the workflow has never run. Status: IMPLEMENTED, not VALIDATED.

Note also that the matrix targets 3.11/3.12 while this build ran on **3.13.0**. The package declares requires-python = ">=3.11" and works on 3.13 locally, but 3.13 is not in the CI matrix — a small inconsistency worth fixing.

83.5 What would be needed for a real deployment

requirement	present
image builds	not verified
image scanned for vulnerabilities	no
multi-stage build to shrink the image	no (single stage, build-essential retained)
pinned dependency versions (lockfile)	no — pyproject.toml uses lower bounds
artifact provenance / SBOM	no
readiness vs liveness probes distinguished	no (one healthcheck)
horizontal scaling / resource limits	no
secrets management	not applicable (no secrets)

PART XVII — SCIENTIFIC AUDIT

84. Final scientific audit

Phase M produced ten audit reports (reports/11–20) plus two Phase L audits (09, 10). This section consolidates what they found.

84.1 The twelve audits and their verdicts

#	audit	central question	finding
09	Zero-price exclusion	is dropping 26.6% of raw rows defensible?	Yes, with a documented tilt. 1,851,380 zero-price rows, only 677 with sales; 73.4% are leading/trailing runs. Rule kept; the sample now skews to continuously stocked series (21.0 vs 3.1 mean weekly units).
10	Cost leakage	does any future cost information reach a decision?	No. 4,671,333 rows, 0 mismatches ; poisoning changed nothing at or before the poisoned week and did change the next; 0 of 19 recommendations changed. Proves <i>temporal availability</i> , not economic correctness.
11	Constraint attribution	model or rules?	Rules. 2.9% of final recommendations come from an interior model optimum; 57.1% are guardrail corners; the $\pm 10\%$ cap binds first in 44.6%.
12	Model-value ablation	do rules reproduce the decisions?	Largely yes. A no-model rule matches within one 5¢ step 82.5% of the time and the decision state 90.0%. No profit comparison made — it would be circular.
13	Guardrail ablation	where does the pricing signal stop mattering?	At the change cap. Median price variation across four elasticity scenarios: 79.8% research $\rightarrow$ 6.6% support band $\rightarrow$ 0.0% with the cap.
14	Shrinkage audit	is the empirical-Bayes estimator correct?	The formula was right; the inputs were wrong. Mean weight 0.954 $\rightarrow$ 0.780 once panel-robust SEs replaced HC1. τ^2 0.906 $\rightarrow$ 0.766.
15	Inference audit	are the standard errors right?	They were too small by 16×. Two-way clustered SE 0.1058 vs classical 0.0065; per-UPC median inflation 3.4×; share of products significant at 5% 90.1% $\rightarrow$ 75.2% . Estimators validated against statsmodels.
16	Out-of-time price response	which method predicts real price changes best?	shrunk, in every split. 154,899 unseen episodes; non-promotion WAPE 0.560 vs 0.588 native ML, 0.594

#	audit	central question	finding
			pooled, 0.715 null. Naturalistic, not causal.
17	Elasticity stability	is elasticity durable?	Category yes, product no. Max pairwise z between disjoint windows 1.36 (not significant); median product rank correlation +0.19 .
18	Eligibility funnel	why do so few products get their own estimate?	489 → 239 , every transition counted: 104 not in training weeks, 42 too few observations, 58 too little price variation, 24 wrong-signed, 9 too imprecise.
19	Decision-state audit	can the policy layer contradict itself?	No. 0 invariant violations across 13,964 contexts × 2 default profiles. The 4,717 under <i>aggressive</i> are by design and are reported, not suppressed.
20	Claim audit	does the repository over-claim?	0 UNSUPPORTED — across 43 files / 162 occurrences before this report, and still 0 across 49 files / 363 occurrences with it included; --strict makes it enforceable.

84.2 The three findings that changed the project's own description

- 114. **The engine is rule-bounded, not model-driven** (11, 12, 13). The README, STATUS.md and every downstream description were rewritten to say "*rule-bounded pricing with a learned direction*".
- 115. **The published precision was overstated by an order of magnitude** (15), which in turn meant the shrinkage was not working (14). Both numbers were corrected everywhere and the old ones removed.
- 116. **Product-level elasticity is not a durable asset** (17), so it must not be described as one.

84.3 The audits that confirmed rather than corrected

Audits 09, 10, 16, 18, 19 and 20 all **confirmed** the existing design. That matters: an audit suite that only ever finds problems is probably not testing the things that work, and one that never finds problems is not testing anything.

84.4 The pattern worth naming

Three of the four *corrections* originated in **inference**, not in modelling: the standard errors were wrong, which broke the shrinkage, which flattered the per-product estimates. None of that would have surfaced from cross-validation, a better model or more data. It surfaced from re-deriving an estimator and asking what its inputs assumed.

85. What the model actually contributes

If guardrails dominate most recommendations, why do we need the pricing model at all?

This is the hardest question the project faces, and the evidence-based answer has five parts.

85.1 Part 1 — The honest concession

On magnitude, for the median context, under production guardrails: it does not contribute much.

- 2.9% of final recommendations are set by an interior model optimum.
- A no-model rule matches the engine within one 5-cent step 82.5% of the time.
- Median price variation across four different elasticities is 0.0%.

Any claim that "the ML model sets the price" is **false** for this system as configured, and the project says so in its own README.

85.2 Part 2 — Direction

R4 ("always take the maximum allowed increase") can only ever propose an increase. The engine proposes a **cut** whenever the estimated elasticity puts the optimum below the current price. Under the **shrunk** response, 8.0% of actionable recommendations are decreases (91.96% are increases, \$44.1); under the **m1** response, 26.7%.

A rule that only ever raises prices is not a pricing policy — it is a price increase with extra steps. The model is what makes the direction a *decision*.

85.3 Part 3 — Eligibility and refusal

The model layer is what allows the engine to say "I don't know":

refusal	contexts (week 399, standard)
screened out before optimisation (insufficient history / price variation / no cost)	3,528 (25.3%)
risk-gated to REVIEW_REQUIRED	1,353 (9.7%)
kept because the estimated gain is immaterial	716 (5.1%)
total non-actioned	5,597 (40.1%)

R4 has no concept of any of these. It would move every price it was allowed to move. **40% of the engine's output is a refusal**, and the evidence for each refusal is a model artefact: observation counts, distinct prices, elasticity provenance, extrapolation distance, estimated uplift.

85.4 Part 4 — Magnitude in the tail

The median is 0.0% but the **p90 price variation across elasticity scenarios is 14.5%** under the standard profile. One context in ten still has its price materially determined by the estimated response — and those are disproportionately the well-identified, high-volume series where the estimate is most trustworthy and the money is.

85.5 Part 5 — The counterfactual economics

Even when the price comes from a guardrail corner, **the model supplies the dollar figures**: predicted units at both prices, expected revenue, expected gross profit, the estimated uplift, the whole demand curve. A reviewer looking at \$20.27 → \$21.74 and a demand curve is making a different decision from one looking at "go to the cap because the rule says so".

This is a real contribution and it is also the one most vulnerable to §64: those dollar figures are model-internal estimates.

85.6 The synthesis

The model contributes direction, eligibility, refusal and the economics of the decision. The rules contribute magnitude.

That is a defensible architecture given an observational elasticity, a 46%-WAPE forecaster and no way to measure realised outcomes. It is not the architecture a press release would describe, and the project describes it accurately instead.

85.7 What would change the answer

change	effect on the model's contribution
run the experiment (§98) and obtain a causally identified elasticity	the guardrails could be widened with evidence, and the interior-optimum share would rise
widen the change cap from 10% to 25%	the model would set magnitude far more often — and would need to be right
add cross-price elasticities	the decision becomes multi-dimensional and no simple rule can approximate it (§101)
calibrate the risk bands against realised error	refusal becomes quantitative rather than heuristic
model promotion depth, not just its presence	the largest error segment (WAPE 0.656) would shrink, improving both forecast and response

86. The weakest scientific component

Counterfactual policy evaluation. The backtest's policy economics are circular by construction and cannot be repaired offline.

86.1 Why this is the weakest, ranked against the alternatives

candidate	why it is <i>not</i> the weakest
causal identification	genuinely absent, but it is honestly and comprehensively disclosed , quantified (the naive-to-FE swing), and the fix is designed in docs/PRICING_EXPERIMENT.md. A known, bounded, disclosed limitation.
incomplete promotion information	real and material (WAPE 0.656 on promotion weeks), but measured — the promotion/non-promotion split is reported everywhere it matters.
no substitution / cross-price effects	arguably the largest <i>missing mechanism</i> , but it is a scope decision, clearly stated, and it does not corrupt any number that is reported.
the heuristic risk layer	uncalibrated, but labelled as heuristic in the config, the code, the API, the dashboard and every report.
counterfactual policy evaluation	the only one that produces a number people will quote.

86.2 The specific defect

The backtest reports **+10.45% model-internal estimated gross profit** for the ML policy versus historical pricing, and reports/07 reports **+8.30%** for the portfolio. Both are produced by the same fitted price-response model that proposed the prices being scored (§64).

This is the weakest component because:

117. **It is the number a reader will remember.** "+10%" is the headline; the circularity is a caveat. Caveats lose to headlines.
118. **The bias cannot be signed.** It depends on the direction of the model's elasticity error, which is unknown (§64.4).

119. **It is unfixable offline.** Cross-validation, held-out sets and off-policy estimators all fail for the reasons in §64.3.
120. **Everything downstream inherits it.** Any ROI calculation, business case or CV bullet built on those numbers inherits the same circularity.

86.3 How the project mitigates it — and why mitigation is not a fix

mitigation	in place
the metric is <i>named</i> <code>model_internal_estimated_*</code>	☒
<code>realisable_profit_uplift_pct</code> is forced to 0 for non-actionable rows	☒
disclaimers in the API schema, every dashboard page and every report	☒
the model-value ablation refuses to compare profits	☒
<code>audit_claims.py --strict</code> blocks "actual uplift" wording	☒
§64 exists as a standalone chapter	☒
the out-of-time validation partly escapes circularity (§45)	☒
an unbiased estimate of policy value	✗ impossible offline

Every one of these makes the number harder to *misread*. None makes it correct.

86.4 What would fix it

Only §98 — a randomised pricing experiment producing a difference-in-differences estimate of gross profit per store-week with clustered standard errors. Nothing short of intervention.

87. The strongest scientific component

The Phase M audit suite itself — and within it, the inference-and-shrinkage correction chain.

87.1 Why the audit suite, and not the model

The demand model is competent and unremarkable: a ridge log-log with good features, beating a boosted model by 2.2% relative WAPE. Any capable practitioner would build something similar.

What is **rare** is a project that:

121. built a system,
122. built a second, independent apparatus to measure how much of the answer that system actually supplies,
123. found the answer was **2.9%**,
124. and published that as the headline.

87.2 The single strongest chain: inference → shrinkage → policy

Follow it end to end.

Step 1 — a question about assumptions. Are HC1 standard errors right on a panel with within-store persistence and category-wide weekly shocks? Obviously not, once asked.

Step 2 — measure it properly. Seven covariance assumptions, estimators **validated against statsmodels to machine precision**, reported side by side. Result: two-way clustered SE is **16.24×** the classical one; the per-UPC median inflation is **3.38×**.

Step 3 — a statistical consequence. The share of products significant at 5% falls from **90.1% to 75.2%**. One in six "significant" products was not.

Step 4 — a modelling consequence nobody would have predicted. Understated SEs inflate $w = \tau^2 / (\tau^2 + se^2)$ from *both* directions, so the empirical-Bayes shrinkage was **silently doing nothing**. Mean weight 0.954, median 0.989 — piled against the ceiling.

Step 5 — isolate the cause. Seven estimator variants on identical data show that changing the τ^2 estimator or the prior mean moves the mean weight by < 0.001 , while changing the standard errors alone moves it from 0.955 to 0.780. **The formula was correct; the inputs were wrong.**

Step 6 — correct it and propagate. τ^2 0.906 $\rightarrow$ 0.766, mean weight 0.954 $\rightarrow$ **0.780**, and the old figure was **removed from every document** rather than left alongside the new one.

Step 7 — encode the failure mode as a test. `tests/test_shrinkage.py` now asserts that understated standard errors silently disable the shrinkage.

Step 8 — check the scientific conclusion survives. The elasticity-stability audit re-runs the whole stack on five windows *with the corrected inference* and finds the category elasticity stable (max $z = 1.36$). Under the old standard errors that test would have declared violent, significant drift. **The covariance assumption changed the scientific conclusion, not just the interval.**

87.3 Runner-up: the constraint-attribution methodology

The leave-one-out design over 13,964 contexts $\times$ 11 pipeline runs, made feasible by a 450 $\times$ -faster replica that is trusted **only because 69 tests pin it against the real optimizer across 60 parameter combinations and 400 real contexts with zero mismatches.**

The methodological point is that the audit instrument was itself audited before its results were believed — and in the process it **found a genuine bug in the production code** (the over-reported `PRICE_CHANGE_LIMIT` binding, DECISIONS #48).

87.4 Runner-up: the out-of-time price-response validation

154,899 realised price-change episodes on weeks the estimator never saw, with a **null model included**, results split by promotion state, the winner made the default on measurement rather than inheritance, and the causal limitation stated in a box at the top of its own report.

87.5 What these have in common

All three follow the same pattern: **state a claim precisely, build an independent instrument to test it, report what the instrument says even when it is unflattering, and encode the finding as a test so it cannot silently regress.**

That is the transferable skill, and it is worth more than any individual number in this report.

PART XVIII — LIMITATIONS

Every item below is drawn from `KNOWN_LIMITATIONS.md`, `STATUS.md` or an audit artifact. Nothing here is speculative. The consolidated register with IDs is in **Appendix K**.

88. Data limitations

88.1 Historical data (1989–1997)

The panel is US grocery retail from three decades ago. Price levels, category dynamics, competitive structure, promotion mechanics, shopper behaviour and the retail landscape have all changed. Dominick's itself was acquired by Safeway in 1998 and the banner was retired in 2013.

88.2 The zero-price exclusion removes 26.56% of raw rows

1,753,521 rows (plus 97,859 already removed for `ok = 0`). Defensible (\$10) but consequential: the surviving sample tilts toward continuously stocked, higher-volume series — **21.0 mean weekly units vs 3.1** for heavily excluded series — and **no canonical row has zero sales**, so intermittent demand is out of scope by construction.

88.3 Promotion coding is incomplete

94.55% of raw rows carry no `sale` code. Two undocumented codes (`G`, 11,075 rows; `L`, 1 row) appear. Display and feature advertising are absent entirely. Any estimated "price effect" partly absorbs the whole marketing bundle (\$11.4, \$33.3).

88.4 AAC is not economic cost

The implied unit cost comes from an accounting margin on Average Acquisition Cost, which lags replacement cost. Within-series AAC is itself volatile (median CV **0.088**, p90 0.153) and **falls 9.16% on price cuts**. The cost-leakage audit proves temporal availability, **not** economic correctness (\$8.3, \$23.4).

88.5 No customer dimension

The grain is store × week. There is no basket composition, no traffic normalisation, no loyalty data, no customer identity. Personalised pricing is not merely unimplemented — it is **unimplementable** on this data, which is a stronger guarantee (\$94.3).

88.6 Stock-outs are invisible

Unavailability and demand collapse look identical in scanner data. A suppressed sales week caused by an empty shelf is read as weak demand.

88.7 No competitor prices

Rival stores' prices are entirely unobserved, so competitive response cannot be modelled or controlled for.

88.8 Extreme values retained by design

10.07% of week-over-week price-change events exceed $\pm 50\%$ (77,152 events); the maximum unit price is \$26.02 (a T-shirt in the Cereals commodity file); 6 UPCs have negative total observed gross profit; the maximum weekly demand is 18,688 units. All kept and flagged rather than trimmed (DECISIONS #7), and bounded downstream by the extrapolation guardrail.

88.9 Elasticity regressions use a 1.2M-row subsample

Seeded (`seed=42`) for tractability. The one full-sample check gives **-2.0443** against **-2.0289** on the subsample — a difference far inside the standard error — so subsampling is not driving the result. Full-panel estimates for the other specifications were not run.

89. Modelling limitations

89.1 Point predictions only

No predictive intervals, so the optimizer maximises an **expectation** without weighing dispersion. A candidate price with a high expected profit and enormous variance is indistinguishable from a safe one. A quantile or distributional model would let the objective become risk-adjusted.

89.2 Weekly WAPE ≈ 0.46 on the test window

0.364–0.560 across backtest weeks. **Any counterfactual difference smaller than that error deserves scepticism** — which is the entire rationale for the materiality threshold.

89.3 Systematic under-forecast (test bias -3.883 units)

Concentrated on promotional weeks (bias -37.24 there, against -1.13 elsewhere), and driven partly by Jensen's inequality on the log target. **No smearing correction was applied** — a fair criticism (\$41.4).

89.4 A prediction cap is applied

5× the maximum training demand, to stop the log-target model exploding on extreme inputs. Recorded in the artifact rather than hidden.

89.5 The promotion flag is assumed known at decision time

Realistic, but an assumption (\$20.4). Impact bounded to 7.35% of rows, with error reported separately by promotion state.

89.6 Only 54.08% of series are eligible

19,707 of 36,443. The rest lack history or price variation and always receive keep-current.

89.7 The boosted model's price response is unsafe

M2's implied elasticity is **positive in 7% of contexts** and ranges from -10.1 (p10) to -0.39 (p90). Monotonic constraints (`monotonic_cst`) exist in scikit-learn and were **not used** — an obvious extension that was not taken.

89.8 No price × context interaction in the shipped response

The hybrid applies one elasticity per UPC. It cannot differ between a promotion week and a normal week for the same product — even though the promotion-split regressions show those elasticities differ by **0.83** (-2.662 vs -1.830).

89.9 The constant-elasticity form is imposed, not learned

$Q(p) = \hat{Q}(p_0)(p/p_0)^\epsilon$ cannot represent kinks, reference-price effects, threshold behaviour or asymmetric responses to increases and decreases — all of which are documented phenomena in pricing research.

90. Econometric limitations

90.1 No causal identification

The central limitation. $P(Q \mid \text{price}, X) \neq P(Q \mid \text{do}(\text{price}))$. Prices were set by the retailer, not randomised. The naive-to-fixed-effects swing ($-0.348 \rightarrow -2.419$) demonstrates how contaminated the raw association is, and nothing guarantees the controlled estimate is causal either — only that it is the least contaminated of the four (§33).

90.2 Shrinkage fixes variance, not bias

If the pooled elasticity is biased by promotion contamination, **every shrunk product estimate inherits a share $(1 - w_i)$ of that bias** — and the products with the least own information inherit the most.

90.3 Product-level elasticity does not reproduce across time

Median rank correlation between disjoint training windows: **+0.19**. Sign stability is 100% and the *category* estimate is stable (max pairwise z 1.36), but a durable product-specific elasticity is not supported. And the product-level conclusion itself rests on only **34 common products** across the three disjoint windows.

90.4 133 of 372 products have no usable own elasticity

100 too thin, 24 wrong-signed, 9 too imprecise. They are priced with the pooled fallback — **35.8% of week-399 decision contexts**.

90.5 Two-way clustering is used only at the pooled level

Inside a single product the Cameron-Gelbach-Miller subtraction is unstable, so the per-UPC standard error is the **max** of the store-clustered, week-clustered and HC1 estimates — conservative, but not the two-way estimator.

90.6 Wrong-signed products are rejected, which is itself a selection

24 products with positive coefficients are removed before shrinkage. That is the right conservative choice (§30.6), but it means the surviving distribution is **truncated at zero**, which biases the estimated τ^2 and the shrunk distribution toward the elastic side by an amount nobody has quantified here.

90.7 The $\sin 52 / \cos 52$ seasonality is a single harmonic

One annual sine-cosine pair captures the dominant annual cycle and nothing else — no semi-annual harmonic, no holiday indicators, no promotion-calendar structure.

91. Optimization limitations

91.1 Business rules, not the model, set the magnitude of most recommendations

2.9% interior model optima; 57.1% guardrail corners; a no-model rule matches within one 5¢ step **82.5%** of the time (§54, §56). This is the headline limitation and it is stated as such in the README.

91.2 The demand forecast does not choose the price at all

$\hat{Q}(p_o)$ cancels from the arg-max. Verified numerically: scaling the base model by 10× leaves every recommended price unchanged.

91.3 The guardrails neutralise the elasticity assumption

Median price variation across four elasticities: 79.8% → 6.6% → **0.0%** as layers are added. Robustness to being wrong is also insensitivity to being right (\$55.5).

91.4 Materiality and the risk bands are asserted, not fitted

1% materiality, 150/15/2% for LOW, 80/8/5% for MEDIUM. Calibrating them would mean fitting realised out-of-sample error against the bands, which was not done.

91.5 The 5-cent grid quantises the answer

The grid optimum sits a median of **\$0.025** (p90 \$0.043) from the continuous optimum inside the same interval — about 1% of a typical price.

91.6 Single-period, single-product optimization

No multi-week policy, no cross-price effects, no inventory constraint, no category-level margin target, no competitor reaction.

91.7 No absolute price ceiling is configured

`MAX_PRICE` is present in **0** contexts. For a discretionary category like cereal this is defensible; for essential goods it would be a serious gap (\$94.2).

91.8 Batch optimization is per-context

3,000 contexts take ~105 s. `simulation.simulate_many` provides the fully vectorised path but the batch script does not use it.

92. Business and deployment limitations

92.1 Not production ready

No orchestration, no model registry, no live monitoring store, no alerting, no serving SLOs, no authentication, no rate limiting, no CI on the real data (by design — the licence forbids redistributing it).

92.2 Docker never built

Daemon unavailable in this environment. Status **NOT TESTED**, not PASS.

92.3 CI never executed and the repository has no commits

No GitHub remote; `git log` reports no commits; `git_commit` in the model metadata is the literal string "HEAD". Model artifacts cannot be tied to a code revision.

92.4 The dashboard is not a deployment pattern

It loads the ~4.7M-row feature panel into memory. First page load takes ~40 s. Fine on a workstation; not a serving architecture.

92.5 REVIEW_REQUIRED is a queue nobody staffs

1,353 items per week under the standard profile (9.7% of 13,964). There is no prioritisation layer — no sorting by value at risk, product importance or confidence. Producing the queue is the easy half.

92.6 Policy profiles are DEMO settings

Chosen for illustration, not derived from industry benchmarks or from this retailer's operating constraints.

92.7 Notebooks are thin wrappers

`notebooks/01_pricing_eda.ipynb` and `02_elasticity.ipynb` wrap the scripts; the scripts are the source of truth for every number.

92.8 Documentation drift existed, and was closed in v1.0

Quantified instances found while writing this report (full list in the sources file and in Appendix J.24): the README's demo elasticity (-3.016 vs the applied -2.940) and uplift ($+6.51\%$ vs $+7.23\%$), two of the three method-agreement percentages ($87.3\%/89.0\%$ vs the artifacts' $86.7\%/91.7\%$), the stale interview answer in `docs/INTERVIEW_RED_TEAM.md` Q2, and several stale test counts.

This section proposed "a `make regenerate-readme` step that rebuilt the README's numeric blocks from artifacts". The v1.0 engineering closure did exactly that, and added a second, broader mechanism: a cross-document numeric audit that checks every hand-authored statement of a headline metric against the artifact that owns it. Both are enforced by tests, so this limitation is closed for headline numeric results. It is **not** closed for prose claims - those are covered by the separate claim audit (`reports/20_CLAIM_AUDIT.md`), which classifies rather than verifies.

92.9 A concurrent agent session wrote into the repository during the build

Recorded in `STATUS.md` (open issue 11) and `reports/VALIDATION_SUMMARY.md` (open issue 8). The tree was re-verified end to end afterwards, and this report independently re-ran the tests, the lint, the demo, the API and the dashboard.

93. External validity

93.1 What this data is

US grocery retail, one chain, one metropolitan area, one category, 1989–1997. 93 stores, 489 UPCs, 366 weeks.

93.2 What would need to change for modern e-commerce

dimension	Dominick's 1989–97	modern e-commerce	consequence
price change cost	shelf tags, staff time, weekly cadence	near zero , continuous	the $\pm 10\%$ weekly cap is far too coarse; the decision becomes near-continuous and a bandit framing starts to make sense (§102)
competitor prices	unobserved	scraped hourly	competitor price becomes a first-class feature and a source of endogeneity
customer identity	none	full session and purchase history	personalisation becomes <i>possible</i> , which makes §94's ethics binding rather than moot
assortment	fixed shelf	effectively unlimited	cross-price and cannibalisation effects dominate (§101)
demand signal	weekly units	clicks, add-to-carts, impressions, conversion	far richer, and available before purchase
experimentation	in-store, expensive, slow	cheap, fast, continuous	the causal problem is solvable; there is no excuse

dimension	Dominick's 1989–97	modern e-commerce	consequence
			for observational elasticity
inventory	store shelf	warehouse, drop-ship, backorder	price interacts with fulfilment cost and availability
stock-outs	invisible	observable	the \$88.6 confound disappears

The single most important change: **experimentation becomes cheap**. The project's central limitation — no causal identification — is a *data-collection* limitation, and modern e-commerce collects the right data by default.

93.3 What would need to change for other retail sectors

sector	the binding difference
fashion / apparel	markdown optimization over a finite selling season with perishing inventory. The objective becomes dynamic and inventory-constrained; a single-period model is the wrong tool.
fresh / perishable groceries	spoilage makes price a function of remaining shelf life. Requires a time-to-expiry state variable.
consumer electronics	steep intertemporal price decay, strong reference-price effects, and consumers who strategically wait. Constant elasticity is badly wrong.
airlines / hotels	fixed capacity, perishable inventory, booking curves. Revenue management, not price optimization.
B2B / contract pricing	negotiated, volume-tiered, relationship-dependent. Not a shelf-price problem at all.
pharmaceuticals / regulated goods	price caps, reimbursement schedules, legal constraints dominate. The constraint layer becomes the whole system.

93.4 What transfers unchanged

The **architecture** transfers even when the data does not:

125. separate forecasting from price response;
126. estimate the price response with methods designed for identification, not prediction;
127. use panel-robust inference and shrink noisy product estimates;
128. hold cost fixed across the candidate grid;
129. bound recommendations by explicit, attributable guardrails;
130. gate on evidence quality, not just on constraint feasibility;
131. **audit how much of the answer the model actually supplies;**
132. never report a model-internal counterfactual as a realised outcome.

`docs/USING_YOUR_OWN_DATA.md` documents the porting path.

PART XIX — RESPONSIBLE PRICING

94. Ethics and pricing safety

Price optimization is one of the few analytics applications whose output lands directly on a customer's receipt. That deserves explicit boundaries, and `docs/RESPONSIBLE_PRICING.md` sets them.

94.1 What this engine optimises over

Product, store and time context only: UPC, store, week, seasonality, promotion state, recent demand, recent price, decision-time cost. Nothing else exists in the data.

94.2 What it must never optimise over

- individual customers or households;
- any protected or sensitive attribute — race, ethnicity, religion, disability, age, gender, sexual orientation, immigration status;
- **proxies** for those attributes — neighbourhood income, ZIP-code demographics, device type, browsing history, inferred willingness-to-pay.

94.3 Personalised pricing

Not implemented, and not implementable on this data — there is no customer dimension.

The document states this positively rather than defensively, and the reasoning is worth repeating: "*we could not*" is a weaker guarantee than "*we would not*". Personalised pricing based on inferred ability to pay raises discrimination, transparency and trust problems that a portfolio project has no business hand-waving through.

94.4 Store identity and the discrimination risk it carries

A store is a place, and places correlate with demographics.

Dominick's genuinely used price zones, and this project uses store as a decision dimension because pricing *was* zone-based — the observed median cross-store price spread is 13.6% (p90 27.5%). But zone pricing is precisely where "price by location" can shade into "charge more where a protected group shops".

A real deployment would require:

- an explicit review of price differences across stores **against demographic data**;
- a **cap on cross-store dispersion** for essential goods;
- legal sign-off in every jurisdiction of operation.

None of that is implemented here, and the document says so.

94.5 Price gouging

The guardrails are demo-grade, not a gouging policy. What exists: a maximum change per decision, a minimum-margin floor, an extrapolation guardrail keeping prices inside historically observed levels, and a materiality threshold. These *limit movement*; they do not encode an ethical policy.

What a real system needs, and this one does not have:

requirement	present
hard caps on increases for staple and essential goods	✗ (MAX_PRICE is configured in 0 contexts)
an emergency mode freezing or reducing prices during disasters, supply shocks or declared emergencies — a legal requirement in many jurisdictions	✗

requirement	present
category-level rules — cereal is discretionary, infant formula is not, and no model should treat them identically	✗
monitoring for cumulative increases across consecutive weeks	✗

The last is worth naming: the $\pm 10\%$ cap is **per decision**. Applied for four consecutive weeks it compounds to **+46%**. Nothing in this engine tracks cumulative movement, and a production system would need a rolling cap.

94.6 Customer trust and transparency

- **Price flapping erodes trust as much as price level does.** The backtest measures recommendation stability (36.9% of prices unchanged under the ML policy) precisely because a model that re-prices every week is operationally and reputationally expensive.
- **Every recommendation is auditable** — logged with inputs, constraints, model version, elasticity and its source, risk level and reason codes.
- **The engine can always answer "why this price?"** in plain reason codes rather than "the model said so".

94.7 Legal and regulatory obligations

Not legal advice, but the obligations a real deployment must map:

- unfair or deceptive pricing practices rules;
- price-gouging statutes (often state-level or emergency-triggered);
- unit-pricing and price-accuracy display requirements;
- **competition law — in particular, never use a shared model or signalling mechanism to coordinate prices with competitors;**
- consumer-protection rules on advertised promotions.

The competition-law point deserves emphasis because it is the one an ML team is most likely to walk into unknowingly: a pricing model shared across competing retailers, or one that reads and reacts to competitor prices in a way that converges on tacit coordination, is a genuine antitrust exposure regardless of intent.

95. Human approval

95.1 The intended workflow

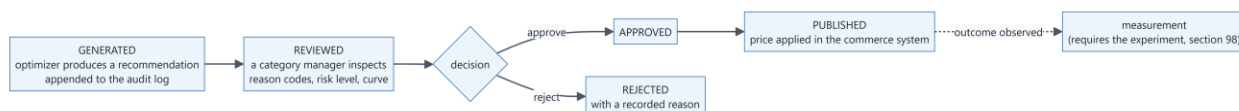


Diagram 8

LIFECYCLE_STATES = ("GENERATED", "REVIEWED", "APPROVED", "REJECTED", "PUBLISHED") in `src/pricing_engine/audit.py`.

95.2 Where the human is mandatory

decision state	human required?	contexts (week 399, standard)
RECOMMEND_CHANGE	not by the engine — but the lifecycle still routes it through REVIEWED → APPROVED before PUBLISHED	8,265 (59.2%)
KEEP_CURRENT	no action needed	4,346 (31.1%)
REVIEW_REQUIRED	yes — the proposal is never applied without approval	1,353 (9.7%)

A HIGH-risk context is never actionable under either default profile. Verified: **0 of 4,760** HIGH-risk contexts auto-changed a price.

95.3 What the reviewer is given

For each queued item: current price, proposed price, final price, decision state, risk level and its factors, the elasticity applied and its source, the feasible interval and every guardrail's value, the full reason-code list, risk notes in plain English, the predicted units / revenue / gross profit at both prices, the demand curve with the feasible band shaded, and the model version.

The dashboard's Recommendation Engine page is that reviewer interface, and its REVIEW_REQUIRED banner states the consequence explicitly: *"this context is HIGH risk, so the proposal is NOT applied automatically. A human must approve it. The final price stays at \$X."*

95.4 What is not built

capability	status
the lifecycle states and their persistence	☒ implemented
a queue UI with assignment, prioritisation and SLAs	✗
approval authentication and authorisation	✗
write-back to a commerce system	✗
capture of rejection reasons as training signal	✗
staffing for 1,353 items a week	✗

The fifth is the most interesting omission. A reviewer who rejects a recommendation knows something the model does not — a competitor promotion, a supply problem, a brand agreement. Capturing that as a labelled signal would be one of the highest-value additions to the system and it is not implemented.

PART XX — PRODUCTION DEPLOYMENT

96. How this would become a real production system

96.1 Target architecture

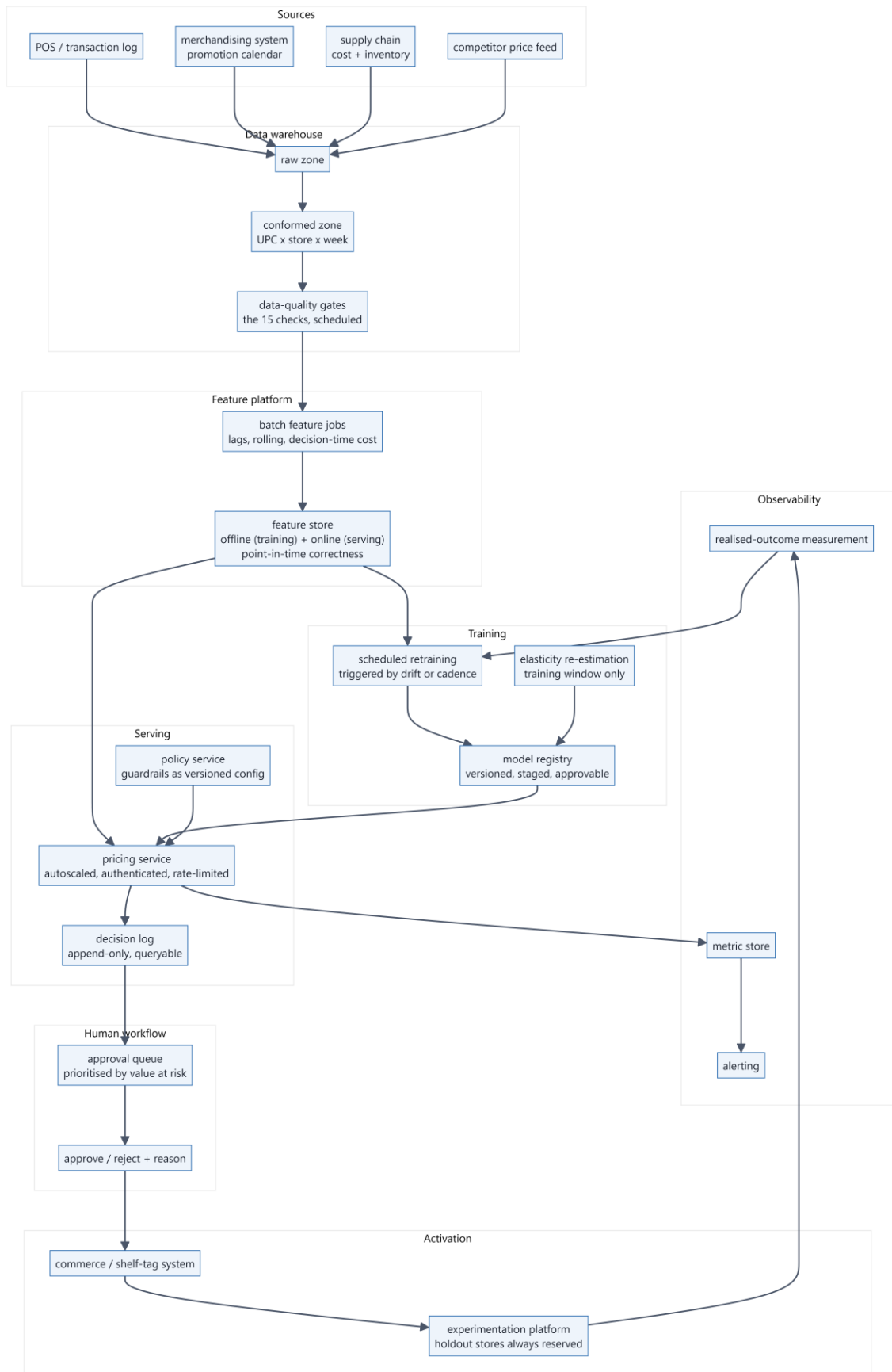


Diagram 9

96.2 What is demo-only today — explicitly

component	today	production
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component	today	production
data ingestion	one-off download of a static archive	scheduled pipelines with SLAs and quality gates
feature computation	a script over a Parquet file	a feature store with point-in-time correctness and online serving
training	manual <code>make train</code>	scheduled and drift-triggered, with approval gates
model storage	a <code>.joblib</code> file plus a JSON	a registry with staging, promotion and rollback
code lineage	none — the repo has no commits	commit SHA + build ID on every artifact
serving	a single uvicorn process, no auth	autoscaled, authenticated, rate-limited, traced
policy config	a YAML file in the repo	versioned, reviewable, independently deployable
approval	a lifecycle enum	a real queue with assignment and SLAs
activation	none	write-back to the commerce system with confirmation
measurement	model-internal only	a permanent holdout and a real experiment
monitoring	an offline script	a metric store, alerting, on-call

96.3 The dependency ordering that matters

The **experimentation platform is not the last box; it is the first prerequisite**. Without a permanent holdout, every economic claim the system makes remains model-internal forever, no matter how good the engineering elsewhere becomes. §64 is not a limitation of this project's maturity — it is a property of observational data.

96.4 Rollout sequence

133. **Shadow mode.** Generate recommendations, log them, apply none. Measure agreement with what the merchant actually did and the size of the proposed moves. Weeks 1–8.
134. **Human-approved pilot.** A handful of stores, `conservative` profile, every recommendation approved by a person. Weeks 9–16.
135. **Randomised experiment** (§98) on a treated store set with a control set. Weeks 17–28.
136. **Staged rollout** conditional on the primary estimate and the guardrail metrics, keeping a **permanent holdout**.
137. **Automation** only for LOW-risk, high-confidence contexts, with the review queue retained for everything else.

97. Scaling

97.1 Where this implementation would break

dimension	today	breaks at
SKUs × stores	36,443 series	~10 ⁵ –10 ⁶ series: the per-context Python loop dominates
batch runtime	3,000 contexts in ~105 s (≈ 35 ms each)	1M contexts ≈ 10 hours in the current loop
feature panel in memory	~4.7M rows	tens of millions: the dashboard and API both load it in-process
elasticity table	372 products	fine — this scales trivially
model artifact	one global model	fine, but a single model over 10 ⁶ heterogeneous SKUs is a modelling question, not an engineering one

97.2 Millions of SKUs

The vectorised path already exists and is unused. `simulation.simulate_many(model, contexts, price_matrix, unit_costs)` scores (`n_contexts × n_candidates`) rows in **one** model call. Switching the batch script to it is the single highest-leverage engineering change in the repository (limitation #18).

Beyond that: partition by store or category and run partitions in parallel; the optimization is embarrassingly parallel because contexts are independent (which is also exactly why cross-price effects are missing — §101).

97.3 Batch versus real-time pricing

	batch (weekly)	real-time
fits	grocery shelf prices — this project	e-commerce, marketplaces
latency budget	hours	milliseconds
what changes	nothing architecturally	features must be served online with point-in-time correctness; the elasticity table becomes a cached lookup; the optimizer must be sub-millisecond
feasibility here	already the case	the grid search over ~15 candidates is already fast enough; the bottleneck is feature retrieval

97.4 Caching

Natural boundaries, in decreasing hit rate: the **elasticity table** (changes only on re-estimation), the **model artifact** (already loaded once at startup), **series statistics** (`n_obs`, distinct prices, CV, price support — change weekly), and the **feasible interval** for a context (a pure function of `po`, cost, support and the policy).

97.5 Distributed training

The current fit is 31 s on 3.2M rows — nowhere near needing distribution. At 10^8 rows: sample (this project's own evidence is that a 1.2M subsample gives -2.0289 against -2.0443 on the full sample), or use out-of-core / incremental learners, or partition by category and fit per-segment models.

97.6 Feature store

The requirement is **point-in-time correctness**: given (`upc, store, week`), return exactly the features that were knowable at that moment. This project achieves it with a batch table and a shift discipline; at scale it needs a purpose-built store, because the failure mode — training/serving skew — is silent and produces confidently wrong prices.

97.7 Model segmentation

One global model is defensible here (489 UPCs, one category, one chain). At scale the options are: per-category models (different demand dynamics), per-region models (different competition), per-velocity models (fast movers versus long tail), or a hierarchical model pooling across levels — which is exactly the empirical-Bayes idea of §31 applied to the demand model rather than to the elasticity.

PART XXI — EXPERIMENTATION

98. Pricing A/B test

The full design is in `docs/PRICING_EXPERIMENT.md`. **No experiment was run and no results are invented.**

98.1 Hypothesis

Applying the constrained optimizer's recommended prices (standard profile, gross-profit objective) to eligible UPC $\times$ store series increases **gross profit per store-week** relative to the retailer's current pricing, without material harm to units or customer complaints.

98.2 Unit of randomisation: store $\times$ category-block $\times$ week

Randomise **whole stores** into treatment and control for a defined block of Cereals UPCs.

Why not randomise individual UPCs within a store. Shoppers substitute heavily within cereal. If Kellogg's Corn Flakes is treated and Post Toasties is not, the "control" product absorbs the treated product's lost or gained volume, and the estimated effect is contaminated by cannibalisation. This is the same mechanism that §101 identifies as the largest missing model component.

Why not a switchback (randomise weeks within a store). Promotion calendars, seasonality and shopper stock-piling create carry-over across weeks; a switchback would need a long washout that eats the sample.

Trade-off accepted: store-level randomisation means **few independent units** and requires clustered inference — which is precisely the lesson §29 taught about this data.

98.3 Design

- **Two arms:** control (current pricing process) and treatment (optimizer recommendations, human-reviewed, `standard` profile).
- **Stratified assignment** on pre-period gross profit, store size and price zone, to reduce imbalance with a small number of clusters.
- **Duration: 12 weeks minimum** — long enough to cover a promotion cycle and its post-promotion dips, short enough to limit drift — plus a **4-week pre-period** used only for stratification and pre-trend checks.
- **Freeze the recommendation policy for the whole test.** No mid-flight tuning.

98.4 Primary metric

Gross profit per store-week for the treated category block.

Chosen because it is the objective the engine optimises, it is measurable weekly, and — unlike revenue or units alone — **it cannot be gamed by shifting volume between prices.**

One primary metric, one primary test.

98.5 Secondary metrics

Revenue per store-week; unit sales per store-week; average selling price; realised gross margin rate; **category share of store sales** (to detect substitution *out of* the category).

Reported with a multiplicity correction (Holm or Benjamini-Hochberg) and labelled exploratory.

98.6 Guardrail metrics with pre-registered stopping rules

- customer complaints per 1,000 transactions

- share of items with a price increase above a defined threshold
- **out-of-stock rate for treated items** — a price cut that empties the shelf is not a win
- returns / refunds where applicable
- basket size and store traffic (store-level spillover)
- any legally sensitive category flags

Breach means the arm is paused, not explained away.

98.7 Sample size and power

For a clustered design the usable sample is the number of **stores**, not store-weeks. With k stores per arm, T weeks, within-store weekly gross-profit standard deviation σ and intra-cluster correlation ρ :

$$SE \approx \sigma \sqrt{((1 + (T-1)\rho)/(kT))}$$

Detecting a small relative effect (2–3% of gross profit) in a category with high weekly volatility typically requires **tens of stores per arm and a multi-week horizon**.

The honest planning step is to compute σ and ρ from the actual chain's pre-period and solve for k — not to assert a number here. The design document does exactly that, and notes that this project's own demand model has a weekly WAPE around 0.46; category-level aggregation reduces noise, but the design must be powered against **measured** variance, not hope.

98.8 Duration and seasonality

12 weeks minimum, spanning at least one full promotion cycle. Week fixed effects in the analysis absorb chain-wide seasonality; weeks with chain-wide promotional events are either excluded or modelled explicitly.

98.9 Spillovers and other threats

threat	handling
spillover between stores (cross-shopping)	randomise at store level; prefer geographically separated stores; measure traffic
within-category substitution	treat whole category blocks , not single UPCs
seasonality	week fixed effects; run across a full promotion cycle
contamination from chain-wide promotions	log promotion calendars; exclude or model those weeks
novelty effects	discard the first week from the primary window (pre-registered)
competitor response	monitor local competitor prices where available; a 12-week window limits exposure
implementation failure	audit that treated shelves actually carry the recommended price — an experiment measures what was <i>executed</i> , not what was recommended

The last is the one most often forgotten and most often fatal.

98.10 Pre-registered analysis plan

- **Primary estimator:** difference-in-differences on store-week gross profit with **store and week fixed effects**, standard errors **clustered by store**.
- **Pre-trend check** on the pre-period, reported whether or not it is flattering.
- One primary metric, one primary test; secondaries corrected for multiplicity and labelled exploratory.
- **Analysis code written and frozen before unblinding.**

98.11 Rollout safety

138. Start with a **small pilot** (few stores, **conservative** profile).
139. Expand only after the guardrails hold and the primary estimate is stable.
140. Keep a **permanent holdout** so the effect can be re-measured after full rollout — models decay, and without a holdout nobody can tell.
141. Keep the human review step throughout.

99. How real business uplift would be established

99.1 The conversion

today	after the experiment
"model-internal estimated portfolio gross-profit uplift +8.30% "	*"prices set by this engine caused a change in gross profit per store-week of X% , 95% CI [L, U], estimated by difference-in-differences with store-clustered standard errors over N stores and 12 weeks"*

99.2 The four things that change

142. **The proposer stops being the judge.** Outcomes are *observed* at the prices the engine chose, not predicted by the model that chose them. §64's circularity is broken by construction.
143. **Randomisation supplies identification.** Treatment assignment is independent of unobserved demand, so $P(Q \mid \text{do}(\text{price}))$ becomes estimable. §33 dissolves.
144. **The estimate acquires an interval.** Not a point from a simulation, but an estimate with a standard error that reflects the actual clustering — the lesson of §29 applied to the outcome rather than the elasticity.
145. **Guardrail metrics become evidence.** Complaints, out-of-stocks and traffic are measured, not assumed away.

99.3 What would still be unknown afterwards

Honesty cuts both ways. Even a clean experiment would leave open:

- **external validity** — the effect in these stores, this category, this quarter; not a universal elasticity;
- **decay** — models degrade (this one loses 0.12 WAPE over four quarters), so the measured effect has a shelf life, which is why the permanent holdout matters;
- **the counterfactual policy** — the experiment measures the engine against *the retailer's current process*, not against R4 or any other rule. Measuring "is the model worth it?" would require a **third arm** running the rule-only policy;
- **long-run customer effects** — 12 weeks cannot measure trust, loyalty or brand perception.

99.4 The third arm, and why it is the interesting one

Given §56's finding — a no-model rule matches the engine within one 5¢ step 82.5% of the time — the highest-value experiment is not "engine vs status quo". It is **three arms**:

arm	policy
A — control	the retailer's current process
B — rule only	R4: always take the maximum allowed increase, through the same guardrails
C — full engine	the hybrid response, constrained optimizer and risk gate

The A-vs-C contrast answers *"is this worth deploying?"*. The **B-vs-C contrast** answers **"is the model worth building?"** — the question §85 can only answer argumentatively today.

That comparison is the single most valuable extension of this project, and it costs one extra arm.

PART XXII — FUTURE WORK

100. Cross-price elasticity

100.1 What is missing

Every elasticity in this project is **own-price**:

$$\varepsilon_{ii} = (\partial \log Q_i) / (\partial \log p_i)$$

A complete demand system needs the cross-price terms:

$$\varepsilon_{ij} = (\partial \log Q_i) / (\partial \log p_j), \quad i \neq j$$

with $\varepsilon_{ij} > 0$ for **substitutes** (raising Post's price increases Kellogg's volume) and $\varepsilon_{ij} < 0$ for **complements**.

100.2 Why this matters most in exactly this category

Cereal is a substitution-heavy category. A shopper who finds their usual brand at \$3.66 instead of \$3.35 does not necessarily leave without cereal — they frequently buy a different box. The engine models that lost volume as **demand destroyed**; in reality much of it is **volume redirected within the same category, in the same store, in the same week**.

The direction of the error is knowable even if its size is not: **own-price elasticity estimated without cross-price terms overstates the category-level volume loss from a price increase**. So the engine is, if anything, *too conservative* about price increases at the category level — while simultaneously being blind to the fact that re-pricing many substitutes at once is not the sum of the individual effects.

100.3 Why it was not done

obstacle	detail
dimensionality	489 UPCs $\Rightarrow 489^2 = 239,121$ elasticities. With 372 products in the training window and a median of ~5,400 usable rows per product, the parameters vastly outnumber the identifying variation.
identification	own-price elasticity is already unidentified (§33). Cross-price terms inherit every confounder and add more: competing brands promote in coordinated waves, so their prices move together for reasons unrelated to substitution.
the shrinkage problem gets worse	133 of 372 products already lack a usable <i>own</i> estimate. Their cross-price terms would be pure noise.

100.4 How it would be done

approach	idea	cost
category aggregation	model demand for brand $\times$ segment aggregates (e.g. "adult ready-to-eat", "kids' pre-sweetened") rather than UPCs	loses per-UPC precision; makes the problem tractable
structural demand systems (AIDS, nested logit, BLP)	impose economic structure — adding-up, homogeneity, Slutsky symmetry — to reduce the free parameters	strong functional-form assumptions; still needs identification
regularised full matrix	estimate ε_{ij} with a sparsity penalty and a hierarchical prior by brand/segment	the empirical-Bayes idea of §31 in matrix form
price-index approach	add a <i>competitor price index</i> (the mean price of the product's nearest substitutes) as one extra regressor per	the cheapest useful step: one parameter per product, directly interpretable, and immediately usable in the hybrid

approach	idea	cost
	product	response

The **price-index approach** is the right next step for this repository. It is tractable, it needs no new data, and it would convert a completely unmodelled mechanism into a measured one.

101. Cannibalisation and multi-product optimization

101.1 The structural problem

The engine solves 13,964 **independent** single-product problems:

$$\max_{(p_i \in \mathcal{F}_i)} (p_i - c_i)Q_i(p_i) \quad \text{for each } i \text{ separately}$$

The business problem is a **joint** one:

$$\max_{(p \in \mathcal{F})} \sum_i (p_i - c_i)Q_i(p_1, \dots, p_n)$$

These coincide only if $\partial Q_i / \partial p_j = 0$ for all $i \neq j$ — i.e. only if no substitution exists. In cereal, that is false.

101.2 The concrete failure

Suppose the engine recommends +9% on the twelve largest cereal brands in a store simultaneously. Each recommendation is individually sound under its own model. Jointly, the shopper's substitution options have all become more expensive at once, so the aggregate volume response is **different from — and probably smaller than — the sum of the individual responses**. The engine cannot see this and reports the sum.

The reverse case is worse: cutting one brand's price cannibalises its substitutes, so the *category* gain is smaller than the item-level gain the engine reports.

101.3 What a joint optimizer would need

ingredient	status
a cross-price matrix (\$100)	✗
a category-level objective	✗ (the objective is per-item)
a joint feasible set (category margin targets, price-ladder constraints, brand-tier ordering)	✗
a multi-dimensional solver — the grid becomes k^n and is infeasible	✗ (would need a mathematical program or coordinate descent)
price-architecture constraints (e.g. the 18 oz box must cost more than the 12 oz box)	✗ not even single-product-implementable today

The last is worth naming separately: **price-ladder consistency** (larger sizes cost more per pack but less per ounce; premium tiers price above value tiers) is a hard business requirement in real grocery pricing and is entirely absent here. Two independently optimised sizes of the same product could easily end up inverted.

101.4 Why this is the largest missing mechanism

KNOWN_LIMITATIONS.md #6 calls it "arguably the largest missing mechanism", and that is right. Unlike the causal problem, which is a *data* limitation nothing in this repository can fix, cannibalisation is a **modelling scope** decision that could be addressed with the data already in hand — at the cost of a much harder estimation problem.

102. Dynamic pricing, bandits and RL

102.1 The honest framing first

These are not automatically better, and for this problem they would currently be worse. The reasons are specific.

102.2 Why they do not fit *this* problem

requirement	contextual bandit / RL	this setting
online interaction	the learner must <i>act</i> and observe the result	offline historical panel; no ability to act
fast feedback	rewards within seconds to hours	weekly ; a 12-week experiment yields 12 observations per store
many cheap trials	thousands of arms pulled per day	~45,000 price decisions per week, but each is a real customer-facing action with reputational cost
acceptable exploration cost	random exploration is cheap	a randomly-priced shelf is a real revenue loss and a trust cost
stationarity within a learning horizon	reasonably stable	eight years of drift; promotion calendars dominate
a well-defined state	observable	inventory, competitor prices and shopper stock-piling are all unobserved

A contextual bandit run on weekly grocery prices would spend most of its budget exploring, at real cost, and would converge slowly against a non-stationary target.

102.3 When they *would* become relevant

condition	why
e-commerce with continuous repricing	feedback in hours, exploration cost near zero, price changes invisible to most shoppers
a category where exploration is cheap	long-tail SKUs with low volume and low visibility
an existing experimentation platform	the bandit is then a <i>more efficient</i> experiment, not a replacement for one
a genuinely sequential objective	markdown optimization over a finite season, where today's price changes tomorrow's inventory — a real MDP rather than a repeated single-period problem
inventory as an explicit state	perishables, fashion, event tickets

102.4 The intermediate step that is actually available

Thompson sampling over a small, safe price ladder within an approved experiment. Rather than full RL:

- restrict the action space to 3–5 prices inside the existing feasible interval;
- run it only on LOW-risk, well-identified series;
- keep the guardrails and the risk gate untouched;
- treat it as an **adaptive experiment** that allocates more of the sample to promising prices, not as an autonomous pricer.

That would improve on §98's fixed two-arm design without abandoning identification, and it is the version of "dynamic pricing" this project could justify.

102.5 What the guardrail evidence implies

§55 shows the recommended price is insensitive to the elasticity assumption under production guardrails. **A learning algorithm operating inside those same guardrails would have almost nothing to learn**, because the feasible interval is narrow and the answer is usually its edge. Any serious bandit deployment would first require widening the guardrails — which requires the causal evidence of §98. **The ordering is: experiment → identification → wider guardrails → adaptive pricing**. Not the reverse.

103. Better causal identification

Ranked by strength.

103.1 Randomised prices — the gold standard

Assign prices exogenously (§98). Identification by construction; nothing else comes close. **Cost**: real revenue at risk, operational complexity, executive sponsorship.

103.2 Instrumental variables

Need an instrument Z that shifts price without affecting demand except through price:

$$\text{Cov}(Z, p) \neq 0, \quad \text{Cov}(Z, u) = 0$$

Candidate instruments in retail pricing:

instrument	rationale	available here?
wholesale / commodity cost shocks (grain, packaging, energy)	shift the retailer's cost, hence price; plausibly unrelated to local demand	✗ not in the Cereals file
distance-to-warehouse × fuel price	shifts delivered cost differentially by store	✗ no store geography
manufacturer trade-deal calendars	exogenous timing of promotional funding	✗ not recorded
prices of the same UPC in geographically distant stores of the same chain (a Hausman instrument)	captures common cost shocks, excludes local demand	partially — the data supports it, but the exclusion restriction fails if chain-wide demand shocks exist, which §29 shows they do (week clustering inflates SEs 12×)

The last row is instructive: the one feasible instrument is invalidated by a correlation structure this project has already measured.

103.3 Natural experiments and regression discontinuity

Need a documented policy change, a zone reassignment, a discrete pricing rule or a threshold. **None identified** in the Cereals extract. Note that the broader Dominick's research programme *did* include in-store pricing experiments in several categories — but the Cereals movement file carries no experiment assignment or window that could be verified, so this project claims none of its estimates are experimental (§33.4).

103.4 Difference-in-differences

Needs a treated and a control group with credible parallel pre-trends. There is no exogenous treatment to define here.

103.5 Causal ML — and the trap it represents

Double machine learning, causal forests and orthogonal learning are **estimation** technologies. They relax functional-form assumptions and give valid inference under **unconfoundedness given observables**. They do **not** manufacture identification.

In this data the confounders — planned promotions, display, feature ads, competitor prices, anticipated demand — are **unobserved**. Applying double ML here would produce a more flexible, equally confounded estimate, dressed in causal vocabulary. That would be worse than the current honest observational estimate, because it would *sound* identified.

This is worth carrying into any interview: **the question "have you tried causal ML?" is often a test of whether you know the difference between estimation and identification.**

103.6 What is actually worth doing with the data in hand

step	value
add a competitor price index (\$100.4)	converts one unmodelled confounder into a measured control; cheapest real improvement
model promotion depth, not just presence	the biggest error segment (WAPE 0.656) and the biggest identification threat are the same rows
richer promotion controls — interact the flag with product and season	partially separates the marketing bundle from the price
UPC × store × season fixed effects	absorbs more time-varying heterogeneity, at the cost of precision
a Hausman-style instrument with an explicit test of the exclusion restriction	worth attempting <i>and reporting the failure</i> — the attempt itself is informative

PART XXIII — BUSINESS CASE

104. Business interpretation

104.1 Who uses this and how

A **category manager** for Cereals, owning ~45,000 weekly price decisions across 93 stores and 489 UPCs. Their week looks like this:

Monday — the batch runs. 13,964 decision contexts scored under the **standard** profile:

outcome	contexts	share	what the manager does
RECOMMEND_CHANGE	8,265	59.2%	review in bulk, spot-check, approve
KEEP_CURRENT	4,346	31.1%	nothing
REVIEW_REQUIRED	1,353	9.7%	the queue — individual judgement required

Tuesday — work the queue. For each of the 1,353, the dashboard shows the proposal, the final price (unchanged), the risk level with its factors, the elasticity and its provenance, the feasible interval, the reason codes and the demand curve. The manager approves, rejects or overrides.

Wednesday — publish. Approved prices flow to shelf tags. (Not implemented here — \$95.4.)

Ongoing — investigate. When a buyer asks "*why did we raise Cap'n Crunch by 9%?*", the answer is in the audit log: model version, elasticity -2.940 from the product's own estimate with weight 0.92, cost \$2.56, feasible range \$3.02–\$3.69, the optimum was \$3.88 but the $\pm 10\%$ cap bound, estimated gross profit \$20.27 → \$21.74.

104.2 What the manager gains

gain	mechanism	evidence
triage	40.1% of contexts are actively <i>not</i> recommended for change	3,528 screened out + 1,353 gated + 716 immaterial
direction	a defensible answer to "up or down?" per item	8.0% of actionable recommendations are cuts under shrunk
consistency	the same evidence produces the same decision every week	deterministic given the artifacts
explainability	reason codes and a feasible interval, not a black box	every recommendation carries both
auditability	every decision reconstructible after the fact	34-column append-only log
an explicit risk boundary	HIGH-risk contexts never move automatically	0 of 4,760

104.3 What the manager must supply

Judgement the engine cannot have: competitor activity, supplier relationships, brand agreements, planogram constraints, price-ladder consistency across sizes, category strategy, and the knowledge that a particular SKU is a traffic driver whose price is a strategic signal rather than a profit lever.

104.4 What the manager must not be told

That the engine will deliver +8.30% gross profit. That number is a model-internal estimate produced by the model that proposed the prices. The honest statement is:

"Under the fitted price-response model, applying these recommendations would produce 8.30% more gross profit than current pricing on the scored contexts. Whether it actually would is unknown until it is tested (§98)."

105. Example end-to-end pricing decision

The complete trace for one real recommendation, from raw context to final decision. Verified live on 2026-08-19.

105.1 The context

field	value
product	CAPN CRUNCH JUMBO CR (UPC 3000006560)
store	86
decision week	399 (1997-05-01)
current effective unit price	\$3.35
decision-time unit cost (lagged, forward-filled AAC)	\$2.5571
current gross margin	23.7%
observed price support	\$1.50 – \$3.79, over 359 weeks, 56 distinct prices

105.2 Recent observed history

week	price	units	revenue	gross profit	promotion
392	\$3.35	18	\$60.30	\$15.39	NONE_RECORDED
393	\$3.35	12	\$40.20	\$10.11	NONE_RECORDED
394	\$3.35	12	\$40.20	\$10.11	NONE_RECORDED
395	\$1.50	17,824	\$26,736.00	-\$18,768.67	NONE_RECORDED
396	\$3.19	465	\$1,483.35	\$805.90	Simple price reduction
397	\$3.35	25	\$83.75	\$47.32	NONE_RECORDED
398	\$3.35	11	\$36.85	\$8.72	NONE_RECORDED
399	\$3.35	12	\$40.20	\$9.52	NONE_RECORDED

Week 395 is a spectacular event: **17,824 units at \$1.50** — 1,485× the surrounding weeks — with an observed gross profit of **-\$18,769**, and **no recorded promotion code**. It is a textbook illustration of §11.4 (absence of a code proves nothing) and §18.5 (extremes are retained, not trimmed).

105.3 Elasticity

field	value
training observations for this UPC	20,125
distinct prices	215
raw per-UPC estimate	-3.0178
HC1 standard error	0.0398
panel-robust standard error	0.2557 (6.4× inflation)
τ^2	0.7661
shrinkage weight	0.9214
pooled prior	-2.0289
applied elasticity	-2.9401
source	shrunk_product

105.4 The economics on a grid

Predicted units, revenue and gross profit at candidate prices (model-internal):

candidate	predicted units	revenue	gross profit	margin rate
\$2.51	59.74	\$149.94	-\$2.81	-1.9%
\$2.71	47.68	\$129.22	\$7.29	5.6%
\$2.91	38.68	\$112.54	\$13.65	12.1%
\$3.11	31.81	\$98.93	\$17.59	17.8%
\$3.35 ← current	25.56	\$85.64	\$20.27	23.7%
\$3.51	22.29	\$78.23	\$21.24	27.1%
\$3.66 ← recommended	19.71	\$72.13	\$21.74	30.1%
\$3.81	17.51	\$66.72	\$21.94	32.9%
\$3.88 ← unconstrained optimum	≈16.9	≈\$65.6	≈\$22.0	≈34.1%
\$4.11	14.01	\$57.60	\$21.76	37.8%

Revenue falls monotonically. Gross profit peaks at the closed-form optimum $\$2.5571 \times 2.9401/1.9401 = \3.875 and then falls.

105.5 The guardrails

bound	source	value	binds?
lower	ABSOLUTE_PRICE_FLOOR	\$0.0100	
lower	OUTSIDE_EXTRAPOLATION_RANGE	\$1.4250	
lower	COST_FLOOR	\$2.5571	
lower	MIN_MARGIN (5%)	\$2.6916	
lower	PRICE_CHANGE_LIMIT (-10%)	\$3.0150	☒
upper	OUTSIDE_EXTRAPOLATION_RANGE	\$3.9795	
upper	PRICE_CHANGE_LIMIT (+10%)	\$3.6850	☒
	feasible interval	[\$3.015, \$3.685]	

The unconstrained optimum (\$3.875) is outside the feasible interval. This is one of the 7,967 guardrail-corner contexts.

105.6 The decision

field	value
proposed candidate price	\$3.66 (+9.25%)
final recommended price	\$3.66
decision state	RECOMMEND_CHANGE
actionable	true
risk level	LOW
predicted units	25.56 → 19.71 (-22.9%)
expected revenue	\$85.64 → \$72.13 (-15.78%)
expected gross profit	\$20.27 → \$21.74 (+7.23%)
reason codes	PRICE_CHANGE_LIMIT, PROFIT_UPLIFT_POSITIVE
risk notes	none

105.7 What a manager should take from this

146. **The engine wanted +15.7% and was allowed +9.25%.** The guardrail did the last third of the work.
147. **This is a volume-for-margin trade:** -22.9% units, -15.8% revenue, +7.2% gross profit. If the mandate were share or volume, this recommendation would be **wrong** — and the engine supports a revenue objective for exactly that case.
148. **The elasticity is 92% this product's own**, which is why this context is LOW risk and actionable rather than queued.
149. **+7.23% is a model-internal estimate.** If the true elasticity is the pooled -2.029 rather than -2.940, the optimum would be $\$2.5571 \times 2.029/1.029 = \5.04 — far above the cap — so the recommendation at \$3.66 would be *unchanged*, but the predicted volume loss would be smaller and the estimated profit gain **larger**. The guardrail makes the price robust; it does not make the *estimate* robust.
150. **Week 395 is a warning.** This series has carried \$1.50 with 17,824 units. The extrapolation guardrail's lower bound of \$1.425 reflects that. A model asked about \$1.50 here would not be extrapolating — and would predict a very different world.

CAPN CRUNCH JUMBO CR (3000006560), store 86, week 399 (1997-05-01) - elasticity -2.940 (shrunk_product); shaded band = feasible interval 3.02 – 3.69. MODEL-INTERNAL estimates.

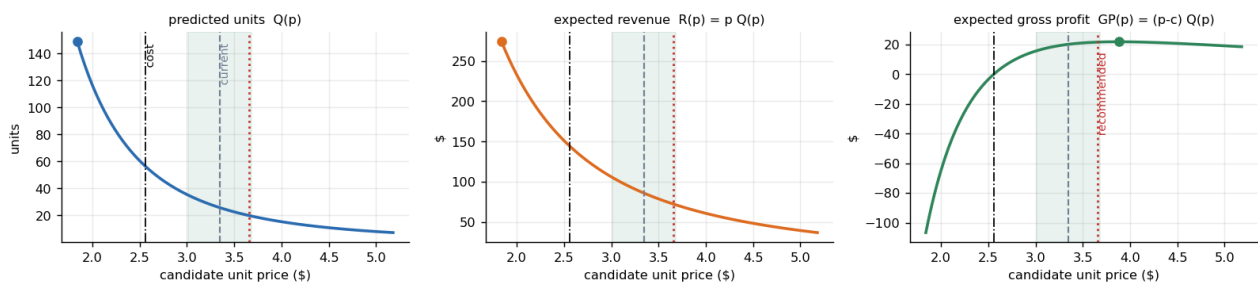


Figure 23 — The same figure as §2.3, repeated here because it is the decision. Shaded band = the feasible interval; grey dashed = current; red dotted = recommended; black dash-dot = cost.

106. Portfolio-level recommendation summary

106.1 The 3,000-context batch (reports/07)

Week 399, standard profile, shrunk response, gross-profit objective.

category	contexts	share
actionable (RECOMMEND_CHANGE)	1,766	58.87%
keep current (KEEP_CURRENT)	923	30.77%
review required (REVIEW_REQUIRED)	311	10.37%
high-risk blocked (HIGH risk, never auto-actioned)	1,039 HIGH-risk, 0 actionable	—

statistic	value
median absolute price change	8.39%
share of changes that are increases	91.28%
median internal uplift among actionable	13.33%
risk mix	LOW 1,175 · MEDIUM 786 · HIGH 1,039
elasticity source	shrunk_product 1,904 · pooled_fallback 1,096
median elasticity used	-2.029
portfolio expected gross profit, current	\$37,580.82
portfolio expected gross profit, if applied	\$40,700.96
model-internal estimated uplift	+8.30%

statistic	value
runtime	105.3 s

106.2 The full week (13,964 contexts, reports/11/19)

category	contexts	share
actionable	8,265	59.19%
keep current	4,346	31.12%
review required	1,353	9.69%
HIGH risk	4,760	34.09%
HIGH risk that auto-changed a price	0	0.00%

106.3 Why the two batches differ slightly

58.87% vs 59.19% actionable, 10.37% vs 9.69% review. The 3,000-context batch is a **sample** of the 13,964; the differences are sampling variation of the expected order. This report quotes both and always names which is which — a practice worth adopting generally, because "58.9% actionable" and "59.2% actionable" appearing in the same document without labels is exactly how a metrics dictionary drifts.

106.4 What the portfolio view is for

question	answer from the batch
how much work is this?	8,265 approvals + a 1,353-item queue per week
how aggressive is the engine?	median +8.4%, 91% increases, 31% no change
is it safe?	0 of 4,760 HIGH-risk contexts auto-changed
how much is model-driven?	2.9% interior optima (\$54)
what is it worth?	unknown — the +8.30% is model-internal (\$64)

The last two rows are the ones a serious reviewer reads. Everything above them describes activity; those two describe evidence.

PART XXIV — PROJECT EVOLUTION

107. How the project changed

This chapter exists because the project's value is in the **iteration**, not in the first architecture. `DECISIONS.md` records **52 numbered decisions** across three phases, with alternatives and rationale for each.

107.1 Phase A–K — the initial build (decisions 1–30)

Official data acquisition; the canonical panel with a counted exclusion ledger; the decision-time feature contract with a twelve-channel leakage audit; the elasticity ladder; six models on a chronological split; a native ML price response; a grid optimizer with guardrails; a risk layer; a FastAPI service and a nine-page dashboard.

The architecture at the end of this phase: the ML demand model answered price counterfactuals **directly**. The optimizer swept candidate prices through the fitted forecaster and took the arg-max.

107.2 The problem discovered

`scripts/price_response.py` measured the fitted model's own price response by central finite difference over 300 real contexts:

source	elasticity
native ML implied	–3.102
controlled econometric, + UPC × store FE	–2.419
controlled econometric, + promo/season/trend	–1.909
pricing pooled, training weeks only	–2.029, 95% CI [–2.236, –1.821]

The forecaster's implied elasticity was **outside the confidence interval of every controlled estimate**. Using it for counterfactuals would bake a systematic pessimism about price increases into every recommendation (§42.2).

Crucially, the demand curves themselves were **fine** — 100% monotone decreasing, 0% flat, 0% negative predictions. The model was not broken. It was *well-calibrated for forecasting and wrong for intervention*, which is a much harder failure to notice.

107.3 Phase L — pricing-science hardening (decisions 31–42)

change	decision
separate forecasting from price response: $Q(p) = \hat{Q}(p_0)(p/p_0)^{\epsilon}$	#31
default method shrunk: per-UPC empirical-Bayes elasticity	#32
keep the native ML response as a benchmark, do not delete it	#33
elasticities fitted on training weeks only (2–257), script aborts otherwise	#34
reject wrong-signed and imprecise product estimates before shrinking	#35
invert the risk scale so HIGH means risky; bands stricter than eligibility	#36
three decision states + an actionable flag	#37
HIGH risk is never auto-actionable under default profiles	#38
rename uplift to <code>model_internal_estimated_*</code> everywhere	#39
keep the zero-price rule but evidence it (audit reports/09)	#40
stamp the reference price explicitly; a missing one raises	#41
score accuracy with the base model, policies with the pricing model	#42

Two of these deserve emphasis as *failures corrected*:

#36 — the risk-scale inversion. The scale used to read backwards: "HIGH" meant *high confidence* and sat on actionable recommendations. A reviewer scanning a queue would have read the most dangerous label as the safest one.

#41 — the silent no-op. During candidate scoring the row's price *is* the candidate, so a silent fallback for a missing reference price would make $p/p_0 = 1$, the elasticity term would vanish, and the engine would quietly return baseline demand for every candidate — a system that appeared to work and recommended nothing meaningful. It now raises.

107.4 Phase M — the final scientific audit (decisions 43–52)

Ten audit reports. What they found is in §84. What changed as a result:

change	decision
report constraint dominance as the headline architectural fact , not an appendix	#43
per-UPC standard errors become panel-robust , not HC1	#44
τ^2 by REML , measured around the prior the shrinkage targets	#45
mean shrinkage weight 0.954 → 0.780 ; the old figure removed everywhere	#46
split recommended_price into proposed and FINAL ; force realisable_uplift to 0 when not actionable	#47
fix build_bounds to report constraints binding at the final interval, not every constraint that tightened a running bound	#48
audits use an exact replica of the optimizer, pinned by 69 tests	#49
shrunk retained on out-of-time evidence , not inheritance	#50
remove "no forecast skill is lost" repository-wide; replace with the precise statement	#51
add <code>audit_claims.py --strict</code> as an enforceable build check	#52

107.5 The four corrections that changed a published number

#	correction	before	after
1	standard errors: HC1 → two-way clustered (pooled)	SE 0.0065	SE 0.1058 (16.2×)
2	mean empirical-Bayes shrinkage weight	0.954	0.780
3	τ^2 (REML, panel-robust inputs)	0.906	0.766
4	share of products significant at 5%	90.1%	75.2%

And two corrections that changed a **claim** rather than a number:

- "no forecast skill is lost" → "the hybrid preserves the forecaster's baseline prediction at the reference price, not necessarily its accuracy at counterfactual prices";
- "ML pricing" → "rule-bounded pricing with a learned direction".

107.6 The arc

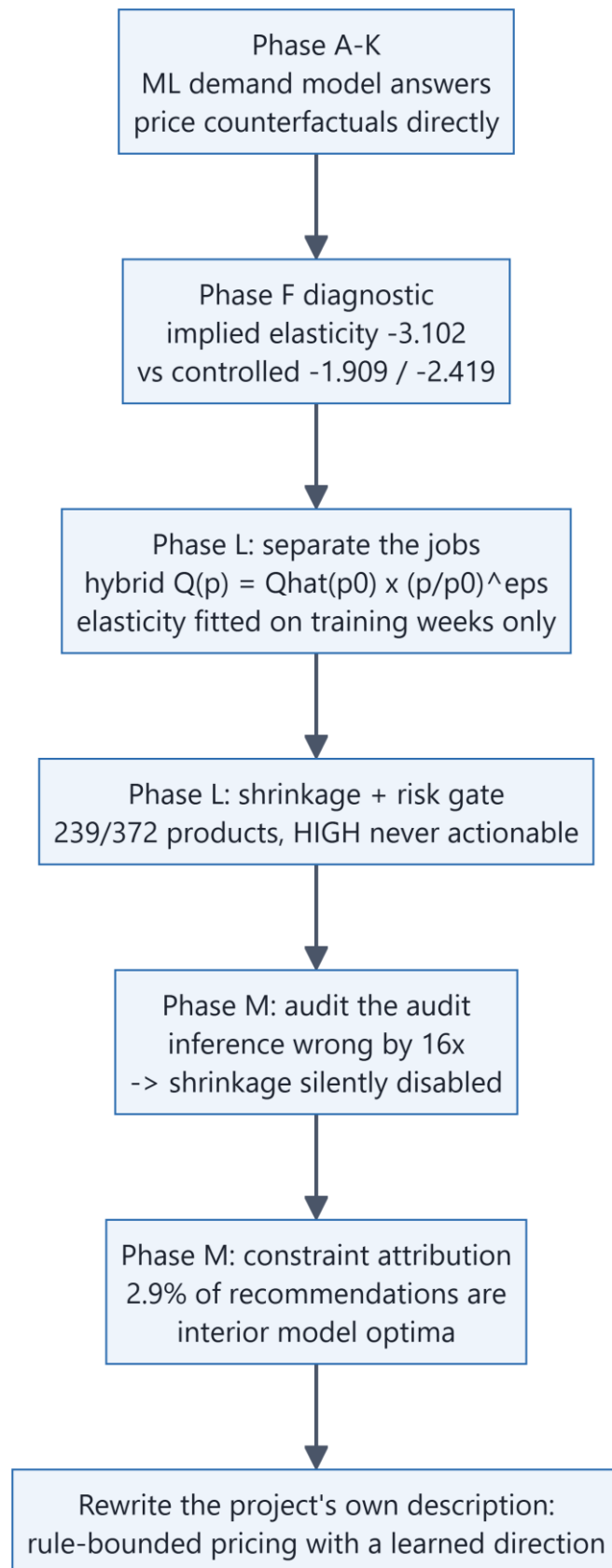


Diagram 10

107.7 What this demonstrates

Not that the first architecture was good — it was not. It demonstrates:

151. **Building a diagnostic for a component nobody else measures.** The Phase F price-response check is not standard practice. It is what found the problem.
152. **Responding architecturally rather than cosmetically.** The response was not "clip the elasticity"; it was to separate two jobs that had been conflated.
153. **Keeping the rejected alternative as a measured benchmark,** so the decision remains auditable and reversible.
154. **Auditing the fix.** Phase M found that the Phase L shrinkage — the centrepiece of the correction — was not working, because of an assumption in its *inputs*.
155. **Publishing the least flattering finding as the headline (#43),** and rewriting the project's own description to match the evidence.
156. **Encoding each finding as a test** so it cannot silently regress.

A perfect first implementation would be less informative than this record.

PART XXV — INTERVIEW PREPARATION

108. The thirty most important questions

Each entry gives a **deep answer**, a **short answer** (for when you have 30 seconds), the **common trap**, and the **repository evidence**.

Q1. What does this project do?

Deep. It recommends a shelf price for a given product, store and week that maximises expected gross profit subject to explicit business and scientific guardrails, on the Dominick's Finer Foods Cereals scanner panel — 4,707,776 UPC × store × week observations, 489 products, 93 stores, 366 weeks, 1989–1997. The pipeline runs from an official download through a validated canonical table, a decision-time feature contract, a demand forecaster, a separately estimated elasticity, a hybrid price response, a constrained grid optimizer, a risk gate, an append-only audit log, a FastAPI service and a nine-page dashboard. It was covered by 519 tests at Phase N and twelve audit reports that measure how much of the answer the model actually supplies.

Short. A price-recommendation engine on real retail scanner data, with the scientific limits measured rather than asserted.

Trap. Saying "I used ML to optimise prices." That claim is false for this system — 2.9% of final recommendations come from an interior model optimum.

Evidence. `README.md`, `reports/11_CONSTRAINT_ATTRIBUTION.md`.

Q2. Why Dominick's rather than a Kaggle retail dataset?

Deep. It is one of very few publicly documented retail scanner panels that carries store-level weekly **price**, **unit movement**, **promotion coding** and **gross margin** simultaneously. The margin field is decisive: without a cost you cannot build a gross-profit objective, and revenue optimization on an elastic category prices toward cost and loses money. The store dimension makes zone pricing (and therefore endogeneity) visible instead of hidden. And it is academically documented, so its quirks can be justified rather than invented.

Short. It is the only public panel with price, units, promotions **and** cost — and cost is what makes gross profit possible.

Trap. Answering "it's big" or "it's real". Size is not the reason.

Evidence. `docs/DATA_SOURCE.md`, `DECISIONS.md` #1.

Q3. What is `effective_unit_price` and why not just use `price`?

Deep. `price` is the price of the **bundle** scanned; `qty` is how many items the bundle contains; `move` counts **individual units**. So `price × move` is wrong whenever `qty > 1`. The correction is `price / qty`. Only 0.105% of rows are bundles (4,946 of 4,707,776), but for those the unit price would be overstated by **2–4×**, and a 4× price error propagates into the elasticity, the implied cost, the objective and the recommendation.

Short. `price` is a bundle price; dividing by `qty` gives the price a customer pays per unit. It affects 0.1% of rows by a factor of 2–4×

Trap. Dismissing it as negligible. The frequency is low; the magnitude is not, and it is a silent error.

Evidence. `docs/DATA_DICTIONARY.md`, validation check `formula_effective_unit_price` (max deviation 0.000e+00), `DECISIONS.md` #3.

Q4. Why is the cost column called `estimated_unit_aac`?

Deep. `profit` is a gross-margin percentage computed on **Average Acquisition Cost** — an inventory-accounting measure of what the retailer paid on average for stock on hand. It is not necessarily the economically relevant replacement cost: forward buying, inventory ageing and trade-fund accounting all drive a wedge. In this panel the median within-series AAC coefficient of variation is **0.088**, and AAC falls **9.16%** on price cuts — so it moves, and it moves with price. Calling it `true_unit_cost` would make every downstream gross-profit number sound more authoritative than the evidence supports.

Short. It is an *accounting* average cost, not a replacement cost, and it lags. Naming it precisely stops the gross-profit figures being over-read.

Trap. Treating cost as a known constant. It is an estimate with measured volatility, and the optimum is linear in it.

Evidence. `DECISIONS.md #4`, `artifacts/metrics/eda_summary.json` (`aac_stability`, `margin_mechanics`).

Q5. You dropped 26.6% of the raw data. Justify it.

Deep. Those rows have `price = 0`, so no unit price can be derived and the decision variable of the whole project is undefined. The exclusion was audited rather than assumed: of **1,851,380** zero-price rows, only **677** — 0.037% — recorded any sales; **73.4%** sit in leading or trailing runs (the item was not yet carried, or had been delisted); and there is **not a single row** with a positive price and zero sales. Imputing a price would fabricate the optimised variable. The resulting selection tilt is measured and published: the retained series sell 21.0 mean weekly units against 3.1 for heavily excluded series.

Short. No price means no unit price; 99.96% of them had no sales anyway; 73% are before-first-listing or after-delisting. The tilt toward continuously stocked series is measured and documented.

Trap. Saying "they were bad data". You must know the counts and the positional evidence, and you must volunteer the selection bias.

Evidence. `reports/09_ZERO_PRICE_AUDIT.md`, `artifacts/metrics/zero_price_audit.json`.

Q6. How do you know there is no leakage?

Deep. Twelve enumerated channels, each with a control and a **failing-if-broken** test: outcome columns are excluded by a module-level constant; rolling windows are computed on the already-shifted series; the reference price is an expanding median of shifted prices; the split is chronological; rows without history are dropped rather than imputed; one shared `recompute_price_features` implementation prevents stale price features; cost is held fixed across the grid; selection uses validation only; the backtest scores only post-training weeks; and the elasticity is fitted on training weeks only, with the script aborting otherwise. On top of the unit tests, a **poisoning audit on the real 4.7M-row panel** corrupts future costs and asserts nothing changes at or before the decision — and that it *does* change the following week, so the test has teeth.

Short. Twelve channels, each with a test that fails if it breaks, plus a poisoning audit on the real data: 4,671,333 rows compared, 0 mismatches, and 0 of 19 real recommendations changed by poisoned future costs.

Trap. Answering "I used a time-based split". That is one channel of twelve.

Evidence. `docs/DATA_LEAKAGE_AUDIT.md`, `tests/test_features.py`, `tests/test_cost_leakage.py`, `reports/10_COST_LEAKAGE_AUDIT.md`.

Q7. Why WAPE rather than MAPE or RMSE?

Deep. MAPE is undefined at zero demand and explodes on small denominators — fatal on a panel whose median weekly demand is 8 units. RMSE is dominated by promotional spikes (the RMSE/MAE ratio here is 8.5,

and one backtest week has RMSE 168.8 against MAE 8.2). WAPE — total absolute error divided by total actual units — is volume-weighted, so it reflects the store-weeks that matter commercially, and it is well defined when individual weeks are zero. All five metrics are reported; WAPE is the headline and the selection criterion.

Short. MAPE breaks at low volume, RMSE is dominated by promotion spikes. WAPE is volume-weighted and well defined. All five are reported anyway.

Trap. Not knowing what the number means in business terms. **0.4565 means total absolute error equals 45.65% of total actual units.**

Evidence. docs/METRICS.md, artifacts/metrics/evaluation.json.

Q8. Why did the linear model beat gradient boosting?

Deep. Five reasons, in order. (a) The gap is small — 0.4135 vs 0.4226, a 2.2% relative difference; the honest first statement is that they are roughly equal. (b) Retail demand is close to multiplicative, so $\log(1+Q) \sim \log(P) + \log\text{-lags}$ is near the correct functional form rather than an approximation. (c) The feature set already encodes the interactions a tree would discover — `price_vs_last_week`, `price_vs_series_reference` and `price_vs_recent_mean` are ratios, i.e. explicit price $\times$ history interactions, and `upc_demand_prior` encodes the product-level scale. (d) Ridge's uniform shrinkage across the collinear price block is close to the right prior on a very noisy panel. (e) There is no evidence of strong threshold effects in this mature category. And the tie-breaker that mattered more than accuracy: the boosted model's implied elasticity is **positive in 7% of contexts** and spans -10.1 to -0.39 .

Short. Feature engineering transferred the booster's edge into the linear model — and the booster's price response points the wrong way 7% of the time, which is disqualifying for a pricing engine.

Trap. "Linear models are more interpretable." True but secondary; lead with the functional form and the price-response safety.

Evidence. artifacts/metrics/model_metrics.json (implied_elasticity per model), reports/04_MODEL_COMPARISON.md.

Q9. Your naive elasticity is -0.35 and your controlled one is -2.42 . Explain.

Deep. Composition. Different cereals sit at different price *levels* for reasons unrelated to elasticity — brand, size, ingredient cost. Pooling them, the regression reads between-product differences as a price response. Adding UPC fixed effects moves it from -0.348 to **-2.289** (a factor of 6.6); adding store fixed effects to **-2.419** . Then adding promotion, seasonality and trend moves it **back** to -1.909 , because promotion bundling biases in the opposite direction. The naive R^2 is **0.0126** — log price alone explains 1.3% of the variance in log demand, so the naive coefficient is a slope through a cloud.

Short. The naive number is composition across products and stores. Fixed effects remove it ($-0.35 \rightarrow -2.42$); promotion controls then pull back (-1.91). The two biases have opposite signs.

Trap. Only naming one bias. The direction of the M4 correction is the part that shows you understand the mechanism.

Evidence. reports/03_ELASTICITY_ANALYSIS.md, artifacts/metrics/elasticity.json.

Q10. Is your elasticity causal?

Deep. No. The engine estimates $P(Q \mid \text{price}, X)$; pricing needs $P(Q \mid \text{do}(\text{price}))$. Prices were set by the retailer in response to demand conditions, not randomised. The evidence that they differ is internal to the project: a seven-fold change in the headline coefficient produced purely by adding controls. The specific threats are promotion bundling (-2.66 on promotion weeks vs -1.83 without), incomplete promotion coding (94.55% of rows carry no code, plus two undocumented codes), retailer anticipation, seasonality, zone pricing, product

heterogeneity including 5.1% of UPCs with a *significant positive* coefficient, invisible stock-outs and unobserved competitor prices. The experiment that would settle it is fully designed and was not run.

Short. No. It is observational, the confounders are named and quantified, and the experiment that would fix it is designed in `docs/PRICING_EXPERIMENT.md`.

Trap. Hedging with "well, sort of, with fixed effects". Fixed effects remove time-invariant confounders, not anticipation or unobserved promotion.

Evidence. `docs/CAUSAL_LIMITATIONS.md`, `docs/PRICING_EXPERIMENT.md`.

Q11. Why did you separate forecasting from price response?

Deep. The fitted forecaster's own implied elasticity, measured by central finite difference over 300 real contexts, is **-3.102** — outside the two-way clustered 95% interval of every controlled estimate (`[-2.236, -1.821]` for the pooled pricing estimate). If the truth is `-2.0` and the model believes `-3.1`, then at a 10% price increase it expects to lose 26.5% of volume where the truth is 17.4% — systematic pessimism about price increases, baked into every recommendation and invisible to any accuracy metric. So the two jobs were split: the ML model answers "where is demand now, given all context?" (test WAPE 0.4565), and a separately estimated elasticity answers "how does demand move with price?".

Short. The best forecaster is not necessarily the best intervention model. Its implied elasticity was `-3.10` against a controlled `-2.03`, so the price counterfactual was moved to an estimate whose provenance and precision are known.

Trap. Framing it as "the model was bad". It was the *best* forecaster in the comparison — that is exactly the point.

Evidence. `reports/05_PRICE_RESPONSE_VALIDATION.md`, `src/pricing_engine/models/hybrid.py`, `DECISIONS.md` #31.

Q12. Write the hybrid equation and say precisely what it preserves.

Deep. $Q(p) = \hat{Q}(p_0) \cdot (p/p_0)^\epsilon$, where p_0 is the reference (current) price stamped from the observed row before any candidate is substituted, $\hat{Q}(p_0)$ is the ML forecaster's prediction with **every** price-dependent feature recomputed at p_0 , and ϵ is the separately estimated elasticity. At $p = p_0$ the ratio is 1, so the hybrid reproduces the base model's prediction **exactly** — pinned by 15 parametrised tests over 5 elasticities $\times$ 3 price levels. What it does **not** claim: accuracy at any other price. Away from p_0 the curve is the elasticity model's, not the forecaster's, and its counterfactual accuracy is a separate empirical question, measured out of time in `reports/16`.

Short. $Q(p) = \hat{Q}(p_0)(p/p_0)^\epsilon$. It preserves the forecaster's baseline prediction **at the reference price** — not its accuracy at counterfactual prices.

Trap. Saying "no forecast skill is lost". That claim was **removed from this repository** in Phase M precisely because it is a statement about all prices and the equality holds at one.

Evidence. `src/pricing_engine/models/hybrid.py` docstring, `DECISIONS.md` #51, `tests/test_phase_1_pricing.py`.

Q13. Explain the empirical-Bayes shrinkage.

Deep. A normal-normal hierarchical model: $\epsilon_i \mid \theta_i \sim N(\theta_i, se_i^2)$ and $\theta_i \sim N(\mu, \tau^2)$. The posterior mean is $\hat{\theta}_i = w_i \epsilon_i + (1-w_i) \mu$ with $w_i = \tau^2 / (\tau^2 + se_i^2)$. τ^2 is the between-product variance of *true* elasticities, estimated by REML around the pooled prior; μ is the pooled controlled elasticity **-2.029**, which is also the fallback for products with no usable estimate, so the two are consistent. Shipped values: $\tau^2 = 0.766$, τ

= 0.875, median robust SE = 0.390, **mean weight 0.780**, 239 usable products of 372. No product has a weight above 0.99 or below 0.25.

Short. Precision-weighted blending of each product's own noisy estimate with the category estimate: $w = \tau^2 / (\tau^2 + se^2)$. $\tau^2 = 0.766$, mean weight 0.780 across 239 of 372 products.

Trap. Reciting the formula without knowing that **the standard errors are half of it**. See Q14.

Evidence. `src/pricing_engine/economics/elasticity_store.py`, `reports/14_SHRINKAGE_AUDIT.md`, `tests/test_shrinkage.py`.

Q14. Something went wrong with that shrinkage. What was it?

Deep. The mean shrinkage weight was **0.954** — essentially no shrinkage. The audit ran seven estimator variants on identical data and found that changing the τ^2 estimator (moment $\rightarrow$ REML) or the prior mean (sample $\rightarrow$ pooled) moved the mean weight by **less than 0.001**, while changing the standard errors alone moved it from 0.955 to 0.78. **The formula was correct; the inputs were wrong.** HC1 standard errors on a panel with within-store persistence and category-wide weekly shocks were too small by a median factor of **3.79**, and understated se inflates $w = \tau^2 / (\tau^2 + se^2)$ from *both* directions — se^2 shrinks directly, and τ^2 is estimated as observed dispersion minus average sampling variance, so it inflates too. Corrected: τ^2 0.906 $\rightarrow$ **0.766**, mean weight 0.954 $\rightarrow$ **0.780**. The failure mode is now a regression test.

Short. The estimator was fine; its standard errors were understated $\sim 3.8\times$, which silently pushed every weight toward 1. Fixing the inference moved the mean weight from 0.954 to 0.780.

Trap. Not knowing that under-stated SEs inflate w from both directions. That is the part that shows you understand the estimator.

Evidence. `reports/14_SHRINKAGE_AUDIT.md` (the seven-variant table), `reports/15_ELASTICITY_INFERENCE_AUDIT.md`.

Q15. Why do standard errors need clustering here, and by how much?

Deep. Three correlated dimensions exist structurally: within-store persistence (local demand shocks last months), category-wide weekly shocks (holidays, weather, the chain's promotion calendar), and within-product persistence. Classical SEs assume none of them. Measured on the pooled specification: HC1 is $1.78\times$ classical; UPC $\times$ store panel clustering $2.35\times$; store $4.67\times$; week $12.46\times$; UPC $12.84\times$; **two-way (UPC, week) $16.24\times$** . The published interval goes from $[-2.042, -2.016]$ to **$[-2.236, -1.821]$** . At the product level the median inflation is $3.38\times$ and the share significant at 5% falls from 90.1% to **75.2%**. The estimators were validated against statsmodels to machine precision.

Short. Store persistence and category-wide weekly shocks. Two-way clustered SEs are **$16\times$** the classical ones; one product in six loses its significance.

Trap. Saying "I used robust standard errors". HC1 is robust and is still $1.78\times$ too small here — the clustering is the point, not the heteroskedasticity.

Evidence. `reports/15_ELASTICITY_INFERENCE_AUDIT.md`, `src/pricing_engine/economics/inference.py`.

Q16. Why hold cost fixed across the candidate grid?

Deep. The tempting shortcut is to reuse the observed gross-margin *percentage* at a new price. That makes the implied cost scale with the price: $c(p) = p(1-m)$, so $GP(p) = p \cdot m \cdot Q(p)$ — gross profit becomes proportional to revenue, the gross-profit objective silently collapses into a revenue objective, and the model never sees the margin expansion a real price increase produces. Every recommendation would be wrong and the numbers would look plausible. The correct method is $GP(p) = (p - c) \cdot \hat{Q}(p)$ with c a decision-time estimate held fixed.

Economically this is right too: within a week's pricing decision the acquisition cost is already determined — the shelf price does not change what was paid. It is enforced by a test.

Short. Reusing the margin percentage mechanically manufactures profit and turns the objective into revenue. Cost is a constant of the decision, held fixed and tested.

Trap. Not noticing that this failure produces plausible-looking numbers.

Evidence. `src/pricing_engine/simulation/counterfactual.py`,
`tests/test_simulation.py::test_cost_is_held_fixed_across_price_grid`, `DECISIONS.md` #21.

Q17. Why a grid search instead of `scipy.optimize`?

Deep. Six reasons. Retail prices are genuinely discrete, so optimising on the grid the price will actually take avoids rounding a continuous answer outside a constraint. The architecture supports a tree model whose surface is piecewise constant, so gradients are zero almost everywhere. Materiality, the risk gate, rounding and charm pricing are thresholds and state machines, not smooth penalties. A grid makes every candidate's score available, which is what makes reason codes and "why not \$3.71?" answerable. The problem is one-dimensional over a bounded interval, so exhaustive search is **globally optimal by construction** — no local minima, no initialisation, no convergence criterion. And the whole grid is one batched model call. The cost is measured: the grid optimum sits a median of **\$0.025** from the continuous optimum in the same interval, about 1% of price.

Short. Discrete prices, a possibly-step-function model, non-smooth business rules, and full auditability — for a one-dimensional bounded problem where exhaustive search is globally optimal and costs a median \$0.025 of quantisation.

Trap. Implying grids are a shortcut. Here they are strictly better, and the quantisation cost is quantified.

Evidence. `DECISIONS.md` #19, `artifacts/metrics/constraint_attribution.json`
(`median_rounding_distance_dollars`).

Q18. How do you know the optimizer is correct?

Deep. Closed-form ground truth. For linear demand $Q = a - bp$ the profit-maximising price is $(a+bc)/(2b)$ and the revenue-maximising price is $a/(2b)$. On $Q = 100 - 10p$, $c = \$2.50$ the fixtures give \$6.25 and \$5.00, and both are recovered within one \$0.05 grid step. Their difference — \$1.25 — is exactly $c/2$, the theoretical gap. Additional fixtures cover the inelastic case (optimum at the upper guardrail, `PRICE_CHANGE_LIMIT` raised), starting at the optimum, sub-materiality, the hybrid reproducing $Q_0(p/p_0)^\epsilon$ exactly, and the shrinkage weights matching the closed form. What these do **not** prove is that the demand curve is right — that is a different problem.

Short. Against closed-form optima: $(a+bc)/(2b) = 6.25$ and $a/(2b) = 5.00$ on a linear fixture, both recovered within one grid step, plus the inelastic boundary case.

Trap. Saying "I wrote unit tests". Say *what the ground truth was*.

Evidence. `tests/test_optimizer.py`, `src/pricing_engine/optimization/objective.py`.

Q19. How much of the final recommendation comes from the model?

Deep. 2.9%. Over all 13,964 week-399 decision contexts under the default policy: 2.9% of final recommendations come from an interior optimum of the estimated price response; **57.1%** sit on a guardrail corner; 25.3% are screened out before optimisation; 9.7% are risk-gated; 5.1% fail materiality. The constraint that binds first is the $\pm 10\%$ price-change cap, in **44.6%** of contexts. The unconstrained optimum wants a median **+49.0%** price change; the system recommends **+3.7%**. This is a rule-bounded pricing system with a learned direction, and the README says so.

Short. 2.9%. The rest are guardrail corners, screened-out contexts, risk gates and materiality. This is rule-bounded pricing with a learned direction.

Trap. Being defensive. This finding is the project's strongest evidence of scientific honesty — lead with it.

Evidence. `reports/11_CONSTRAINT_ATTRIBUTION.md`, `artifacts/metrics/constraint_attribution.json`.

Q20. Then why build the model at all?

Deep. Five parts. (1) Concede the magnitude point honestly. (2) **Direction** — a no-model rule can only ever raise prices; the engine proposes cuts whenever the estimated optimum is below current (8.0% of actionable recommendations under `shrunk`, 26.7% under `m1`). (3) **Eligibility and refusal** — 40.1% of contexts are actively *not* actioned (3,528 screened out, 1,353 risk-gated, 716 immaterial), and every refusal is justified by a model artefact. (4) **The tail** — p90 price variation across elasticity scenarios is still 14.5% under the standard profile, concentrated in the well-identified, high-volume series. (5) **The economics** — the model supplies predicted units, revenue, gross profit and the whole demand curve, which is what a reviewer actually reads.

Short. The model contributes direction, eligibility, refusal and the economics of the decision. The rules contribute magnitude. 40% of the engine's output is a justified refusal, which no rule can produce.

Trap. Over-claiming after conceding. The concession is what makes the defence credible.

Evidence. `reports/12_MODEL_VALUE_ABLATION.md`, `reports/13_GUARDRAIL_ABLATION.md`, `STATUS.md` open issue 1.

Q21. A rule with no model matches your engine 82.5% of the time. Is the model useless?

Deep. "Always take the maximum allowed increase" lands within one 5-cent grid step of the engine's final price in **82.5%** of contexts and agrees on the decision state in **90.0%**. That is the same fact as Q19 seen from the other side: the model's unconstrained optimum sits above the change cap in 86.7% of contexts, so the constrained answer is "go to the cap" — which is what R4 does by construction. But R4 can never propose a **cut**, has no concept of eligibility, risk or materiality, and produces no economics. And critically: the ablation compares **decisions**, never profits, because scoring a rule with the model's own demand curve and declaring the model better would be circular. Nobody knows which earns more; that requires the three-arm experiment.

Short. It matches the price 82.5% of the time and the decision 90%. It cannot cut prices, cannot refuse, and cannot say why. Which earns more is untested — deliberately, because measuring it offline would be circular.

Trap. Claiming the model earns more. The repository explicitly refuses to make that comparison.

Evidence. `reports/12_MODEL_VALUE_ABLATION.md` §1 ("Not the question").

Q22. What does the +8.30% uplift mean?

Deep. It means: *if the fitted price-response model is correct*, applying the recommended prices instead of the current ones on 3,000 week-399 contexts would produce 8.30% more gross profit — **as computed by that same model**. Portfolio expected gross profit \$37,580.82 → \$40,700.96. It is **circular**: the same fitted response both proposes and scores the candidate prices. This is not fixable offline — cross-validation validates predictions at observed prices; a held-out set contains the retailer's prices, not the policy's; off-policy estimators need a known behaviour policy with positive propensity over the evaluated actions, and the retailer's rule is unknown and effectively deterministic. The bias cannot even be signed, because it depends on the direction of the model's elasticity error.

Short. A model-internal estimate: the model that chose the prices also scored them. Circular by construction, unfixable offline, and named `model_internal_estimated_profit_uplift_pct` so the caveat cannot be dropped.

Trap. Quoting it as a business result. Never do this.

Evidence. `artifacts/metrics/recommendations.json`, `reports/06_BACKTEST.md`, §64 of this report.

Q23. What *can* you verify offline?

Deep. Demand accuracy on a chronologically held-out window (test WAPE 0.4565 on 749,040 rows, scored once); the 37.7% relative improvement over the best naive baseline; accuracy decay through time (0.409 → 0.528 across five quarters); optimizer correctness against closed-form optima; constraint behaviour across 13,964 contexts × 11 runs; decision-state invariants (0 violations); absence of leakage including a poisoning audit on real data; the elasticity estimation window; and the out-of-time price-response ranking on 154,899 realised price-change episodes. What cannot be verified: true counterfactual demand at any unobserved price, the uplift figures, the causal effect, whether the model beats the rule, and whether the risk bands are calibrated.

Short. Everything about the machinery. Almost nothing about the economics. The one partial exception is the out-of-time validation on 154,899 real price changes.

Trap. Blurring the two categories. The distinction *is* the answer.

Evidence. §65 of this report; `reports/VALIDATION_SUMMARY.md`.

Q24. What is your out-of-time validation and why does it not solve the causal problem?

Deep. 154,899 real price-change episodes in weeks 258–399 — weeks the elasticity estimator never saw. For each, the contextual baseline is the ML model's prediction at the *previous* price using week t 's context, and each candidate response method predicts the demand that followed. `shrunk` has the lowest WAPE in every split: 0.585 all episodes, **0.560** non-promotion, 0.590 promotion — against native ML 0.604/0.588/0.607, pooled 0.625/0.594/0.631, and a no-price-response null of 0.809/0.715/0.826. That evidence is why `shrunk` is the default. But it is **not causal**: weeks in which the retailer changed price are not a random sample of weeks, and 73.3% of episodes carry a promotion code in at least one of the two weeks. A method can win here by correctly predicting a confounded association.

Short. 154,899 real price changes on unseen weeks; `shrunk` wins every split. It validates a *predictor* of the association, not a causal effect, because price changes are not randomly assigned.

Trap. Presenting it as causal validation. Its own report opens with a boxed warning that it is not.

Evidence. `reports/16_OUT_OF_TIME_PRICE_RESPONSE.md`.

Q25. Is elasticity stable over time?

Deep. Category yes, product no. Re-estimating the whole stack on five windows inside the training period: the pooled elasticity ranges over 0.420 between the most extreme windows, but the largest pairwise z-statistic between **disjoint** windows is **1.36** — not significant once panel clustering is admitted. So a stable category elasticity around -2 is defensible. Product-level ordering is not: the median Spearman rank correlation of `shrunk` product elasticities between disjoint windows is **+0.19**, with median absolute differences of 0.70–0.95 — roughly the size of τ itself. Sign stability is 100%. Note that under classical standard errors the category test would have shown violent significant drift; the covariance assumption changed the scientific conclusion.

Short. Category elasticity is stable (max pairwise z 1.36). Product-level ordering is not (rank correlation +0.19). So the category number is an asset; a product-specific elasticity is not.

Trap. Claiming product-level elasticities are a durable asset. They are not, and the project says so.

Evidence. `reports/17_ELASTICITY_STABILITY.md`.

Q26. How does the risk layer work, and is it calibrated?

Deep. Three bands over observation count, distinct prices and extrapolation distance: LOW needs ≥ 150 observations, ≥ 15 distinct prices and $\leq 2\%$ extrapolation; MEDIUM needs ≥ 80 , ≥ 8 and $\leq 5\%$; anything else is HIGH. Two additional demotions: a series where $>50\%$ of weeks carry a promotion code, or a product using the pooled elasticity fallback, cannot be LOW. Risk is assessed twice — before optimisation at the current price, and again at the proposed price. HIGH is never actionable under either default profile: **0 of 4,760** HIGH-risk contexts auto-changed a price. **It is a heuristic, not calibrated confidence.** The thresholds are asserted; calibrating them would mean fitting realised out-of-sample error against the bands, which was not done — and it is listed as an open issue.

Short. A transparent rule over history, price variation and extrapolation distance, assessed twice, with HIGH never auto-actionable. Explicitly a heuristic — the thresholds are asserted, not fitted.

Trap. Calling it "confidence". The project deliberately does not.

Evidence. `src/pricing_engine/optimization/risk.py` docstring, `configs/config.yaml`, `STATUS.md` open issue 7.

Q27. What are the three decision states and why three?

Deep. `RECOMMEND_CHANGE` (actionable — the final price is the proposal), `KEEP_CURRENT` (leave it alone), `REVIEW_REQUIRED` (a proposal exists but a human must approve; the final price stays where it is). Three rather than two because a binary split throws information away: a HIGH-risk context can deserve a human's attention, and collapsing it into "keep" discards a real signal while collapsing it into "change" acts on evidence that does not support it. Phase M also split `recommended_price` into `proposed_candidate_price` and `final_recommended_price`, and forces `realisable_profit_uplift_pct` to 0 when not actionable — so a consumer that forgets to read the `actionable` flag cannot display an unauthorised price or sum an unrealisable uplift. Six invariants are asserted over all 13,964 contexts: **0 violations** under both default profiles.

Short. Change, keep, and review-required. The third exists because "the model found something but the evidence is thin" is real information. Proposal and final price are separate fields, with zero invariant violations across 13,964 contexts.

Trap. Not knowing why the proposal/final split exists.

Evidence. `reports/19_DECISION_STATE_AUDIT.md`, `DECISIONS.md` #47, `tests/test_decision_states.py` (246 tests).

Q28. How would you prove this makes money?

Deep. A randomised experiment. Unit of randomisation: **store × category block**, not individual UPCs — because shoppers substitute within cereal and a "control" product would absorb the treated product's volume. Stratified assignment on pre-period gross profit, store size and price zone. 12 weeks minimum plus a 4-week pre-period, policy frozen throughout. Primary metric: **gross profit per store-week** for the treated block, one primary test. Guardrails with pre-registered stopping rules: complaints per 1,000 transactions, out-of-stock rate, traffic, basket size. Primary estimator: difference-in-differences with store and week fixed effects and **standard errors clustered by store**. And the most interesting design is **three arms** — control, rule-only (R4), full engine — because the B-vs-C contrast is what answers "is the model worth building?".

Short. Randomise whole stores into treatment and control for a category block, 12 weeks, primary metric gross profit per store-week, DiD with store-clustered SEs. Add a rule-only third arm to test whether the model is worth it.

Trap. Randomising products within a store — cannibalisation destroys the comparison.

Evidence. docs/PRICING_EXPERIMENT.md.

Q29. What is the weakest part of this project?

Deep. Counterfactual policy evaluation. The backtest's policy economics are circular by construction — the same fitted response proposes and scores the prices — and it is unfixable offline. It is weaker than the causal limitation because the causal limitation is *disclosed and bounded*, whereas the uplift figure is **the number a reader will quote**. Everything downstream — any ROI calculation, business case or CV bullet — inherits the circularity. The project mitigates it comprehensively (the metric is named `model_internal_estimated_*`, disclaimers travel in the API schema and every dashboard page, the model-value ablation refuses to compare profits, and `audit_claims.py --strict` blocks "actual uplift" wording) but mitigation is not a fix. Only an experiment is.

Short. Counterfactual policy evaluation. The uplift figures are self-scored and circular, and that is the number people remember. Only an experiment fixes it.

Trap. Naming something trivial, or naming the causal limitation without explaining why the circularity is worse.

Evidence. §64 and §86 of this report; KNOWN_LIMITATIONS.md #4, #27.

Q30. What is the strongest part?

Deep. The audit suite, and within it the inference → shrinkage → policy chain. The demand model is competent and unremarkable. What is rare is building a second, independent apparatus to measure how much of the answer the first one supplies, finding that it is **2.9%**, and publishing that as the headline. The single strongest thread: question whether HC1 standard errors are right on a panel with within-store persistence and category-wide weekly shocks → measure seven covariance assumptions, validated against statsmodels → find two-way clustering is **16×** classical → find the share of significant products falls from 90.1% to 75.2% → discover that the understated SEs had **silently disabled the empirical-Bayes shrinkage** → isolate the cause with seven estimator variants showing the formula was right and the inputs wrong → correct it (0.954 → 0.780) and remove the old figure everywhere → encode the failure mode as a regression test → re-run the stability audit with corrected inference and find the scientific conclusion *changes*.

Short. The audit apparatus. Especially the chain where an inference assumption turned out to have silently disabled the shrinkage, and fixing it changed both the numbers and a scientific conclusion.

Trap. Naming the model. It is the least distinctive part of the project.

Evidence. reports/14, reports/15, reports/17, reports/11.

Bonus — five rapid-fire

question	answer
What is $p^* = c \cdot \epsilon / (1 + \epsilon)$?	the profit-maximising price under constant elasticity, defined only for $\epsilon < -1$; for $-1 < \epsilon < 0$ profit rises without bound and the answer is set by constraints
Why does the demand forecast not choose the price?	$\hat{Q}(p_0)$ is constant across candidates, so it cancels from $\arg\max (p - c)Q(p)$. Verified: scaling the base model 10× leaves every recommended price unchanged
Why 3 policy profiles?	to make the sensitivity of the output to policy explicit and measurable; aggressive exists to demonstrate what the risk gate prevents (4,717 HIGH-risk auto-changes)
What is the biggest missing mechanism?	cross-price / cannibalisation. The engine prices each UPC × store independently, so the category effect of re-pricing many substitutes is not the sum of the individual effects
What would you do next?	(1) add a competitor price index; (2) model promotion <i>depth</i> , not just presence; (3) switch the batch script to

question	answer
	simulate_many; (4) commit the repository so artifacts have code lineage; (5) run the three-arm experiment

109. Five-minute project explanation

I built an end-to-end price-recommendation engine on the Dominick's Finer Foods Cereals scanner panel — real store-level retail data from the Kilts Center at Chicago Booth. Ninety-three stores, 489 products, 366 weeks, 1989 to 1997. About 6.6 million raw rows became 4.7 million validated observations at a UPC-by-store-by-week grain, with every exclusion counted in an audit ledger and fifteen formula and structural checks passing.

The question the system answers is: given a product, a store, a week and a feasible price range, what price maximises expected gross profit without taking risks the evidence cannot support?

On the modelling side, *I predict weekly units — never revenue or profit, because those are functions of the target and using them is leakage. Money is derived after prediction, which is also what makes the counterfactual coherent. The split is strictly chronological: train weeks 2 to 257, validate 258 to 342, test 343 to 399. I compared four naive baselines, a ridge log-log regression and a gradient-boosted Poisson model. Ridge won on validation WAPE, 0.4135 against 0.4226, and scored 0.4565 on the test window — about a 38% relative improvement over the best naive baseline.*

Then I found the interesting problem. *I measured the fitted model's own price response by finite difference across 300 real decision contexts. Its implied elasticity was minus 3.1. My controlled econometric estimates were minus 1.9 to minus 2.4, and the pooled pricing estimate was minus 2.03 with a two-way clustered confidence interval of minus 2.24 to minus 1.82. The forecaster's price response was outside the interval of every controlled estimate. The demand curves were fine — a hundred percent monotone decreasing — the model just had the slope wrong. That is the lesson: the best forecaster is not necessarily the best intervention model.*

So I separated the two jobs. *The pricing layer uses a hybrid: $Q(p)$ equals the ML model's baseline prediction at the current price, times the price ratio raised to a separately estimated elasticity. The forecaster supplies context — seasonality, promotion state, recent demand. The elasticity supplies the price response, and it is fitted on training weeks only, so an elasticity used to price week 399 never saw week 399's outcome. Per-product estimates are shrunk toward the category estimate by empirical Bayes, with weights τ^2 over τ^2 plus the squared standard error. 239 of 372 products get their own estimate; the rest use the category number and are flagged.*

The optimizer *is a grid search over discrete candidate prices — because retail prices are discrete, the constraints are not smooth, and I wanted every candidate's score available for reason codes. It is validated against closed-form optima: for linear demand the profit optimum is $(a+bc)/(2b)$, and*

it recovers 6.25 on a fixture where the truth is 6.25. Six guardrails are intersected into one feasible interval, and a heuristic risk layer gates anything with thin evidence: HIGH-risk contexts never auto-change a price — zero of 4,760 did.

Then I audited the whole thing, and this is the part I would lead with. I asked: when this engine recommends a price, does the model choose it, or is it just the nearest point the guardrails allow? I ran all 13,964 decision contexts of the final week through the pipeline eleven times, leave-one-out over every constraint. **Two point nine percent** of final recommendations come from an interior optimum of the estimated price response. Fifty-seven percent are guardrail corners. A rule that never looks at demand — "always take the maximum allowed increase" — lands within one five-cent grid step of my engine's price 82.5% of the time.

So I rewrote the project's own description. This is **rule-bounded pricing with a learned direction**, not ML pricing. The model contributes direction, eligibility, refusal — 40% of contexts are actively not actioned — and the economics of the decision. The rules contribute magnitude.

Two more corrections mattered. I was using HC1 standard errors on a panel with within-store persistence and category-wide weekly shocks. Two-way clustering by product and week made them **sixteen times** larger, and the share of products significant at 5% fell from 90 to 75 percent. That correction also revealed that my empirical-Bayes shrinkage had been silently doing nothing — understated standard errors push every weight toward one from both directions. Mean weight went from 0.954 to 0.780. An inference bug had become a modelling bug.

What I will not claim. Nothing here is causal. Prices were set by the retailer, not randomised, and my naive elasticity of minus 0.35 versus minus 2.42 with fixed effects shows exactly how contaminated the raw association is. The uplift figures — plus 8.3 percent portfolio gross profit — are **model-internal estimates**: the same fitted response proposes and scores the prices. That is circular and it is unfixable offline. The metric is literally named `model_internal_estimated_profit_uplift_pct` so the caveat cannot be dropped from a slide, and a strict claim-audit script fails the build if unsupported wording reappears.

The one thing that partly escapes it: I validated the price-response methods against 154,899 **real** price-change episodes in weeks the estimator never saw. The shrunk product elasticity has the lowest error in every split. That is why it is the default — on measurement, not inheritance.

It ships as a FastAPI service and a nine-page Streamlit dashboard, with 493 tests, clean lint, a Dockerfile and a CI workflow. Every recommendation is logged with its model version, elasticity, provenance, constraints, risk level and reason codes. And the whole thing is honest about what it is: a decision-support tool with measured limits, not an autonomous pricer.

110. One-minute project explanation

I built a price-recommendation engine on real retail scanner data — the Dominick's Cereals panel, 4.7 million validated store-week observations.

A ridge log-log demand model forecasts weekly units on a strict chronological split, reaching test WAPE 0.4565 against 0.73 for the best naive baseline. But I found the model's own implied price elasticity was minus 3.1 while every controlled econometric estimate said minus 1.9 to minus 2.4 — so I separated forecasting from price response. The pricing layer is a hybrid: the ML baseline at the current price, scaled by a separately estimated, empirical-Bayes-shrunk elasticity fitted on training weeks only. A grid optimizer validated against closed-form optima maximises expected gross profit inside six guardrails, and a risk gate routes thin-evidence contexts to human review — zero of 4,760 high-risk contexts ever auto-changed a price.

*Then I audited how much the model actually contributes: **2.9%** of final recommendations come from an interior model optimum; 57% are guardrail corners; a rule with no demand model matches my price 82.5% of the time. So I describe it as rule-bounded pricing with a learned direction, not ML pricing.*

Nothing is causal — prices were not randomised — and the uplift figures are model-internal by construction. That is stated in the metric name, the API schema, every dashboard page and an enforceable claim-audit script.

It shipped as a FastAPI service and a nine-page dashboard with 519 tests at Phase N.

111. Thirty-second version

I built a retail price-optimization engine on 4.7 million rows of real Dominick's scanner data: a demand model, a separately estimated price elasticity with panel-robust inference and empirical-Bayes shrinkage, a constrained optimizer validated against closed-form optima, and a risk gate that routes thin-evidence decisions to a human.

*The most interesting result is an audit I ran on my own system: only **2.9%** of its recommendations are actually set by the model — the rest are guardrail corners. So I call it rule-bounded pricing with a learned direction, not ML pricing.*

Nothing in it is causal, and I never report the uplift as a business result — the metric is literally named "model-internal estimated" so the caveat cannot be dropped.

PART XXVI — CLAIMS

The repository maintains an enforceable claim audit (`scripts/audit_claims.py --strict`): **0 UNSUPPORTED** — across 43 files and 162 occurrences before this report, and still 0 across 49 files and 363 occurrences once this report is included in the scan. This part restates the boundary in exact wording.

112. Claims that are fully supported

Each of these can be quoted verbatim.

Data

- `"Built a validated canonical panel of 4,707,776 UPC × store × week observations from 6,602,582 raw rows, with every exclusion counted and 15 / 15 formula and structural checks passing."`
- `"Coverage: 489 UPCs × 93 stores × 366 weeks, 1989-09-14 to 1997-05-01, totalling $262,008,582 observed revenue and $40,091,143 observed gross profit (15.30% blended margin)."`
- `"The 26.56% zero-price exclusion was audited: of 1,851,380 zero-price rows, only 677 recorded any sales and 73.4% sit in leading or trailing runs."`
- `"0 duplicate UPC × store × week rows; the metadata join is verified many_to_one with the row count asserted before and after."`

Modelling

- `"Selected model M1 ridge log-log, chosen on a rule fixed in advance — lowest validation WAPE among price-aware models — validation WAPE 0.4135, test WAPE 0.4565 (MAE 8.503, RMSE 72.064, bias −3.883) on 749,040 held-out rows scored once."`
- `"A 37.7% relative WAPE improvement over the best naive baseline (rolling-mean-4, test WAPE 0.7326)."`
- `"Model persistence is lossless: reloading the saved artifact and re-scoring the test window gives a WAPE difference of 0.00e+00."`
- `"Error is reported by segment: WAPE 0.374 on non-promotion rows versus 0.656 on recorded-promotion rows, with bias −1.13 versus −37.24."`

Leakage

- `"Twelve enumerated leakage channels, each with a control and an enforcing test."`
- `"Decision-time cost leakage audited on the real panel: 4,671,333 rows compared, 0 mismatches; a future-cost poisoning test changed nothing at or before the poisoned week and did change the following week; 0 of 19 real recommendations changed."`

Econometrics

- `"Observational elasticity ladder: naive pooled −0.348 → UPC fixed effects −2.289 → UPC × store fixed effects −2.419 → plus promotion, seasonality and trend −1.909."`
- `"Pooled pricing elasticity −2.029, two-way clustered (UPC, week) 95% interval [−2.236, −1.821], fitted on training weeks 2–257 only."`
- `"Two-way clustered standard errors are 16.24× the classical ones; the share of products significant at 5% falls from 90.1% to 75.2%. Estimators validated against statsmodels."`
- `"Empirical-Bayes shrinkage with REML  $\tau^2 = 0.766$ , mean weight 0.780 across the 239 of 372 products with a usable own estimate."`
- `"Category elasticity is stable across disjoint training windows (largest pairwise  $z = 1.36$ , not significant); product-level ordering is not (median rank correlation +0.19)."`

Optimization and policy

- "The optimizer recovers the closed-form optima $(a+bc)/(2b) = 6.25$ and $a/(2b) = 5.00$ within one \$0.05 grid step on a linear-demand fixture."
- "Over all 13,964 week-399 contexts, **2.9%** of final recommendations come from an interior optimum of the estimated price response; **57.1%** are guardrail corners; the $\pm 10\%$ change cap binds first in **44.6%**."*
- "A rule with no demand model matches the engine's final price within one 5-cent grid step in **82.5%** of contexts and agrees on the decision state in **90.0%**."*
- "Median recommended-price variation across four elasticity scenarios falls from **79.8%** (cost floor only) to **0.0%** once the $\pm 10\%$ change cap is added."*
- "**0 of 4,760** HIGH-risk contexts auto-changed a price; **0** decision-state invariant violations across 13,964 contexts under both default profiles."*

Validation

- "On **154,899** unseen price-change episodes in weeks 258–399, the shrunk product elasticity has the lowest WAPE in every split (**0.560** on non-promotion episodes versus 0.588 native ML, 0.594 pooled, and 0.715 for a no-price-response null). This is naturalistic prediction, not causal identification."*

Engineering

- "**519 tests pass at Phase N**, 0 failed, 0 skipped; ruff check . clean; all 9 dashboard pages render; the FastAPI service and the batch pipeline share one pricing model."* (Verified 2026-08-19.)
- "Every recommendation is written to an append-only audit log with its model version, elasticity and provenance, constraints, risk level and reason codes."

113. Claims that require qualification

Each of these is defensible **only** with its qualifier attached. The qualifier is not optional.

claim	required qualification
"portfolio gross-profit uplift of +8.30% "*	" model-internal estimated " — the same fitted price-response model proposed and scored the prices; it is an internal simulation, not a realised or unbiased policy value, and it is unfixable offline
"the ML policy delivers +10.45% gross profit versus historical pricing"*	" model-internal estimated ", on 400 contexts × 10 weeks, with the same circularity
"price elasticity of −2.029 "*	" observational estimate ", fitted on training weeks only, with a two-way clustered 95% interval of [−2.236, −1.821]; not a causal parameter
"the shrunk product elasticity is the best price-response method"	" lowest WAPE on 154,899 out-of-time price-change episodes " — predictive, not causal; 73.3% of episodes carry a promotion code
"239 products have their own elasticity"	...and 133 do not and use the pooled fallback — 35.8% of decision contexts
"risk level LOW / MEDIUM / HIGH"	" heuristic risk score, not calibrated statistical confidence "; the thresholds are asserted, not fitted
"the demand model achieves WAPE 0.4565"	at observed prices, on the test window; it says nothing about accuracy at counterfactual prices
"the hybrid preserves the forecaster's prediction"	at the reference price only — not at any other price
"the engine recommends a price"	only when the decision state is RECOMMEND_CHANGE ; REVIEW_REQUIRED and KEEP_CURRENT ship the current price
"the pipeline is validated"	15 structural checks and 519 tests pass at Phase N; " validated " refers to the machinery, not to the economics
"54.08% of series are eligible"	eligible for price optimization under the configured screen;

claim	required qualification
	the remaining 45.92% always receive keep-current
"the decision-time cost is leakage-safe"	temporal availability is proven; economic correctness of the AAC proxy is not
"Docker and CI are implemented"	the image was never built (daemon unavailable) and the workflow has never run (no remote, no commits)
"3,000 contexts scored"	a sample ; the full week is 13,964 contexts and the percentages differ slightly

114. Claims that must not be made

These are false, unsupported, or both, for this project as it stands.

✗ forbidden claim	why it is wrong	what to say instead
"delivers a 7% (or 8.3%, or 10.45%) profit increase"	the figures are self-scored by the model that chose the prices; nothing was applied or realised	"model-internal estimated uplift of +8.30% on the scored contexts"
"proves that lowering/raising prices causes demand to change by X"	prices were not randomised; $P(Q price) \neq P(Q do(price))$	"the observational estimate is -2.029, with the confounders named"
"a production-ready autonomous pricing system"	no orchestration, no registry, no monitoring store, no alerting, no auth, no serving SLOs, Docker never built, CI never run	"a decision-support demo with measured limits"
"finds the optimal price"	it finds the arg-max of an estimated demand curve on a discrete grid inside policy constraints ; the true optimum is unknown	"recommends the feasible price that maximises estimated gross profit"
"ML-driven pricing"	2.9% of final recommendations come from an interior model optimum; a no-model rule matches 82.5% of the time	"rule-bounded pricing with a learned direction"
"validated on real business outcomes"	only demand accuracy at observed prices was validated against outcomes	"demand accuracy validated on a held-out window; policy economics are model-internal"
"the model is 95% accurate"	meaningless for regression; WAPE 0.4565 means 45.65% total absolute error relative to total units	quote WAPE and say what it means
"no forecast skill is lost by the hybrid"	removed from this repository in Phase M ; the equality holds at the reference price only	"preserves the baseline prediction at the reference price"
"our elasticities are stable and reusable per product"	median rank correlation between disjoint windows is +0.19	"the category elasticity is stable; product-level ordering is not"
"the risk score is a confidence level"	it is an uncalibrated heuristic over history, price variation and extrapolation distance	"heuristic risk score; HIGH means risky and is gated"
"the system handles millions of SKUs"	the batch path is a per-context loop; 3,000 contexts take ~105 s	"a vectorised path exists (<code>simulate_many</code>) but the batch script does not use it yet"
"the data is clean"	28.7% of raw rows were excluded, 10.07% of price changes exceed $\pm 50\%$, and a T-shirt appears in the Cereals file	"every exclusion is counted and every anomaly is flagged rather than trimmed"

114.1 The enforcement mechanism

`scripts/audit_claims.py --strict` scans the repository for ten watched phrases, classifies each occurrence, recognises disclaimers as disclaimers, and **fails the build** if an UNSUPPORTED claim reappears. Current state: **0 UNSUPPORTED** across 49 files and 363 occurrences — a scan that includes this report.

The 38 "needs qualification" hits are uses of *validated*, *guarantees* or *profit uplift* either as pipeline-stage labels (****Status:**** VALIDATED, make features # modelling table with decision-time guarantees) or carrying the qualifier §113 requires. They are surfaced rather than whitelisted — the right default for a scanner of this kind.

PART XXVII — CV / GITHUB POSITIONING

115. Three CV bullets

Using **final, actual** numbers only.

AI Pricing & Revenue Optimization Engine — *Python, pandas, scikit-learn, statsmodels, FastAPI, Streamlit, Docker*

* Built an end-to-end retail price-recommendation system on **4.7M validated UPC × store × week observations** (Dominick's Finer Foods, Kilts Center / Chicago Booth): demand forecasting on a strict chronological split (**test WAPE 0.4565**, a **37.7% relative improvement** over the best naive baseline), a separately estimated price elasticity with **two-way clustered inference** and **REML empirical-Bayes shrinkage** (pooled **-2.029**, 95% CI **[-2.24, -1.82]**; 239 of 372 products with a usable own estimate), a constrained grid optimizer validated against **closed-form optima**, and a risk gate under which **0 of 4,760 high-risk contexts ever auto-changed a price**.

* Diagnosed that the best-performing forecaster's **implied elasticity (-3.10)** lay outside the confidence interval of every controlled econometric estimate, and re-architected the system to **separate forecasting from price response** ($Q(p) = \hat{Q}(p_0) \cdot (p/p_0)^{\epsilon}$); selected the shipped response method on **154,899 out-of-time price-change episodes**, where it achieved the lowest WAPE in every split (0.560 vs 0.715 for a no-price-response null).

* Built an independent **audit suite** measuring how much of each recommendation the model actually supplies — finding that only **2.9%** come from an interior model optimum and that a no-model rule reproduces the engine's price within one 5-cent step **82.5%** of the time — and discovered that understated standard errors had **silently disabled** the empirical-Bayes shrinkage (mean weight 0.954 → **0.780**). Shipped with **519 tests at Phase N**, a FastAPI service, a 9-page Streamlit dashboard and 22 generated analysis reports.

115.1 Alternative shorter set (for a one-page CV)

* **Retail price optimization on 4.7M real scanner observations** — demand model (test WAPE **0.4565**), panel-robust elasticity estimation with empirical-Bayes shrinkage, constrained optimizer validated against closed-form optima, FastAPI + Streamlit, **519 tests at Phase N**.

* **Found and fixed a price-response defect**: the best forecaster's implied elasticity (-3.10) contradicted every controlled estimate (-1.91 to -2.42), so forecasting and price response were separated and the replacement was selected on **154,899 out-of-time price changes**.

* **Audited the system's own contribution**: only **2.9%** of recommendations are set by the model; a no-model rule matches 82.5% of the time. Published as the headline finding rather than an appendix.

115.2 What these bullets deliberately avoid

No "increased profit by X%". No "causal". No "production-ready". No "optimal prices". Every number is traceable to an artifact in this report's sources file.

116. GitHub repository description

Short (the GitHub "About" field, ≤ 350 characters):

Retail price-optimization engine on 4.7M real Dominick's scanner observations: demand forecasting, panel-robust elasticity with empirical-Bayes shrinkage, constrained optimization with reason codes, risk gating, FastAPI + Streamlit. Includes an audit showing only 2.9% of recommendations are set by the model. Extensively tested. Nothing causal is claimed.

Long (README opening):

AI Pricing & Revenue Optimization Engine

A price-recommendation system built end to end on **real retail scanner**

data: the Dominick's Finer Foods Cereals panel from the Kilts Center, University of Chicago Booth (93 stores, 489 UPCs, 366 weeks, 1989–1997).

> *Given a product, store, week, commercial context and a feasible price range, what price should we recommend to maximise expected gross profit, while respecting business and scientific guardrails?*

Every number in this repository was produced by running the pipeline. Nothing is hardcoded, and every counterfactual figure is labelled a **model-internal estimate**: demand at prices that were never charged was never observed, and the same fitted price-response model both proposes and scores candidate prices.

What makes it different from a demand-forecasting notebook: a twelve-report scientific audit that measures how much of each recommendation the model actually supplies. The answer is **2.9%** — so the project describes itself as rule-bounded pricing with a learned direction, not ML pricing.

Suggested topics: pricing · price-optimization · elasticity · econometrics · causal-inference · demand-forecasting · empirical-bayes · retail-analytics · fastapi · streamlit · mlops · scikit-learn · panel-data

117. Skills demonstrated

Mapped to actual implementation, with the file that evidences each.

skill	evidence in this repository
Python	17,370 lines across a src/ package, 30 CLI scripts, an API, a dashboard and 14 test files; dataclasses, StrEnum, protocols-by-convention, context managers, typed signatures, custom exception hierarchies
pandas	a 4.7M-row panel; grouped shifted lags and rolling windows; transform with per-group closures; expanding medians; validate="many_to_one" joins with row-count assertions; categorical dtypes; Parquet with explicit column projection

skill	evidence in this repository
numpy	vectorised economics throughout; <code>np.errstate</code> guards around every division; manual within-group demeaning; <code>keep_independent_columns</code> for rank-deficient designs; batched price-grid construction
statistics	WAPE / MAE / RMSE / bias / sMAPE; PSI and KS drift statistics; percentile and CV summaries; a normal-normal hierarchical model with REML estimation of τ^2
econometrics	log-log elasticity; fixed-effects absorption by within transformation; HC1, one-way and two-way clustered covariance validated against statsmodels; formal z-tests between disjoint windows; explicit treatment of endogeneity and identification
machine learning	ridge on a log target with a <code>ColumnTransformer</code> pipeline; <code>HistGradientBoosting</code> with Poisson loss ; smoothed target-encoding prior fitted on training rows only; a documented prediction cap; model selection on a pre-declared rule
time-series validation	chronological split by week index; no shuffling anywhere; twelve enumerated leakage channels with tests; poisoning audits on the real panel; quarterly performance decay measured
optimization	constrained grid search; six guardrails intersected into one interval with provenance; closed-form validation $((a+bc)/(2b), a/(2b), c \cdot \epsilon/(1+\epsilon))$; leave-one-out constraint attribution over 13,964 contexts $\times$ 11 runs
causal reasoning	the naive-to-fixed-effects ladder as evidence of confounding; an explicit $P(Q \text{price}) \neq P(Q \text{do}(\text{price}))$ treatment; a full randomised-experiment design with clustered inference and pre-registered stopping rules; an explicit refusal to present causal ML as identification
FastAPI	lifespan model loading; strict Pydantic validation including a cross-field validator and a grid-size cap; typed responses; disclaimers as schema fields; correct 404/422/503 semantics; 15 contract tests
Streamlit	9 pages driven entirely by artifacts; explicit <code>cache_data/cache_resource</code> ; Plotly dual-axis charts; missing-artifact handling; a smoke test that renders every page
testing	493 tests: unit, property/invariant (246 decision-state), analytical (closed-form), leakage/poisoning, replica-equivalence (69), integration, and API contract — plus two tests-of-the-tests
MLOps	config-driven thresholds with env override; dataset SHA-256 and content fingerprints; model artifacts carrying split, metrics, environment and data fingerprint; atomic writes; append-only audit log with lifecycle states; drift/schema/performance monitoring; Dockerfile with a non-root user; a CI matrix
software engineering	<code>src/</code> layout, one shared implementation for every price-dependent feature, a single model interface across two families, domain-specific exceptions, ruff with bugbear, 100-char lines, and a fast audit replica pinned by tests to the production code
business reasoning	gross profit chosen over revenue with a derivation of why they differ by $c/2$; guardrails justified operationally; a review queue quantified at 1,353 items/week and its staffing cost named; ethics and legal obligations documented; three CV bullets that refuse to claim uplift
scientific communication	22 generated reports, 52 recorded decisions with alternatives, a claim audit that fails the build, and a project that publishes its least flattering finding as its headline

PART XXVIII — CONCLUSION

118. Final project assessment

PORTFOLIO READY WITH CLEAR LIMITATIONS

This matches the repository's own self-assessment in `STATUS.md`, and this report — having independently re-run the tests, the lint, the demo, the API, the dashboard and every figure — concurs.

118.1 Why "portfolio ready"

criterion	evidence
real, licensed, documented data	4,707,776 validated observations from official Kilts Center sources, with provenance hashes
the whole pipeline runs	make all from download to reports; every number in every report is generated
rigorous validation	15 structural checks, 12 leakage channels with enforcing tests, poisoning audits on the real panel, closed-form optimizer validation, 0 invariant violations over 13,964 contexts
defensible econometrics	fixed effects, panel-robust and two-way clustered inference validated against statsmodels, REML empirical-Bayes shrinkage, formal stability tests
a real scientific finding, honestly handled	the forecaster's implied elasticity contradicted the controlled estimates; the architecture was changed in response and the rejected alternative kept as a measured benchmark
an audit of its own contribution	2.9% interior model optima, published as the headline rather than buried
working software	FastAPI (5 endpoints, model loaded once), Streamlit (9 pages), 519 tests at Phase N, clean lint, an audit log, a Dockerfile, a CI workflow
calibrated communication	0 UNSUPPORTED claims across 49 files (this report included), enforced by a script that fails the build

118.2 Why "with clear limitations"

Ranked by how much they should change what is claimed:

157. **Business rules, not the model, set the magnitude of most recommendations** (2.9% interior optima; a no-model rule matches 82.5% of the time).
158. **No causal identification.** Every price effect is observational.
159. **Cross-price and substitution effects are not modelled** — arguably the largest missing mechanism.
160. **Counterfactual policy economics are circular** and unfixable offline.
161. **Product-level elasticity does not reproduce across time** (rank correlation +0.19).
162. **133 of 372 products have no usable own elasticity** — 35.8% of decision contexts.
163. **The risk layer is a heuristic**, not calibrated confidence.
164. **REVIEW_REQUIRED creates a 1,353-item weekly queue** that nothing staffs.
165. **The AAC cost proxy** lags true replacement cost; only its temporal availability is proven.
166. **Docker was never built; CI never ran; the repository has no commits.**

118.3 Why not "NOT READY"

Every limitation above is **known, measured, documented and disclosed**, most of them by instruments this project built specifically to find them. The pipeline runs end to end on real data; the tests pass; the claims are audited. A project is not unready because it has limits — it is unready when it does not know what they are.

118.4 Why not "PORTFOLIO READY" without qualification

Because the guardrail-dominance finding is genuinely material to what the system is, because nothing here is causal, and because the headline economic figure is circular. A reviewer discovering any of those unprompted would rightly discount everything else. Stating them first is both more honest and more persuasive.

118.5 Who this is ready for

audience	verdict
data science / ML interview portfolio	strongly ready — the audit work is unusually rare and highly differentiating
pricing / revenue-management interview	strongly ready — the elasticity, inference and guardrail-attribution work is domain-credible
ML engineering interview	ready — clean architecture, a real test suite, a service, and one honest gap (no commits, Docker untested)
an academic write-up	not without the experiment — the causal question is open by construction
production deployment	not ready , and the repository says so in six places

118.6 The three things to fix first

167. **Commit the repository.** `git add -A && git commit` immediately gives every future artifact a code lineage and lets the CI workflow actually run. Cost: minutes.
168. **Switch the batch script to `simulate_many`.** The vectorised path already exists and is tested; using it is the single highest-leverage engineering change. Cost: hours.
169. **Regenerate the README's numeric blocks from artifacts.** Three quantified drift instances were found while writing this report; a `make regenerate-readme` step would close the class of problem permanently. Cost: hours.

119. Final lessons

119.1 The best forecaster is not necessarily the best intervention model

The selected model won the accuracy comparison **and** had a price response outside the confidence interval of every controlled econometric estimate (-3.10 vs $[-2.24, -1.82]$). Predictive accuracy is measured on the distribution you trained on; an intervention needs the response surface to be right in a direction that distribution barely explores. **They are different objectives and no accuracy metric can tell them apart.**

119.2 Measure how much of the answer your model actually supplies

The single most valuable artefact in this project is the constraint-attribution audit, and its answer was **2.9%**. Almost no ML system publishes this number, and almost every one would benefit from knowing it. The technique generalises: run your system with each constraint relaxed in turn, compare against the unconstrained optimum, and count.

119.3 Inference assumptions are modelling assumptions

Understated standard errors did not merely produce over-confident intervals. They **silently disabled the empirical-Bayes shrinkage**, because $w = \tau^2 / (\tau^2 + se^2)$ is pushed toward 1 from both directions when `se` is too small. An inference bug became a modelling bug, and it was invisible from any accuracy metric. **Ask what your standard errors assume, and whether anything downstream consumes them.**

119.4 Name the thing accurately and the caveat cannot be dropped

`model_internal_estimated_profit_uplift_pct` is a mouthful, and that is the point. A caveat in a footnote is lost the first time somebody copies a number into a slide; a caveat in the field name travels with the value through the API, the log, the dashboard and the spreadsheet.

119.5 Keep the alternative you rejected, and measure against it

The native ML price response was not deleted when the hybrid replaced it. It remained as `--method ml`, was compared on identical contexts, and was tested head-to-head on 154,899 out-of-time episodes. That is what makes the decision auditable — and reversible if the evidence changes.

119.6 Refusal is a feature

40% of this engine's output is a refusal: 25.3% screened out, 9.7% risk-gated, 5.1% immaterial. A system that always finds a better price is not a system anybody should trust. Building the ability to say "*the evidence does not support acting here*" — and justifying each refusal with a model artefact — is harder and more valuable than building the recommendation.

119.7 Poisoning tests beat assertions, and tests-of-tests beat both

A test asserting "the lag equals the previous row" verifies the code you wrote. A test that corrupts the future and asserts nothing changes verifies the code you *did not* think about. And a poisoning test that also asserts the poison **did** land somewhere is the only version that proves the test itself works.

119.8 Guardrails are the right response to a well-characterised uncertainty

The guardrails are not covering for a weak model. Given an observational elasticity, a 46%-WAPE forecaster, product-level estimates that do not reproduce across windows, and no way to observe outcomes at unobserved prices, a system that let the model move prices by 49% would be reckless. Robustness to being wrong is also insensitivity to being right — and that trade is the correct one here. It stops being correct the day an experiment supplies identification.

119.9 Publish the least flattering finding as the headline

The constraint-attribution result made the project sound less impressive and made every other claim in it more credible. A reviewer who finds a limitation you disclosed reads it as rigour; a reviewer who finds one you concealed discounts everything else you wrote.

119.10 Offline evaluation has a hard ceiling, and knowing where it is *is* the skill

Everything about the machinery can be verified. Almost nothing about the economics can. The one thing this engine exists to claim — that these prices would earn more money — cannot be established by any offline method, and requires an experiment. Recognising that boundary, stating it, building verification for everything on the near side of it, and designing the experiment for the far side, is what separates a pricing *project* from a pricing *demo*.

PART XXIX — POST-FREEZE ENGINEERING

120. Phase O addendum (added Thursday, 2026-08-27)

Everything above this section describes the system as it stood at the v1.0.0 freeze (Phase N, verified 2026-08-19) and is unchanged by what follows. On **Thursday, 2026-08-27**, a post-freeze engineering phase ("Phase O") added three items from `FUTURE_WORK.md` §3 ("Product and operations — no new science"): a recommendation review/approval workflow, monitoring metric history with local threshold alerting, and a batch scale-testing harness. None of it touches an estimator, model, feature, guardrail or decision rule — every scientific finding in Parts I–XXVIII, including the headline **2.9%** interior-optimum attribution, stands exactly as reported.

What was added:

- **Review/approval workflow** — every logged recommendation now carries a `rec_id`; a new append-only `recommendation_transitions.csv` records `GENERATED` → `REVIEWED` → `APPROVED/REJECTED` → `PUBLISHED` transitions (an illegal edge raises rather than silently no-opping); a `scripts/review.py` CLI and a tenth Streamlit dashboard page ("Review queue") let a human work the `REVIEW_REQUIRED` backlog. This is the tool a reviewer would use, not a reviewer, an SLA, or a notification path.
- **Monitoring history + local alerting** — `scripts/monitor.py` now appends one JSON line per run to `artifacts/metrics/monitoring_history.jsonl` (previously a single overwritten snapshot) and evaluates the run against config-driven local thresholds (`configs/config.yaml` `monitoring:` block), exiting non-zero on a breach. These thresholds are loosely calibrated against this project's own reported numbers, not agreed with a business or fitted against realised out-of-sample error.
- **Batch scale harness** — `scripts/benchmark_scale.py` sweeps `optimize_price_batch` up to 30,000+ contexts by tiling the same real sampled pool `benchmark_batch.py` already proved equivalent to the per-context path. Sizes above the base sample are that sample **replicated**, not independent catalogue growth or a scaling claim to other hardware.

What this section does not claim: no realised staffing of the review queue, no business-calibrated monitoring thresholds, no metric-store database or paging integration, no proof of scaling to a larger real catalogue, and no GitHub remote was created or pushed to. Full evidence — real command output for every item above, including the one-time archival of the pre-Phase-O `recommendation_log.csv` that adding `rec_id` triggered, and one disclosed (not fabricated) residual test-count inconsistency inside this very report — is in `reports/22_POST_FREEZE_ENGINEERING.md`.

APPENDICES

Appendix A — Full dataset dictionary

A.1 Canonical table — `data/processed/dominicks_cereals.parquet`

Grain: one valid `upc` × `store` × `week` observation (uniqueness asserted). 4,707,776 rows × 31 columns.

Raw fields (as documented in the Kilts Center manual)

column	type	meaning
<code>upc</code>	<code>int64</code>	Universal Product Code of the item
<code>store</code>	<code>int32</code>	Dominick's store number
<code>week</code>	<code>int32</code>	Dominick's week index (1 = week starting 1989-09-14)
<code>move</code>	<code>float</code>	number of individual units sold in that store-week — the prediction target

column	type	meaning
price	float	price of the bundle actually scanned (may cover several units)
qty	float	number of items in the bundle
sale	string	promotion code: B Bonus Buy, C Coupon, S simple price reduction (also observed: undocumented G, L)
profit	float	gross-margin percentage , based on Average Acquisition Cost
ok	int8	1 = valid, 0 = suspect / trash per the manual (only ok = 1 is kept)

Raw fields intentionally **not** loaded: PRICE_HEX, PROFIT_HEX (redundant binary encodings of price and profit).

Product metadata (joined from upccer.csv, validate="many_to_one")

column	meaning
com_code	commodity code (Cereals)
descrip	product description, e.g. CAPN CRUNCH JUMBO CR
size	package size string, e.g. 12 OZ
case	units per case
nitem	internal item number

Derived economics

column	formula	meaning
effective_unit_price	$\text{price} / \text{qty}$	price paid per individual unit — the decision variable
revenue	$\text{effective_unit_price} \times \text{move}$	observed revenue for that store-week-UPC
gross_margin_rate	$\text{profit} / 100$	margin as a fraction, not a percentage
estimated_unit_aac	$\text{effective_unit_price} \times (1 - \text{gross_margin_rate})$	unit cost implied by the accounting margin — never called "true cost"
gross_profit	$\text{revenue} \times \text{gross_margin_rate} = (\text{p} - \text{aac}) \times \text{move}$	observed gross profit

Calendar fields

column	meaning
week_start_date	1989-09-14 + (week - 1) × 7 days; store weeks run Thursday → Wednesday
year, month, quarter, week_of_year	derived from week_start_date

Promotion fields

column	meaning
recorded_promotion_flag	1 if a documented code (B/C/S) is present
recorded_promotion_type	Bonus Buy / Coupon / Simple price reduction / UNKNOWN_CODE / NONE_RECORDED

NONE_RECORDED means *no code was recorded*, **not** *no promotion happened*.

Quality flags (kept; never used to silently drop rows)

column	definition	share
margin_implausible_flag	gross_margin_rate outside (-1, 1)	0.0%
aac_nonpositive_flag	implied unit AAC ≤ 0	0.0%
zero_move_flag	zero units sold that week	0.0%
bundle_flag	qty > 1	0.1051%
has_metadata_flag	UPC matched the metadata file	100%

A.2 Feature table — data/processed/dominicks_cereals_features.parquet

4,707,776 rows; **4,671,333** usable for training (36,443 series-opening rows lack lags). 29 engineered features.

Price-dependent (5) — recomputed for every candidate price

feature	formula	availability
effective_unit_price	p	KNOWN AT DECISION TIME
log_price	log(p)	KNOWN AT DECISION TIME
price_vs_last_week	p / lag_price_1 - 1	KNOWN AT DECISION TIME
price_vs_series_reference	p / series_reference_price - 1	KNOWN AT DECISION TIME
price_vs_recent_mean	p / roll_mean_price_4 - 1	KNOWN AT DECISION TIME

Context (24) — held fixed while a candidate price varies

feature	formula / source	availability	missing
store, upc, com_code	identity (categorical)	BEFORE DECISION	0%
week_of_year, month, quarter	from week_start_date	BEFORE DECISION	0%
time_index	the Dominick's week index (int32)	BEFORE DECISION	0%
recorded_promotion_flag	binary code presence	REQUIRES ASSUMPTION	0%
lag_move_1	y_{t-1} within series	BEFORE DECISION	0.774%
lag_move_2	y_{t-2}	BEFORE DECISION	1.536%
lag_move_3	y_{t-3}	BEFORE DECISION	2.282%
lag_move_4	y_{t-4}	BEFORE DECISION	3.017%
roll_mean_move_4/8/13	mean of the shifted series, min_periods=2	BEFORE DECISION	1.536%
roll_std_move_4	sd of the shifted series, min_periods=2	BEFORE DECISION	1.536%
lag_price_1, lag_price_2	p_{t-1}, p_{t-2}	BEFORE DECISION	0.774% / 1.536%
roll_mean_price_4	mean of shifted prices, min_periods=1	BEFORE DECISION	0.774%
series_reference_price	expanding median of shifted prices	BEFORE DECISION	0.774%
decision_time_unit_cost	estimated_unit_aac_{t-1}, forward-filled	REQUIRES ASSUMPTION	0.774%
lag_promotion_1	promotion flag at t-1	BEFORE DECISION	0.774%
series_age_weeks	groupby(upc, store).cumcount()	BEFORE DECISION	0%
package_size_oz	parsed from size (LB → ×16)	BEFORE DECISION	0.705%

Model-only derived (added inside DemandModel.prepare)

feature	formula
sin52, cos52	sin(2 π ·week_of_year/52), cos(2 π ·week_of_year/52)

feature	formula
upc_demand_prior	$(\sum \log 1p(y) + 50 \cdot \bar{y}) / (n + 50)$ per UPC, fitted on training rows only

Never used as features: revenue, gross_profit, profit, gross_margin_rate (week t), estimated_unit_aac (week t), the raw sale code of week t, ok, qty, price (bundle form).

A.3 Elasticity table — artifacts/models/elasticity_table.csv

372 product rows + 23 meta_* columns.

column	meaning
upc	product
n_obs, n_stores, n_weeks, n_distinct_prices, price_cv, price_min, price_max	identifying variation available
elasticity_raw	the per-UPC OLS coefficient with store FE absorbed
std_error	panel-robust — max(store-clustered, week-clustered, HC1)
std_error_hc1, std_error_cluster_store, std_error_cluster_week	components
se_inflation_vs_hc1	$\text{std_error} / \text{std_error_hc1}$
t_stat, ci_low, ci_high, r_squared, significant_negative	inference
usable, usable_for_pricing, reject_reason	eligibility
shrinkage_weight	$w_i = \tau^2 / (\tau^2 + \text{se}_i^2)$
elasticity_shrunk	$w_i \cdot \epsilon_i + (1 - w_i) \cdot \mu$
elasticity_final	after clipping to $ \epsilon \in [0.2, 6.0]$ — the value applied
elasticity_source	shrunk_product (239) or pooled_fallback (133)
clipped	whether clipping bound (0 rows)
meta_*	pooled estimate, its several SEs, τ^2 , prior mean, method, n used, estimation window, clip band

A.4 Recommendation audit log — artifacts/recommendation_log.csv

34 columns, append-only, 3,002 rows currently. Columns: logged_at_utc, lifecycle_state, upc, store, product_description, decision_week, decision_week_start_date, objective, policy_profile, model_version, price_response_method, elasticity_used, elasticity_source, current_price, proposed_candidate_price, final_recommended_price, proposed_price_change_pct, price_change_pct, decision, actionable, predicted_units_current, predicted_units_recommended, expected_revenue_current, expected_revenue_recommended, expected_gross_profit_current, expected_gross_profit_recommended, model_internal_estimated_profit_uplift_pct, model_internal_estimated_revenue_uplift_pct, realisable_profit_uplift_pct, unit_cost_used, risk_level, reason_codes, risk_notes, constraints_json.

Appendix B — Formula reference

B.1 Data derivations

$$\text{effective_unit_price} = \text{price} / \text{qty}$$

$$\text{revenue} = \text{effective_unit_price} \times \text{move} = (\text{price} \times \text{move}) / \text{qty}$$

$$\text{gross_margin_rate} = \text{profit} / 100$$

$$\text{estimated_unit_aac} = \text{effective_unit_price} \times (1 - \text{gross_margin_rate})$$

$$\text{gross_profit} = \text{revenue} \times \text{gross_margin_rate} = (p - aac) \times \text{move}$$

$$\text{week_start_date} = 1989-09-14 + (\text{week} - 1) \times 7 \text{ days}$$

B.2 Economics

$$Q = Q(p, X) \quad R(p) = pQ(p) \quad GP(p) = (p - c)Q(p)$$

$$\text{unit gross margin} = p - c \quad \text{margin rate} = (p - c)/p \quad \text{price change} = p_{\text{new}}/p_{\text{old}} - 1$$

B.3 Elasticity

$$E = (\% \Delta Q) / (\% \Delta P) = (\partial \log Q) / (\partial \log P)$$

Arc (midpoint):

$$E_{\text{arc}} = ((Q_2 - Q_1) / ((Q_1 + Q_2) / 2)) / ((P_2 - P_1) / ((P_1 + P_2) / 2))$$

Log-log:

$$\log Q = \beta_0 + \beta_1 \log P + \sum_k \gamma_k X_k + u, \quad \beta_1 = E$$

Within (fixed-effects) transformation:

$$\log Q_{it} - \text{mean}(\log Q_i) = \beta_1(\log P_{it} - \text{mean}(\log P_i)) + u_{it}$$

Seasonality controls:

$$\sin_{52} = \sin((2\pi w)/52), \quad \cos_{52} = \cos((2\pi w)/52), \quad \text{trend} = w/100$$

B.4 Empirical-Bayes shrinkage

$$\hat{\epsilon}_i \mid \theta_i \sim N(\theta_i, \text{se}_i^2), \quad \theta_i \sim N(\mu, \tau^2)$$

$$\hat{\theta}_i = w_i \hat{\epsilon}_i + (1 - w_i) \mu, \quad w_i = \tau^2 / (\tau^2 + \text{se}_i^2)$$

Method of moments:

$$\tau_{\text{mom}}^2 = \max(\text{mean}((\epsilon_i - \mu)^2) - \text{mean}(\text{se}_i^2), 0)$$

REML fixed point:

$$w_i = 1/(\tau^2 + \text{se}_i^2), \quad \tau^2 \leftarrow (\sum_i w_i^2 [(\epsilon_i - \mu)^2 - \text{se}_i^2 + 1/(\sum_j w_j)]) / (\sum_i w_i^2)$$

B.5 Price response

Hybrid:

$$Q(p) = \hat{Q}(p_0)(p/p_0)^\epsilon, \quad Q(p_0) = \hat{Q}(p_0)$$

Native ML implied elasticity (central finite difference):

$$\epsilon = (\log \hat{Q}(p(1+r)) - \log \hat{Q}(p(1-r))) / (\log(1+r) - \log(1-r)), \quad r = 0.02$$

B.6 Optimization

Linear demand $Q = a - bp$:

$$p_{\text{GP}}^* = (a + bc)/(2b), \quad p_{\text{R}}^* = a/(2b), \quad p_{\text{GP}}^* - p_{\text{R}}^* = c/2$$

Constant elasticity $Q = k p^{-\{\epsilon\}}$:

$$p_{\text{GP}}^* = c\epsilon/(1+\epsilon) \quad (\epsilon < -1), \quad p_{\text{GP}}^* \rightarrow \infty \quad (-1 < \epsilon < 0)$$

p_{R}^* : revenue is monotone; the optimum is a boundary of $\mathcal{F}$

Constraints (intersected):

$$\ell = \max\{p_{\min}, p_0(1-\delta), p^{\text{hist}}_{\min}(1-t), c, c/(1-m)\}$$

$$h = \min\{p_{\max}, p_0(1+\delta), p^{\text{hist}}_{\max}(1+t)\}$$

Materiality:

$$(V(p^*) - V(p_0))/(|V(p_0)|) \geq \theta$$

B.7 Metrics

$$\text{WAPE} = (\sum_i |y_i - \hat{y}_i|) / (\sum_i y_i) \quad \text{MAE} = 1/n \sum_i |y_i - \hat{y}_i| \quad \text{RMSE} = \sqrt{(1/n \sum_i (y_i - \hat{y}_i)^2)}$$

$$\text{bias} = 1/n \sum_i (\hat{y}_i - y_i) \quad \text{sMAPE} = 1/n \sum_i (|y_i - \hat{y}_i|) / ((|y_i| + |\hat{y}_i|)/2)$$

Population Stability Index:

$$\text{PSI} = \sum_b (p_b^{\text{cur}} - p_b^{\text{ref}}) \ln(p_b^{\text{cur}} / (p_b^{\text{ref}}))$$

B.8 Experiment design

$$\text{SE} \approx \sigma \sqrt{((1 + (T-1)\rho)/(kT))}$$

Appendix C — Model hyperparameters (actual final values)

C.1 M1 ridge log-log — the selected model

component	parameter	value
target transform	—	log1p(y), inverted with expm1
numeric imputer	strategy	median
scaler	with_mean / with_std	true / true
one-hot encoder	handle_unknown	ignore
one-hot encoder	min_frequency	50
one-hot encoder	sparse_output	true
Ridge	alpha	1.0
Ridge	fit_intercept	true
Ridge	solver	auto
Ridge	tol	1e-4
Ridge	random_state	42
ColumnTransformer	remainder	drop
ColumnTransformer	sparse_threshold	0.3
model	PREDICTION_CAP_MULTIPLE	5.0 (cap = 5 × max training demand)
model	UPC_PRIOR_SMOOTHING	50.0
model	LINEAR_EXCLUDED_FEATURES	("time_index",)
model	LOGIP_FEATURES	the 8 demand-level lag/rolling features
	numeric features used	24 (26 total minus 2 categorical)
	categorical features	store, com_code
	fit time	31.0 s on 3,206,437 rows
	version	ridge_loglog-20260818-132847

C.2 M2 HGB Poisson

parameter	value
-----------	-------

parameter	value
loss	poisson
learning_rate	0.06
max_iter	500
max_leaf_nodes	63
min_samples_leaf	40
l2_regularization	1.0
early_stopping	false
max_bins	255
max_depth	None
max_features	1.0
categorical_features	from_dtype
monotonic_cst	None (not used — see §89.7)
random_state	42
fit time	181.9 s

C.3 Elasticity estimation

parameter	value
min_obs_per_upc	200
min_distinct_prices_per_upc	10
max_se_for_product_estimate (panel-robust)	1.5
shrinkage	empirical_bayes
shrinkage_tau2_estimator	reml
shrinkage_prior_mean	pooled
pooled subsample size	1,200,000 (seed 42)
controls	recorded_promotion_flag, sin52, cos52, trend
absorbed	UPC × store fixed effects
per-UPC SE	max(store-clustered, week-clustered, HC1)
pooled SE	two-way clustered (UPC, week)
sensitivity_scenarios	-1.5, -1.9, -2.4, -3.1
clip band	$ \epsilon \in [0.2, 6.0]$
runtime	35.0 s

C.4 Optimization

parameter	value
objective	gross_profit
price_step	\$0.05
price_rounding	\$0.01
charm_pricing	false
random seed	42

Appendix D — Configuration profiles

D.1 Policy profiles (DEMO settings)

parameter	conservative	standard ☒	aggressive (DEMO ONLY)
max_price_change_pct	0.05	0.10	0.20

parameter	conservative	standard ☒	aggressive (DEMO ONLY)
extrapolation_tolerance_pct	0.02	0.05	0.10
min_gross_margin_rate	0.10	0.05	0.00
materiality_threshold_pct	0.02	0.01	0.005
allow_below_cost	false	false	false
high_risk_action	keep_current	review_required	⚠ recommend
medium_risk_max_price_change_pct	0.03	0.05	0.20

D.2 Behaviour over all 13,964 week-399 contexts

profile	RECOMMEND_CHANGE	KEEP_CURRENT	REVIEW_REQUIRED	median change	p90 change	LOW	MEDIUM	HIGH	HIGH auto-changed
conservative	45.50%	54.50%	0%	0.0%	+4.40%	7,163	2,041	4,760	0
standard	59.19%	31.12%	9.69%	+3.75%	+9.27%	5,511	3,693	4,760	0
aggressive	70.79%	29.21%	0%	—	—	4,032	1,830	8,102	⚠ 4,717

D.3 Eligibility screen

parameter	value
min_observations	40
min_distinct_prices	5
min_price_cv	0.05
series passing	19,707 / 36,443 (54.08%)

D.4 Risk bands (deliberately stricter than eligibility)

band	min observations	min distinct prices	max extrapolation
LOW	150	15	2%
MEDIUM	80	8	5%
HIGH	(meets neither)		

Plus two demotions from LOW to MEDIUM: `promotion_share > 0.5`, or `elasticity_source == "pooled_fallback"`. No support at all $\Rightarrow$ HIGH.

D.5 Price-response methods

method	ϵ used	status
ml	none — the base model answers counterfactuals directly	benchmark, retained
pooled	−2.029 for every row	supported
shrunk	per-UPC empirical Bayes, pooled fallback	shipped default

Appendix E — API schemas

E.1 Endpoints

method	path	request model	response model
GET	/health	—	HealthResponse
GET	/model/info	—	ModelInfoResponse

method	path	request model	response model
POST	/predict-demand	PredictDemandRequest	PredictDemandResponse
POST	/simulate-prices	SimulatePricesRequest	SimulatePricesResponse
POST	/recommend-price	RecommendPriceRequest	RecommendPriceResponse

E.2 Requests

```

class PredictDemandRequest(BaseModel):
    upc: int
    store: int = Field(..., ge=1)
    week: int | None = Field(None, ge=1)
    price: float | None = Field(None, gt=0)

class SimulatePricesRequest(BaseModel):
    upc: int
    store: int
    week: int | None = Field(None, ge=1)
    min_price: float = Field(..., gt=0)
    max_price: float = Field(..., gt=0)
    step: float = Field(0.05, gt=0, le=5.0)
    unit_cost: float | None = Field(None, gt=0)

    @model_validator(mode="after")
    def check_range(self):
        if self.max_price < self.min_price:
            raise ValueError("max_price must be greater than or equal to min_price")
        if (self.max_price - self.min_price) / self.step > 500:
            raise ValueError("requested grid exceeds 500 candidate prices; widen step or narrow range")
        return self

class RecommendPriceRequest(BaseModel):
    upc: int
    store: int
    week: int | None = Field(None, ge=1)
    objective: Literal["gross_profit", "revenue"] = "gross_profit"
    policy_profile: Literal["conservative", "standard", "aggressive"] = "standard"
    unit_cost: float | None = Field(None, gt=0)

```

python

E.3 RecommendPriceResponse — the 30 fields

field	type	note
upc, store	int	
decision_week, decision_week_start_date	int, str	
product_description	str None	
objective, policy_profile	str	
model_version	str None	
price_response_method	str None	required to interpret the uplift
elasticity_used	float None	
elasticity_source	str None	shrunk_product / pooled_fallback / ml_native
current_price	float	
proposed_candidate_price	float	not a business price on its own
final_recommended_price	float	what would be charged

field	type	note
proposed_price_change_pct	float	
price_change_pct	float	of the final price; 0.0 when not actionable
decision	Literal["RECOMMEND_CHANGE", "KEEP_CURRENT", "REVIEW_REQUIRED"]	
actionable	bool	the single flag consumers must respect
predicted_units_current / _recommended	float	
expected_revenue_current / _recommended	float	
expected_gross_profit_current / _recommended	float None	None when no cost
model_internal_estimated_profit_uplift_pct	float None	model-internal
model_internal_estimated_revenue_uplift_pct	float None	
realisable_profit_uplift_pct	float None	forced to 0.0 when not actionable
unit_cost_used	float None	
risk_level	Literal["LOW", "MEDIUM", "HIGH"]	heuristic
reason_codes	list[str]	sorted, de-duplicated
risk_notes	list[str]	plain-English caveats
constraints	dict	bounds + every guardrail's value
disclaimer	str	default value; travels with the response

E.4 Error semantics

condition	status	detail
model not loaded	503	"Model is not loaded yet."
unknown UPC × store in the served window	404	names the served week range
unknown week for a known series	404	lists the served weeks
invalid payload (Pydantic)	422	field-level errors
empty or oversized price grid	422	explicit message
optimizer refusal	422	the <code>OptimizerError</code> message

Appendix F — Reason codes

code	meaning	fires when
KEEP_CURRENT_OPTIMAL	leave the price alone	attached to every KEEP_CURRENT outcome
LOW_CONFIDENCE	the evidence does not support automatic action	with HIGH-risk gating
INSUFFICIENT_HISTORY	fewer than 40 observations	eligibility screen

code	meaning	fires when
INSUFFICIENT_PRICE_VARIATION	fewer than 5 distinct prices, or price CV < 0.05	eligibility screen
OUTSIDE_EXTRAPOLATION_RANGE	the extrapolation guardrail is an active edge	constraint binding
COST_UNAVAILABLE	no decision-time cost for a gross-profit objective	eligibility screen
MARGIN_CONSTRAINT	the cost floor or minimum-margin floor is an active edge	constraint binding
PRICE_CHANGE_LIMIT	the maximum-change window is an active edge	constraint binding
PROFIT_UPLIFT_POSITIVE	the proposal beats current on estimated gross profit by $\geq$ materiality	gross-profit objective
REVENUE_UPLIFT_POSITIVE	as above, revenue objective	revenue objective
NO_FEASIBLE_PRICE	the constraint intersection is empty, or no candidate survives	constraints
NON_MATERIAL_UPLIFT	the estimated gain is below the materiality threshold	materiality check
DEMAND_CURVE_NOT DECREASING	the simulated curve is not monotone decreasing	curve diagnostic (never fires under the hybrid)
HIGH_RISK_REVIEW_REQUIRED	HIGH risk under high_risk_action: review_required	risk gate
HIGH_RISK_KEEP_CURRENT	HIGH risk under high_risk_action: keep_current	risk gate
MEDIUM_RISK_CONSERVATIVE	MEDIUM risk; the change cap has been tightened	risk adjustment
POOLED_ELASTICITY_FALLBACK	no usable product-specific elasticity	elasticity provenance

Observed frequencies (3,000-context batch, week 399, standard, `shrunk`):

code	count
PRICE_CHANGE_LIMIT	2,216
PROFIT_UPLIFT_POSITIVE	2,077
KEEP_CURRENT_OPTIMAL	923
INSUFFICIENT_PRICE_VARIATION	668
INSUFFICIENT_HISTORY	567
POOLED_ELASTICITY_FALLBACK	523
OUTSIDE_EXTRAPOLATION_RANGE	465
MEDIUM_RISK_CONSERVATIVE	416
LOW_CONFIDENCE	311
HIGH_RISK_REVIEW_REQUIRED	311
MARGIN_CONSTRAINT	208
NON_MATERIAL_UPLIFT	136
NO_FEASIBLE_PRICE	37
DEMAND_CURVE_NOT DECREASING	0
COST_UNAVAILABLE	0
REVENUE_UPLIFT_POSITIVE	0 (gross-profit run)
HIGH_RISK_KEEP_CURRENT	0 (standard profile uses review_required)

Appendix G — Test inventory

519 tests at Phase N, 0 failed, 0 skipped — verified 2026-08-19, 7.05 s.

file	tests	coverage
tests/test_decision_states.py	246	risk × decision × actionable × proposal × final-price invariants over every combination and all three profiles; audit-log schema rotation
tests/test_attribution.py	69	the audit replica reproduces optimize_price exactly across 60 parameter combinations; the baseline demand level cannot change the recommended price; closed-form unconstrained optimum; constraint intervals
tests/test_phase_1_pricing.py	40	hybrid mechanics; shrinkage maths; per-UPC estimator; risk gating across all three profiles; uplift naming; elasticity provenance and window; $Q_{\text{hybrid}}(p_o) == Q_{\text{ML}}(p_o)$ across 5 elasticities × 3 price levels
tests/test_optimizer.py	23	analytical optima $(a+bc)/(2b)$ and $a/(2b)$; every constraint; keep-current paths; objective errors; risk levels; inelastic boundary case
tests/test_data_pipeline.py	17	derived formulas; zero-qty safety; promotion coding; week decoding; exclusion rules; metadata join cardinality; grain uniqueness
tests/test_economics.py	16	revenue/margin identities; price-variation summary; arc-elasticity edge cases; log-log recovery on synthetic data; fixed-effects confounding demonstration
tests/test_simulation.py	16	price-grid construction; only price features move; cost held fixed across the grid; single batched call; curve diagnostics
tests/test_api.py	15	health; model info; model loaded once (not per request); prediction contract; validation rejections; decision states; policy profiles
tests/test_features.py	13	lag alignment; shifted rolling windows; leakage poisoning; decision-time cost; price-feature recomputation (both directions); temporal split
tests/test_shrinkage.py	12	four limiting cases; monotonicity in se; [0,1] bounds; prior-mean handling; that understated SEs silently disable the shrinkage
tests/test_repo_and_downloader.py	10	licensed data git-ignored; official URLs only (host asserted); zip-slip and absolute-path rejection; archive-member selection
tests/test_monitoring.py	9	PSI; KS; schema checks; drift ranking; performance windows
tests/test_cost_leakage.py	5	decision-cost identity; future poisoning and that the test has teeth ; recommendation invariance; missing-cost refusal
tests/test_integration.py	2	raw → processed → features → model → simulation → optimization → audit log

file	tests	coverage
total	493	

Appendix H — Repository tree (actual, at report time)

183 tracked-eligible files (excluding `.git`, caches, `data/`, `*.egg-info` and the 46 PNG figures).

```

.
├── .dockerignore
├── .env.example
├── .github/workflows/ci.yml
├── .gitignore
├── AI_Pricing_Revenue_Optimization_Claude_Codex_Master_Prompt.txt
├── Claude Code Prompt – Ultra-Detailed Full Project Report.md
├── DECISIONS.md           # 52 recorded decisions, alternatives, rationale
├── Dockerfile
├── KNOWN_LIMITATIONS.md
├── Makefile
├── README.md
├── ROADMAP.md
├── STATUS.md
├── docker-compose.yml
├── pyproject.toml
├── api/
│   ├── __init__.py
│   ├── main.py           # FastAPI app, lifespan model loading
│   ├── schemas.py        # Pydantic request/response models
│   ├── state.py          # AppState: model + serving context slice
│   └── routes/pricing.py  # predict-demand / simulate-prices / recommend-price
├── artifacts/
│   ├── figures/          # 22 pipeline PNGs
│   ├── report_figures/   # 24 PNGs generated for this report (14 figures + 10 screenshots)
│   └── metrics/          # 33 JSON/CSV artifacts
│       ├── raw_audit.json      build_audit.json      data_validation.json
│       ├── dataset_fingerprint.json feature_build.json eda_summary.json
│       ├── elasticity.json     elasticity_estimation.json elasticity_by_upc.csv
│       ├── elasticity_funnel.json elasticity_inference.json elasticity_stability.json
│       ├── elasticity_per_upc_inference.csv elasticity_stability_by_upc.csv
│       ├── model_metrics.json  evaluation.json      monitoring.json
│       ├── price_response.json price_response_comparison.json
│       ├── out_of_time_price_response.json price_change_episodes.csv
│       ├── recommendations.json backtest.json        shrinkage_audit.json
│       ├── constraint_attribution.json/.csv model_value_ablation.json
│       ├── guardrail_ablation.json decision_state_audit.json claim_audit.json
│       ├── zero_price_audit.json cost_leakage_audit.json
│       └── price_variation_upc_store.csv
│   └── models/
│       ├── demand_model.joblib
│       ├── demand_model_metadata.json
│       ├── elasticity_table.csv
│       ├── recommendation_log.csv
│       └── recommendation_log_schema_pre_phase_m.csv
├── configs/config.yaml    # the single source of every threshold
├── dashboard/app.py       # Streamlit, 9 pages
├── docs/                  # 15 documents
│   ├── DATA_SOURCE.md     DATA_DICTIONARY.md     FEATURE_AVAILABILITY.md
│   ├── DATA_LEAKAGE_AUDIT.md CAUSAL_LIMITATIONS.md PRICING_SCIENCE.md
│   ├── MODEL_CARD.md       METRICS.md                TECHNICAL_DESIGN.md
│   ├── BUSINESS_CASE.md    INTERVIEW_GUIDE.md    INTERVIEW_RED_TEAM.md
│   └── RESPONSIBLE_PRICING.md PRICING_EXPERIMENT.md USING_YOUR_OWN_DATA.md
├── notebooks/            # thin wrappers; the scripts are the source of truth
│   ├── 01_pricing_eda.ipynb
│   └── 02_elasticity.ipynb
├── reports/              # 22 generated reports + this one
│   ├── 01_DATA_AUDIT.md .. 20_CLAIM_AUDIT.md
│   ├── MONITORING_DESIGN.md
│   └── VALIDATION_SUMMARY.md

```

```

├── AI_PRICING_REVENUE_OPTIMIZATION_FULL_REPORT.md      ← this document
├── AI_PRICING_REVENUE_OPTIMIZATION_REPORT_SOURCES.md
├── REPORT_QA.md

├── scripts/ # 30 CLIs, one per pipeline stage
│   ├── download_dominicks.py build_dataset.py validate_data.py
│   ├── build_features.py run_eda.py run_elasticity.py
│   ├── train.py evaluate.py price_response.py
│   ├── estimate_elasticity.py compare_price_response.py
│   ├── optimize.py backtest.py monitor.py
│   ├── run_demo.py smoke_dashboard.py
│   ├── audit_zero_price.py audit_cost_leakage.py
│   ├── audit_constraints.py audit_model_value.py audit_guardrails.py
│   ├── audit_shrinkage.py audit_elasticity_inference.py
│   ├── audit_out_of_time_response.py audit_elasticity_stability.py
│   ├── audit_eligibility_funnel.py audit_decision_states.py
│   ├── audit_claims.py _phase_m.py
│   └── make_report_figures.py ← added for this report

├── src/pricing_engine/ # 34 modules
│   ├── config.py audit.py
│   ├── data/ schema · loader · cleaning · validator
│   ├── economics/ metrics · elasticity · inference · elasticity_store
│   ├── features/ build
│   ├── models/ baselines · demand_model · hybrid · metrics
│   ├── simulation/ price_grid · counterfactual
│   ├── optimization/ objective · constraints · risk · optimizer · policies · attribution
│   ├── monitoring/ drift
│   └── utils/ io

├── tests/ # 14 files, 519 tests (at Phase N)
│   ├── conftest.py
│   ├── test_data_pipeline.py test_repo_and_downloader.py
│   ├── test_features.py test_economics.py
│   ├── test_simulation.py test_optimizer.py
│   ├── test_phase_1_pricing.py test_cost_leakage.py
│   ├── test_api.py test_monitoring.py
│   ├── test_integration.py test_attribution.py
│   └── test_shrinkage.py test_decision_states.py

```

text

Appendix I — Reproduction commands

```

# ---- environment -----
python -m pip install -e ".[api,dashboard,dev]" # Python 3.11+

# ---- data (official Kilts Center URLs only) -----
python scripts/download_dominicks.py # or: make download
python scripts/build_dataset.py # → reports/01_DATA_AUDIT.md
python scripts/validate_data.py # 15 checks
python scripts/build_features.py # → 4,671,333 usable rows

# ---- analysis -----
python scripts/run_eda.py # → reports/02 + 6 figures
python scripts/run_elasticity.py # → reports/03

# ---- models -----
python scripts/train.py # → reports/04, model artifact
python scripts/evaluate.py # reload + re-score
python scripts/price_response.py --n-contexts 300 # → reports/05

# ---- pricing layer -----
python scripts/estimate_elasticity.py # training weeks only
python scripts/compare_price_response.py --n-contexts 300 # → reports/08
python scripts/optimize.py --batch 3000 --profile standard # → reports/07
python scripts/optimize.py --upc 3000006560 --store 86 # one context
python scripts/backtest.py --weeks 10 --contexts-per-week 400 # → reports/06
python scripts/monitor.py # drift / schema / performance

# ---- audits -----
python scripts/audit_zero_price.py # → reports/09
python scripts/audit_cost_leakage.py --n-series 25 # → reports/10
python scripts/audit_constraints.py # → reports/11

```

```
python scripts/audit_model_value.py # → reports/12
python scripts/audit_guardrails.py # → reports/13
python scripts/audit_shrinkage.py # → reports/14
python scripts/audit_elasticity_inference.py # → reports/15
python scripts/audit_out_of_time_response.py # → reports/16
python scripts/audit_elasticity_stability.py # → reports/17
python scripts/audit_eligibility_funnel.py # → reports/18
python scripts/audit_decision_states.py # → reports/19
python scripts/audit_claims.py --strict # → reports/20, fails on UNSUPPORTED

# ---- demo, quality gates, applications -----
python scripts/run_demo.py # one real UPC x store
python -m pytest # 519 passed at Phase N
python -m ruff check . # All checks passed
python scripts/smoke_dashboard.py # 9 pages render
python -m uvicorn api.main:app --host 127.0.0.1 --port 8000 --reload
python -m streamlit run dashboard/app.py

# ---- this report's figures -----
python scripts/make_report_figures.py # → artifacts/report_figures/

# ---- everything -----
make all
```

bash

Example API call

```
curl -X POST localhost:8000/recommend-price \
-H "content-type: application/json" \
-d '{"upc": 3000006560, "store": 86, "objective": "gross_profit", "policy_profile": "standard"}'
```

bash

Appendix J — Final consolidated metrics table

This is the authoritative metrics dictionary for this report. Every number used anywhere above appears here with its source artifact. Where the repository's prose disagrees with an artifact, the artifact value is used and the discrepancy is flagged.

J.1 Data

metric	value	source
raw movement rows	6,602,582	raw_audit.json
raw exact duplicate rows	0	raw_audit.json
raw duplicate grain rows	0	raw_audit.json
raw UPCs / stores / weeks	490 / 93 / 367	raw_audit.json
ok = 1 / ok = 0 rows	6,461,297 / 141,285	raw_audit.json
sale codes: NA / B / S / G / C / L	6,242,568 / 254,261 / 91,259 / 11,075 / 3,418 / 1	raw_audit.json
raw zero-price rows	1,851,380 (agrees across both artifacts)	raw_audit.json (price.n_zero), zero_price_audit.json
raw zero-move rows	1,850,703	raw_audit.json (move.n_zero)
canonical rows	4,707,776	build_audit.json
rows removed	1,894,806 (28.698%)	build_audit.json
removed by ok_flag_zero	141,285 (2.1398%)	build_audit.json
removed by non_positive_price	1,753,521 (26.5581%)	build_audit.json
canonical UPCs / stores / weeks	489 / 93 / 366	build_audit.json
date range	1989-09-14 ... 1997-05-01	build_audit.json
observed total units	90,766,941	build_audit.json
observed total revenue	\$262,008,582.30	build_audit.json
observed total gross profit	\$40,091,143.49	build_audit.json

metric	value	source
blended gross margin	15.3015%	eda_summary.json
mean / median effective unit price	\$3.1157 / \$3.15	eda_summary.json
mean gross-margin rate per row	17.419%	eda_summary.json
share of rows with a recorded promotion	7.3484%	eda_summary.json
share of bundle rows (qty > 1)	0.10506% (4,946 rows)	eda_summary.json
validation checks	15 / 15 pass	data_validation.json
Parquet SHA-256	51f9148b...761a9	dataset_fingerprint.json
DataFrame fingerprint	e9d26f2c...22d0	dataset_fingerprint.json

J.2 Zero-price audit

metric	value	source
zero-price rows	1,851,380 (28.0402% of raw)	zero_price_audit.json
...of which recorded any sales	677 (0.0366%)	zero_price_audit.json
rows with price > 0 and zero sales	0	zero_price_audit.json
leading / trailing / interior share	15.73% / 57.69% / 26.21%	zero_price_audit.json
leading + trailing	73.42%	derived
series never priced	6,868 (0.371%)	zero_price_audit.json
interior gaps with sales	660	zero_price_audit.json
UPCs above 80% zero	112 of 490	zero_price_audit.json
mean units: heavy-zero vs light-zero series	3.053 vs 20.999	zero_price_audit.json
mean price: heavy-zero vs light-zero series	\$2.717 vs \$3.228	zero_price_audit.json

J.3 EDA

metric	value	source
UPCs generating 80% of revenue	121 of 489	eda_summary.json
top-10 revenue / gross-profit share	14.147% / 12.677%	eda_summary.json
UPCs with negative total gross profit	6	eda_summary.json
UPC × store series	36,443	eda_summary.json
median obs / distinct prices per series	78 / 8	eda_summary.json
median within-series price CV	0.0791	eda_summary.json
median price-change rate	13.803%	eda_summary.json
series eligible for pricing	19,707 (54.076%)	eda_summary.json
median distinct prices per UPC	51	eda_summary.json
median cross-store price spread	13.627% (p90 27.452%)	eda_summary.json
median within-series AAC CV	0.0884 (p90 0.1528)	eda_summary.json
corr(price, margin rate)	-0.0143	eda_summary.json
price-change events	766,384 (42.59% cuts)	eda_summary.json
mean AAC change on a price cut	-9.159%	eda_summary.json
mean units with / without recorded promo	54.46 / 16.49	eda_summary.json
mean price with / without recorded promo	\$2.544 / \$3.161	eda_summary.json
WoW price jumps > 50%	10.067% (77,152)	eda_summary.json
max / min observed unit price	\$26.02 / \$0.05	eda_summary.json

J.4 Features and split

metric	value	source
rows in the feature table	4,707,776	feature_build.json
usable for training	4,671,333	feature_build.json
dropped for no history	36,443	feature_build.json
engineered features	29	feature_build.json
distinct weeks	365	feature_build.json
train weeks / rows	2–257 / 3,206,437	feature_build.json
validation weeks / rows	258–342 / 715,856	feature_build.json
test weeks / rows	343–399 / 749,040	feature_build.json
train / valid / test dates	1989-09-21...1994-08-11 / 1994-08-18...1996-03-28 / 1996-04-04...1997-05-01	feature_build.json
feature-table fingerprint	b4b3928a...53ae	feature_build.json
build time	73.4 s	feature_build.json

J.5 Demand models

model	valid WAPE	valid MAE	valid RMSE	valid bias	test WAPE	test MAE	test RMSE	test bias
M0a last-week naive	0.80071	15.251	63.889	+0.075	0.77352	14.407	88.014	+0.141
M0b rolling-mean(4)	0.74710	14.230	51.926	+0.219	0.73256	13.644	70.211	+0.227
M0c series historical mean	0.76070	14.489	47.196	+1.971	0.74412	13.860	62.329	+1.443
M0d seasonal naive (52w)	0.79201	15.086	56.413	+0.794	0.82074	15.287	76.395	+1.041
M1 ridge log-log ☒	0.41346	7.875	39.597	–3.706	0.45649	8.503	72.064	–3.883
M2 HGB Poisson	0.42264	8.050	38.378	+0.363	<i>not scored</i>	—	—	—

Source: `model_metrics.json`, `evaluation.json`.

metric	value	source
selected model	M1 ridge log-log	model_metrics.json
model version	ridge_loglog-20260818-132847	demand_model_metadata.json
test sMAPE	0.40003	evaluation.json
reload WAPE difference	0.00e+00	evaluation.json
M1 fit time / M2 fit time	31.0 s / 181.9 s	model_metrics.json
M1 raw log_price coefficient	–0.013495	model_metrics.json
M1 implied elasticity (valid, at training)	median –3.3223	model_metrics.json
M1 implied elasticity (test window)	median –3.1884	evaluation.json
M2 implied elasticity	median –4.5122, share negative 93.02%	model_metrics.json
relative WAPE improvement over best baseline (test)	37.7%	derived

Segments (M1, test window)

segment	n	units	MAE	RMSE	WAPE	bias
no recorded promotion	691,836	9,868,496	5.336	56.089	0.37408	–1.125

segment	n	units	MAE	RMSE	WAPE	bias
recorded promotion	57,204	4,082,965	46.800	173.070	0.65569	-37.238
price CV ≤ 0.05	73,077	1,066,404	5.840	18.149	0.40017	-1.955
price CV 0.05–0.10	243,380	3,356,144	4.952	28.601	0.35911	-1.437
price CV 0.10–0.15	351,047	7,368,028	9.903	67.390	0.47181	-4.834
price CV > 0.15	81,536	2,160,885	15.460	159.434	0.58334	-8.819

J.6 Elasticity — analysis ladder (Phase D, full panel, 1.2M subsample)

specification	elasticity	SE	95% CI	R ²
M1 naive pooled	-0.34800	0.00321	[-0.35430, -0.34170]	0.01260
M2 + UPC FE	-2.28941	0.00721	[-2.30354, -2.27527]	0.13154
M3 + UPC $\times$ store FE	-2.41875	0.07115	[-2.55819, -2.27930]	0.19662
M4 + promo/season/trend	-1.90930	0.06899	[-2.04452, -1.77407]	0.22677
promotion weeks only	-2.66170	0.00742	—	0.29993
non-promotion weeks	-1.83004	0.00777	—	0.09874

Source: `elasticity.json`.

metric	value
arc elasticity: usable pairs / undefined	716,858 / 84.104%
arc median / p25 / p75	-2.8219 / -8.5057 / +0.2560
per-UPC (Phase D): median / p10 / p90	-2.3580 / -4.0944 / -0.2399
share of UPCs elastic / wrong-signed significant	80.845% / 5.070%
per-store: median / min / max	-0.3855 / -0.9470 / +0.2971

J.7 Elasticity — the pricing table (training weeks 2–257)

metric	value	source
pooled controlled elasticity	-2.028860	elasticity_estimation.json
pooled SE (clustered by UPC, via <code>fit_loglog</code>)	0.082764	elasticity_estimation.json
pooled SE, two-way (UPC, week)	0.105846	elasticity_estimation.json
pooled 95% CI, two-way	[-2.23631, -1.82141]	elasticity_estimation.json
pooled SE, HC1	0.011614	elasticity_estimation.json
UPC $\times$ store FE elasticity	-2.221718 (SE 0.083980)	elasticity_estimation.json
pooled controlled, full training sample	-2.044259	elasticity_inference.json
τ^2 (REML)	0.766135	elasticity_estimation.json
τ	0.875291	elasticity_estimation.json
τ^2 (method of moments)	0.673069	elasticity_estimation.json
prior mean (pooled) / freely estimated	-2.028860 / -1.933409	elasticity_estimation.json
mean shrinkage weight	0.779742	elasticity_estimation.json
median / min / max weight	0.834102 / 0.269277 / 0.983396	shrinkage_audit.json
median per-product robust SE	0.390359	shrinkage_audit.json
median per-UPC SE inflation vs HC1	3.79280	elasticity_estimation.json
products / usable	372 / 239	elasticity_estimation.json
rejected: thin / wrong sign / imprecise	100 / 24 / 9	elasticity_estimation.json
clipped	0	elasticity_estimation.json
final elasticity: median / p10 / p90	-2.028860 / -2.789781 / -1.169093	elasticity_estimation.json
shrunk-only distribution: mean / median	-1.9544 / -1.8980 / 0.7692 / -4.0578 /	elasticity_table.csv

metric	value	source
/ sd / min / max	-0.3306	
runtime	35.0 s	elasticity_estimation.json

Weight bands: $<0.25 \rightarrow 0 \cdot 0.25-0.5 \rightarrow 25 \cdot 0.5-0.75 \rightarrow 48 \cdot 0.75-0.9 \rightarrow 96 \cdot 0.9-0.99 \rightarrow 70 \cdot >0.99 \rightarrow 0$.

Phase L (pre-correction) comparison: τ^2 0.905865, median SE 0.101426, mean weight **0.955169**.

J.8 Inference audit (pooled controlled, 1.2M subsample, coefficient -2.02886)

covariance	SE	ratio to classical	95% CI	clusters
classical	0.006519	1.00	[-2.04164, -2.01608]	—
HC1	0.011614	1.78	[-2.05162, -2.00610]	—
clustered by UPC × store panel	0.015334	2.35	[-2.05891, -1.99881]	26,413
clustered by store	0.030455	4.67	[-2.08855, -1.96917]	86
clustered by week	0.081213	12.46	[-2.18803, -1.86969]	255
clustered by UPC	0.083690	12.84	[-2.19289, -1.86483]	369
two-way (UPC, week)	0.105846	16.24	[-2.23631, -1.82141]	—

Per-UPC (n = 262): median HC1 SE 0.10714, store-clustered 0.14451, week-clustered 0.38321; median robust/HC1 ratio **3.3817** (p90 6.8282); share significant at 5%: HC1 **90.08%** → robust **75.19%**.

J.9 Eligibility funnel and coverage

stage	UPCs
in the processed dataset	489
in the modelling table	476
observed in training weeks	372
≥ 200 training observations	330
≥ 10 distinct prices	272
correctly signed	248
panel-robust SE ≤ 1.5	239

Week-399 decision contexts: **13,964** — **shrunk_product** **8,962 (64.18%)**, **pooled_fallback** **5,002 (35.82%)**.

J.10 Elasticity stability

window	weeks	rows	pooled	SE (two-way)	FE	τ^2	mean w	usable
W1 early	2–87	979,559	-1.9032	0.2232	-2.2792	1.0156	0.6626	85/259
W2 middle	88–129	545,034	-2.3230	0.2131	-3.2805	1.2273	0.6200	87/221
W3 late	130–257	1,681,844	-2.2404	0.1254	-2.4863	1.0698	0.8370	218/297
W4 expanding	2–129	1,524,593	-1.9173	0.1658	-2.3036	1.0445	0.6819	146/285
W5 full	2–257	3,206,437	-2.0289	0.1058	-2.2217	0.7661	0.7797	239/372

Pairwise z (disjoint windows): W1–W2 **+1.3605**, W1–W3 +1.3168, W2–W3 -0.3345. **Max |z| = 1.3605** (not significant). Rank correlations: -0.0536 / +0.3498 / **+0.1867**; median **+0.1867**; sign stability **100%**; 34 common products.

J.11 Native ML price-response validation (300 contexts, week 399, ±30%)

metric	value
share monotone decreasing	100.0%
share with any increasing segment	0.0%

metric	value
share flat	0.0%
share with negative predictions	0.0%
median local implied elasticity	-3.10199
p10 / p90	-3.48204 / -2.66715
median relative demand span	1.83175
profit optimum below current	26.667%
revenue optimum below current	100.0%
profit optimum at a grid edge	7.667%

J.12 Price-response method comparison (300 contexts, week 399, standard)

metric	m1	pooled	shrunk
share actionable	77.333%	70.000%	74.667%
share KEEP_CURRENT	10.333%	16.667%	11.667%
share REVIEW_REQUIRED	12.333%	13.333%	13.667%
median absolute change	4.842%	8.361%	7.538%
share increases	73.276%	94.762%	91.964%
median internal uplift	8.277%	17.572%	13.913%
mean recommended price	\$3.1810	\$3.2681	\$3.2679

pair	mean abs gap	p90	% of price	identical	same decision state	opposite direction	corr
m1 vs pooled	\$0.0870	\$0.300	2.738%	63.00%	79.67%	4.00%	0.9808
m1 vs shrunk	\$0.0898	\$0.302	2.870%	64.67%	86.67%	7.67%	0.9750
pooled vs shrunk	\$0.0318	\$0.100	0.953%	86.67%	91.67%	2.00%	0.9892

Sensitivity: median candidate-price spread across the four scenarios — **wide guardrails 21.505%, production guardrails 0.000%**.

J.13 Out-of-time price-response validation

154,899 episodes, weeks 258–399; 3 dropped; $|\Delta p| \in [5\%, 60\%]$; 73.331% carry a promotion code; median observed implied elasticity **-2.4524** (all), **-1.8623** (non-promotion).

split	n	null	pooled	shrunk	native ML
all episodes — WAPE	154,899	0.80860	0.62488	0.58531	0.60420
all episodes — MAE		45.431	35.109	32.885	33.947
all episodes — bias		-37.263	-29.349	-24.990	-26.068
no promotion — WAPE	41,310	0.71516	0.59439	0.55957	0.58774
no promotion — MAE		24.048	19.987	18.816	19.763
no promotion — bias		-18.333	-14.765	-12.757	-13.288
promotion — WAPE	113,589	0.82635	0.63068	0.59020	0.60732
sign accuracy (all / no-promo / promo)			81.742% / 68.315% / 86.625%	same	same

J.14 Batch recommendations (3,000 contexts, week 399, standard, shrunk)

metric	value
RECOMMEND_CHANGE	1,766 (58.867%)

metric	value
KEEP_CURRENT	923 (30.767%)
REVIEW_REQUIRED	311 (10.367%)
median absolute price change	8.389%
share increases	91.280%
median internal uplift among actionable	13.328%
risk: LOW / MEDIUM / HIGH	1,175 / 786 / 1,039
HIGH-risk actionable	0
elasticity source: shrunk / pooled	1,904 / 1,096
median elasticity used	-2.028860
portfolio expected GP current → if applied	\$37,580.82 → \$40,700.96
model-internal estimated uplift	+8.3025%
runtime	105.3 s

J.15 Full-week decision states (13,964 contexts)

profile	RECOMMEND_CHANGE	KEEP_CURRENT	REVIEW_REQUIRED	HIGH risk	HIGH auto-changed	invariant violations
standard	8,265 (59.19%)	4,346 (31.12%)	1,353 (9.69%)	4,760	0	0
conservative	6,354 (45.50%)	7,610 (54.50%)	0	4,760	0	0
aggressive (DEMO)	9,885 (70.79%)	4,079 (29.21%)	0	8,102	4,717	4,717 (by design)

HIGH-risk split under **standard**: KEEP_CURRENT 3,407, REVIEW_REQUIRED 1,353.

J.16 Constraint attribution (13,964 contexts, week 399, standard, shrunk)

determinant	contexts	share
guardrail corner	7,967	57.05%
screened out before optimisation	3,528	25.26%
risk gate	1,353	9.69%
materiality threshold	716	5.13%
learned signal (interior optimum)	400	2.86%

first binding constraint	contexts	share
MAX_PRICE_CHANGE	6,222	44.56%
HISTORICAL_SUPPORT	3,373	24.16%
NONE	1,924	13.78%
EXTRAPOLATION	1,877	13.44%
MATERIALITY	565	4.05%
RISK_GATE	3	0.02%

metric	value
unconstrained optimum inside bounds	968 (6.93%)
proposal matches it	638 (4.57%)
final price matches it	635 (4.55%)
median rounding distance	\$0.0250 (p90 \$0.0427)
median unconstrained price change	+48.997% (p10 -4.68%, p90 +174.61%)
median final price change	+3.746%
unconstrained optimum above current	86.680%

J.17 Guardrail ablation

layer	median change	share elasticity changes price	median variation across scenarios	p90 variation	constraint moves optimum
A research (cost floor only)	+49.129%	99.993%	79.772%	80.386%	0.0%
B extrapolation only	+10.811%	61.021%	6.596%	38.523%	74.606%
C + price-change cap	+7.538%	30.622%	0.000%	16.949%	90.275%
D standard	+3.746%	21.348%	0.000%	14.528%	95.904%
E conservative	0.000%	20.166%	0.000%	4.094%	98.489%

J.18 Model-value ablation

rule	price differs > 1 grid step	within one grid step	mean abs diff	median abs diff %	same decision state
R0 keep current	59.08%	40.92%	\$0.1406	4.332%	30.71%
R1 hold historical margin	51.53%	48.47%	\$0.2154	4.225%	90.05%
R2 cost-plus 25%	34.93%	65.07%	\$0.1070	1.007%	88.83%
R3 nearest modal price	49.47%	50.53%	\$0.1789	2.462%	78.89%
R4 max allowed increase	17.45%	82.55%	\$0.0605	0.615%	89.96%

Model median actionable change: **+8.361%**.

J.19 Backtest (weeks 390–399)

metric	value
mean weekly WAPE	0.47770
min / max weekly WAPE	0.36385 / 0.56025
mean weekly bias	−4.661

policy	expected revenue	expected gross profit	mean price	share unchanged	GP vs historical	rev vs historical
HistoricalPricePolicy	\$170,709	\$37,679	\$3.1805	100%	0.00%	0.00%
SimpleMarginPolicy	\$173,228	\$38,546	\$3.1340	0%	+2.300%	+1.475%
ElasticityBaselinePolicy	\$161,290	\$42,834	\$3.3359	6.44%	+13.683%	−5.518%
MLPricingPolicy	\$163,340	\$41,617	\$3.3202	36.91%	+10.451%	−4.317%

All "vs historical" figures are model-internal estimates (\$64).

J.20 Monitoring

metric	value
schema check	passed (749,040 rows)
prediction drift PSI / KS	0.05556 (stable) / 0.09502
prediction mean: reference → current	16.691 → 15.001
largest feature PSI	time_index 12.412 (by construction)
largest <i>informative</i> feature PSI	series_reference_price 0.363 (\$2.886 → \$3.247)
WAPE 1996Q2 → 1997Q2	0.4091 → 0.5278

J.21 Cost-leakage audit

check	result
identity: rows compared / mismatches	4,671,333 / 0
share equal to same-week AAC	55.44%
poisoning: series / from week / rows before poison	25 / 398 / 5,537
unchanged at or before poison	true
changed the next week (test has teeth)	true
recommendation-level: series / changed	19 / 0
all passed	true

J.22 Quality gates (verified 2026-08-19)

gate	result
pytest	519 passed, 0 failed, 0 skipped, 7.05 s (at Phase N)
ruff	All checks passed
dashboard smoke	9/9 pages render (independently re-verified by live capture)
API live	/health, /model/info, /recommend-price all responded
claim audit (re-run with this report included)	0 UNSUPPORTED, 38 needs-qualification, 325 safe, 49 files
Docker build	NOT TESTED (daemon unavailable)
CI workflow	implemented, never executed (no remote, no commits)

J.23 The worked example (CAPN CRUNCH JUMBO CR 3000006560, store 86, week 399)

field	value
current price	\$3.35
decision-time unit cost	\$2.5570550
observed support	\$1.50 – \$3.79, 359 weeks, 56 distinct prices
raw elasticity / robust SE / weight	–3.017796 / 0.255658 / 0.921394
applied elasticity	–2.9400592 (shrunk_product)
feasible interval	[\$3.0150, \$3.6850]
unconstrained CE optimum	\$3.8748
proposed = final price	\$3.66 (+9.2537%)
predicted units	25.5647 → 19.7077
expected revenue	\$85.6418 → \$72.1302 (–15.777%)
expected gross profit	\$20.2714 → \$21.7365 (+7.2274%)
decision / risk	RECOMMEND_CHANGE / LOW
reason codes	PRICE_CHANGE_LIMIT, PROFIT_UPLIFT_POSITIVE

J.24 Documentation drift found while writing this report — and closed in v1.0

Every row below was a genuine defect: a number edited in one document and not in another. All of them were corrected during the v1.0 engineering closure, at the source rather than by editing this report, and the class of defect was removed as well (see J.24b).

location	stated at the time	artifact value → now published	status
README.md §3 demo block	elasticity –3.016 (shrunk_product)	applied elasticity –2.9401 (–3.0178 is the raw estimate)	corrected
README.md §3 demo block	model-internal uplift +6.51%	+7.2274% (live run_demo.py)	corrected

location	stated at the time	artifact value → now published	status
		and API)	
README.md §10, KNOWN_LIMITATIONS.md #28, VALIDATION_SUMMARY.md row 15	ml vs shrunk decision agreement 87.3%	86.67%	corrected
same	pooled vs shrunk decision agreement 89.0%	91.67%	corrected
docs/INTERVIEW_RED_TEAM.md Q2	2.1% interior optima, 56.3% guardrail corners, 81.2% rule match	2.9% / 57.1% / 82.5%	corrected
KNOWN_LIMITATIONS.md #37, DECISIONS.md #32	Phase L τ^2 1.101 / 1.10	the audit's Phase L reconstruction gives 0.9059	both reported, with the reason (§31.7)
README.md §8 make test lint comment	"121 tests"	the live collected count	corrected, and the count is now generated
docs/TECHNICAL_DESIGN.md	"121 tests"	the live collected count	corrected
STATUS.md, VALIDATION_SUMMARY.md, this report	493 tests	the live collected count	corrected

J.24b Why this class of defect cannot recur silently

Two mechanisms were added in v1.0, both of which fail the build rather than warn:

170. **Generated README metrics.** `scripts/update_readme_metrics.py` writes every headline number in README §2 straight from the artifacts, between `<!-- BEGIN GENERATED METRICS -->` markers. `tests/test_readme_metrics.py` fails when the block and the artifacts disagree, and separately when the published test count differs from the count pytest actually collects in that run.
171. **Cross-document numeric audit.** `scripts/audit_metric_consistency.py` reads each headline metric from the single artifact that owns it and checks every sentence in the hand-authored documents that states it, writing `artifacts/metrics/metric_consistency.json`. `tests/test_metric_consistency.py` fails on any mismatch. Lines that record a *correction* ("0.954 → 0.780") are recognised and kept: superseded values are history, not drift, and this project does not delete them.

Appendix K — Final limitations register

Consolidated and de-duplicated from `KNOWN_LIMITATIONS.md`, `STATUS.md` and the audit artifacts. **S** = scientific, **M** = modelling, **E** = econometric, **O** = optimization, **B** = business/engineering, **L** = this report's additions.

id	limitation	severity	evidence
S1	No causal identification. Prices were not randomised; $P(Q price) \neq P(Q do(price))$	critical	<code>docs/CAUSAL_LIMITATIONS.md</code> ; −0.348 → −2.419 swing
S2	Counterfactual policy economics are circular — the same fitted response proposes and scores	critical	§64; <code>reports/06</code>
S3	Cross-price / substitution effects unmodelled — the category effect of re-pricing many substitutes $\neq$ the sum	high	§101
S4	Promotion coding incomplete — 94.55% of rows carry no code; G/L undocumented; display and feature absent	high	<code>raw_audit.json</code> ; −2.66 vs −1.83 split

id	limitation	severity	evidence
S5	AAC is not economic cost; median within-series CV 0.088; falls 9.16% on price cuts	medium	eda_summary.json
S6	No customer dimension	medium	store × week grain
S7	Stock-outs invisible	medium	scanner-data property
S8	Historical data 1989–1997	medium	\$93
S9	No competitor prices	medium	not in the file
M1	Point predictions only; no intervals; the objective is risk-neutral	medium	\$89.1
M2	Weekly WAPE $\approx$ 0.46 (0.364–0.560 across backtest weeks)	medium	evaluation.json, backtest.json
M3	Systematic under-forecast (bias –3.883; –37.24 on promotion weeks); no smearing correction	medium	\$41.4
M4	Prediction cap at 5× max training demand	low	demand_model_metadata.json
M5	Promotion flag assumed known at decision time	medium	docs/DATA_LEAKAGE_AUDIT.md
M6	Only 54.08% of series are eligible	medium	eda_summary.json
M7	The boosted model's price response is positive in 7% of contexts; monotonic_cst not used	medium	model_metrics.json
M8	No price × context interaction in the shipped response	medium	\$89.8
M9	Constant-elasticity form imposed, not learned	medium	\$89.9
E1	Shrinkage fixes variance, not bias — pooled bias propagates with weight $(1-w_i)$	high	KNOWN_LIMITATIONS.md #34
E2	Product-level elasticity does not reproduce across windows (rank corr +0.19); only 34 common products	high	reports/17
E3	133 of 372 products use the pooled fallback — 35.8% of contexts	high	reports/18
E4	Two-way clustering used only at the pooled level	low	DECISIONS.md #44
E5	Wrong-signed products rejected $\Rightarrow$ a truncated distribution whose effect on τ^2 is unquantified	medium	\$90.6
E6	Elasticity regressions use a seeded 1.2M subsample	low	full-sample check: –2.0443 vs –2.0289
E7	Single annual harmonic for seasonality	low	\$90.7
O1	Business rules, not the model, set the magnitude of most recommendations (2.9% interior optima)	critical	reports/11, reports/12
O2	The demand forecast does	high	tests/test_attribution.py

id	limitation	severity	evidence
	not choose the price ($\hat{Q}(p_o)$ cancels)		
O3	Guardrails neutralise the elasticity assumption (79.8% → 0.0% variation)	high	reports/13
O4	Materiality and risk bands asserted, not fitted	medium	configs/config.yaml
O5	The 5-cent grid quantises the answer (median \$0.025 from the continuous optimum)	low	constraint_attribution.json
O6	Single-period, single-product; no inventory, no competitor reaction	high	\$91.6
O7	No absolute price ceiling configured (MAX_PRICE present in 0 contexts)	medium	constraint_attribution.json
O8	Batch optimization is per-context; <code>simulate_many</code> exists but is unused	medium	KNOWN_LIMITATIONS.md #18
O9	The ±10% cap is per decision; four consecutive weeks compounds to +46%, and nothing tracks cumulative movement	medium	\$94.5
B1	Not production ready — no orchestration, registry, metric store, alerting, auth, SLOs	high	STATUS.md
B2	Docker image never built (daemon unavailable)	medium	VALIDATION_SUMMARY.md row 36
B3	CI never executed	medium	no remote
B4	Dashboard loads the full 4.7M-row panel; ~40 s first load	medium	KNOWN_LIMITATIONS.md #19
B5	REVIEW_REQUIRED produces 1,353 items/week that nothing staffs and nothing prioritises	high	reports/19
B6	Policy profiles are DEMO settings	medium	configs/config.yaml
B7	Notebooks are thin wrappers	low	KNOWN_LIMITATIONS.md #23
B8	A concurrent agent session wrote into the repository during the build	low	STATUS.md open issue 11
L14	(new) The dashboard's Reports tab lists only reports 01–10 + 2 — the ten Phase M audits, containing the project's most important findings, are unreachable from the app	medium	dashboard/app.py page_methodology
L15	(new) The repository has no commits. <code>git_commit</code> in the model metadata is the literal string "HEAD"; artifacts have no code lineage; CI cannot run	high	git log; demand_model_metadata.json
L16	Documentation drift: quantified instances between	low	Appendix J.24, J.24b

id	limitation	severity	evidence
	the README/limitations prose and the current artifacts — closed in v1.0 (generated README metrics + a cross-document numeric audit, both test-enforced); prose claims remain classified, not verified		
L17	(new) The CI matrix targets Python 3.11/3.12 while this build ran on 3.13.0	low	.github/workflows/ci.yml

Appendix L — Glossary

term	definition
AAC	Average Acquisition Cost — an inventory-accounting measure of what the retailer paid on average for stock on hand. Not necessarily the replacement cost relevant to a marginal pricing decision. In this project the derived column is <code>estimated_unit_aac</code> and is never called "true cost".
Actionable	a recommendation whose decision state is <code>RECOMMEND_CHANGE</code> ; the only state in which <code>final_recommended_price</code> differs from <code>current_price</code> .
Arc elasticity	a two-point elasticity using midpoint denominators, symmetric in the direction of the price change. Undefined for 84.1% of consecutive week pairs in this panel.
Bias (forecast)	mean signed error, $\text{mean}(\hat{y} - y)$. Negative means systematic under-forecast. This project's test bias is -3.883 units.
Candidate price	one price on the grid being scored. Not a recommendation.
Causal effect	the change in demand caused by <i>setting</i> a price, $P(Q \mid \text{do}(\text{price}))$. Not estimated anywhere in this project.
Clustered standard error	a covariance estimator allowing arbitrary correlation within clusters (store, week, product). Two-way clustering here is 16.24× the classical SE.
Counterfactual	an outcome at a price that was never charged. Never observed; only estimated.
Decision state	<code>RECOMMEND_CHANGE</code> / <code>KEEP_CURRENT</code> / <code>REVIEW_REQUIRED</code> .
Decision-time unit cost	the latest AAC known before the priced week, lagged one week and forward-filled within the series.
Effective unit price	$\text{price} / \text{qty}$ — the price paid per individual unit, and the decision variable of this project.
Elasticity (own-price)	$E = \partial \log Q / \partial \log P$. Elastic if $ E > 1$.
Empirical Bayes	estimating the prior from the data, then using it to shrink noisy individual estimates: $\theta_i = w_i \epsilon_i + (1-w_i) \mu$.
Endogeneity	correlation between a regressor and the error term, here because price is chosen in response to demand conditions.
Extrapolation	evaluating the model at a price outside the series' observed price support. Bounded here by the extrapolation guardrail.
Fixed effects	group-specific intercepts (UPC, store) absorbing time-invariant heterogeneity, implemented by a within transformation.
Guardrail	a business or scientific constraint bounding the feasible price. Six are intersected into one interval.
HC1	a heteroskedasticity-robust covariance estimator with a small-sample correction. Robust to heteroskedasticity, not to clustering — 1.78× classical here, still 9× too small versus two-way.

term	definition
Hybrid price response	$Q(p) = \hat{Q}(p_0) \cdot (p/p_0)^{\epsilon}$, separating the ML forecaster's contextual baseline from a separately estimated elasticity.
Materiality threshold	the minimum estimated relative gain required to recommend a change (1% under the standard profile).
Model-internal estimated uplift	a counterfactual profit or revenue gain computed by the same fitted model that proposed the price. Circular by construction; never a realised or causal outcome.
PSI	Population Stability Index, $\sum (p_{cur} - p_{ref}) \cdot \ln(p_{cur}/p_{ref})$; a distribution-shift measure.
Panel data	repeated observations of the same units over time — here UPC $\times$ store series across 366 weeks.
Price support	the interval of prices a UPC $\times$ store series has actually carried.
Proposed candidate price	what the optimizer proposed. Not a business price unless the decision state is actionable.
REML	Restricted Maximum Likelihood — the estimator used for τ^2 , correcting for having estimated the prior mean.
Reason code	a machine-readable label explaining a decision (e.g. PRICE_CHANGE_LIMIT). Testable, unlike a feature attribution.
REVIEW_REQUIRED	a proposal exists but is not applied automatically; a human must approve. 9.7% of contexts under the standard profile.
Risk level	a heuristic LOW/MEDIUM/HIGH score over history, price variation, extrapolation distance, promotion entanglement and elasticity provenance. Not calibrated confidence. HIGH = risky.
Shrinkage	pulling a noisy individual estimate toward a pooled estimate in proportion to its imprecision. Mean weight 0.780 here.
sMAPE	symmetric mean absolute percentage error, $\text{mean}(y - \hat{y} / ((y + \hat{y})/2))$.
τ^2 (tau squared)	the between-product variance of <i>true</i> elasticities in the hierarchical model. 0.766 (REML) here.
UPC	Universal Product Code — the item identifier.
WAPE	Weighted Absolute Percentage Error, $\sum y - \hat{y} / \sum y$. Volume-weighted and well defined at zero. 0.4565 on the test window.
Within transformation	subtracting group means to absorb fixed effects.
Zero-price row	a raw row with price = 0 — 28.04% of the raw file; excluded because no unit price can be derived.

End of report.